



Network Specifications - Authorization

October 2023

GLOBAL NETWORK SERVICES

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Summary of Changes

This table shows the changes to the *Network Specifications - Authorization* manual.

Summary of Changes

Subject	Revision	First Publication Date
Updated Definitions	The following definitions have been updated throughout the <i>Network Specifications - Authorization</i> manual: • Bit 50 Currency Code, Reconciliation • Bit 54 Amount, Additional The following definition has been updated throughout Chapter 5, Authorization Messages: • Bit 5 Amount, Reconciliation	August 2023 <i>Network Participant Updates</i>
No Stand-In on page 2-12 Prepaid Card Authorization on page 2-18 No Stand-In on page 2-31	These sections in the Authorization Processing chapter were modified to support the language specific to Partial Authorization.	August 2023 <i>Network Participant Updates</i>
POS Authorization Request (1100) Message Layout on page 5-1	The EMV Tag for Application PAN Sequence Number has been corrected from 5F3 to 5F34.	October 2023 <i>Network Specifications - Authorization</i> manual
Programming Notes for POS Authorization Request (1100) on page 5-4	Updates have been made to the following messages to support Variable Authorizations: • Bit 24 Function Code Notes for Estimated Authorizations Valid values 183 and 184, Incremental Authorization Valid Values 185 and 186, and Prepaid Balance Inquiry Authorization Valid Values 181 and 182 have been updated.	August 2023 <i>Network Participant Updates</i>
Programming Notes for POS Authorization Response (1110) on page 5-39	Updates have been made to the following messages to support Partial Authorizations: • Bit 4 Amount, Transaction optionality has been updated and the note was removed. • Bit 5 Amount, Reconciliation optionality has been updated and the note was removed. • Bit 30 Amount, Original definition and optionality have been updated. • Bit 39 Action Code note has been updated. • Bit 54 Amount, Additional optionality and notes have been updated.	August 2023 <i>Network Participant Updates</i>

Summary of Changes (continued)

Subject	Revision	First Publication Date
Programming Notes for POS Authorization Advice (1120) on page 5-53	Updates have been made to the following message to add three (3) new reserved values in positions 6-7: • Bit 48 Stand-In Parameter Information for DPP Requests and Non-DPP Requests.	October 2023 <i>Network Specifications - Authorization</i> manual
	Updates have been made to the following message to support Partial Authorizations: • Bit 30 Amount, Original definition has been updated.	August 2023 <i>Network Participant Updates</i>
Programming Notes for POS Authorization Reversal Advice (1420) on page 5-79	Updates have been made to the following message to support Partial Authorizations: • Bit 30 Amount, Original definition has been updated.	August 2023 <i>Network Participant Updates</i>
Programming Notes for POS Authorization Reversal Advice Response (1430) on page 5-88	Update has been made to the following message to support Variable Authorization: • Bit 30 Amount, Original definition has been updated.	August 2023 <i>Network Participant Updates</i>
Programming Notes for ATM Reversal Advice (1420) on page 5-113	Updates have been made to the following messages to support Partial Authorizations: • Bit 30 Amount, Original definition and notes have been updated.	August 2023 <i>Network Participant Updates</i>
Programming Notes for POS Authorization Adjustment Financial Advice (1220) on page 5-138	Updates have been made to the following message to support Partial Authorizations: • Bit 30 Amount, Original definition and optionality have been update and the note has been removed.	August 2023 <i>Network Participant Updates</i>
Programming Notes for POS Authorization Adjustment Financial Advice Response (1230) on page 5-145	Updates have been made to the following messages to support Partial Authorizations: • Bit 30 Amount, Original definition has been updated. • Bit 54 Amount, Additional optionality and note have been updated.	August 2023 <i>Network Participant Updates</i>

1 Introduction

In This Introduction

Overview

The *Network Specifications - Authorization* manual contains:

- Authorization Processing
- ATM Processing
- Authorization message layouts and programming notes

These messages send data between Participants and the American Express® Global Network.

Who Must Use This Manual

Participants (Acquirers, Issuers) and Authorized Processors must know the function of each message, format, and data element definition used on the Network.

About This Document

Publication Cycle

The Network Specifications manuals and *Codes Reference Guide* will be published two times each year, in April and October.

The *Network Participant Updates* (NPU) will be published on the second Friday following the second full business week of the month, and will include technical and code changes.

Summary of Changes

Important changes are in the [Summary of Changes on page v](#).

Style and Symbols

Words that are capitalized in this manual have specific meaning in the context of the *Business and Operational Policies* and Network Specifications manuals. Refer to the Definitions chapter in *Network Specifications - Instructions* manual.

References to this manual and other manuals are printed in *Italics*. The use of **Bold** is for visual emphasis.

All changes to the Network Specifications manuals and *Codes Reference Guide* will be indicated with a change bar. A change bar is a vertical line in the left margin that visually identifies new or revised text. **Only** substantive changes will be listed in the Summary of Changes tables in each manual. Substantive changes are defined as those changes having impact to Participants.



A trash can icon has been placed in the margin next to any significant deletion of text, including sections, tables, paragraphs, notes, and bullet points. An example of this icon appears in the margin next to this paragraph. If text has been deleted and changed, a change bar will appear next to the trash can icon.

Note: A “Note” indicates supplemental information. This paragraph is an example of a Note.

Technical Change Notification

GNS Participants are told about technical changes at different times.

Type of Change	When Participants are Told
Mandatory system/technical change	Six (6) months before the April or October effective date
Optional system/technical change	As necessary
Necessary legal change	The notification and effective dates are aligned with the legal and regulatory requirements
Reminders/information only	When necessary

2 Authorization Processing

Authorization Data Flow

Overview

The Global Network Services Authorization System (GNSAS) sends mandatory and optional messages between Acquirers and Issuers (or Authorized Processors or platforms that interface with their system) for real-time Point of Sale (POS) and Automated Teller Machine (ATM) Transaction Authorization Requests and Responses.

Authorization Data Flow

The data flow in [Figure 2-1 Authorization Request/Response Flow](#) applies to these Authorization messages:

- POS Authorization Request (1100)
- POS Authorization Response (1110)
- ATM Financial Transaction Request (1200)
- ATM Financial Transaction Response (1210)

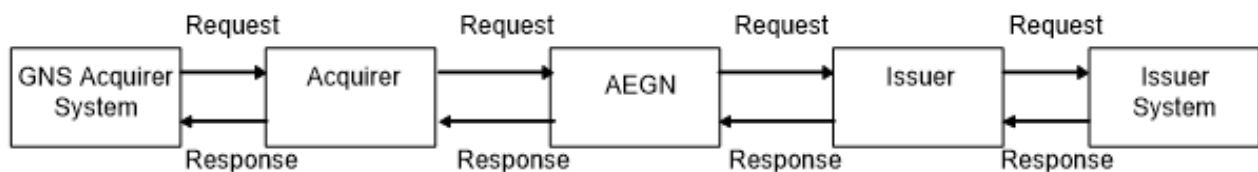


Figure 2-1. Authorization Request/Response Flow

More Instruction

Refer to the *Network Connectivity* manual on Knowledge Base for more instruction about communication with the Network.

POS Functionality Process

Requesting Authorization

An Acquirer sends a real-time POS Authorization Request (1100) message to request authorization of a Cardmember's Transaction at the point of sale in a Service Establishment (S/E).

Responding to Authorization Requests

An Acquirer receives a real-time POS Authorization Response (1110) from the Cardmember's issuing bank with an authorization decision (approve or deny).

Procedure to Receive and Send

Table 2-1 shows a POS Transaction that starts with a Card swipe.

Table 2-1. POS Sending/Receiving Procedure

Step	Who	Does What
1	Cardmember	Swipes their Card at an S/E to purchase goods or services
2	POS Device	Sends the Transaction data to the S/E's Acquirer
3	Acquirer	Creates a POS Authorization Request (1100) message and sends it to the Network
4	Network	Sends the POS Authorization Request (1100) message to the Cardmember's Issuer
5	Issuer	Uses internal Authorization rules and Cardmember data to make a POS Authorization Response (1110) to send to the Network
6	Network	Sends the POS Authorization Response (1110) to the Acquirer
7	Acquirer	Sends the POS Authorization Response (1110) to the S/E. The S/E completes or cancels the Transaction based on the decision of the Cardmember's Issuer

Functionality Exception

Issuers who contract with AEGNS to decision the POS Authorization Request (1100) messages of their Cardmembers **do not receive or send** the real-time POS Authorization Request (1100) messages described in this manual.

The Network sends Transaction data to these Issuers for settlement and Transaction detail reports.

The Role of the Transaction Identifier

Overview

The Transaction Identifier (TID) is a unique Network-generated number that follows a Transaction during its authorization life cycle.

Authorizations (POS)

These describe the TID authorization life cycle for POS authorizations:

- The Network creates and assigns a unique TID when it receives a POS Authorization Request (1100) from an Acquirer. All Acquirers must certify to receive the TID.
- The TID is only passed to the Issuer if the Issuer is certified for the TID.

Authorizations (ATM)

These describe the TID authorization life cycle for ATM authorizations:

- The Network creates and assigns a unique TID to the ATM Financial Transaction Request (1200) if the Acquirer is certified or not certified for the TID.
- The TID is passed to certified Issuers if the Acquirer is certified or not certified.

- The Network always returns the TID to certified Acquirers, regardless of the Issuer's certification status.

TID Assignment - Point of Sale Transactions

TID Assignment and Send

The Network sends the TID in POS Authorization messages to certified Acquirers and Issuers.

Note: Acquirers must certify to receive the TID.

Table 2-2. POS TID Assignment and Send Procedure

Step	When	TID Assignment
1	An Acquirer generates a POS Authorization Request (1100)	Network generates a TID if the Acquirer is certified.
2	The Network sends the Authorization Request to the Issuer.	The Network sends the TID to the Issuer if the Issuer and Acquirer are certified.
3	The Issuer sends a POS Authorization Response (1110) to the Network.	If the Issuer receives the TID from the Network, the Issuer echoes it in Bit 31.
4	The Network sends the POS Authorization Response (1110) to the Acquirer.	If the Acquirer is certified, the Network passes the TID.
POS Reversal Advice/Response (1420/1430)		
1	An Acquirer generates POS Reversal Advice (1420) to tell the Issuer of a change in the disposition of the Transaction.	If the Acquirer is certified, the Acquirer echoes the TID if it was received in the 1110.
2	The Network sends the POS Reversal Advice to the certified Issuer.	The Network sends the TID to a certified Issuer if Issuer and Acquirer are certified.
3	The Issuer sends a POS Reversal Response (1430) to the Network.	If the certified Issuer receives the TID from the Network, the Issuer echoes it in Bit 31.
4	The Network sends the POS Authorization Reversal Advice Response (1430) to the Acquirer.	If the Acquirer is certified, the Network passes the TID.

TID and Late Arriving Responses

When the Network receives a late arriving POS Authorization Response (1110) with an Action Code of 000 (Approved) from an Issuer, it returns a TID in the Network-generated POS Authorization Advice (1120) and the POS Reversal Advice (1420) message to the certified Issuer.

TID Assignment - ATM Transactions

TID Assignment

The Network sends the TID to certified Issuers in ATM request and reversal advice messages whether the Acquirer is certified or not certified for the TID.

A certified Acquirer receives a TID in these ATM message responses, regardless of the Issuer's certification status:

- ATM Financial Transaction Response (1210)
- ATM Reversal Advice Response (1430)

Table 2-3. ATM TID Assignment/Send Procedure

ATM Financial Transaction Request/Response (1200/1210)	
1. An Acquirer generates an ATM Financial Transaction Request (1200) for authorization.	The Network generates a TID.
2. The Network sends the ATM Authorization Request to the certified Issuer.	The Network sends the TID to a certified Issuer if the Acquirer is certified or not certified.
3. The Issuer sends an ATM Financial Transaction Response (1210) to the Network.	If the certified Issuer receives the TID from the Network, the Issuer echoes it in Bit 31.
4. The Network sends the ATM Financial Transaction Response (1210) to the Acquirer.	If the Acquirer is certified, the Network sends the TID.
ATM Reversal Advice/Response (1420/1430)	
1. An Acquirer generates an ATM Reversal Advice (1420) to tell the Issuer of a change in the disposition of the Transaction.	If the Acquirer is certified, the Acquirer echoes the TID if it was received in the 1210.
2. The Network sends the ATM Reversal Advice to the certified Issuer.	The Network sends the TID to a certified Issuer if the Acquirer is certified or not certified for the TID.
3. The Issuer sends an ATM Reversal Response (1430) to the Network.	If the certified Issuer receives the TID from the Network, the Issuer echoes it in Bit 31.
4. The Network sends the ATM Reversal Response (1430) to the Acquirer.	If the Acquirer is certified, the Network passes the TID.

Failure to Dispense

If there is a failure to dispense from an ATM, the ATM generates an ATM Reversal Advice (1420) and returns the TID from the ATM Financial Transaction Response (1210) in this message.

More Instruction

Refer to [ATM Processing on page 3-1](#) for instructions about ATM processing.

Currency Conversion

Overview

The Network changes the Acquirer's currency to the Issuer's selected currency during the currency conversion procedure.

Conversion

The Network converts the amount stored in Bit 4 (Amount, Transaction) to an equivalent amount in the Issuer's requested currency type before it sends Authorization POS or ATM messages to the Issuer. This converted amount is stored in Bit 5 (Amount, Reconciliation).

Note: The Issuer's currency type is requested during Participant Setup. This currency code is in Bit 50 (Currency Code, Reconciliation) on all Authorization Request, response, and advice messages.

Bit 4 (Amount, Transaction) is sent, as it is received from the Acquirer, to the Issuer in the 1100.

Bit 49 (Currency Code, Transaction) is sent, as it is received from the Acquirer, to the Issuer in the 1100.

Authorization Reversals

Overview

An Authorization Reversal message reverses the effect of a previous authorization message. The Network has two types of reversals:

- [System Generated Authorization Reversals](#)
- [S/E Initiated Authorization Reversals](#)

System Generated Authorization Reversals

[Table 2-4](#) shows how the Network, Acquirers, and S/Es use system generated Authorization Reversals.

Table 2-4. How to Use System Generated Authorization Reversals

Who	Sends	Why
The Network	POS Authorization Reversal Advice (1420) with a Processing Code of 004000	To reverse an authorization approval that an Issuer sends after the Network does Stand-In. Refer to Reversal Flow #1 on page 2-6 for more instruction.
An Acquirer or S/E		To reverse an authorization approval if the Transaction is not completed at the point of sale. For example, the Acquirer can not send the approval to the S/E or a device error prevents the receipt of the approval. Refer to Reversal Flow #2 on page 2-7 for more instruction.

S/E Initiated Authorization Reversals

[Table 2-5 on page 2-6](#) shows how S/Es use S/E initiated Authorization Reversals.

Important: Using Processing Code 024000 to identify S/E Initiated Reversals is optional for Acquirers and Issuers.

Table 2-5. How to Use S/E Initiated Reversals

Who	Sends	Why
The S/E	POS Authorization Reversal Advice (1420) with a Processing Code of 024000	To reverse an authorization approval after the Transaction is completed. For example, a Cardmember requests cancellation of a Transaction that is already approved. Refer to Reversal Flow #2 on page 2-7 for more instruction.

Acquirer Certification for Processing Code 024000

Certification is not necessary for Acquirers, but the Network Test Simulator has test conditions for Acquirers. It is not necessary for AEGNS to review test results from an Acquirer.

Issuer Certification for Processing Code 024000

Issuers must do Phase I, Unattended Testing on the Network Test Simulator. AEGNS must review the test results before activating the S/E Initiated Reversal function.

Note: Issuers who do not certify for Processing Code 024000 - Merchant Initiated Reversal receive Processing Code 004000 - System Generated Reversal on all POS Authorization Reversal Advice (1420) messages.

Issuer Requirements

When an Issuer receives a POS Authorization Reversal Advice (1420) message, the Issuer must respond with a POS Reversal Advice Response (1430) message with Action Code 400 (Reversal Accepted). If the Issuer does not respond, the Network sends the message two more times. If the Network does not receive a response after three attempts, the Network does not try to send the message again. The Network sends the next Transaction in the queue. This lets the Network deliver queued messages more efficiently.

Issuers must try to match the POS Authorization Reversal Advice (1420) message to the first POS Authorization Request (1100) and release any funds held on the Cardmember account.

Refer to the Authorizations Reversals section in the *Business and Operational Policies* manual for more instruction.

Reversal Flow #1

Table 2-6 and Figure 2-2 on page 2-7 applies to System Generated Authorization Reversals from the Network.

Table 2-6. Reversal Flow #1

Step	Who	Does What
1	Acquirer	Sends a POS Authorization Request (1100) message to the Network
2	Network	Sends the POS Authorization Request (1100) message to the Issuer, but receives no response
3	Network	Does Stand-In for the Issuer and sends a POS Authorization Response (1110) message to the Acquirer ¹
4	Issuer	Responds to the first request with a POS Authorization Response (1110) message to the Network

Table 2-6. Reversal Flow #1 (continued)

Step	Who	Does What
5	Network	Sends a POS Authorization Reversal Advice (1420) message to the Issuer to reverse the Issuer's late-arriving approval
6	Issuer	Sends a POS Authorization Reversal Advice Response (1430) message to the Network.

1. The Stand In process includes the POS Authorization Advice (1120) and the POS Authorization Advice Response (1130).

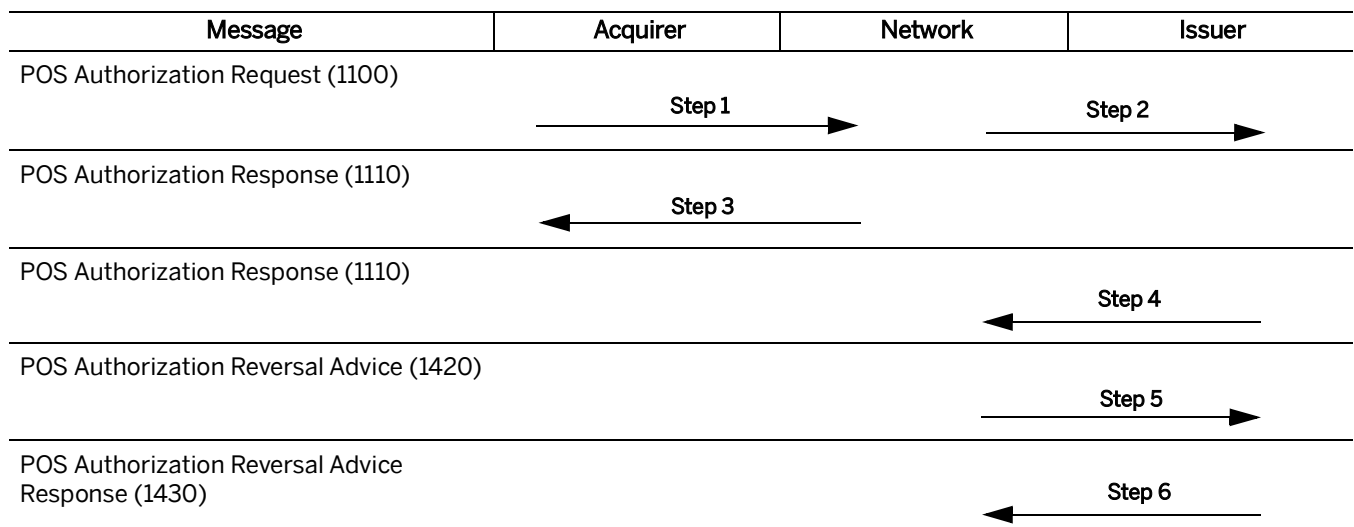


Figure 2-2. Reversal Flow #1

Reversal Flow #2

Table 2-7 and Figure 2-3 on page 2-8 applies to:

- System Generated Authorization Reversals from an Acquirer or S/E
- S/E Initiated Reversal

Table 2-7. Reversal Flow #2

Step	Who	Does What
1	Acquirer	Sends a POS Authorization Request (1100) to the Network
2	Network	Sends the POS Authorization Request (1100) to the Issuer
3	Issuer	Sends a POS Authorization Response (1110) to the Network
4	Network	Sends the POS Authorization Response (1110) to the Acquirer
5	Acquirer	Sends a POS Authorization Reversal Advice (1420) to the Network

Table 2-7. Reversal Flow #2 (continued)

Step	Who	Does What
6	Network	Sends a POS Authorization Reversal Advice Response (1430) to the Acquirer
7	Network	Sends the POS Authorization Reversal Advice (1420) to the Issuer
8	Issuer	Sends a POS Authorization Reversal Advice Response (1430) to the Network

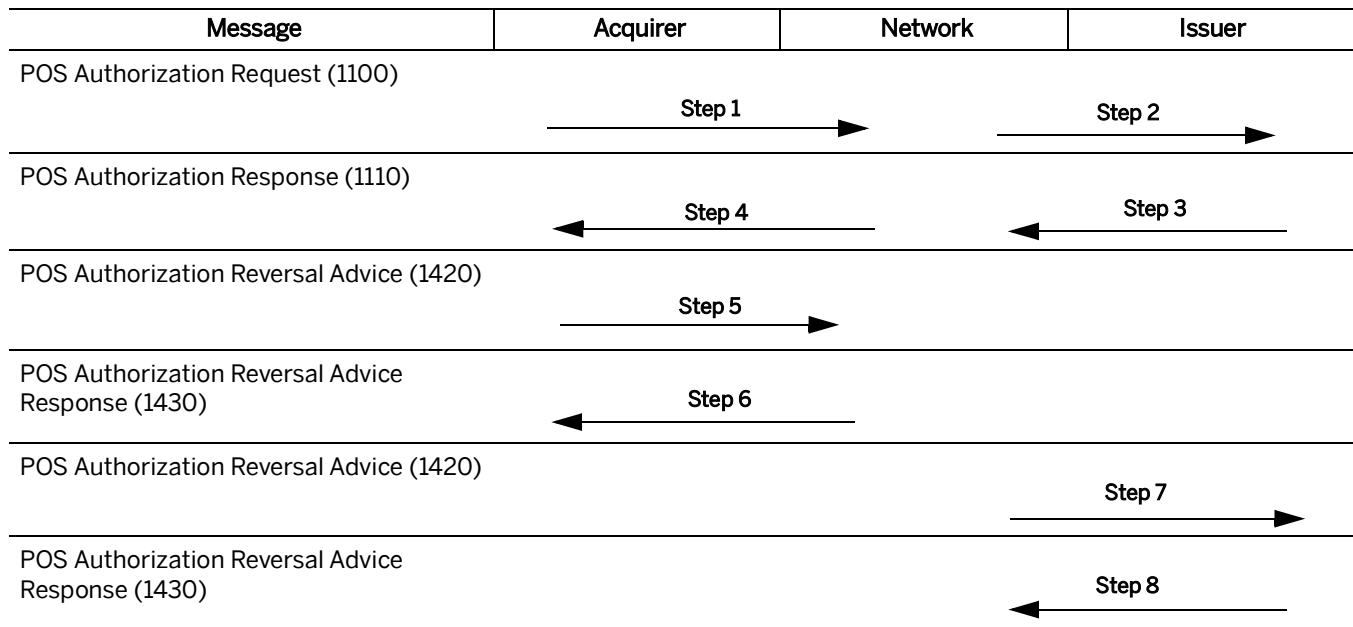


Figure 2-3. Reversal Flow #2

Authorization Stand-In

What Is Stand-In?

The Network offers Stand-In Authorization to decision POS requests for Issuers. Stand-In occurs when an Issuer is offline, does not respond before timeout (an 8 second timeframe), experiences a communications failure, or when there are scheduled Issuer outages. In the above instances, the Network uses Stand-In parameters supplied by the Issuer to approve or deny POS Transactions.

Establishing Stand-In Parameters

During Participant Setup, Issuers supply the decision criteria for the Network to use during Stand-In. Stand-In decisions are based on values defined for:

- Single Charges
- CSC validation (3CSC, 4CSC, or 5CSC)
- Velocity by Card BIN range
- Cardmember negative status

This criteria is in the Stand-In Parameter Tables. Refer to the *Business and Operational Policies* manual for instructions about Stand-In tables.

To change their Stand-In parameters, Issuers must contact their AEGNS Representative.

Stand-In Table Limits

The Network approves requests that fall below the Stand-In table limits. If directed to do so by the Issuer, the Network can decline authorizations that are more than the Stand-In table limit.

The Network can decline the Transaction if negative Card data is in the Network Exception Database. Refer to [Network Exception Database on page 2-13](#) for more instruction.

Scheduled Stand-In

AEGNS recommends that Issuers request scheduled Stand-In before they do maintenance or for scheduled system outages. [Table 2-8](#) gives the procedure to start and end scheduled Stand-In.

Table 2-8. Scheduled Stand-In Procedure

Step	Who	Does What
1	Issuer	Starts scheduled Stand-In by sending a Network Management Request (1804) with a Function Code of 802 (System Condition/Signoff) to sign off from the Network. The Issuer must also put a code in Bit 48 to tell the Network to use a special Stand-In table.
2		Ends scheduled Stand-In by sending a Network Management Request (1804) with a Function Code of 801 (System Condition/Signon) to show readiness to return online
3		May send a response to an 1804 with a Network Management Response (1814), using a Function Code of 800 (Accepted) to discontinue Stand-In processing

Unscheduled Stand-In

[Table 2-9](#) describes the process for unscheduled Stand-In.

Table 2-9. Unscheduled Stand-In Procedure

Step	Who	Does What
1	Network	Stands in when a response is not received to an Authorization Request within eight seconds. The eight second time frame includes the roundtrip time from when the request is routed to the time the switch receives the Issuer's response. As part of the eight seconds, the Participant has two seconds to respond to the Authorization Request. The two second time frame starts when the Authorization Request leaves the Network and ends when the response is received at the Network. Refer to the <i>Network Connectivity</i> manual for information about connecting with the Network.
2	Network	Tell the AEGNS Help Desk if GNSAS stands in for an Issuer who has not invoked a Stand-In request
3	AEGNS Help Desk	Tell Participants when there is too much unscheduled Stand-In, and resolves issues in a timely manner

Store and Forward Process

After Stand-In activity, the Network transmits POS messages to Issuers with the Store and Forward procedure. Refer to [Table 2-10](#) for Store and Forward instruction.

Table 2-10. Store and Forward Procedure

Step	Who	Does What
1	Network	Creates and stores POS Authorization Advice (1120) messages for each Cardmember Authorization Request that is decisioned
2	Network	Puts the messages in a queue in chronological order and includes the data about how the request was resolved
3	Issuer	Tells the Network that it is available for Authorization Request decisioning
4	Network	Sends the stored messages, one by one, to the Issuer
5	Issuer	Responds to each POS Authorization Advice (1120) with a POS Authorization Advice Response (1130) with a 900 (Advice Accepted) or 904 (Format Error) Action Code ¹ Note: If the Issuer does not acknowledge an 1120 message with a correct response (1130), the Network will send the message two more times. If the Network does not receive a correct response after the third delivery, the Network discards the message and sends the next Transaction in the queue. This lets the Network send queued messages efficiently.
6	Issuer	Adjusts Cardmembers' accounts as necessary

1. POS Authorization Advice messages responded to by the Issuer with a 904 Action Code are reported on the Authorization Transaction Detail report.

Accumulation Database

The Network uses data accumulated for all Transactions (Issuer resolved and Stand-In) to build a stored set of Transactions to use as a reference during Stand-In. Refer to [Table 2-4 on page 2-11](#).

Note: Accumulation of all Transactions will occur for Issuers on Release 2000.2 or greater. Earlier releases only store Transactions accumulated during Stand-In.

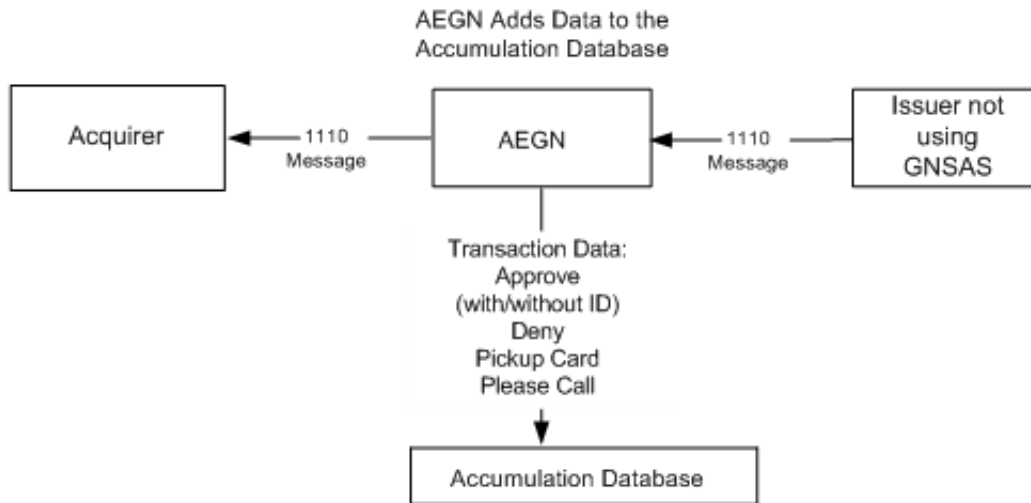


Figure 2-4. Updating the Accumulation Database

The Network uses the data in the accumulation database, in conjunction with velocity Stand-In tables defined by the Issuer, for decisions during Stand-In, as depicted in [Figure 2-5](#).

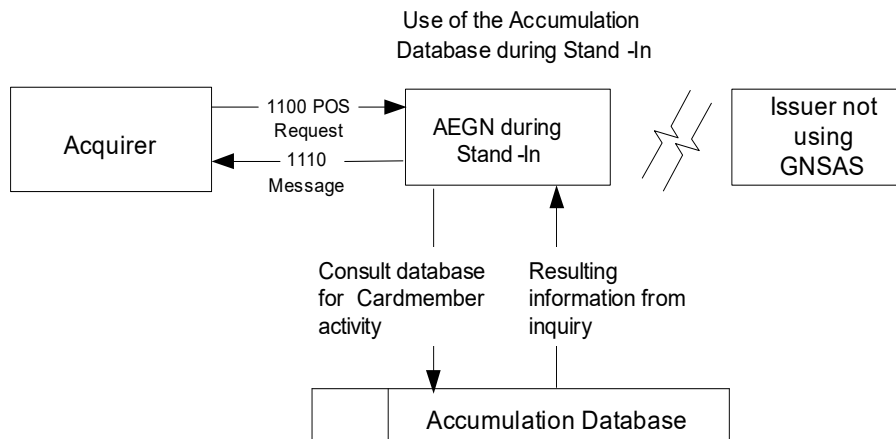


Figure 2-5. Using the Accumulation Database During Stand-In

The Network stores select Transaction data during processing and Stand-In. During Stand-In, the Network uses this data to show the Cardmember's spending and to make good decisions.

Receiving Decision Notification

To receive notification of Stand-In decisions made by the Network, Issuers must be certified to receive the POS Authorization Advice (1120) and send the POS Authorization Advice Response (1130).

Protection Device

During Stand-In, the Network suspends Cards used for multiple duplicate requests during a 24-hour period. The suspension is in effect until midnight Mountain Standard Time (UTC-7), at which time the Network revalidates the Card. This controls too much repeat Card activity and protects Issuers and GNSAS from this type of Transaction pattern.

No Stand-In

There is no Stand-In for these request types:

- Transactions with Issuer-based Deferred Payment Plan (DPP) data in Bit 48. The Network responds to these requests with Action Code 911 (Card Issuer Timed Out) in the POS Authorization Response (1110) message.
- Automated Address Verification requests without an Authorization Request. The Network responds to these requests with Action Code 000 (Approval) and a Bit 44 AAV code R (Retry) in the POS Authorization Response (1110) message.
- ATM Authorization Requests. The Network does not respond to the Acquirer. (The ATM can issue a time out or cancel the request.)
- American Express Verification Value (AEVV) validation where the Issuer generated the AEVV.
- Partial Authorization, Balance Inquiry, and Balance Return.

Global Authorization Network Alternate Routing (GAN AR)

Overview

When an Issuer is not available to process Authorization Requests from an Acquirer, Global Network Services Authorization System (GNSAS) supplies Stand-In Authorization services. The optional Stand-In Authorization servicing lets Issuers set Transaction criteria used by the GNSAS to decision an Authorization Request for them.

With the introduction of the GAN AR capability, POS Authorization Request (1100) and POS Authorization Response (1110) messages can bypass the GNSAS when the GNSAS is not available for processing. The GAN AR capability lets Issuers receive and decision POS Authorization Request (1100) messages that can otherwise have been decisioned by Host or Remote Stand-In Authorization rules.

Transaction Types

These Transaction types are **not** available for alternate routing. These Transactions use existing Remote Stand-In rules:

- ATM Transactions
- Cash Disbursement Transactions
- Deferred Payment Plan Transactions and Plan-N

If GNSAS is not available, the Network will not give Fraud Indicators in Bit 62 on the POS Authorization Request (1100) message.

Other Transaction types are not eligible based on specific Transaction scenarios.

In addition, Transactions must originate from an Acquirer who submits POS Authorization Request (1100) messages through the GAN.

Certification

All certified Issuers can participate in GAN AR. No more certification is necessary. Contact your AEGNS Representative for more instruction.

Exception Processing

Overview

Issuers can supply more Cardmember data that is used during Authorization Stand-In and other Network services.

The Network maintains two tables that Issuers can update:

- Network Exception Database (Refer to the [Network Exception Database on page 2-13](#) for instructions)
- Network Preferred/Delinquent File (Refer to [Network Preferred/Delinquent File on page 2-17](#) for instructions)

Issuers can use the messages and files described in [Table 2-11](#) to update the data in the Network Exception Database and Network Preferred/Delinquent Database.

Table 2-11. Message Used to Update Issuer Decision Reference Files

The Message or File...	Uses...	To Update This...
Batch Exception Update Request (9344)	Batch mode update, with Network response to requests in the batch file	Network Exception Database
Real Time Exception Request (1304)	Single online update and response for each message	
Exception File	Batch mode update	
Enhanced Exception File	Batch mode update	
Preferred/Delinquent File ¹	Batch mode update	Network Preferred/Delinquent Database

1. This file can be sent only by Insourced Issuers who use GNSAS to decision their Transactions.

Network Exception Database

Overview

The Network Exception Database contains the Cardmember account numbers that get special rules during Authorization Stand-In and the Recurring Billing Cancellation Service.

How To Update

The Network Exception Database gets data from these formats:

- Batch Exception Update Request (9344). Refer to the *Network Specifications - Financial/Non-Financial* manual for message layouts and programming notes.
- [Real Time Exception Request \(1304\) on page 5-130](#)
- [Exception File on page 5-153](#)
- [Enhanced Exception File on page 5-158](#)

Exception Status

Issuers use Exception and Message Reason Codes to give a status to a Card account on the Network Exception Database. Issuers can add, update, or delete a status from a Card account at any time. Each status has a priority value.

Multiple status codes can be added to a single account, however, only one status code with priority 1 or 2 can be added to a single account. Only one “Inspect”, “Stolen”, or “Lost” exception status can be applied to a single card account at a time.

Table 2-12. Exception and Message Reason Code Priority

Priority Value	Exception Status	Exception Code	Message Reason Code	Meaning
1	Inspect	04	3702	Known Card fraud
2	Stolen	05	3001	Card was reported stolen
2	Lost	03	3000	Card was reported lost
Not Applicable	Non-received/ Undelivered Card	11	3002	Card was reported undelivered (non-received)
Not Applicable	Collect	22	3700	Card identified as a high credit risk (collection)
Not Applicable	Bank Negative	91	3701	Miscellaneous code
Not Applicable	VIP Cardmember	18	3703	Very Important Person (VIP) Cardmembers
Not Applicable	Recurring Billing Cancellation	BZ ¹	3704 ²	Cardmember has withdrawn consent for certain recurring billing Transactions

1. Exception Code BZ is valid for the Enhanced Exception File only. The Recurring Billing Cancellation Service is not supported on the Exception File.
2. Message Reason Code 3704 is valid for the Batch Exception Update Request (9344) only. The Recurring Billing Cancellation Service is not supported on the Real Time Exception Request (1304).

An Exception Status that is “Not Applicable” can be used alone or with any other Exception Statuses. A priority 1 must be deleted before adding a priority 2.

Refer to [Table 2-12](#) for more instruction.

Example 1: Cardmember account A has an Exception Status of Lost. An Issuer sends a message with an Exception Status of Inspect. Inspect is a higher priority than Lost. Inspect replaces Lost on the Network Exception Database.

Example 2: Cardmember account B has an Exception Status of Stolen. An Issuer sends a message with an Exception Status of Lost. The Priority Value of Stolen and Lost is the same. The Network Exception Database is not changed. To replace Stolen with Lost, an Issuer must send a message to delete Stolen, then send a message to add Lost.

Example 3: Cardmember account C has an Exception Status of Lost. An Issuer sends a message with an Exception status of Non-received. Non-received has a priority of Not Applicable. The Card will be blocked with the Lost and Non-received statuses.

Example 4: Cardmember account E has an Exception Status of Bank Negative. An Issuer sends a message with an Exception Status of Non-received. Bank Negative and Non-received statuses have a priority of Not Applicable. The Card will be blocked with the Bank Negative and Non-received statuses.

Stand-In

When the Network does Stand-In authorization for an Issuer, the Network does not approve an Authorization Request for a Cardmember account with a negative exception status in the Network Exception Database. A negative status is Inspect, Stolen, Lost, Non-Received, Collect, and Bank Negative. The Network uses the VIP Stand-In tables, with Issuer specified parameters, for accounts that have a VIP Exception Status.

Reports

Refer to the Reports chapter in the *Network Specifications - Instructions* manual for the Exception Detail Report and the Recurring Billing Cancellation Records Reports.

Recurring Billing Cancellation Service

Overview

The Network can block recurring Transactions on behalf of Issuers from AEGNS designated markets when a Cardmember removes their consent. The Network determines which Transactions to block by comparing Authorization and Presentment messages against the Recurring Billing Cancellation status in the Network Exception Database.

Note: Issuers participating in the Recurring Billing Cancellation Service must use the Enhanced Exception File. Refer to the [Enhanced Exception File on page 5-158](#).

Blocking Procedure

These conditions must exist in order for an Authorization Request or First or Second Presentment to be blocked by the Network.

- The Issuer must be certified for the Recurring Billing Cancellation Service.
- Position 5 of the Transaction Point of Service Data Code must equal 9 - Cardmember not present, recurring billing.
- The Transaction must match an active Recurring Billing Cancellation status on the Network Exception Database. The Network uses the data elements in [Table 2-13 on page 2-16](#) to match a Transaction with a Recurring Billing Cancellation status.

Table 2-13. Matching Criteria for Recurring Billing Cancellation Service

Data Element Name	Position in Authorization Request (1100)	Position in First or Second Presentment (1240)	Must be	Value in Network Exception Database
Primary Account Number	Bit 2	Field 3	Equal to	Primary Account Number
Card Acceptor Identification Code	Bit 42	Field 35	Equal to	Card Acceptor Identification Code
Transaction Currency Code	Bit 49	Field 68	Equal to	Transaction Currency Code
Transaction Amount	Bit 4	Field 5	Greater than or equal to	Transaction Amount, Minimum
			Less than or equal to	Transaction Amount, Maximum

Matching Authorization Request

If a matching Authorization Request is found:

- The Network sends a POS Authorization Response (1110) message to the Acquirer with Bit 39 Action Code = 100 (Deny)
- The Network sends a POS Authorization Advice (1120) message to the Issuer with Bit 39 Action Code = 198 (Deny Recurring Billing Cancellation)

Note: The Recurring Billing Cancellation Service is not only for Stand-In processing. The Network blocks matching Transactions when the Issuer is available.

The Daily Authorization Reporting supplies the Issuer with a record of POS Authorization Advice messages sent to the Issuer with Action Code 198 = Deny Recurring Billing Cancellation.

Matching First or Second Presentment

If the Network finds a matching First or Second Presentment the Network rejects the Presentment with Reject Reason Code 1039 (Recurring Billing Cancellation). Acquirers must not send rejected Recurring Billing Cancellation Transactions to the Network again.

Certification

Issuers must certify for the Recurring Billing Cancellation Service. Issuers must use the Batch Exception Update Request (9344) or the Enhanced Exception File to send a Recurring Billing Cancellation status to the Network.

The Network rejects a Recurring Billing Cancellation status from Issuers who are not certified with Reject Reason Code 1047 (Participant not certified for feature).

Refer to the *Business and Operational Policies* manual for more instruction on the Recurring Billing Cancellation Service.

Network Preferred/Delinquent File

Overview

Insourced Issuers use this file to supply data to the Network on exposure and delinquency amounts and Cardmembers or VIP data to be used during the Authorization process.

How To Submit

Issuers who use GNSAS to decision Authorization Requests supply Cardmember data through the Preferred/Delinquent File. The Network uses this file to update the Network Preferred/Delinquent File.

How The Network Uses This File

The Network consults the Network Preferred/Delinquent File for decisioning when it receives a Cardmember Authorization Request. This file interacts with the Criteria Table (CRIT), which is established during Participant Setup. The CRIT is the primary means of managing risk for Card portfolios. The amounts in the CRIT table are expressed in U.S. Dollars.

Update Frequency

Issuers typically send a new file each day to update delinquent and preferred Cardmember data.

Credit Code Preferred

The Preferred/Delinquent File contains a field called the Credit Code Preferred field. This field lets Issuers override selected settings in the CRIT. Each code represents a series of indexes. Each index can override a different type of limit in the CRIT. This lets Issuers designate preferred Cardmembers with higher limits or delinquent Cardmembers with lower limits.

For each preferred Cardmember, the Network first consults the CRIT, compares the limit for the industry, then multiplies that limit value by the multiplier from the Credit Code Preferred field. The Transaction can be subject to other decision criteria, for example, have multiple charges occurred in a short time period, the number of small charges, etc. [Table 2-14](#) supplies the types of limits that can be applied to Cardmember spending.

Table 2-14. Credit Code Preferred Index Terms

Abbreviation	Term	Description
Hex	Hexadecimal	Hexadecimal representation of the Credit Code
Dec	Decimal	Decimal representation of the Credit Code
Max Exp Index	Maximum Exposure	Index for the Account's Total Due plus amounts that are not billed
Sgl Chg Index	Single Charge	Index for the amount of a single charge
All Chg Index	All Charges	Index for the sum of amounts requested at a particular S/E type for the designated accumulation period
Small Chg Index	Small Charges	Index for the sum of small charges, regardless of S/E type

Table 2-14. Credit Code Preferred Index Terms (continued)

Abbreviation	Term	Description
Tot Exp	Total Exposure	The rule to apply for the Credit Code, to find out how the account exposure is calculated. If the Total Exposure is set to "Yes" then the calculated total exposure (approved accumulations on all supplementary cards plus total due plus Transaction amount plus Sign and Travel ¹) is compared to the adjusted CRIT Overall Governor. If it is set to "No," the standard overall governor calculation (approved accumulations on this plastic plus Transaction amount) is compared to the adjusted CRIT Overall Governor.

1. Sign and Travel is a GNSAS specific program that lets some charges be paid on a revolving basis. The full amount is not payable when a Cardmember receives a monthly statement.

Recurring Billing and Standing Authorization Transactions

What Is It?

Recurring billing are Transactions where the S/E has a recurring contracted billing amount with the Cardmember. These Transactions are usually requested at the same time each month for the same dollar amount.

Standing authorization Transactions are Transactions where the S/E has a "preferred" or "frequent" Cardmember's Card data on file. The time and amount of the Transactions can be different.

Point of Service Data Code Values

S/Es and Acquirers must use "4" (Cardmember not present, standing authorization) or "9" (Cardmember not present, recurring billing) in Position 5 of Bit 22 Point of Service Data Code to show the Transaction is a recurring billing or standing authorization.

Issuers use the Point of Service Data Code received from the S/E to decision the credit Authorization Request. Bit 14 Card Expiration Date is optional and can not always be populated in recurring billing or standing Authorization Transactions.

Prepaid Card Authorization

Prepaid Card Overview

The Prepaid Card Authorization feature is designed to give S/Es and American Express Prepaid Cardmembers redemption capabilities unique to Prepaid Cards at the Point of Sale.

Facilitating Prepaid Card Authorizations



It is not necessary for S/Es to know which American Express Cards are Prepaid Cards, and S/Es may submit, a normal Authorization Request to the Issuer for approval.

Prepaid Card Stand-In

Decisioning for Prepaid Card Authorizations will be based on an Issuer's existing Stand-In limit.

Partial Authorization and Balance Return

Partial Authorization Overview

- Partial Authorization is an optional functionality that allows S/Es to obtain an Authorization for less than the requested purchase amount.
- Issuers choosing to support Partial Authorization can approve an Authorization Request for a partial amount.
- The Cardmember then has the option to either pay for the outstanding amount of the purchase by other means or cancel the Transaction.

Balance Return Overview

- When an Authorization is requested by an S/E with a Card type that supports an available balance, the available amount of funds in the Card currency remaining on a Card may be returned by the Issuer in the Authorization Response.
- Balance return is not permitted on Debit Cards.

Facilitating Partial Authorizations and/or Balance Return

- S/Es choosing to support Partial Authorizations and/or balance return must submit an Authorization Request using the appropriate Function Code to either request both a Partial Authorization and a balance return, or just a balance return.
- A balance return may be requested on its own using a unique Function Code, but all Partial Authorizations automatically include balance return.

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Partial Authorizations and/or Balance Return Handling

As an S/E does not need to differentiate the Card product type prior to Authorization, the same Function Code may be used.

- The same balance return Function Code of 182 may be used for a Prepaid Card as well as a Charge, Credit, or Debit Card.
- AEGNS will only send Function Codes to Issuers certified to receive them.
- If an Issuer is not certified for a particular Function Code, AEGNS will convert the Function Code to the next level supported by the Issuer or downgrade it to a basic Authorization.

Example:

- A certified Acquirer sends an Authorization Request with Partial Authorization and balance return (Function Code 181).
- If the Issuer does not support both Partial Authorization and balance return, AEGNS will convert Function Code 181 to Function Code 182 (balance return only).
- If the Issuer does not support balance return Function Code 182, then it will be converted to Function Code 100 (Basic Authorization Request) and sent to the Issuer.
- Issuers must not return available balance for Debit Cards in Bit 54 - Amount, Additional in the POS Authorization Response (1110) message. Bit 54 will be dropped by AEGNS and not sent to the Acquirer if the available balance is returned from the Issuer on Debit Card Transactions.

Partial Authorizations and/or Balance Return Examples

Table 2-15. Partial Authorization and Balance Return Example

Prepaid Card Balance = \$100	POS Authorization Request (1100)		POS Authorization Response (1110)				
	Bit 4	Bit 24	Bit 4	Bit 24	Bit 30	Bit 39	Bit 54
	Amount, Transaction	Function Code	Amount, Transaction	Function Code	Amount, Original	Action Code	Amount, Additional
181 - Full Approval	75	181	75	181	N/A	000	25
181 - Partial Approval	150	181	100	181	150	002	0

Table 2-16. Balance Return Example

Prepaid Card Balance = \$100	POS Authorization Request (1100)		POS Authorization Response (1110)				
	Bit 4	Bit 24	Bit 4	Bit 24	Bit 30	Bit 39	Bit 54
	Amount, Transaction	Function Code	Amount, Transaction	Function Code	Amount, Original	Action Code	Amount, Additional
182 - Full Approval	75	182	75	182	N/A	000	25
182 - Decline	150	182	150	182	N/A	100	100

Partial Authorizations and/or Balance Return Certification

Acquirer

For an Acquirer's S/Es to participate in Partial Authorization and/or balance return, Acquirers must certify to send the appropriate Function Code.

Issuer

For an Issuer to offer Partial Authorizations and/or balance return, they must certify to:

- Receive the appropriate Function Code
- Return the remaining Card balance in Bit 54 - Amount, Additional in the POS Authorization Response (1110)

If an Issuer has Cards that are valid for Transactions outside of the issuing country, the Issuer must convert the Transaction currency to their preferred currency when calculating Partial Authorizations and Balance Inquiries and then convert it back to the currency of the Transaction currency for the Authorization Response.

Balance Inquiry - Prepaid

Balance Inquiry Overview

S/Es may request the remaining Prepaid Card balance.

Prepaid Balance Inquiry Handling

Balance Inquiries are only permitted on Prepaid Cards.

- For Prepaid Balance Inquiries, the POS Authorization Request (1100) message must be sent with:
 - Function Code 181 and/or 182
 - A zero Transaction Amount in Bit 4
- Only balance inquiries for Prepaid Card BIN ranges will be forwarded to an Issuer. Balance inquiries for all other Card types will be rejected back to the Acquirer with Action Code 110 (Invalid Amount).
- If available, the Issuer will return the following in the POS Authorization Response (1110) message.
 - A space filled Approval Code in Bit 38
 - An Action Code of 000 in Bit 39
 - The available balance in Bit 54

Balance Inquiry Stand-In

Stand-In is not supported for balance inquiry. If an Issuer is not available, a balance inquiry sent by an Acquirer will receive a 'Please Call' Action Code 107 in the POS Authorization Response (1110) message.

Balance Inquiry Examples

Table 2-17. Prepaid Card Balance Return Inquiry Example

Prepaid Card Balance = \$100	POS Authorization Request (1100)		POS Authorization Response (1110)					
	Bit 4	Bit 24	Bit 4	Bit 24	Bit 30	Bit 38	Bit 39	Bit 54
	Amount, Transaction	Function Code	Amount, Transaction	Function Code	Amount, Original	Approval Code	Action Code	Amount, Additional
181 Balance Inquiry	0	181	0	181	N/A	spaces	000	100
182 Balance Inquiry	0	182	0	182	N/A	spaces	000	100

POS and ATM 5CSC Validation

What Is It?

5CSC (Card Security Code) validation of POS and ATM Financial Transactions lets the Network do a 5CSC validation for all Authorization Requests when track data is present, and then send the results to certified Network Issuers.

Important: Issuers can certify to have the Network deny only those Transactions with invalid 5CSC values.

5CSC POS Process

Table 2-18 on page 2-22 shows 5CSC validation for POS Transactions.

Table 2-18. 5CSC Validation For Authorizations

Decisioning Situation	Acquirer Steps	Network Steps	Issuer Steps
Issuer Decisioned Transactions ¹	1. Send POS Authorization Request (1100) to Network.	2. Do 5CSC validation. Send the POS Authorization Request (1100) with the CSC flag in Bit 62 to certified Issuer. 3. Pass the POS Authorization Response (1110) to the Acquirer. Bit 62 is not returned to the Acquirer in the 1110 message.	4. If certified, use the CSC flag to assist in decisioning the Transaction. 5. Return POS Authorization Response (1110).
Network Decisioned Transactions ²	1. Send POS Authorization Request (1100) to Network.	2. Do 5CSC validation. Decision the Transaction only if the validation results are invalid or zero/blank-filled. A POS Authorization Response (1110) is sent to the Acquirer with an Action Code 100 (Deny). Note: If the validation results are not invalid or zero/blank-filled then the Transaction is passed to the certified Issuer for decisioning. 3. Create a POS Authorization Advice (1120) message and pass to the Issuer to tell them of the disposition of the Transaction. Bit 62 will only be included if the Issuer is certified.	4. Return a POS Authorization Advise Response (1130) to the Network, acknowledging receipt of the 1120.

1. Available for ATM Transaction processing, using the ATM Financial Transaction Request (1200) and the ATM Financial Transaction Response (1210)
2. Not available for ATM Transaction processing

Emergency Replacement Cards

For Emergency Card Replacement (ECR), the American Express hardware key is used. If the hardware key fails, then verification is attempted with the American Express software key. If the software key fails, then verification is attempted with the American Express 5CID.

Special Programs

Overview

Special Programs with specific requirements exist to facilitate unique processing environments. These programs let Participants give specialized service to their S/Es and Cardmembers. Participants can decrease costs associated with processing Transactions on the Network.

Special Programs include the following:

- ZIP Code Verification (ZCV) - The ZCV program lets participating S/Es submit the Cardmember's billing address ZIP code as part of the POS Authorization Request (1100) message for Issuer validation as a fraud prevention measure. For more instruction, refer to the *Network Specifications - Financial/Non-Financial* manual.
- Printed Card Security Code (PCSC) Validation (Card Present Transactions only) - The PCSC Validation program lets participating S/Es use the 3- and 4-digit Card Security Code or 4DBC as a verification tool during the Authorization process. For more instruction, refer to the *Network Specifications - Financial/Non-Financial* manual.

Refer to the *Business and Operational Policies* manual for more instruction.

Deferred Payment Plan (DPP)

What Is It?

A DPP is an optional Issuer/Acquirer feature that lets a Cardmember pay for a purchase over time. The Issuer or Acquirer divides the Transaction amount into small amounts that are billed to the Cardmember in installments.

Certification

Certification is required if the Issuer or Acquirer selects this feature.

Cardmember Options

Cardmembers can request a DPP if their Issuer is certified.

If the Acquirer for the Service Establishment (S/E) has a DPP also, the Cardmember can choose which plan to use at the time of the Transaction. The S/E shows this choice as the Plan Type and includes it in the Authorization Request.

Bit 48

Bit 48 holds DPP data (payment options, installment amount, etc.) for these Authorization messages:

- Issuer DPP pre-authorization (1100/1110)
- Issuer DPP authorization (1100/1110)
- Acquirer DPP authorization (1100/1110)
- Advice of Stand-In decisions made by the Network for Acquirer-based DPP requests

The DPP data includes the principal amount (Transaction Amount), DPP plan type, number of installments, and the amount of each installment. The Network validates the data for the correct Plan Type.

The Network does not do logic checks or edit the data for content before sending it from the Acquirer to the Issuer. The Network does not edit DPP response data received from the Issuer before sending it to the Acquirer.

Example: A DPP Authorization Request can have a total Transaction amount of \$1000 and request 2 payments of \$100. Although the total Transaction amount does not equal the 2 payments, the Network does not reject the message, but sends it to the Issuer for resolution. The Issuer can deny the Authorization or generate a referral response. It is not necessary for the Issuer to generate Bit 48 response data.

Note: Bit 48 exchanges data not defined in the ISO 8583 message standard.

Issuer-Based Deferred Payment Plan

Important

The Network rejects DPP pre-authorization or DPP authorization messages for Issuers who are not certified for DPP.

Issuer-Based Plan Process

An Issuer-based DPP has 2 Transactions:

- A Pre-Authorization message to request the terms for the available plans
- An Authorization message to get approval for the Transaction

When a DPP is Issuer-based, use the procedure in [Table 2-19](#).

Table 2-19. Issuer-Based DPP Procedure

Step	If...	Who...	Does What...
1		Acquirer	Sends a POS Authorization Request (1100) message as an inquiry with a Function Code of 108 (Inquiry)
2	Acquirer and Issuer support DPP	Network	Sends the POS Authorization Request (1100) message to the Issuer
	Acquirer and/or Issuer do not support DPP	Network	Sends a POS Authorization Response (1110) to the Acquirer with an Action Code of 115 (Requested Function Not Supported)
3	Issuer is online and responds	Issuer	Sends the Network a POS Authorization Response (1110) that contains the payment choices that are available to the Cardmember. This data replaces the request data that had been sent by the Acquirer.
4		Network	Sends the Issuer's POS Authorization Response (1110) to the Acquirer
5		Acquirer	Sends the Issuer's POS Authorization Response (1110) to the S/E
6		Cardmember	Accepts one of the Issuer's payment plans
7		Acquirer	Sends a second POS Authorization Request (1100) with a Function Code of 100 (Authorization Request) to show the payment choice of the Cardmember
8		Issuer	Sends a POS Authorization Response (1110) approving or denying the request

Stand-In/Issuer Time Out

Important: The Network does not do Stand-In for Issuer-based DPPs.

If the Network does Stand-In for an Issuer or an Issuer times out when the Network attempts to send the POS Authorization Request (1100), the Network generates a POS Authorization Response (1110) using Action Code 911 (Card Issuer Timed Out) or 912 (Host Unavailable). The Network does not send a POS Authorization Advice (1120) to the Issuer for these requests. The S/E and Cardmember must then decide how to continue, since the Issuer must be available to approve Issuer-based DPP Authorization Request.

Late Arriving Issuer Response

The Network creates and sends a POS Authorization Reversal Advice (1420) to the Issuer if an Issuer sends an approved late-arriving POS Authorization Response (1110) to the first DPP Authorization Request.

Acquirer-Based Deferred Payment Plan

Acquirer-Based Plan Process

An Acquirer-based DPP has one Transaction; an Authorization message to get approval for the Transaction.

No pre-authorization is sent.

Use this procedure when a DPP is Acquirer-based.

Table 2-20. Acquirer-Based DPP Procedure

Step	If...	Who...	Does What...
1	Cardmember requests deferred payment	Acquirer	Sends a POS Authorization Request (1100) message with a Function Code of 100 (Authorization Request)
2	Acquirer and Issuer support DPP	Network	Sends the message to the Issuer
	Acquirer and/or Issuer do not support DPP	Network	Returns a POS Authorization Response (1110) to the Acquirer with an Action Code of 115 (Requested Function Not Supported)
3	Issuer is online	Issuer	Sends a POS Authorization Response (1110) approving or denying the request. The response confirms the payment data, and includes a description of the plan.

Stand-In

If an Issuer has requested that the Network decision POS Authorization Request messages containing Acquirer-based DPP detail in Bit 48, the Network uses the current Issuer single charge and velocity values and the number of installments.

The number of installments in Bit 48 of the request must be less than or equal to the number of installments specified by the Issuer. If the number of installments is greater than specified by the Issuer, the Network rejects the request.

The Network sends a POS Authorization Advice (1120) to the Issuer when the Issuers get online. Bit 48 of the 1120 gives the DPP data of the Transaction approved by the Network and the number of installments from the first request.

Note: This feature is effective during scheduled and unscheduled Stand-In.

Stand-In Limitations

DPP Stand-In is host-based only. Acquirers must be certified for DPP in order to have DPP Stand-In.

Remote Stand-In does not give approvals for DPP Transactions.

Voice Referral Handling Service

What Is It?

Referrals occur when an S/E calls the Acquirer Market Authorization Center for a Transaction decision. This telephone contact typically occurs if the first Transaction was returned by the Issuer with Action Code 107 (Please Call Issuer).

The Acquirer Market Authorization Center collects Transaction data through the Voice Response Unit originated communications, and contacts the AEGNS Authorizing Center. Using contact data, the AEGNS Authorizing Center calls the correct people (Acquirer, Card Acceptor, Cardmember, and Issuer) to get more data or Cardmember verification to resolve the referral Transaction.

Certification

Issuers who will use the Voice Referral Handling Service for their Authorization Request must complete Phase I testing (using the Network Test Simulator).

Certification is not necessary for Acquirers.

Valid Action Codes

Issuers certified for the Voice Referral Handling Service who respond to voice-originated Transactions (1100 with a "V" in position 7 of Bit 22) through the referral process can use one of five Action Codes in the POS Authorization Response (1110), Issuer to Network.

If the Issuer sends a different Action Code in response to a voice Transaction, the Network changes the Issuer's Action Code to 100 (Deny) and sends the Issuer a POS Authorization Advice (1120) advising them of the decline.

Table 2-21 shows all valid Action Codes for the Voice Referral Handling Service.

Table 2-21. Action Codes for 1110 Referral Response

Action Code	Description	Usage
000	Approve	The Transaction is approved as requested. An Approval Code is returned to the Acquirer in Bit 38.
001	Approve with ID	The Issuer requests that identification be established through voice contact between the acquiring center's authorizer and the S/E.
100	Deny	The Issuer refuses to authorize the Transaction. The S/E must decline the sale, or can request a different form of payment.
107	Voice Referral	The Issuer requests voice communication between an acquiring authorizer and the issuing bank's authorizer, S/E or Cardmember (CM) during an Authorization Request.

Table 2-21. Action Codes for 1110 Referral Response (continued)

Action Code	Description	Usage
200	Pick Up	The Issuer requests the Card be captured at the S/E.

Note: Participants certified for Voice Referral Handling Service before October 2008 must use Action Code 191 as Action Code 107 is not valid for these Participants.

Important Rules

The Issuer must follow the published standard in creating the POS Authorization Response (1110) message and must observe these rules:

- If GNSAS receives a POS Authorization Response (1110) message with an invalid Action Code, it generates a “Deny” response to the Acquirer and sends a POS Authorization Advice (1120) message to the Issuer to tell them of the denial. Refer to [Valid Action Codes on page 2-26](#).
- If the Issuer returns a POS Authorization Response (1110) message with an “Approve with ID” Action Code, the Approval Code (Bit 38) *must* be blank.

Process

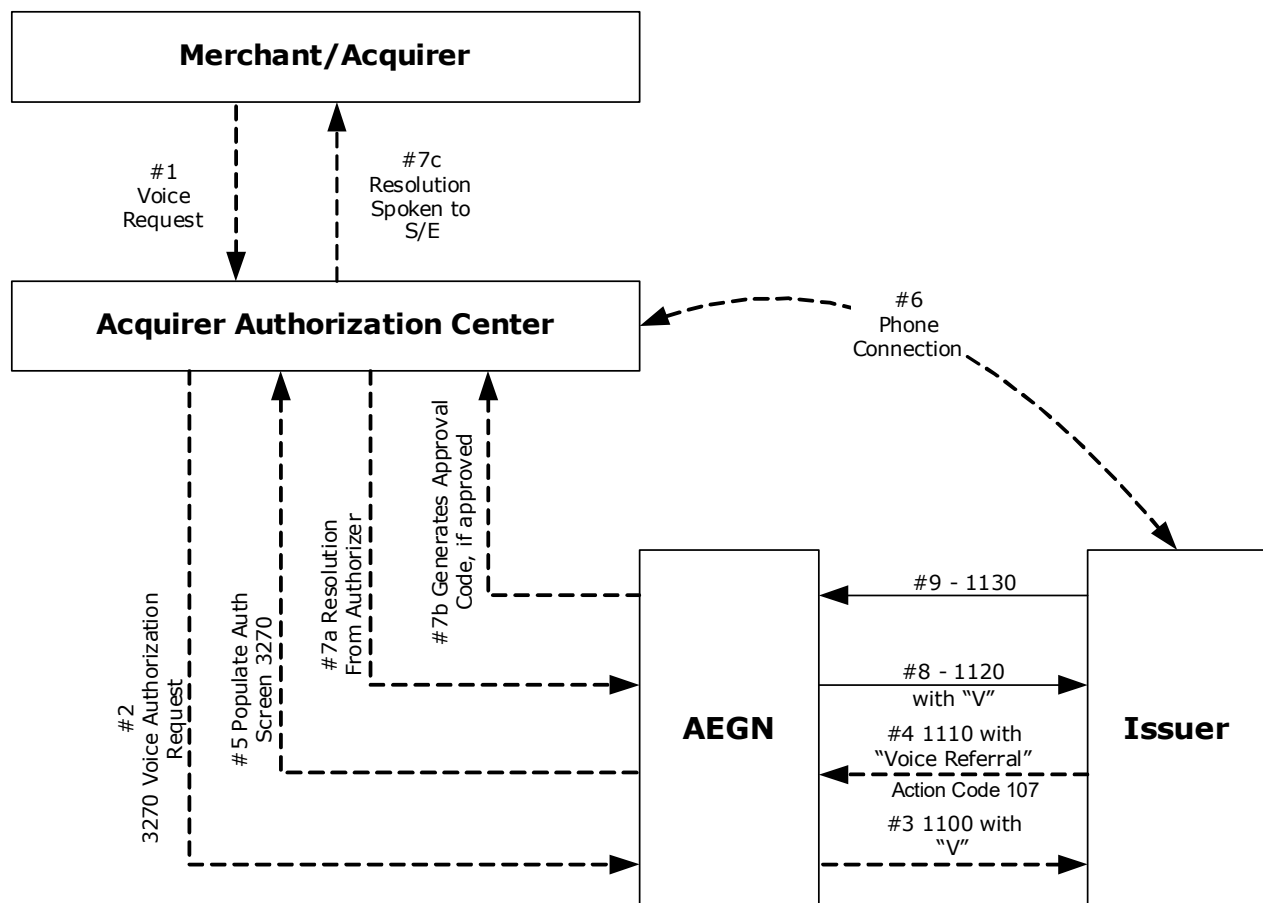
[Table 2-22](#) describes the procedure for the Voice Referral Handling Service.

Table 2-22. Voice Referral Handling Service Procedure

Step	Who	Does What
1	Acquirer’s authorizer	Keys a Transaction into the Voice Response Unit/3270 terminal when an S/E telephones their acquiring market authorization center.
2	Network	Converts the keyed Transaction into a POS Authorization Request (1100) message with a voice (“V”) indicator in the Point of Service Data Code (Bit 22). This is sent to the Issuer. Refer to Transaction #2 in Figure 2-6. Voice Referral Handling Service Transaction Flow on page 2-28 .
3	Issuer	Responds with a POS Authorization Response 1110 message with one of the five valid Action Codes. Refer to Transaction #4 in Figure 2-6. Voice Referral Handling Service Transaction Flow on page 2-28 if Action Code 107 was used. Specific, semi-custom Issuer data, such as Issuer name and telephone number, stored in GNSAS will populate the Acquirer authorizer’s 3270 terminal, enabling him/her to telephone the Issuer authorizer. Note: Issuers can respond to the 1100 with voice indicator with an 1110 message that contains Action Code 000, 001, 100, 107, or 200 when they are certified for the Voice Referral Handling Service. All certified Issuers will receive a “V” in the 1100 message Point of Service Data Code for all voice Transactions. If an Action Code other than the five listed is used in the 1110 message, the Network will decline the Transaction.

Table 2-22. Voice Referral Handling Service Procedure (continued)

Step	Who	Does What
4	Acquirer's authorizer	- Can start a three-way conversation or send data between the Issuer and the Service Establishment and/or Cardmember. This gives the Issuer sufficient data to decision the Transaction. - Authorizers use the decision of the Issuer to set a resolution code. - GNSAS generates an approval code and supplies it to the Acquirer. - Acquirer supply the Approval Code to the Card Acceptor.
5	Network	Sends a POS Authorization Advice 1120 message with 'V' in Bit 7 to the Issuer to apprise them of the results of the Transaction (approved or denied). If the Transaction was approved, the system generated Approval Code is included. Refer to Transaction #8 in Figure 2-6 .
6	Issuer	Returns a POS Authorization Advice Response (1130) message to acknowledge the Transaction.



NOTE: Step 7 has 3 parts (7a, 7b, and 7c) that occur concurrently

Figure 2-6. Voice Referral Handling Service Transaction Flow

Important

The POS Authorization Advice (1120) message contains the Approval Code generated by the Network if the Transaction was approved.

Automated Address Verification (AAV)

What Is It?

Acquirers can verify a Cardmember's address on POS Authorization Request (1100) messages and decrease fraud losses with the AAV feature. AAV validates the Cardmember account data for "Card not present" Transactions. The address verification can be sent in the first POS Authorization Request (1100) message using Processing Code 004800 or separately (for example, AAV Only) using Processing Code 174800. Refer to Chapter 2, Business Policies and Rules in the *Business and Operational Policies* manual for AAV policy.

Acquirers can verify a Cardmember's address on POS Authorization Request (1100) messages and decrease fraud losses with the AAV feature. AAV validates the Cardmember account data for "Card not present" Transactions. The address verification can be sent in the POS Authorization Request (1100) message using Processing Code 004800, 200800, or separately (for example, AAV Only) using Processing Code 174800. Refer to Chapter 2 Business Policies and Rules in the *Business and Operational Policies* manual for AAV policy.

Certification

Acquirers and Issuers must be certified for this service.

The Network does not send AAV requests **from** Acquirers that are not certified. The Network does not send AAV requests **to** Issuers that are not certified. The POS Authorization Response (1110) shows that one of the Participants is not certified for AAV.

AAV with Credit Authorization Request

Table 2-23 shows the procedure in which the submitted credit authorization Transaction includes AAV.

Table 2-23. Automated Address Verification Procedure with Credit Authorization Request

Step	Who	Does What
1	Acquirer	Requests verification of an address by sending the alphanumeric or numeric piece of the postal code and address. This request for address verification can be in the credit Authorization Request (Processing Code 004800 or 200800), in which Bit 63 Automated Address Verification contains the Cardmember's address data. The flow of POS Authorization Requests (1100) and POS Authorization Responses (1110) stays the same.

Table 2-23. Automated Address Verification Procedure with Credit Authorization Request

Step	Who	Does What
2	Issuer	Sends a code to show the results in matching the address alphanumeric or numeric data using a POS Authorization Response (1110) message, in which: • Bit 38 will contain the Approval Code, if approved, to the credit Authorization Request. • Bit 39 will contain the Action Code that shows the Issuer resolution (i.e., approve, deny, call, etc.) for the credit Authorization Request. • Bit 44 Additional Response Data, Position 3 contains the AAV Response Code that shows the result of matching the address supplied by the Acquirer in Bit 63 of the 1100 message with the Issuer's Cardmember records. Refer to the <i>Codes Reference Guide</i> for valid responses.

AAV Only Process

Table 2-24 shows the procedure to submit an AAV request separately from the credit Authorization Request.

Table 2-24. Automated Address Verification Only Procedure

Step	Who	Does What
1	Acquirer	Requests verification of an address by sending the alphanumeric or numeric piece of the postal code and address using a POS Authorization Request (1100) for AAV Only, in which: • Bit 3 Processing Code requires a value of 174800 (AAV only). • Bit 4 Amount, Transaction must be zero-filled for AAV. • Bit 14 Card Expiration Date cannot be present in AAV only requests. • Bit 63 Automated Address Verification contains the Cardmember's address data.
2	Issuer	Sends a code to show the results in matching the address alphanumeric or numeric data using a POS Authorization Response (1110) for AAV, in which: • Bit 3 Processing Code echoes the value of 174800 (AAV only) from the 1100 message. • Bit 38 Approval Code must be space-filled for AAV • Bit 39 Action Code requires a value of 000 for AAV • Bit 44 Additional Response Data, Position 3 contains the AAV Response Code to show the result of matching the address supplied by the Acquirer in Bit 63 of the 1100 message with the Issuer's Cardmember records.

Stand-In

AAV is not available to Acquirers when the Network does Stand-In processing for an Issuer. The Network does Stand-In for POS Authorization Requests with AAV data in Bit 63. The Network then instructs the Acquirer to try the address verification.

The Network sends POS Authorization Advice messages received during Stand-In when the Issuer's system is online. No POS Authorization Advice is sent for AAV-only requests.

Remote Stand-In

What Is It?

This is an optional Issuer feature that lets a Remote Host do Stand-In if the Acquirer cannot transmit to the Host.

Note: If an Issuer does not request Remote Host-based Stand-In from the Network, the Remote Host responds with a "Please Call Issuer."

Certification

Issuers who select the Remote Stand-In feature must be certified.

What Does The Remote Host Check?

The Remote Host uses the same established Issuer-supplied VIP limits or single charge limits as a Host-based Stand-In.

Issuer's exception records are checked. Cardmembers with a negative status in the Exception Database are denied by the Remote Host. Refer to [Cardmember Very Important Person \(VIP\) Status on page 2-38](#) for instruction about processing Remote Stand-In for VIPs.

Refer to the *Business and Operational Policies* for more instruction.

No Stand-In

This functionality is not supported in Remote Stand-In:

- Transactions with Issuer-based Deferred Payment Plan (DPP) data in Bit 48. The Network responds to these requests with Action Code 107 (Please Call Issuer) or Action Code 115 (Function Not Supported) in the POS Authorization Response (1110) message.
- Automated Address Verification requests without an Authorization Request. The Network responds to these requests with Action Code 000 (Approval) and a Bit 44 AAV code R (Retry) in the POS Authorization Response (1110) message.
- ATM Authorization Requests. The Network does not respond to the Acquirer. (The ATM can issue a time out or cancel the request.)
- American Express Verification Value (AEVV) validation where the Issuer generated the AEEV.
- AEIPS and Expresspay Cryptogram Validation.
- Partial Authorization with Balance Return.
- Balance Inquiry and balance return.
- Transaction blocking as part of the Recurring Billing Cancellation Service.

Product Differentiation

The Remote Host uses eight-digit product differentiation to decision Transactions with the Stand-In parameters of the Issuer.

Chip Card Transaction Processing

What Is It?

Chip Cards contain a programmable chip with memory. Chip Card Transactions are secure and significantly decrease some types of fraud (such as counterfeit cards).

These Transactions can also decrease communication costs. The Chip Card reader keeps data about Transactions. This lets S/Es send this data in a batch.

Certification

Participants must be certified to use this feature.

Chip Card Procedure

At the Point of Service, the Chip Card reader activates the Card electronics to read the Card's stored data and transmit it to the Acquirer.

When the Acquirer receives the Chip Card Transaction, it formats and transmits the data in Bit 55 to the Network. The Network transmits the Transaction to the Issuer to be decisioned.

Bit 55

Bit 55 contains the Transaction Certificate (authentication data), encoded using a cryptogram. The Chip Card's Transaction Certificate authenticates that the Cardmember is using the Card by verifying the PIN or other data about the Cardmember. This data is sent to the Issuer in its first format.

Issuers must accept and decrypt the Authentication data to decision the Transaction.

The Issuer sends a response and other data, which can include new programming instructions, to the Network. The Network sends it to the Acquirer, the Acquirer sends it to the S/E, and then to the Chip Card.

Other Data

Chip Card Transactions change other data fields in the authorization message. The changed fields are:

Table 2-25. Changed Data

Bit	Bit Name	Comments
Bit 2	Primary Account Number	Supplied from the Chip Card chip, rather than magnetic stripe.
Bit 22	Point of Service Data Code	This bit will contain different values to identify that the card and/or terminal have chip and PIN capability.
Bit 35	Track 2 Data	Supplied from the Chip Card chip, rather than magnetic stripe.
Bit 45	Track 1 Data	Supplied from the Chip Card chip, rather than magnetic stripe.

Stand-In

During Host Stand-In, the Network can decline Transactions with an invalid AEIPS and Expresspay Cryptogram Validation. Key exchange between AEGNS and the Issuer is necessary for this service. AEIPS and Expresspay Cryptogram Validation is not supported during Remote Stand-In. Refer to [Remote Stand-In on page 2-31](#) and [AEIPS and Expresspay Cryptogram Validation on page 2-43](#) for more instruction.

Contactless (Expresspay)

Overview

Cardmembers can start a Transaction with a wave, tap, or touch of their Contactless device at a contactless reader attached to a Point of Sale terminal.

Contactless Network functionality allows issuance and acceptance of Expresspay Contactless Cards on the Network.

The Expresspay specifications and American Express ICC Payment Specification (AEIPS) are related and are based on the EMV specifications. These specifications have a similar EMV Transaction at a terminal. Expresspay devices start Contactless Transactions in Europay, Mastercard/Visa (EMV) and non-EMV markets.

Important

Card readers and POS terminals must be enabled for Expresspay.

Certification

Certification is required if the Issuer or Acquirer selects this feature.

Reference Materials

[Table 2-26](#) lists the reference materials associated with Expresspay. The Acquirer and Issuer Chip Card Implementation Guides describes how to have a successful chip card migration. The AEIPS documents give instruction about integrated circuit technology. The AEIPS documents are on <https://network.americanexpress.com/globalnetwork/v4/secure/knowledge-base>. Use the Search function to locate the AEIPS documents.

Table 2-26. Reference Materials

Title	Overview
<i>Acquirer Chip Card Implementation Guide</i>	The business and technical steps necessary to have a successful Chip Card migration, with: Migration Planning Details, Terminal Requirements by Transaction Step, and Certification. Identifies best practices for addressing typical challenges that occur with the acquiring of Chip Card Transactions
<i>AEIPS Chip Card Specifications</i>	Give the mandatory and optional data elements and functionality for American Express, all American Express entities, Issuers, and Global Network Services (AEGNS) Card Issuers when implementing payment Chip Cards using integrated circuit technology

Table 2-26. Reference Materials (continued)

Title	Overview
<i>AEIPS Terminal Technical Manual</i>	Assists Terminal Vendors, S/Es, Resellers, or Authorized Processors with implementing American Express EMV acceptance on a Terminal, using the American Express Integrated Circuit Card Payment Specification (AEIPS)
<i>Expresspay Card Specification</i>	Defines the requirements for a payment application that resides on a card which only supports the contactless interface
<i>Expresspay Terminal Specification</i>	Defines the requirements for a contactless payment application implemented on a Terminal
<i>Issuer Chip Card Implementation Guide</i>	Describes the business and technical steps necessary to have a successful Chip Card migration. This guide also describes the American Express requirements for Chip Cards over and above those which are mandated in the EMV specifications (emvco.com).
<i>Online PIN Processing Implementation Guide</i>	A resource for Participants to understand American Express requirements for implementing Online PIN as a primary Cardholder Verification Method (CVM) for POS Transactions

Transit Capabilities

Table 2-27 on page 2-35 shows Expresspay authorization capabilities for the transit industry.

Table 2-27. Expresspay Authorization Features

Feature	Description
Transit Access Terminal (TAT) Indicator	The TAT indicator is used to identify Expresspay Transactions that occur at Transit Access Terminals. The Network supports a TAT indicator in the form of a new Point of Service Data Code value. Transactions that occur at transit-associated payment kiosks, retail ticketing offices, or online ticketing websites will not contain a TAT indicator. AEGNS requires Transit S/Es and their Acquirers to identify Expresspay Transit Access Terminal (TAT) Transactions in Authorization Requests, with Account Status Checks, and First and Second Presentments. Authorizations Transit S/Es and their Acquirers must send all of these data elements in the POS Authorization Request (1100) message, with Account Status Checks, in order to identify Expresspay TAT Transactions: • Point of Service Data Code - Bit 22, Position 4 (Operating Environment) = "Z" (Transit Access Terminal - TAT) • Point of Service Data Code - Bit 22, Position 6 (Card Present) - "X" (Contactless Transaction) • Card Acceptor Business Code - Bit 26, Valid Merchant Category Code Values are: • 4111 - Local and Suburban Commuter Passenger Transportation, including Ferries • 4112 - Passenger Railways • 4131 - Bus Lines • 4784 - Tolls and Bridge Fees • 7523 - Parking Lots and Garages The above codes will be passed through the Network (not validated or edited) to the Issuer. Transit S/Es and their Acquirers are required to pass complete and accurate data (as identified above) to correctly identify Expresspay TAT Transactions to the Issuer. Optionality and Certification Issuer: Mandatory for Expresspay Issuers (certification is required) Acquirer: Conditional¹ (certification is required) 1. This feature is conditional by default. When an Acquirer is preparing to process Expresspay Transactions originating from Transit Access Terminals (TATs) with a Pay As You Go fare model, this feature becomes mandatory.

Table 2-27. Expresspay Authorization Features (continued)

Feature	Description
Account Status Check	The Account Status Check lets Issuers identify POS Authorization Request (1100) messages where a transit S/E is requesting an Account Status Check only. Account Status Checks in the POS Authorization Request (1100) message show a Function Code of 190 - Account Status Check in Bit 24. Three important characteristics of the Account Status Check Authorization Request are: • The Transaction amount in the Account Status Check Authorization Request can be different (nominal amount value for example) from the amount up to which a transit S/E can send a subsequent submission. • A deny response from the Issuer to the Authorization Request can put the Cardmember on a transit S/E deny list and prevent the Cardmember's access into the transit system. • An Account Status Check POS Authorization Request will not always contain a Transit Access Terminal (TAT) indicator in the form of a Point of Service Data Code - Bit 22, Position 4 (Operating Environment) = Z (Transit Access Terminal - TAT). Some transit Transactions cannot occur at Transit Access Terminals. The Network does regular Stand-In processing for Account Status Check Authorization Requests for all Issuers who have supplied Stand-In tables. Note: Issuers must send Card exception data to the Network using the Authorization Exception process as described in Exception Processing on page 2-13. Complete and correct Card account exception data is necessary for the Network Stand-In procedure to make accurate decisions. Optionality and Certification Issuer: Mandatory for Expresspay Issuers (certification is required) Acquirer: Conditional¹ (certification is required) 1. This feature is conditional by default. When an Acquirer is preparing to process Expresspay Transactions originating from Transit Access Terminals (TATs) with a Pay As You Go fare model, this feature becomes mandatory.

Table 2-27. Expresspay Authorization Features (continued)

Feature	Description
Application Transaction Counter (ATC) Synchronization	ATC Synchronization lets Acquirers and their S/Es send Expresspay-enabled Card/device ATC values to Expresspay Issuers. Function Code 191 - ATC Synchronization must be used in Bit 24 of the POS Authorization Request (1100) message. Important characteristics (in the POS Authorization Request (1100) message) of the ATC Synchronization Authorization Request are: • Bit 4 - Amount, Transaction must contain a small amount (one unit of currency or less, for example: \$1.00 or £1.00 or €1.00). • Bit 55 - ICC System Related Data, Subfield 10 (Amount Authorized) contains a zero (0.00) amount. Important characteristics (in the POS Authorization Response (1110) message) of the ATC Synchronization Authorization Request are: • Bit 39 - Action Code must contain 900 - Advice accepted. • It is not necessary for Expresspay Issuers to calculate or decision open to buy for ATC Synchronization requests. The Network does not do Stand-In processing for ATC Synchronization Authorization Requests. If an Expresspay Issuer is unable to respond to an ATC Synchronization Authorization Request, the S/E will receive an Action Code of 900 - Advice Accepted in Bit 39 of the POS Authorization Response (1110) message. Optionality and Certification Issuer: Optional for Expresspay Issuers (certification is required) Acquirer: Conditional¹ (certification is required) 1. This feature is conditional by default. When an Acquirer is preparing to process Expresspay Transactions originating from Transit Access Terminals (TATs) with a Pay As You Go fare model, this feature becomes mandatory.
Expresspay Translation (PAN Request)	Expresspay Translation lets Acquirers and their S/Es request and receive the Primary Account Number (PAN) of an Expresspay-enabled Card/device from Expresspay Issuers. Certified Acquirers and their S/Es must use Function Code 194 - Expresspay Translation (PAN Request) or 196 - Expresspay Translation - PAN Request Extended in Bit 24 of the POS Authorization Request (1100) message. Expresspay Translation (PAN request) feature is limited to Transit S/Es using Merchant Category Codes (MCCs) of 4111, 4112, 4131, 4784, and 7523. Optionality and Certification Issuer: Conditional. The Expresspay Translation (PAN request) feature is optional to Expresspay Issuers. It only applies to Issuers who issue contactless products with Alias Account Number or Device Primary Account Number which are different from the PAN (Primary Account Number) on the Card. All Expresspay Issuers with a product that uses Alias or must certify for Function Code 194 - Expresspay Translation (PAN request) Expresspay Issuers must provide the PAN in Bit 34 - Primary Account Number, Extended of the POS Authorization Response (1110) message. Acquirer: Conditional¹ (certification is required). This feature is conditional by default. When an Acquirer is preparing to process Expresspay Transactions originating from Transit Access Terminals (TATs) with a Pay As You Go fare model, this feature becomes mandatory.

Table 2-27. Expresspay Authorization Features (continued)

Feature	Description
Negative Account File - Expresspay Only Accounts	The negative Card account file contains only Expresspay accounts. This customized file can be made available to Acquirers and their transit S/Es. Acquirers can contact their AEGNS Representatives to get this optional file for their transit S/Es. Optionality and Certification Issuer: N/A (this feature is not applicable to Issuers) Acquirer: Optional (certification is required)

Cardmember Very Important Person (VIP) Status

What Is It?

This feature lets the Network approve the Transactions of Issuer-identified VIP Cardmembers during host and remote Stand-In. The Network gives the same level of service, during Stand-In, that an Issuer gives to VIP Cardmembers.

VIP Cardmember status is applied at the Card level.

Certification

Issuers who opt to use the VIP Status feature must be certified.

Communicating VIP Status

Issuers can communicate VIP Cardmember status with one of these messages:

- Exception File Batch Update
- Real Time Exception Update Request (1304)

Note: An Issuer can use one of these messages to override a Cardmember's VIP status.

VIP Declines

VIP Transaction are declined because:

- The VIP Transaction is more than the single-charge limit¹.
- The Account number is listed in the Network Exception File with a negative code.
- It failed 5CSC validation.

Global Personal Identification Number (PIN) Management

What Is It?

Global PIN Management is an optional feature that lets a certified Issuer's Cardmembers unblock or change their PIN on Chip Cards at certified Acquirers' ATMs.

1. Single charge limits are defined by Issuer BIN range (a six-digit range assigned to a Participant). Velocity checks are not used for VIP Accounts during host-based or remote Stand-In.

Certification

Certification is required for Issuers and Acquirers.

Eligibility

To be eligible for Global PIN Management at the ATM, the Card must be issued with a “201” Service Code (available for all Transactions internationally used in an American Express EMV/AEIPS-compliant acceptance environment, integrated circuit present on the Card).

Refer to the *Codes Reference Guide* for a list of Service Codes.

PIN Change Requirements

A PIN change requires online messaging to the Issuer for verification and transportation of the new PIN. The Cardmember’s chosen PIN or hexadecimal representation must align with the PIN data on the Issuer’s PIN database.

Rejected Transactions

An attempted Global PIN Management Transaction is rejected if the Issuer or Acquirer is not certified or is not available. There is no Stand-In.

PIN Management Process

Table 2-28 shows a typical, **successful** PIN Management Transaction that starts at an ATM.

Table 2-28. Successful Pin Management Procedure

Step	Who	Does What
1	Network	Sends an ATM Financial Transaction Request (1200) message to the Issuer. PIN Change: The new PIN is encrypted in Bit 53 (Security Related Control Information). The ATM Financial Transaction Request (1200) must have a Bit 3, Processing Code of PIN Change (960000). PIN Unblock: The ATM Financial Transaction Request (1200) must have a Bit 3, Processing Code of PIN Unblock (970000).
2	Issuer	PIN Change: Echoes back the Processing Code in an ATM Financial Transaction Response (1210) and sends an Issuer Script command in Bit 55 (ICC System Related Data). PIN Unblock: Echoes back the Processing Code in an ATM Financial Transaction Response (1210) and sends an Issuer Script command in Bit 55 (ICC System Related Data).

Table 2-28. Successful Pin Management Procedure (continued)

Step	Who	Does What
3	ATM Device	PIN Change: Writes the new PIN to the Chip using an Issuer Script command in Bit 55 (ICC System Related Data) of the ATM Financial Transaction Response (1210). If the Chip cannot be updated, an ATM Reversal Advice (1420) is sent to the Issuer. The Issuer sends an ATM Reversal Advice Response (1430) with a Processing Code of 960000 (PIN Change) in Bit 3, only if a PIN Change fails. PIN Unblock: The chip will unblock the PIN using an Issuer Script command in Bit 55 (ICC System Related Data) of the ATM Financial Transaction Response (1210). The Issuer returns an ATM Reversal Advice Response (1430) with a Processing Code of 970000 (PIN Unblock) in Bit 3, only if a PIN Unblock fails.

Table 2-29 shows a typical PIN Management Transaction that starts at an ATM and is not successful.

Table 2-29. Unsuccessful PIN Management Procedure

Step	Who	Does What
1	Network	Sends an ATM Financial Transaction Request (1200) message to the Issuer. PIN Change: The new PIN is encrypted in Bit 53 (Security Related Control Information). The ATM Financial Transaction Request (1200) must have a Bit 3, Processing Code of PIN Change (960000). PIN Unblock: The ATM Financial Transaction Request (1200) must have a Bit 3, Processing Code of PIN Unblock (970000).
2	Issuer	PIN Change: Echoes back the Processing Code in an ATM Financial Transaction Response (1210) with an Action Code. Refer to the <i>Codes Reference Guide</i> for a list of Action Codes. PIN Unblock: Refer to the <i>Codes Reference Guide</i> for a list of Action Codes.

Zero Dollar Transactions

All PIN change and PIN unblock Transactions must have a zero amount in these bits:

- Bit 4, Amount, Transaction of the 1200/1210 messages
- Bit 55, ICC System Related Data of the 1210 message
- Bit 4, Amount, Transaction of the 1420 message because it is a non-cash Transaction

Cash Disbursement

Overview

Cash Disbursement functionality lets participating Cash Disbursement Providers (a Participant bank) give cash to a Cardmember. Cash Disbursement Transactions are attended, card present, signature-based, or Chip and PIN-based point of sale Transactions, processed with Merchant Category Code (MCC) 6010 - Financial Institutions - Manual Cash Disbursements and Processing Code 014008 - Cash Disbursement Debit.

Certification

Issuers and Acquirers must certify. Issuers who are new to the Acquirer role for Cash Disbursement Transactions must complete Acquirer certification.

Depending on the role of the Participant, the requirements for certification will vary as shown in [Table 2-30](#).

Table 2-30. Certification for Cash Disbursement

Who	For	Does What
Acquirers ¹	Authorization Messages	Submit and/or receive Transactions with Processing Code 014008 together with MCC 6010 using the Network Test Simulator.
	Clearing and Settlement Messages	Submit and/or receive Transactions with Processing Code 014008 or 220000 together with MCC 6010 using the Network Test Simulator. Submit and/or receive Transactions through the Retrieval Request Notification and Fulfillment Notification messages with the Function Code 640 using the Network Test Simulator. Submit and/or receive Transactions with Message Reason Code 2700 using the Network Test Simulator.
Issuers ²	Authorization Messages	Submit and/or receive Transactions with Processing Code 014008 together with MCC 6010 using the Network Test Simulator.
	Clearing and Settlement Messages	Submit and/or receive Transactions with Processing Code 014008 or 220000 together with MCC 6010 using the Network Test Simulator. Submit and/or receive Transactions through the Retrieval Request Notification and Fulfillment Notification messages with the Function Code 640 using the Network Test Simulator. Submit and/or receive Transactions with Message Reason Code 2700 using the Network Test Simulator.

1. When authorization and clearing testing are complete, Acquirers are required to tell their local AEGNS Representative.
2. When authorization and clearing testing are complete, Issuers are required to tell their local AEGNS Representative.

Credit Processing

If a credit is required, a POS Authorization Reversal Advice (1420) or a POS Authorization Advice (1120) must be submitted with the applicable Processing Code. The Cash Disbursement Credit Processing Code 22000 is not used in the authorization system.

Note: The same Processing Code used in the 1100 POS Authorization Requests for a Cash Disbursement Transaction must be used for subsequent Cash Disbursement messages (e.g., 1110, 1120, 1130).

Stand-In

Participants can request Stand-In for Cash Disbursement Transactions received with MCC 6010. The Issuer must supply Stand-In parameters specific to Cash Disbursement Transactions using the process described in the *Business and Operational Policies* manual. Default Stand-In parameters are set to zero for Cash Disbursement Transactions if not supplied by Network Participants.

ZIP Code Verification

Overview

The ZIP Code Verification (ZCV) program lets participating S/Es submit a Cardmember billing address ZIP code as part of the POS Authorization Request (1100) message for Issuer validation as a fraud prevention measure.

For instruction about the clearing process for ZCVs, see [Special Programs on page 2-22](#).

American Express SafeKey

Overview

Electronic (Internet) commerce has experienced significant growth and continues to grow. With more electronic commerce Transaction volume, fraud and Chargebacks for Card Not Present (CNP) Transactions can increase. American Express SafeKey is an industry-standard Authentication that gives more security for Internet Transactions when used by Issuers, Cardmembers, Acquirers, and Internet S/Es.

American Express SafeKey has the capability to authenticate Cardmembers online. American Express SafeKey is based on the industry-standard 3-D Secure™ protocol. American Express SafeKey is a deterrent to fraud and offers a Fraud Liability Shift (FLS) from the S/E to the Issuer in some circumstances.

Gross fraud is decreased when Cardmembers use a private and undisclosed value (American Express SafeKey) for online purchases.

For instruction about American Express SafeKey and the requirements for implementation, go to www.amexsafekey.com.

Certification

Acquirer

Acquirers must certify to send and receive American Express SafeKey related data to the Network in the POS Authorization Request (1100) message, the POS Authorization Response (1110), and the POS Authorization Advice (1120) message. Acquirers must also certify to send the American Express SafeKey values in the POS First Presentment (1240) message.

If the Acquirer is not certified and submits American Express SafeKey data, the Network will process the Transaction without the American Express SafeKey data and will not apply the American Express SafeKey FLS.

For instruction or to schedule certification, contact your AEGNS Representative.

Issuer

Issuers must certify to receive and send American Express SafeKey related data to the Network in the POS Authorization Request (1100), the POS Authorization Response (1110), and the POS Authorization Advice (1120) messages. Issuers can use the Network Test Simulator to make sure they can receive SafeKey values in the POS First Presentment (1240) message.

If the Issuer is not certified to receive American Express SafeKey data on the POS Authorization Request (1100) message from the Acquirer, the Network removes the American Express SafeKey data from that message before it is sent to the Issuer. In designated American Express SafeKey markets, the Network will continue to send the Program Indicator on the POS First Presentment (1240) message to tell Issuers when the American Express SafeKey FLS applies.

For instruction or to schedule certification, contact your AEGNS Representative.

AEIPS and Expresspay Cryptogram Validation

Overview

During the authorization process, Issuers validate the cryptogram of their Chip Cards as part of the decisioning process.

AEGNS offers an AEIPS and Expresspay Cryptogram Validation service. The Network validates the cryptogram on behalf of the Issuer. The Network sends the results in Bit 62 of the POS Authorization Request (1100) or the ATM Financial Transaction Request (1200). The Issuer must certify for Bit 62 of one or both messages to receive the validation results.

For instruction about the AEIPS and Expresspay Cryptogram Validation service, see the Reporting Fraudulent Transactions chapter in the *Network Specifications - Instructions* manual or contact your AEGNS Representative.

Authorization Adjustment Capability

Overview

Authorization Adjustment is an optional feature that lets Acquirers send the [POS Authorization Adjustment Financial Advice \(1220\)](#) message to Issuers when the last Transaction amount is less than the first authorized amount.

Authorization Adjustment Procedure

Acquirers send the POS Authorization Adjustment Financial Advice (1220) message only if the last Transaction amount is less than the first authorized amount. Issuers should respond with the POS Authorization Adjustment Financial Advice Response (1230) message. The [POS Authorization Adjustment Financial Advice \(1220\)](#) supplies the last amount of the Transaction when the sale is complete. The last amount sent on the POS Authorization Adjustment Financial Advice (1220) message tells the Issuer to release all funds held above the last Transaction amount. It is not necessary for Issuers to wait for settlement to release the hold on available funds.

Example

An Acquirer sends an Authorization Adjustment for a Transaction where the first authorized amount is \$100 and the last Transaction amount is \$75. Refer to [Table 2-31 on page 2-44](#).

Table 2-31. Authorization Adjustment Capability Example

Step	Who	Does What
1	Acquirer	Sends a POS Authorization Request (1100) message to the Network Amount, Transaction (Bit 4) = \$100
2	Network	Sends the POS Authorization Request (1100) message to the Issuer
3	Issuer	Sends a POS Authorization Response (1110) message to the Network
4	Network	Sends the POS Authorization Response (1110) message to the Acquirer
5	Acquirer	Sends a POS Authorization Adjustment Financial Advice (1220) message to the Network Amount, Transaction (Bit 4) = \$75 Amount, Original (Bit 30) = \$100
6	Network	Sends the POS Authorization Adjustment Financial Advice (1220) message to the Issuer
7	Issuer	Sends a POS Authorization Adjustment Financial Advice Response (1230) message to the Network and adjusts the Cardmember's account.
8	Network	Sends the POS Authorization Adjustment Financial Advice Response (1230) message to the Acquirer.

Stand-In

The Network supports Stand-In for Issuers certified for the POS Authorization Adjustment Financial Advice (1220) message. During Stand-In, Network stores and sends authorization adjustment financial advice data in a POS Authorization Advice (1120) message. Refer to [POS Authorization Advice \(1120\) on page 5-52](#) for instructions.

Participant Certification

This is an optional feature that requires certification by Participants.

Online PIN Cardholder Verification Method

Overview

Online Personal Identification Number (PIN) validation is a Cardholder Verification Method (CVM) used to authenticate the Cardmember at the Point of Sale (POS). Issuers and Acquirers can use the optional Online PIN method as an acceptable CVM to complete a Card Present Transaction at a PIN-capable POS. The Acquirer must send a POS Authorization Request (1100) message to the Network which contains encrypted PIN data entered by the Cardmember at the POS. The Network sends the POS Authorization Request (1100) to the Issuer.

The Network uses the Acquirer's encryption key to decrypt the PIN from the Acquirer. The Network uses the Issuer's encryption key to encrypt the PIN value when the Network sends the POS Authorization Request (1100) message to the Issuer.

The Issuer validates the accuracy of the PIN and uses it to make a Transaction decision. The Issuer sends the Action Code 000 (Approved) in the POS Authorization Response (1110) message if the Transaction is approved.

Refer to the *Online PIN Processing Implementation Guide* for Online PIN requirements.

Feature Availability

Contact your AEGNS Representative for market availability.

Host and Remote Stand-In

The Network does not provide Host or Remote Stand-In services for Online PIN Transactions. The POS Authorization Response (1110) message sends Action Code 912 (Host Unavailable) if Stand-In is requested. A normal Stand-In process occurs if the Cardmember PIN is not in the POS Authorization Request (1100) message.

Certification

Acquirers

Acquirers must certify.

Issuers

Issuers must certify.

3 ATM Processing

Section 1 ATM Authorization Processing

What Is ATM Authorization Processing?

ATM Authorization processing involves:

1. A Cardmember using a Card to get cash at a participating Automated Teller Machine (ATM) location
2. The Issuer validating the Card and approving the ATM Authorization Request
3. The ATM dispensing the requested cash

Multiple Settlement Currencies

Two Settlement Currencies

Issuers can use two settlement currencies:

- The first for settling Transactions from ATMs in the Issuer's country
- The second for settling Transactions from ATMs outside the Issuer's country

Issuers choose these currencies during AEGNS Participant Setup.

Local/Non-Local ATM Transactions

The Network finds out if an ATM Financial Transaction Request (1200) or an ATM Reversal Advice (1420) is local by comparing the Issuer's Country Code in the Participant Setup records to the Country Code in the message. If the Country Codes are the same, the Transaction is considered local and the Network uses the Issuer's chosen local settlement currency to reconcile the Transaction. If the Country Codes are different, the Transaction is considered non-local and the Network uses the Issuer's chosen non-local settlement currency to reconcile the Transaction.

Sending and Receiving ATM Financial Messages

ATM Process

This table shows a typical ATM Transaction.

Table 3-1. ATM Transaction Cycle

Step	Who	Does What
1	Cardmember	Swipes or puts their Card into an ATM to get cash
2	Acquirer	Sends an ATM Financial Transaction Request (1200) to the Network
3	Network	Sends the request to the Cardmember's Issuer

Table 3-1. ATM Transaction Cycle (continued)

Step	Who	Does What
4	Issuer	Consults its internal Authorization rules and Cardmember data, and sends an ATM Financial Transaction Response (1210) to the Network within 8 seconds
5	Network	Sends the response to the Acquirer
6	Acquirer	Returns the response to the ATM, which then dispenses the cash or declines the request

Sending and Receiving ATM Transaction Reversals

When Is A Reversal Used?

An ATM Reversal Advice (1420) message is sent when:

- An ATM fails to dispense or dispenses only part of the cash amount requested.

Note: An ATM Acquirer can support full or partial dispenses of funds. In these two conditions, the amount dispensed is used in the reversal message.

- The Cardmember cancels the Transaction at the ATM before cash is dispensed.
- The ATM times out and denies the Transaction.
- A communications or other outage has caused a timeout during an ATM Financial Transaction Request (1200) message.

Reversal Process

This table shows a typical ATM reversal Transaction.

Table 3-2. ATM Reversal Transaction

Step	Who	Does What
1	Acquirer	Generates an ATM Reversal Advice (1420) message to the Network
2	Network	• Sends an ATM Reversal Advice Response (1430) to the Acquirer. • Sends an ATM Reversal Advice (1420) message to the Issuer to show a change to the Transaction. • The Issuer must accept Reversal Advice messages that show a partial or a complete reversal of the first Transaction.
3	Issuer	Responds with an ATM Reversal Advice Response (1430) and adjusts its Cardmember's account as necessary Note: If the Network does not receive an ATM Reversal Advice Response to an ATM Reversal Advice within 30 seconds, the reversal is considered timed out. The Network attempts to send an ATM Reversal Advice (1420) up to three times.

Outstanding reversals must not prevent the processing of new Transaction requests.

More Instruction

Refer to [POS and ATM 5CSC Validation on page 2-21](#) for instruction regarding 5CSC validation for ATM Transactions.

Refer to the *Network Connectivity* manual for instruction regarding ATM Transaction routing.

Section 2 ATM Reconciliation

ATM Reconciliation Overview

Description

When a Transaction occurs at an ATM, the Network considers the ATM Transaction only as an authorization message. The American Express Funds Access System (FAS) captures the Transaction data, creates the ATM First Presentment (1240) message, and sends the data to the Network. The Network sends the ATM First Presentment (1240) message to the Issuer.

Unmatched Messages

Occasionally, the message the ATM Acquirer sends will not match those received by the Network; therefore, FAS also uses the authorization Transaction data to create the ATM Reconciliation file. The Network sends the reconciliation file to the Acquirer. The file identifies which Transactions the Network is settling with the Acquirer based on what has been presented to the Issuer. The Network settles with the Acquirer based on the Transactions listed on the ATM Reconciliation file and the Daily Activity Report. Acquirers can use the Network ATM Reconciliation file to match against their internal ATM dispense detail report to reconcile the activity of all ATM Transactions through their system.

AEGNS recommends using the Daily Activity Gross Settlement Report in conjunction with the Reconciliation File to support the reconciliation process.

Reconciliation Process

Acquirers can identify these conditions during reconciliation:

- **Dispensed, but not settled/billed** - The ATM Acquirer has been underpaid by the Network. The Acquirer will raise an AEGNS ATM Exception/First Presentment (DR) and Reversal (CR) form. Refer to the Exhibits chapter of the *Business and Operational Policies* manual for an example of this form.
- **No dispense or partial dispense, but settled** - The ATM Acquirer has been paid by the Network and the Cardmember has been billed, although funds were not dispensed or were not fully dispensed. The ATM Acquirer will not take action while waiting for the Issuer to raise a Retrieval Request using the existing policy and communication.

The Network sends a Daily Activity Report to the Acquirer at the end of the day. This shows the total amount settled with the Acquirer for all Transactions with the Network.

The Daily Activity Report shows data from the First Presentment only; a separate daily Settlement Advice faxed to the Participant displays all other Transaction activity.

ATM Reconciliation Record Flow

Figure 3-1 shows the flow of an ATM Transaction Authorization Request. FAS captures the Transaction data and sends the data to the Network. The Network creates the ATM First Presentment record and sends it to the Issuer.

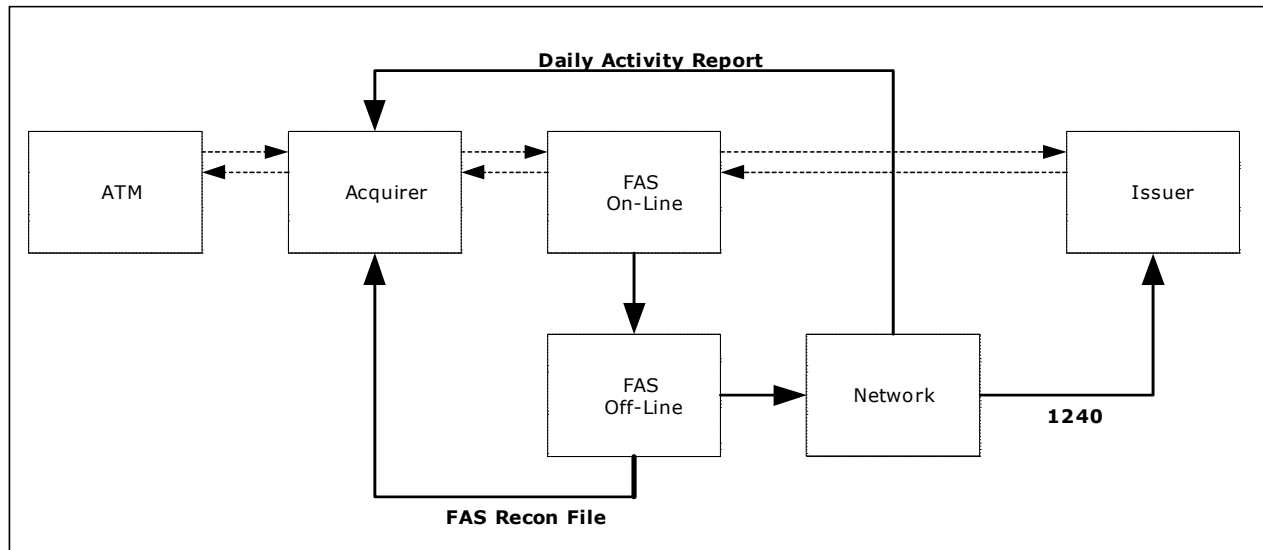


Figure 3-1. ATM Reconciliation Record Flow

ATM Reconciliation Header Record

Description

The File Header is the first data record in the file and serves as the highest level of control in a file. Each file can have only one File Header Record. It designates date, time, and file identification fields that uniquely identify the file from the other files processed through the Network. The forwarding institution for the file creates the File Header Record.

Length of Record

500 bytes

Message Layout

This is the message layout for the ATM Reconciliation Header Record. Refer to the [Programming Notes for ATM Reconciliation Header Record on page 3-5](#) for more instruction.

ATM Reconciliation Header Record Message Layout

Field Number	Format	Positions	Field Name
1	X(1)	1	Record Type
2	X(9)	2-10	Reserved Field
3	X(6)	11-16	Report Batch Number
4	X(1)	17	Filler
5	X(10)	18-27	Run Date
6	X(1)	28	Filler
7	X(8)	29-36	Run Time
8	X(1)	37	Filler
9	X(32)	38-69	Reserved Field
10	X(1)	70	Filler
11	X(430)	71-500	Filler

Programming Notes for [ATM Reconciliation Header Record](#)

Field 1

Definition**Format****Positions****Notes****Record Type**

A unique code that identifies a type of record.

X(1)

1

Required Value: 0 Header

A file that contains the header and trailer is sent to the Participant if no ATM Transaction detail records is submitted.

American Express Global Network Services
Network Specifications - Authorization
ATM Reconciliation Header Record
Field 2 Reserved FieldMessage Layout

Field 2

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(9)
2-10
Alphanumeric Reserved Fields must contain spaces.

Field 3

Definition
Format
Positions

Report Batch Number

A unique number assigned to the daily ATM Reconciliation report.
X(6)
11-16

Field 4

Definition

Format
Positions
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
X(1)
17
Alphanumeric Filler fields must contain spaces.

Field 5

Definition
Format
Positions

Run Date

The local date on which the Network creates the ATM Reconciliation file.
X(10) CCYYMMDD
18-27

Field 6

Definition

Format
Positions
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
X(1)
28
Alphanumeric Filler fields must contain spaces.

Field 7

Definition
Format
Positions

Run Time

The local time when the Network creates the ATM Reconciliation file.
X(8) HH:MM:SS
29-36

Field 8

Definition

Format
Positions
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
X(1)
37
Alphanumeric Filler fields must contain spaces.

Field 9

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(32)
38-69
Alphanumeric Reserved Fields must contain spaces.

Field 10

Definition

Format

Positions

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

X(1)

70

Alphanumeric Filler fields must contain spaces.

Field 11

Definition

Format

Positions

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

X(430)

71-500

Alphanumeric Filler fields must contain spaces.

ATM Record Message

Description

The ATM record contains data pertaining to a single Cardmember Transaction. This includes the account number, reference code, amount, and more data required by the Acquirer to reconcile the account.

Length of Record

500 bytes

Message Layout

This is the message layout for the ATM Record. Refer to the [Programming Notes for ATM Record Message on page 3-9](#) for more instruction.

ATM Record Message Layout

Field Number	Format	Positions	Field Name
1	X(1)	1	Record Type
2	9(5)	2-6	Record Sequence Number
3	X(19)	7-25	Primary Account Number (PAN)
4	X(8)	26-33	Transaction Date
5	X(6)	34-39	Transaction Time
6	X(8)	40-47	System Date
7	X(6)	48-53	System Time
8	X(8)	54-61	Settlement Date
9	X(8)	62-69	Card Acceptor Terminal Identification
10	X(6)	70-75	Systems Trace Audit Number
11	X(3)	76-78	Dispensed Currency
12	9(12)v999	79-93	Amount Requested
13	9(12)v999	94-108	Amount Indicator
14	9(5)v9(7)	109-120	Settlement Conversion Rate
15	X(3)	121-123	Settlement Currency Code
16	9(12)v999	124-138	Settlement Amount Requested
17	9(12)v999	139-153	Settlement Amount Approved
18	9(12)v999	154-168	Settlement Amount Dispensed
19	9(8)v999	169-179	Settlement Network Fee
20	9(8)v999	180-190	Settlement Fee Other
21	9(5)v9(7)	191-202	Reserved Field
22	9(12)v999	203-217	Reserved Field
23	9(12)v999	218-232	Reserved Field
24	9(12)v999	233-247	Reserved Field

ATM Record Message Layout (continued)

Field Number	Format	Positions	Field Name
25	X(3)	248-250	Reserved Field
26	9(12)v999	251-265	Reserved Field
27	9(8)v999	266-276	Reserved Field
28	X(2)	277-278	Terminal Country Code
29	X(2)	279-280	Card Acceptor Country Code
30	X(2)	281-282	Cardmember Billing Country Code
31	X(40)	283-322	Terminal Location
32	X(10)	323-332	Reserved Field
33	X(10)	333-342	Reserved Field
34	X(10)	343-352	Reserved Field
35	X(10)	353-362	Reserved Field
36	X(1)	363	Authorization Status
37	X(1)	364	Transaction Indicator
38	X(3)	365-367	Reserved Field
39	X(6)	368-373	Reserved Field
40	X(4)	374-377	Reserved Field
41	X(3)	378-380	Original Action Code
42	X(4)	381-384	Reserved Field
43	X(6)	385-390	Approval Code
44	X(8)	391-398	Additional Reference Number
45	X(1)	399	Reserved Field
46	9(15)	400-414	Transaction Identifier (TID)
47	X(86)	415-500	Reserved Field

Programming Notes for ATM Record Message

Field 1

Record Type

Definition	A unique code that identifies a type of record.
Format	X(1)
Positions	1
Notes	Required Value: 1 Detail

Field 2

Record Sequence Number

Definition	A sequential number assigned to each record by the file originator. This ascending number starts at 00001 for the first record and is incremented by one for each record in the file.
Format	9(5)
Positions	2-6

Field 3

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
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Format X(19)
Positions 7-25

Field 4 **Transaction Date**

Definition The local date when the Cardmember's original Transaction occurred at a Card Acceptor location or sent by the designated Authorized Processor.

- For telecommunication Transactions, the Transaction Date is the date the bill summary was presented.
- For Mail/Telephone/Internet Transactions, the Transaction Date is the date goods are shipped.

Format X(8) CCYYMMDD
Positions 26-33

Field 5 **Transaction Time**

Definition The local time when a Cardmember's original Transaction occurred at a Card Acceptor location or was sent by the designated Authorized Processor.

On a Second Presentment, this is the time the second presentment Transaction was executed by the Acquirer or its designated Authorized Processor.

Format X(6) HH:MM:SS
Positions 34-39

Field 6 **System Date**

Definition The date the Funds Access System (FAS) accepted the ATM Transaction.

Format X(8) CCYYMMDD
Positions 40-47

Field 7 **System Time**

Definition The time the Funds Access System (FAS) accepted the ATM Transaction.

Format X(6) HH:MM:SS
Positions 48-53

Field 8 **Settlement Date**

Definition The date when the Network calculates the Settlement Amount for one Cardmember Transaction or for one fee. All Transactions transmitted together (e.g., in the same file) by a Participant or all fees from the Network have the same date.

Format X(8) CCYYMMDD
Positions 54-61

Field 9 **Card Acceptor Terminal Identification**

Definition This code identifies a POS terminal at a Card Acceptor location.

Format X(8)
Positions 62-69

Field 10 **Systems Trace Audit Number**

Definition This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format X(6)
Positions 70-75

Field 11 **Dispensed Currency**

Definition The currency code of the currency in which the First Presentment Transaction Amount was executed at the ATM.

Format X(3)
Positions 76-78

Notes Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Field 12 **Amount Requested**

Definition The requested amount of the ATM Transaction in the local currency.
Format 9(12)v999
Positions 79-93

Field 13 **Amount Indicator**

Definition Shows the Participant Net Settlement Amount for a Transaction or group of Transactions as a debit, credit, or zero.
Format 9(12)v999
Positions 94-108

Field 14 **Settlement Conversion Rate**

Definition The rate used to convert the dispensed amount to the settlement amount. If no conversion is required, this value will be "000010000000".
Format 9(5)v9(7)
Positions 109-120

Field 15 **Settlement Currency Code**

Definition The currency code of the Settlement Amount.
Format X(3)
Positions 121-123
Notes Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Field 16 **Settlement Amount Requested**

Definition The amount requested by the Cardmember at the ATM in the Acquirer's settlement currency.
Format 9(12)v999
Positions 124-138

Field 17 **Settlement Amount Approved**

Definition The amount the Issuer approved for the Cardmember ATM Transaction in the Acquirer's settlement currency.
Format 9(12)v999
Positions 139-153

Field 18 **Settlement Amount Dispensed**

Definition The actual amount dispensed to the Cardmember for the ATM Transaction.
Format 9(12)v999
Positions 154-168

Field 19 **Settlement Network Fee**

Definition A fee paid to the Acquirer by the Network for the use of the ATM.
Format 9(8)v999
Positions 169-179

Field 20 **Settlement Fee Other**

Definition A fee charged by the Network for performing the foreign exchange conversion.
Format 9(8)v999
Positions 180-190

Field 21

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
9(5)v9(7)
191-202
Numeric Reserved Fields must contain zeros.

Field 22

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
9(12)v999
203-217
Numeric Reserved Fields must contain zeros.

Field 23

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
9(12)v999
218-232
Numeric Reserved Fields must contain zeros.

Field 24

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
9(12)v999
233-247
Numeric Reserved Fields must contain zeros.

Field 25

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(3)
248-250
Alphanumeric Reserved Fields must contain spaces.

Field 26

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
9(12)v999
251-265
Numeric Reserved Fields must contain zeros.

Field 27

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
9(8)v999
266-276
Numeric Reserved Fields must contain zeros.

Field 28

Definition

Format
Positions
Notes

Terminal Country Code

Identifies where the terminal is located. The Network supplies the alpha country code of the terminal at the Card Acceptor location.
X(2)
277-278
Refer to the Country Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

American Express Global Network Services
Network Specifications - Authorization
ATM Record Message
Field 29 Card Acceptor Country CodeMessage Layout

Field 29

Definition
Format
Positions
Notes

Card Acceptor Country Code

The Country Code of the Card Acceptor location.
X(2)
279-280
Contact your AEGNS representative for the applicable 2-byte code.
Refer to the Country Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Field 30

Definition
Format
Positions
Notes

Cardmember Billing Country Code

The Country Code of the Cardmember based on the issuing BIN.
X(2)
281-282
Refer to the Country Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Field 31

Definition
Format
Positions

Terminal Location

Free form text that describes where the ATM is located.
X(40)
283-322

Field 32

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(10)
323-332
Alphanumeric Reserved Fields must contain spaces.

Field 33

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(10)
333-342
Alphanumeric Reserved Fields must contain spaces.

Field 34

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(10)
343-352
Alphanumeric Reserved Fields must contain spaces.

Field 35

Definition
Format
Positions
Notes

Reserved Field

This field is for AEGNS internal use.
X(10)
53-362
Alphanumeric Reserved Fields must contain spaces.

Field 36

Definition
Format
Positions

Authorization Status

Indicates if the Issuer approved the Authorization request for the Transaction.
X(1)
363

Field 37

Definition

Transaction Indicator

Indicates the status of the Transaction based on the Transaction logs at the time of the authorization request.

Format	X(1)
Positions	364
Notes	Valid Values: • H = Normal Transaction processed without errors and dispensed • E = Transaction had errors and was not able to dispense (not approved, invalid Transactions, etc.) • C = Correction Transaction

Field 38

Definition	Reserved Field This field is for AEGNS internal use.
Format	X(3)
Positions	365-367
Notes	Alphanumeric Reserved Fields must contain spaces.

Field 39

Definition	Reserved Field This field is for AEGNS internal use.
Format	X(6)
Positions	368-373
Notes	Alphanumeric Reserved Fields must contain spaces.

Field 40

Definition	Reserved Field This field is for AEGNS internal use.
Format	X(4)
Positions	374-377
Notes	Alphanumeric Reserved Fields must contain spaces.

Field 41

Definition	Original Action Code A code that echoes the Action Code supplied on the Participant's original file transmission. Refer to the Action Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Format	X(3)
Positions	378-380
Notes	Required Value: 000 Approved

Field 42

Definition	Reserved Field This field is for AEGNS internal use.
Format	X(4)
Positions	381-384
Notes	Alphanumeric Reserved Fields must contain spaces.

Field 43

Definition	Approval Code A unique code assigned by the Card Transaction authorizing institution to indicate the disposition for the Transaction.
Format	X(6)
Positions	385-390
Notes	The Approval Code is mandatory. Do not use spaces.

Field 44

Definition	Additional Reference Number A number assigned by the Acquirer to assist in locating/retrieving source data and documentation.
Format	X(8)
Positions	391-398

Field 45

Definition**Format****Positions****Notes**

Reserved Field

This field is for AEGNS internal use.

X(1)

399

Alphanumeric Reserved Fields must contain spaces.

Field 46

Definition**Format****Positions****Notes**

Transaction Identifier (TID)

A unique number given by the Network to identify a Cardmember Transaction throughout its life cycle.

9(15)

400-414

For instruction about the TID and the authorization life cycle, see [The Role of the Transaction Identifier on page 2-2](#). For instruction about the TID and the clearing life cycle, see The Role of the Transaction Identifier in the *Network Specifications - Financial/Non-Financial* manual.

Field 47

Definition**Format****Positions****Notes**

Reserved Field

This field is for AEGNS internal use.

X(86)

415-500

Alphanumeric Reserved Fields must contain spaces.

ATM Reconciliation Trailer Record

Description

The file trailer is the last data record in the file. It includes control totals used to evaluate the integrity of the file. The forwarding institution creates the file trailer for the file.

Length of Record

500 bytes

Message Layout

This is the message layout for the ATM Reconciliation Trailer Record. Refer to the [Programming Notes for ATM Reconciliation Trailer Record on page 3-16](#) for more instruction.

ATM Reconciliation Trailer Record Message Layout

Field Number	Format	Positions	Field Name
1	X(1)	1	Record Type
2	9(5)	2-6	Record Sequence Number
3	X(4)	7-10	Reserved Field
4	9(5)	11-15	Total Number of Records
5	X(1)	16	Filler
6	9(12).999	17-32	Total Settlement Dispensed Amount
7	X(468)	33-500	Filler

Programming Notes for [ATM Reconciliation Trailer Record](#)

Field 1

[Record Type](#)

Definition	A unique code that identifies a type of record.
Format	X(1)
Positions	1
Notes	Required Value: 9 Trailer

Field 2

[Record Sequence Number](#)

Definition	A sequential number assigned to each record by the file originator. This ascending number starts at 00001 for the first record and is incremented by one for each record in the file.
Format	9(5)
Positions	2-6

Field 3

[Reserved Field](#)

Definition	This field is for AEGNS internal use.
Format	X(4)
Positions	7-10
Notes	Alphanumeric Reserved Fields must contain spaces.

American Express Global Network Services
Network Specifications - Authorization
ATM Reconciliation Trailer Record
Field 4 Total Number of RecordsMessage Layout

Field 4

Definition
Format
Positions

Total Number of Records

Total number of records in the file. The header and trailer are not included in this total.
9(5)
11-15

Field 5

Definition
Format
Positions
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
X(1)
16
Alphanumeric Filler fields must contain spaces.

Field 6

Definition
Format
Positions

Total Settlement Dispensed Amount

Hash total of all Settlement Amount Dispensed fields of all records in the file.
9(12)v999
17-32

Field 7

Definition
Format
Positions
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
X(468)
33-500
Alphanumeric Filler fields must contain spaces.

4 Message Encryption

Full Message Encryption

What Is Full Message Encryption?

Full Message Encryption is the encryption of the full body of the message using encryption keys and Message Authentication Codes (MAC).

Full Message Encryption includes:

- The use of a message header to pass data that uniquely identifies the message and key information for message encryption.
- The use of a message footer to pass the Message Authentication Code (MAC).

Encryption Header

Placement

The full message encryption header is placed after the Transmission Control Protocol/Internet Protocol (TCP/IP) header and associated VLI but before the Message Type Identifier (MTI) of the message being encrypted.

Correlation Identifier (ID)

A unique identifier defined by the sender that enables them to correlate a rejected message to the original message.

Sender

Generate and send a uniquely assigned ID, enabling the sender to receive a rejected message header from the receiver with the same Correlation ID. This allows the sender to match and identify the rejected message header back to the original message.

Receiver

If the receiver is unable to decrypt the message, the Correlation ID must be echoed back to the sender. This allows the sender to match the rejected header to the original message.

Decryption Indicator

Defines if the receiver successfully decrypted or was unable to decrypt the inbound message.

Sender

The creator of the message containing an encrypted message body must send a character space in the Decryption Indicator Field as they are not the receiver and cannot indicate if Decryption was successful or not.

Receiver

If the receiver of a message with an encrypted body is required to respond to the message received, they must include the appropriate Decryption Indicator.

Encryption Footer

Placement

The full message encryption trailer contains the Message Authentication Code (MAC) and is placed after the body of the encrypted message except when a decryption failure occurs.

Decryption Failures

Message Requirements

The following outlines Message Encryption requirements when the receiver of the message is unable to decrypt the message AND is required to respond with a corresponding response message:

1. The response message must only contain an Encryption Header.
 - No encrypted message content should be sent
 - No encryption footer should be sent
2. The Correlation Identifier (ID) - Bit 1 from the Outclear Header message must be echoed back in the corresponding Inclear Header response message.
3. The Decryption Indicator - Bit 2 in the Outclear Header message of an encrypted message must indicate a Decryption Failure and be populated with a value of "1" in the corresponding Inclear Header response message.
4. The Key Identifier (ID) - Bit 14 from the Outclear Header message must be echoed back in the corresponding Inclear Header response message. Passing back the Key ID will allow Issuers to determine the specific key for which an encryption error was encountered.

Example: 1110, 1314, etc.

If a message does not require a corresponding response message, then the business requirements defined by the receiver for failed decryption must be used.

Full Message Encryption Availability

Messages

Full Message Encryption is available for the following messages:

- POS Authorization Request (1100)
- POS Authorization Response (1110)
- POS Authorization Advice (1120)
- POS Authorization Advice Response (1130)
- POS Authorization Reversal Advice (1420)
- POS Authorization Reversal Advice Response (1430)
- POS Authorization Adjustment Financial Advice (1220)
- POS Authorization Adjustment Financial Advice Response (1230)
- ATM Financial Transaction Request (1200)

- ATM Financial Transaction Response (1210)
- ATM Reversal Advice (1420)
- ATM Reversal Advice Response (1430)
- Real Time Exception Request (1304)
- Real Time Exception Response (1314)

Full Message Encryption Outclear

What Is Outclear?

Outclear is the request message transmitted by the sender

Example 1: An Acquirer would Outclear a POS Authorization Request (1100) message to the Network and the Network would outclear the message to the Issuer.

Example 2: An Issuer would Outclear a Real Time Exception Request (1304) message to the Network.

Outclear Header

Outclear Header Message Layout

This table shows the Outclear Header layout for full message encryption. Refer to [Inclear Header Programming Notes on page 4-8](#) for more instruction.

Outclear Header Message Layout

Bit Position	Format	Optionality		Data Element
		A to N	N to I	
N/A	2 Bytes Binary Hex	N/A	M	Header Variable Length Indicator (VLI)
1	20 Bytes Binary Hex	N/A	M	Correlation Identifier (ID)
2	AN(1)	N/A	F	Decryption Indicator
3	7 Bytes Binary Hex	N/A	M	Date and time Creation
4	N(4)	N/A	M	Message Type Identifier
5	AN(1)	N/A	M	Key Management Method
6	AN(1)	N/A	M	PIN Block Format
7	AN(1)	N/A	M	Encryption Algorithm
8	AN(1)	N/A	M	Encryption Mode
9	AN(1)	N/A	M	MAC Mode
10	AN(1)	N/A	F	Reserved Field
11	AN(1)	N/A	F	Reserved Field
12	AN(1)	N/A	F	Reserved Field
13	N(2)	N/A	M	Key Identifier (ID) VLI

Outclear Header Message Layout(continued)

Bit Position	Format	Optionality		Data Element
		A to N	N to I	
14	Binary Hex (15) Max	N/A	M	Key Identifier (ID)
15	N(4)	N/A	M	Encrypted message VLI

Outclear Header Programming Notes

Bit N/A

Definition

Header Variable Length Indicator (VLI)

Header VLI is the total length of the variable data in the Encryption Header (not including the VLI).

Format

2 Bytes, Binary Hex

Optionality

Acquirer to Network: N/A

Network to Issuer: Mandatory

Bit 1

Definition

Correlation Identifier (ID)

Correlation ID is a unique ID, defined by the sender of the outclear header, that enables them to correlate a rejected message to the original message.

Format

20 Bytes, Binary Hex

Optionality

Acquirer to Network: N/A

Network to Issuer: Mandatory

Bit 2

Definition

Decryption Indicator

Indicates if the receiver successfully decrypted or was unable to decrypt the inbound message.

Format

1 Byte, Alphanumeric

Optionality

Acquirer to Network: N/A

Network to Issuer: Filler

Notes

The creator of the message containing an encrypted message body must send a character space within the Decryption Indicator field

Valid Value: ~ (This character represents a space)

Bit 3

Definition

Date and time Creation

Date and time that the message was created by the sender.

Format

7 Byte, Binary Hex YYYYMMDDHHMMSS

Optionality

Acquirer to Network: N/A

Network to Issuer: Mandatory

Bit 4

Definition

Message Type Identifier

This code identifies the type of message sent by the forwarding institution.

Format

4 Bytes, Numeric

Optionality

Acquirer to Network: N/A

Network to Issuer: Mandatory

Bit 5

Definition

Key Management Method

Defines the Key Management Method being used.

Format

1 Byte, Alphanumeric

Optionality

Acquirer to Network: N/A

Notes	Network to Issuer: Mandatory Valid Values: 1 = DUKPT (Derived Unique Key Per Transaction) 2 = DKE (Dynamic Key Encryption) 3 = Reserved for AEGNS
Bit 6	PIN Block Format
Definition	Defines the Pin Block Format being used.
Format	1 Byte, Alphanumeric
Optionality	Acquirer to Network: N/A
Notes	Network to Issuer: Mandatory Valid Values: 0 = Reserved for AEGNS 1 = Reserved for AEGNS 4 = ISO 4 (All AES) ~ = represents a character space indicating no PIN Block is present.
Bit 7	Encryption Algorithm
Definition	Defines the Encryption Algorithm being used.
Format	1 Byte, Alphanumeric
Optionality	Acquirer to Network: N/A
Notes	Network to Issuer: Mandatory Valid Values: 0 = AES 128 1 = Reserved for AEGNS 2 = Reserved for AEGNS 3 = Reserved for AEGNS 4 = Reserved for AEGNS
Bit 8	Encryption Mode
Definition	Defines the Encryption Mode being used.
Format	1 Byte, Alphanumeric
Optionality	Acquirer to Network: N/A
Notes	Network to Issuer: Mandatory Valid Values: 1 = CBC 2 = Reserved for AEGNS
Bit 9	MAC Mode
Definition	Defines the MAC Mode being used.
Format	1 Byte, Alphanumeric
Optionality	Acquirer to Network: N/A
Notes	Network to Issuer: Mandatory Valid Values: 1 = Reserved for AEGNS 2 = AESCMAC

- 3 = Reserved for AEGNS

Bit 10

Definition
Format
Optionality

Note

Reserved Field

This field is for AEGNS internal use.
1 Byte, Alphanumeric
Acquirer to Network: N/A
Network to Issuer: Filler
Alphanumeric Reserved Fields must contain spaces.

Bit 11

Definition
Format
Optionality

Note

Reserved Field

This field is for AEGNS internal use.
1 Byte, Alphanumeric
Acquirer to Network: N/A
Network to Issuer: Filler
Alphanumeric Reserved Fields must contain spaces.

Bit 12

Definition
Format
Optionality

Note

Reserved Field

This field is for AEGNS internal use.
1 Byte, Alphanumeric
Acquirer to Network: N/A
Network to Issuer: Filler
Alphanumeric Reserved Fields must contain spaces.

Bit 13

Definition

Format
Optionality

Key Identifier (ID) (VLI)

Total length of the variable Key ID data (not including VLI). VLI is the total length of the variable data in the KSN (not including VLI).
2 Bytes, Numeric (EBCDIC) LLVAR
Acquirer to Network: N/A
Network to Issuer: Mandatory

Bit 14

Definition
Format
Optionality

Notes

Key Identifier (ID)

Key ID is a key check value that uniquely identifies the key.
15 Bytes Max, Binary Hex
Acquirer to Network: N/A
Network to Issuer: Mandatory
If Key Management Method = DKE then Format = 15 bytes of data comprising PIN, Data and MAC KCV's, each with their own blocks of 5 bytes (10 nibbles Max).
If Key Management Method = DUK then the data will be comprised of the KSN, DID, and TC, each with their own respective blocks of either:

- 10 bytes (80 bits) consisting of 10-5-5 nibbles OR
- 12 bytes (96 bits) divided into three (3) groups of four (4) bytes (32 bits Max).

Bit 15

Definition

Format
Optionality

Encrypted Message VLI

Encrypted Message VLI is the total length of the encrypted variable data following the header (not including the VLI).
4 Bytes Max, Numeric (EBCDIC) LLLLVAR
Acquirer to Network: N/A
Network to Issuer: Mandatory

Outclear Trailer

Outclear Trailer Message Layout

This table shows the Outclear Trailer layout for full message encryption. Refer to [Outclear Trailer Programming Notes](#) for more instruction.

Outclear Trailer Message Layout

Bit Position	Format	Optionality		Data Element
		A to N	N to I	
N/A	4 Bytes Binary Hex	N/A	M	Message Encryption Trailer Length Authentication Code (MAC) VLI
1	58 Bytes Max Binary Hex	N/A	M	Message Encryption Trailer Message Authentication Code (MAC)

Outclear Trailer Programming Notes

Bit N/A

[Message Encryption Trailer Length Authentication Code \(MAC\) VLI](#)

Definition	Header VLI is the total length of the variable data in the Encryption Trailer MAC (not including the VLI).
Format	4 Bytes, Binary Hex
Optionality	Acquirer to Network: N/A Network to Issuer: Mandatory

Bit 1

[Message Encryption Trailer Message Authentication Code \(MAC\)](#)

Definition	The MAC is a cryptographic value used to confirm that the message came from the indicated sender and has not been changed in transit.
Format	58 Bytes, Binary Hex
Optionality	Acquirer to Network: N/A Network to Issuer: Mandatory

Full Message Encryption Inclear

What Is Inclear?

Inclear is the response message transmitted by the receiver.

Example 1: An Issuer would inclear a POS Authorization Response (1110) message to the Network (after receiving the outcleared POS Authorization Request (1100) and the Network would inclear the response message to the Acquirer.

Example 2: The Network would inclear a Real Time Exception Response (1314) message back to the Issuer (after receiving the outcleared Real Time Exception Request (1304) message).

Inclear Header

Inclear Header Message Layout

This table shows the Inclear Header layout for full message encryption. Refer to [Inclear Header Programming Notes](#) for more instruction.

Inclear Header Message Layout

Bit Position	Format	Optionality		Data Element
		I to N	N to A	
N/A	2 Bytes Binary Hex	M	N/A	Header Variable Length Indicator (VLI)
1	20 Bytes Binary Hex	ME	N/A	Correlation Identifier (ID)
2	AN(1)	M	N/A	Decryption Indicator
3	7 Bytes Binary Hex	M	N/A	Date and time Creation
4	N(4)	M	N/A	Message Type Identifier
5	AN(1)	M	N/A	Key Management Method
6	AN(1)	M	N/A	PIN Block Format
7	AN(1)	M	N/A	Encryption Algorithm
8	AN(1)	M	N/A	Encryption Mode
9	AN(1)	M	N/A	MAC Mode
10	AN(1)	F	N/A	Reserved Field
11	AN(1)	F	N/A	Reserved Field
12	AN(1)	F	N/A	Reserved Field
13	N(2)	M	N/A	Key Identifier (ID) (VLI)
14	Binary Hex (15) Max	M	N/A	Key Identifier (ID)
15	N(4)	M	N/A	Encrypted Message VLI

Inclear Header Programming Notes

Bit N/A

[Header Variable Length Indicator \(VLI\)](#)

Definition	Header VLI is the total length of the variable data in the Encryption Header (not including the VLI).
Format	2 Bytes, Binary Hex
Optionality	Issuer to Network: Mandatory Network to Acquirer: N/A

Bit 1

[Correlation Identifier \(ID\)](#)

Definition	Correlation ID is a unique ID, defined by the sender of the outclear header, that enables them to correlate a rejected message to the original message.
Format	20 Bytes, Binary Hex

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Inclear Header
Bit 2 Decryption IndicatorInclear Header Message Layout

Optionality **Issuer to Network:** Mandatory Echo. The Correlation ID from the outclear header must be echoed back in the inclear header.

Network to Acquirer: N/A

Bit 2

Decryption Indicator

Definition Indicates if the receiver successfully decrypted or was unable to decrypt the inbound message.

Format 1 Byte, Alphanumeric

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: N/A

Notes **Valid Value:**

- 0 = Decryption Successful
- 1 = Decryption Failure

Bit 3

Date and time Creation

Definition Date and time that the message was created by the sender.

Format 7 Byte, Binary Hex YYYYMMDDHHMMSS

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: N/A

Bit 4

Message Type Identifier

Definition This code identifies the type of message sent by the forwarding institution.

Format 4 Bytes, Numeric

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: N/A

Bit 5

Key Management Method

Definition Defines the Key Management Method being used.

Format 1 Byte, Alphanumeric

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: N/A

Notes **Valid Values:**

- 1 = DUKPT (Derived Unique Key Per Transaction)
- 2 = DKE (Dynamic Key Encryption)
- 3 = Reserved for AEGNS

Bit 6

PIN Block Format

Definition Defines the Pin Block Format being used.

Format 1 Byte, Alphanumeric

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: N/A

Notes **Valid Values:**

- 0 = Reserved for AEGNS
- 1 = Reserved for AEGNS
- 4 = ISO 4 (All AES)
- ~ = represents a Character Space indicating no PIN Block is present.

Bit 7

Encryption Algorithm

Definition Defines the Encryption Algorithm being used.

Format	1 Byte, Alphanumeric
Optionality	Issuer to Network: Mandatory
Notes	Network to Acquirer: N/A Valid Values: • 0 = AES 128 • 1 = Reserved for AEGNS • 2 = Reserved for AEGNS • 3 = Reserved for AEGNS • 4 = Reserved for AEGNS

Bit 8	Encryption Mode
Definition	Defines the Encryption Mode being used.
Format	1 Byte, Alphanumeric
Optionality	Issuer to Network: Mandatory
Notes	Network to Acquirer: N/A Valid Values: • 1 = CBC • 2 = Reserved for AEGNS

Bit 9	MAC Mode
Definition	Defines the MAC Mode being used.
Format	1 Byte, Alphanumeric
Optionality	Issuer to Network: Mandatory
Notes	Network to Acquirer: N/A Valid Values: • 1 = Reserved for AEGNS • 2 = AESCMAC • 3 = Reserved for AEGNS

Bit 10	Reserved Field
Definition	This field is for AEGNS internal use.
Format	1 Byte, Alphanumeric
Optionality	Issuer to Network: Filler
Note	Network to Acquirer: N/A Alphanumeric Reserved Fields must contain spaces.

Bit 11	Reserved Field
Definition	This field is for AEGNS internal use.
Format	1 Byte, Alphanumeric
Optionality	Issuer to Network: Filler
Note	Network to Acquirer: N/A Alphanumeric Reserved Fields must contain spaces.

Bit 12	Reserved Field
Definition	This field is for AEGNS internal use.
Format	1 Byte, Alphanumeric
Optionality	Issuer to Network: Filler

Bit 13 Key Identifier (ID) (VLI) IncClear Trailer Message Layout
Table

Note **Network to Acquirer:** N/A
Alphanumeric Reserved Fields must contain spaces.

Bit 13 **Key Identifier (ID) (VLI)**
Definition Total length of the variable Key ID data (not including VLI). VLI is the total length of the variable data in the KSN (not including VLI).
Format 2 Bytes, Numeric (EBCDIC) LLVAR
Optionality **Issuer to Network:** Mandatory
Network to Acquirer: N/A

Bit 14 **Key Identifier (ID)**
Definition Key ID is a key check value that uniquely identifies the key.
Format 15 Bytes Max, Binary Hex
Optionality **Issuer to Network:** Mandatory
Network to Acquirer: N/A
Notes If Key Management Method = DKE then Format = 15 bytes of data comprising PIN, Data and MAC KCV's, each with their own blocks of 5 bytes (10 nibbles Max).
If Key Management Method = DUK then the data will be comprised of the KSN, DID, and TC, each with their own respective blocks of either:

- 10 bytes (80 bits) consisting of 10-5-5 nibbles OR
- 12 bytes (96 bits) divided into three (3) groups of four (4) bytes (32 bits Max).

Bit 15 **Encrypted Message VLI**
Definition Encrypted Message VLI is the total length of the encrypted variable data following the header (not including the VLI).
Format 4 Bytes Max, Numeric (EBCDIC) LLLLVAR
Optionality **Issuer to Network:** Filler
Network to Acquirer: N/A
Notes If the receiver of the message is unable to decrypt the message AND is required to respond with a response message (ex 1110, 1314, etc) then that response message must not contain an encrypted message body and Encrypted Message VLI MUST indicate zero (0).

IncClear Trailer

IncClear Trailer Message Layout Table

This table shows the IncClear Trailer layout for full message encryption. Refer to [IncClear Trailer Programming Notes](#) for more instruction.

IncClear Trailer Message Layout

Bit Position	Format	Optionality		Data Element
		I to N	N to A	
N/A	4 Bytes Binary Hex	C	N/A	Message Encryption Trailer Length Authentication Code (MAC) VLI
1	58 Bytes Max Binary Hex	C	N/A	Message Encryption Trailer Message Authentication Code (MAC)

Inclear Trailer Programming Notes

Bit N/A

Message Encryption Trailer Length Authentication Code (MAC) VLI

Definition	Header VLI is the total length of the variable data in the Encryption Trailer MAC (not including the VLI).
Format	4 Bytes, Binary Hex
Optionality	Issuer to Network: C Network to Acquirer: N/A
Notes	If the receiver of the message is unable to decrypt the message AND is required to respond with a response message (ex 1110, 1314, etc) then that response message must not contain an Encryption Trailer.

Bit 1

Message Encryption Trailer Message Authentication Code (MAC)

Definition	The MAC is a cryptographic value used to confirm that the message came from the indicated sender and has not been changed in transit.
Format	58 Bytes, Binary Hex
Optionality	Issuer to Network: C Network to Acquirer: N/A
Notes	If the receiver of the message is unable to decrypt the message AND is required to respond with a response message (ex 1110, 1314, etc) then that response message must not contain an Encryption Trailer.

5 Authorization Messages

POS Authorization Request (1100)

Description

This message is used to request Transaction authorization to purchase goods or services at an S/E.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Request (1100). Refer to the [Programming Notes for POS Authorization Request \(1100\)](#) on page 5-4 for more instruction.

POS Authorization Request (1100) Message Layout

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
	N/A	N(4)		M	M	Message Type Identifier
	N/A	8 bytes, 64 bits for each bitmap Binary (Hex)		M	M	Bitmap Primary
	1	8 bytes, 64 bits for each bitmap Binary (Hex)		C	C	Bitmap Secondary
	2	N(21) max LLVAR		M	M	Primary Account Number (PAN)
	3	N(6)		M	M	Processing Code
	4	N(12)		M	M	Amount, Transaction
	5	N(12)		N/A	M	Amount, Reconciliation
	7	N(10)		O	N/A	Date and Time, Transmission
	11	N(6)		M	M	Systems Trace Audit Number
	12	N(12)		M	M	Date and Time, Local Transaction
	13	N(4)		O	O	Card Effective Date
	14	N(4)		C	C	Card Expiration Date
	18	N(4)		N/A	N/A	Reserved Field
	19	N(3)		M	C	Country Code, Acquiring Institution
	22	AN(12)		M	M	Point of Service Data Code
	23	N(3)		N/A	C	PAN Sequence Number
	24	AN(3)		C	C	Function Code

POS Authorization Request (1100) Message Layout (continued)

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
	26	N(4)		M	M	Card Acceptor Business Code
	31	N(40) max LLVAR		N/A	M	Acquirer Reference Data, Transaction Identifier (TID)
	32	N(13) max LLVAR		M	M	Acquiring Institution Identification (AIN) Code
	34	AN(30) max LLVAR (including length indicator)		N/A	C	Primary Account Number, Extended
	35	ANS(39) max LLVAR		C	C	Track 2 Data
	37	AN(12)		O	O	Acquirer Reference Number (ARN)
	41	ANS(8)		O	O	Card Acceptor Terminal Identification
	42	AN(15)		M	M	Card Acceptor Identification Code
	43	ANS(101) max LLVAR		C	C	Card Acceptor Name and Location
	45	ANS(78) max LLVAR		C	C	Track 1 Data
	48	AN(353) max LLLVAR		N/A	C	Deferred Payment Plan (DPP)
	49	N(3)		M	M	Currency Code, Transaction
	50	N(3)		N/A	M	Currency Code, Reconciliation
	52	Binary (8)		C	C	Personal Identification Number (PIN)
	53	AN(8) max LLVAR		C	C	Security Related Control Information
	55	AN(259) max (includes length indicator) LLLVAR		C	C	ICC System Related Data
		AN(4) EBCDIC	1	M	M	ICC Header Version Name
		(2) Binary (numeric)	2	M	M	ICC Header Version Number
9F26		(8) Binary Hexadecimal	3	M	M	Application Cryptogram

POS Authorization Request (1100) Message Layout (continued)

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
9F10		(33) max LLVAR Binary Hexadecimal (includes length LL indicator)	4	M	M	Issuer Application Data (IAD) (32) Sub-Subfield IAD Data Length (1) Derivation Key Index (1) Cryptogram Version Number (1) Card Verification Results (CVR) (4)
9F37		(4) Binary Hexadecimal	5	M	M	Unpredictable Number
9F36		(2) Binary Hexadecimal	6	M	M	Application Transaction Counter (ATC)
95		(5) Binary Hexadecimal	7	M	M	Terminal Verification Results (TVR)
9A		(3) Binary (numeric)	8	M	M	Transaction Date
9C		(1) Binary (numeric)	9	M	M	Transaction Type
9F02		(6) Binary (numeric)	10	M	M	Amount Authorized
5F2A		(2) Binary (numeric)	11	M	M	Transaction Currency Code
9F1A		(2) Binary (numeric)	12	M	M	Terminal Country Code
82		(2) Binary Hexadecimal	13	M	M	Application Interchange Profile (AIP)
9F03		(6) Binary (numeric)	14	M	M	Amount Other
5F34		(1) Binary (numeric)	15	M	M	Application PAN Sequence Number
9F27		(1) Binary Hexadecimal	16	M	M	Cryptogram Information Data
			17	C	C	ICC Header Version Name
			18	C	C	ICC Header Version Number
			19	C	C	Issuer Authentication Data

POS Authorization Request (1100) Message Layout (continued)

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
	60	AN(210) max LLLVAR		C	C	Market Specific Data
		AN(353) max LLLVAR				Market Specific Data with Additional Authorization Data
	61	4 bytes min, 103 bytes max LLLVAR		C	C	National Use Data
	62	AN(5) min AN(60) max LLLVAR		N/A	C	Fraud Indicators
	63	AN(4) min AN(103) max LLLVAR		C	C	Automated Address Verification
	111	AN(44) max LLLLVAR		C	C	Encryption Data
	112	X(82) LLLVAR		C	C	Payment Account Reference (PAR)
	113	X(18) LLLVAR		O	O	Acceptance Environment Data

Programming Notes for POS Authorization Request (1100)

Bit N/A

Message Type Identifier

Definition

This code identifies the type of message sent by the forwarding institution.

Format

N(4)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Required Value: 1100

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

Format

8 bytes, 64 bits for each bitmap, Binary (Hex)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

If the field is mandatory or optional and the Participant uses that field, its assigned bitmap position must contain a value of 1. If the field is optional and is not used, its assigned bitmap position must contain a value of 0.

Bit 1

Bitmap Secondary

Definition

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

Format

8 bytes, 64 bits for each bitmap, Binary (Hex)

Optionality

Acquirer to Network: Conditional

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Bit 2 Primary Account Number (PAN) Message Layout

Notes	Network to Issuer: Conditional If the field is mandatory or optional and the Participant uses that field, its assigned bitmap position must contain a value of 1. If the field is optional and is not used, its assigned bitmap position must contain a value of 0.
Bit 2	Primary Account Number (PAN)
Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Acquirer to Network: Mandatory
Notes	Network to Issuer: Mandatory A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit. For example, a Cardmember Account Number of 3759-876543-21001 appears as 15375987654321001 Expresspay Transaction This number is supplied by the chip and can be different from the number on the Card, the magnetic stripe, or the contact chip. Payment Token Transaction Acquirer to Network: This bit contains the token. Network to Issuer: This bit contains the PAN.
Bit 3	Processing Code
Definition	This is the service being requested.
Format	N(6)
Optionality	Acquirer to Network: Mandatory
Notes	Network to Issuer: Mandatory Automated Address Verification If either 174800, 004800, or 200800 is used, Bit 63 is required. Only Acquirers certified for the Automated Address Verification services can use these values and send Bit 63. The Network can convert this value to 004000 before it sends a request to a non-certified Issuer or from a non-certified Acquirer to an Issuer. Variable Authorization For Estimated and Incremental Authorizations, either a 004000 or 004800 is required. Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 4	Amount, Transaction
Definition	The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.
Format	N(12)
Optionality	Acquirer to Network: Mandatory
Notes	Network to Issuer: Mandatory Bit 4 Amount, Transaction and Bit 49 Currency Code, Transaction are in the local currency in which the Transaction was executed at the Card Acceptor location. The decimal point is determined by the Currency Code, Transaction (Bit 49). For example, \$100.00 in U.S. Dollars shows as 000000010000. Deferred Payment Plan For Deferred Payment Plan airline Transactions, this amount includes the down payment and boarding fee.

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 Bit 5 Amount, Reconciliation Message Layout

Expresspay

For Expresspay-enabled Card/device Application Transaction Counter (ATC) value requests, Bit 4 must be less than or equal to one unit of currency (for example: \$1.00 or £1.00 or €1.00)

Bit 5

Amount, Reconciliation

Definition

This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.

Format

N(12)

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Mandatory

Notes

Authorization and Presentment can occur on different days. The reconciliation amounts can be different in the Authorization and Presentment messages.

The decimal point is determined by the Currency Code, Reconciliation (Bit 50). For example, \$100.00 in USD shows as 000000010000.

Bit 7

Date and Time, Transmission

Definition

This is the local date and time when the Acquirer transmits the request to the Network.

Format

N(10) MMDDHHMMSS

Optionality

Acquirer to Network: Optional

Network to Issuer: Not Applicable

Notes

The Network does not edit for a valid date and time.

Bit 11

Systems Trace Audit Number

Definition

This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format

N(6)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Acquirer to Network: The Network returns this number in the Network to Acquirer part of the POS Authorization Response (1110) message.

Network to Issuer: This value is assigned by the Network and is a different value than what is generated by the Acquirer.

Bit 12

Date and Time, Local Transaction

Definition

This is the local date and time when a Transaction occurred.

Format

N(12) YYMMDDHHMMSS

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Acquirer to Network: This is the date and time when the Acquirer sent the message to the Network. The value of this field must be a valid date and time.

Network to Issuer:

- **If valid,** this field will contain the date and time as it was sent by the Acquirer. It will be sent to the Issuer as it was received by the Network.
- **If not valid,** this field will contain the date and time when the Network sent the message to the Issuer.

Bit 13

Card Effective Date

Definition

This is the Card Effective Date embossed on the face of the Card.

Format

N(4)

YYMM

Optionality

Acquirer to Network: Optional

Network to Issuer: Optional

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 Bit 14 Card Expiration Date Message Layout

Notes

This field is not required if the message contains either Track 1 or Track 2 data read from the Card's magnetic stripe. Most manual or keyed POS Authorization Requests do not have the Card Effective Date because it is not captured at the point of sale or is not on the Card.

The Network does not supply a default for this value if it is not sent by the Acquirer. If present, Issuers must check for a valid date, as the Network only edits that this field is a valid number.

Expresspay Transactions

This date is supplied by the chip.

Bit 14

Card Expiration Date

Definition

This is the Card Expiration Date embossed on the face of the Card.

Format

N(4) YYYY

Optionality

Acquirer to Network: Conditional.

- This field is required when a Transaction is key-entered and not Recurring Billing or Standing Authorization.
- This bit is required for a Payment Token Transaction

Network to Issuer: Conditional. If this bit is present, it is sent to the Issuer.

Notes

Acquirer to Network: The Network edits this field to make sure it is numeric, non-zero, and 4 digits long.

Network to Issuer: When Track 1 or Track 2 is not present, the Network includes Bit 14 if the Acquirer supplies the Card Expiration Date. S/Es in some countries transpose the month and year values//; thus, Issuers should try to verify the Card Expiration Date in the format MMY.

This field is not present if the message contains either Track 1 or Track 2 data read from the Card's magnetic stripe.

Expresspay Transactions

This date is supplied by the chip.

Payment Token Transaction

Acquirer to Network: This bit contains the token expiration date.

Network to Issuer: This bit contains the PAN expiration date.

Bit 18

Reserved Field

Definition

This field is for AEGNS internal use.

Format

N(4)

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Not Applicable

Bit 19

Country Code, Acquiring Institution

Definition

This is the Country Code of the acquiring institution.

Format

N(3)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Conditional. If this bit is present, it is sent to the Issuer.

Notes

The Network does not edit or verify the field contents.

Refer to the Country Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Bit 22

Point of Service Data Code

Definition

This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).

Format

AN(12)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

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POS Authorization Request (1100)
Bit 23 PAN Sequence Number Message Layout

Example

This shows how the field is populated

0										1		
1	2	3	4	5	6	7	8	9	0	1	2	
2	6	1	1	0	1	2	0	0	1	2	0	

In the above example

Position 1	=	2	Position 5	=	0	Position 9	=	0
Position 2	=	6	Position 6	=	1	Position 10	=	1
Position 3	=	1	Position 7	=	2	Position 11	=	2
Position 4	=	1	Position 8	=	0	Position 12	=	0

Notes

Expresspay TAT Transactions

Position 4 = Z (Transit Access Terminal - TAT)

Position 6 = X (Contactless Transaction)

The Acquirer must echo the Expresspay TAT Transaction data used in the POS Authorization Request (1100) message in the POS First and Second Presentment (1240) message to show that the Transaction started at a Transit Access Terminal - TAT.

Recurring Billing Transactions

Required Value: Position 5 = 9

The Acquirer must echo the Recurring Billing Indicator from the POS Authorization Request (1100) in the POS First and Second Presentment (1240) message.

Online PIN Cardholder Verification Method

Position 8 = 1

Position 9 = 3

Refer to the Point of Service Data Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 23

PAN Sequence Number

Definition

A number distinguishing between separate Cards with the same Primary Account Number.

Format

N(3)

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.

Bit 24

Function Code

Definition

This code shows the reason for a message in its Message Type.

Format

AN(3)

Optionality

Acquirer to Network: Conditional. The Acquirer sends a Function Code if they use a special program (listed below).

Network to Issuer: Conditional.

- Issuers must support Function Codes for estimated and incremental Authorizations.
- Issuers must be certified for the special programs they support.

Notes

Function Code is used to identify these Transaction types:

Variable Authorizations

Estimated Authorizations Valid Values

- 101 - Original Authorization - Amount Estimated
- 183 - Amount Estimated with Partial Authorization and balance return

- 184 - Amount Estimated and balance return
- Incremental Authorizations Valid Values**
- 106 - Supplemental Authorization - Amount Accurate
- 185 - Supplemental Amount Accurate with Partial Authorization and balance return
- 186 - Supplemental Amount Accurate and balance return

DPP Valid Values

- 100 = Authorization Request (for Issuer-based DPP and Acquirer-based DPP Authorization Requests)
- 108 = Inquiry (for DPP pre-Authorization Requests)

Prepaid Balance Inquiry Authorization Valid Values

- 181 = Partial Authorization and balance return
- 182 = Authorization and balance return

Account Status Check

If this field contains a value of 190, the S/E is asking for an Account Status Check. The Network does not send a Function Code 190 unless the Acquirer and Issuer are certified for Contactless (Expresspay) functionality.

ATC Synchronization

Function Code 191 (ATC Synchronization) supplies the Issuer with an ATC value for synchronization. The Network does not send Function Code 191 unless the Acquirer and Issuer are certified to send and receive it.

Expresspay Translation (PAN Request)

Function Code 194 - Expresspay Translation (PAN Only request) and Function Code 196 - Expresspay Translation (PAN request extended - PAN with PAN Expiry Date) is a request by an S/E for an Expresspay Translation from the Network. The Network does not receive and process Function Codes 194 or 196 unless the Acquirer is certified to send it.

Currently, the Network supports Function Code 194 when sending a POS Authorization Request (1100) message to certified Issuers, or when receiving a POS Authorization Response (1110) message from certified Issuers.

For all other Transaction types, Function Code 100 can be used. The use of Function Code 100 on Authorization Requests is optional.

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 26

Definition

Format

Optionality

Notes

Card Acceptor Business Code

Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).

N(4)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Payment Facilitator

This field must contain the MCC of the Sponsored Merchant. The MCC can be different for each transaction dependent on the category of the Sponsored Merchant.

Bill Payment Provider:

This field must contain the MCC of the End Beneficiary. The MCC can be different for each Transaction dependent on the category of the End Beneficiary.

Refer to the Merchant Category Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

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POS Authorization Request (1100)
Bit 31 Acquirer Reference Data, Transaction Identifier (TID) Message
Layout

Bit 31

Acquirer Reference Data, Transaction Identifier (TID)

Definition	This unique number is given by the Network and follows a Transaction during its life cycle.
Format	N(40) max LLVAR
Optionality	Acquirer to Network: Not Applicable
	Network to Issuer: Mandatory. All Issuers must be able to receive the TID.
Notes	Refer to The Role of the Transaction Identifier on page 2-2 for more instruction about the TID and the authorization life cycle.

Bit 32

Acquiring Institution Identification (AIN) Code

Definitions	This is a unique identification code given to an Acquirer by the Network.
Format	N(13) max LLVAR
Optionality	Acquirer to Network: Mandatory
	Network to Issuer: Mandatory
Notes	Acquirer to Network: The variable length indicator is not included in the length of the bit. For example, Acquiring Institution Identification code 45678912345 appears as 1145678912345.
	Network to Issuer: Required Value = 06370100 American Express

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

Bit 34

Primary Account Number, Extended

Definitions	This field is a multipurpose field which provides the Primary Account Number or associated data.
Format	AN(30) Max LLVAR (with length indicator)
Optionality	Acquirer to Network: Not Applicable
	Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.
Notes	This bit is populated by the Network with the Token, Wallet Provider ID, and Token Expiration Date.

Subfield	Format
1	Variable Length Indicator (VLI)
2	Token
3	Wallet Provider ID (ID)
4	Token Expiration Date

Data Example:

Subfield 1 is the VLI and must be 2 bytes. The token immediately follows and is a variable length between 15 and 19 bytes. After the token is a back slash (“\”) delimiter and then a 2 byte ID. After the 2 byte ID is a back slash (“\”) delimiter and then the token expiration date in the format YYMM.

The symbol “~” is used to represent a space.

Example: 23371234567890123\AB\1401

Bit 35

Track 2 Data

Definition	This is the data encoded in a Track 2 magnetic stripe read or the data stored in the Track 2 field on the chip of a Chip Card.
Important	S/Es cannot keep track data from the Authorization Request.
Format	ANS(39) max LLVAR
Optionality	Acquirer to Network: Conditional. This bit must be present if Track 2 data is read at point of sale from magstripe or chip.
	Network to Issuer: Conditional. If this bit is present, it is sent to the Issuer.

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POS Authorization Request (1100)
Bit 37 Acquirer Reference Number (ARN) Message Layout

Notes

Track 2 data contains the EBCDIC digits 0-9, special characters, and field separator value.

Point of Service Data Code

If the Point of Service Data Code shows that track data was captured at point of sale with Position 7 = W, 2, or 5, then Bit 35 or 45 must be present in the 1100.

Track data must be sent as captured. Participants must support both the ANSI track layouts and the ISO 7813 layouts. The Acquirer should strip the beginning and end sentinels and translate from ASCII to EBCDIC.

Expresspay Transactions

This data is supplied by the chip.

Expresspay Transactions do not include the 5CSC value encoded on the magnetic stripe. It is replaced with Expresspay specific data.

Refer to the ISO 7813 Standard Track 2 Layout or ANSI X4.16 Track 2 Layout in *Network Specifications - Instructions* for more instruction.

Bit 37

Acquirer Reference Number (ARN)

Definition

A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

Format

AN(12)

Optionality

Acquirer to Network: Optional

Network to Issuer: Optional

Bit 41

Card Acceptor Terminal Identification

Definition

This code identifies a POS terminal at a Card Acceptor location.

Format

ANS(8)

Optionality

Acquirer to Network: Optional

The Network returns this number in the Network to Acquirer piece of the POS Authorization Response (1110) message.

Network to Issuer: Optional

The Network gives this value. The Network value is different from the Acquirer.

Bit 42

Card Acceptor Identification Code

Definition

The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.

Format

AN(15)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

The Network rejects this message if the S/E number fails the check digit routine.

Bit 43

Card Acceptor Name and Location

Definition

The name of the Card Acceptor as it is known to Cardmembers.

Important

This bit is very important to Participants and Cardmembers and should contain all the available data.

Format

ANS(101) max LLVAR

Optionality

Acquirer to Network: Conditional.

- This bit must be present when an MCC 6010 Financial Institutions Manual Cash Disbursement Transaction is received with a Processing Code of 014008.
- **Payment Facilitator:** Subfields 1 and 4 must contain the Name and Country Code of the Sponsored Merchant.
- **Bill Payment Provider:** Subfield 1 must contain the Name of the End Beneficiary.

Network to Issuer: Conditional. If the Network receives Bit 43 and the Issuer is certified to receive it, the Network sends it to the Issuer.

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POS Authorization Request (1100)
Bit 43 Card Acceptor Name and Location Message Layout

Notes

This bit is free form and Issuers must accept the data as presented by the Acquirers.

Length of Field: 3 bytes minimum, 101 bytes maximum LLVAR with a Variable Length Indicator of 2 bytes EBCDIC, right justified with leading zeros.

Position	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality				Network to Issuer Optionality	Description
				Payment Facilitators	Bill Payment Providers	Manual Cash Disbursement w/ 014008 Processing Code	All Others		
1-2	Variable Length Indicator (VLI)	2 Bytes	Numeric (EBCDIC)	M	M	M	C	C	VLI is the total length of the variable data in Bit 43 (not including VLI).
Subfield 1	Name\Street\City	83 Bytes	Alphanumeric	M	M	M	O	C	Name, Street and City of the Card Acceptor as it is known to Cardmembers. - Payment Facilitator: - Must contain the Name of the Sponsored Merchant. - If Provided, the Street and City must be that of the Sponsored Merchant. - Bill Payment Provider: - Must contain the Name of the End Beneficiary.
Subfield 2	Postal Code	10 Bytes	Alphanumeric	O	O	M	O	C	Postal Code where the Transaction took place. Payment Facilitator: IF provided, Postal Code must be that of the Sponsored Merchant.
Subfield 3	Region	3 Bytes	Alphanumeric	O	O	M	O	C	Region where the Transaction took place. Payment Facilitator: IF provided, Region must be that of the Sponsored Merchant.
Subfield 4	Country Code	3 Bytes	Alphanumeric	M	O	M	O	C	Country Code where the Transaction took place. Payment Facilitator: Must contain the Country Code of the Sponsored Merchant.

Data Examples:

The elements of subfield 1 (subfield 1: name\street\city\) are variable length and use a back slash to identify the end of the element. If the data for an element is missing, a back slash is still used to identify the end of the element (e.g., name\city\, street is missing). Subfields 2, 3, and 4 are fixed format, and a back slash is only used for each of these subfields for which data is not present.

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POS Authorization Request (1100)
Bit 45 Track 1 Data Message Layout

The symbol "~" is used to represent a space.

- Name and City are present: Merchant~One~\Anytown~\~\~\
- Name, City, and Region (State) are present: Merchant~One~\Anytown~\AZ~\
- Name, City, Postal Code, and Region (State) are present:
Merchant~One~\Anytown~\85000~\~\~\~\AZ~\
- Name, City, Postal Code, Region (State), and Country Code are present:
Merchant~One~\Anytown~\85000~\~\~\~\AZ~\840
- No fields are present ~\~\~\~\
- Name and Postal Code are present Merchant~One~\~\85000~\~\~\~\
- Postal Code only present ~\~\85000~\~\~\~\

Cash Disbursement

If Bit 43 is not present, the Authorization Request will be rejected with Action Code 181 Format Error.

Payment Facilitator

Subfield 1 **must** contain the Card Acceptor Name*DBA of the Sponsored Merchant. If street and city is supplied, it must be the street and city of the Sponsored Merchant.

Example: If the Payment Facilitator is "Pay4All" and a Sponsored Merchant is "ezhats" the Card Acceptor Name is sent as: Pay4All*ezhats.

- Subfield 2 - If a postal code is supplied, it must be the postal code of the Sponsored Merchant.
- Subfield 3 - If region is supplied, it must be the region of the Sponsored Merchant.
- Subfield 4 **must** contain the Country Code of the Sponsored Merchant.

Bill Payment Provider:

- Subfield 1 **must** contain the Bill Payment Provider Name*DBA of the End Beneficiary.
Example: If the Bill Payment Provider is "Pay4All" and an End Beneficiary is "ezhats" the Card Acceptor Name is sent as "Pay4All*ezhats."

Bit 45

Definition

Important!

Format

Optionality

Notes

Track 1 Data

This is the data encoded in a Track 1 magnetic stripe read or the data stored in the Track 1 field on the chip of a Chip Card.

S/Es cannot keep track data from the Authorization Request.

ANS(78) max LLVAR

Acquirer to Network: Conditional. Track 1 (Bit 45) must be present if Track 2 (Bit 35) is not present in the message.

Network to Issuer: Conditional. If only Track 1 is present, it is sent to the Issuer.

Track 1 data contains the EBCDIC digits 0-9, special characters, and field separator value.

Point of Service Data Code

If the Point of Service Data Code shows that track data was captured at point of sale with Position 7 = W, 2, or 5, then Bit 35 or 45 must be present in the 1100.

Track data must be sent as captured. Participants must support both the ANSI track layouts and the ISO 7813 layouts. The Acquirer should strip the beginning and end sentinels and translate from ASCII to EBCDIC.

Expresspay Transactions

This data is supplied by the chip.

Expresspay Transactions do not include the 5CSC value encoded on the magnetic stripe. It is replaced with Expresspay specific data.

Refer to the ISO 7813 Standard Track 2 Layout or ANSI X4.16 Track 2 Layout in *Network Specifications - Instructions* for more instruction.

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Network Specifications - Authorization
POS Authorization Request (1100)
Bit 48 Deferred Payment Plan (DPP) Message Layout

Bit 48

Deferred Payment Plan (DPP)

Definition

This is the Deferred Payment Plan data, permitting charges to be billed in equal installments rather than in one (1) total billing.

Format

AN(353) max LLLVAR

Optionality

Acquirer to Network: Conditional. This field contains a value if the Acquirer requests authorization for deferred or extended payments.

The Acquirer to Network piece is not applicable for Legacy DPP.

Network to Issuer: Conditional. If a certified Acquirer sends this bit and the Issuer is certified to receive it, then the Network sends it to the Issuer.

Notes

Legacy DPP

A three-digit variable length indicator is before the Legacy DPP data. The Legacy DPP data is not echoed in the POS Authorization Response (1110) message.

Valid Values:

- 003AXE - Sign and Travel
- 00403NN - Legacy Plan - N – S/E Deferred Payment Plan, where 03 is the Plan N Identifier and NN is the number of installments that varies from 02 to 99. Bit 24 Function Code is not required for the 03 type plan.
- 00405NN - Legacy American Express Deferred Payment Plan, where 05 is the American Express Plan Identifier and NN is number of installments. The number of installments can range from 02 to 99.
 - Bit 24 Function Code = 108 for Pre-Authorization
 - Bit 24 Function Code = 100 for the 05 type plan

Issuer DPP Pre-Authorization

This is a pre-authorization request for deferred or extended payments, including the plan code data.

Bit 24, Function code = 108 Inquiry

Bit Layout (Issuer DPP Pre-Authorization Request)

024DP05bbbb8888999999999999

Valid Values:

Position	Value	Use
1-3	024	A three digit variable length field is before the Variable Length Indicator.
4-7	DP05	Issuer DPP identifier
8-11		Binary (32-bit) bitmap in hexadecimal configuration identifying the fields in the DPP Structure. The bitmap value for Issuer DPP Pre-Authorization Requests is x "30000000." After the bitmap, the layout consists of two subfields of the DPP Structure as described below.

- Subfield 3 N(4) Number of Installments
- Subfield 4 N(12) Amount of Installment
- The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49).

Issuer DPP Authorization

This is the deferred payment plan data for the Issuer Deferred Payment Plan.

Bit 24, Function code = 100 Original Authorization Amount Accurate

Bit Layout (Issuer DPP Authorization Request)

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POS Authorization Request (1100)
Bit 49 Currency Code, Transaction Message Layout

016DP05bbbb77778888

Position	Value	Use
1-3	016	Variable Length Indicator
4-7	DP05	Issuer DPP identifier
8-11		Binary (32-bit) bitmap in hexadecimal configuration identifying the fields in the DPP Structure. The bitmap value for Issuer DPP Authorization Requests is x"60000000." After the bitmap, the layout consists of two subfields of the DPP Structure as described below.

- Subfield 2 AN(4) Plan Type
 - Required Value = 0005 Issuer-sponsored plan
- Subfield 3 N(4) Number of Installments

Acquirer DPP Authorization

This is the deferred payment plan data for the Acquirer Deferred Payment Plan.

Bit Layout (Acquirer DPP Authorization Request)

Valid Values:

028DP03bbbb77778888999999999999

Position	Value	Use
1-3	028	Variable Length Indicator
4-7	DP03	Acquirer DPP identifier
8-11		Binary (32-bit) bitmap in hexadecimal configuration identifying the fields in the DPP Structure. The bitmap value for Acquirer DPP Authorization Requests is x"70000000." After the bitmap, the layout consists of three subfields of the DPP Structure as described below.

- Subfield 2 AN(4) Plan Type
 - Required Value = 0003 Acquirer-sponsored plan
- Subfield 3 N(4) Number of Installments
- Subfield 4 N(12) Amount of Installment
 - The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49).

Bit 49

Definition

Format

Optionality

Notes

Currency Code, Transaction

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

N(3)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition

Format

Optionality

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.

N(3)

Acquirer to Network: Not Applicable

Network to Issuer: Mandatory

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Bit 52 Personal Identification Number (PIN) Message Layout

Notes Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 52 **Personal Identification Number (PIN)**

Definition A four (4) to twelve (12) personal code that lets an Issuer identify the Cardmember for approval of an ATM or POS Transaction.

Format Binary (8)

Optionality **Acquirer to Network:** Conditional. The Acquirer must be certified to send Bit 52 to the Network. Acquirers submitting PIN data in Bit 52 must not submit PIN data in Bit 111 - Encryption Data.

Network to Issuer: Conditional. The Network sends Bit 52 to certified Issuers **only**.

Notes The Network sends Action Code 115 (Requested function not supported) if the Acquirer or Issuer is not certified for Bit 52.

Bit 111 - Encryption Data must not be present when PIN data is sent in Bit 52.

Refer to the *Online PIN Processing Implementation Guide* on Knowledge Base for Online PIN requirements. Use the Search function on Knowledge Base to locate this document.

Bit 53 **Security Related Control Information**

Definition This field contains a variable length indicator and the Card Security Code (CSC) or One-Time Password (OTP) supplied by the Acquirer.

Card Security Code includes the 5CSC or the Printed Card Security Code (PCSC), which includes 3CSC and 4CSC (4DBC).

OTP is a dynamic 4-digit code provided by the Issuer to the Cardmember.

Format AN(8) max LLVAR

Optionality **Acquirer to Network:** Conditional.

This bit must be present if the Transaction is:

- Magnetic stripe read (swiped) with key entered PCSC or OTP data at Point of Service
- Key entered with PCSC or OTP data

Network to Issuer: Conditional. If a certified Acquirer sends this bit and the Issuer is certified to receive it, then the Network sends it to the Issuer.

Notes **Acquirer to Network:**

Only certified Acquirers will receive the PCSC or OTP validation response in Bit 44, Position 4 of the POS Authorization Response (1110) message. Refer to the [Programming Notes for POS Authorization Response \(1110\) on page 5-39](#) for additional instruction. Refer to the *Business and Operational Policies* manual for PCSC validation policy.

Position	Value	Use
1-2	04	Variable Length Indicator
3-6		Keyed Four-Digit Batch Code (4DBC), Keyed Four-Digit Card Security Code (4CSC), or Keyed OTP
		OR
1-2	03	Variable Length Indicator
3-5		Keyed Three-Digit Card Security Code (3CSC)

Network to Issuer:

For the Issuer, this bit can contain an indicator that the Transaction was keyed. This indicator is present only in the Network to Issuer Transaction and should not be supplied by the Acquirer.

These formats can be received and must be validated.

Position	Use
1-2	Variable Length Indicator

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3-6 or 3-7	CSC ¹ , PCSC, or OTP value (which can include a space)								
Examples include:									
<table border="1"> <tr><td>03XXX</td><td>05XXXXX</td></tr> <tr><td>04~XXX</td><td></td></tr> <tr><td>04XXX~</td><td></td></tr> <tr><td>04XXXX</td><td></td></tr> </table>		03XXX	05XXXXX	04~XXX		04XXX~		04XXXX	
03XXX	05XXXXX								
04~XXX									
04XXX~									
04XXXX									
OR									
1-2	Variable Length Indicator								
3	Keyed Indicator "S"								
4-6 or 4-7	PCSC or OTP value (which can contain a space)								
Examples include:									
<table border="1"> <tr><td>04SXXX</td><td>05S~XXX</td></tr> <tr><td>05SXXXX</td><td>05SXXX~</td></tr> </table>		04SXXX	05S~XXX	05SXXXX	05SXXX~				
04SXXX	05S~XXX								
05SXXXX	05SXXX~								

- These positions could contain a 5CSC value.

If the PCSC or OTP is supplied in the Network to Issuer request, a response in Bit 44, Position 4 in the POS Authorization Response (1110) message is required from Issuers.

Bit 44, Position 4 is not used if Bit 53 contains the 5CSC.

Bit 55

Definition

Format

Optionality

Notes

ICC System Related Data

This is the data sent to and from the Integrated Circuit Card and used in an AEIPS or Expresspay Transaction.

AN(259) max (includes length indicator) LLLVAR

Acquirer to Network: Conditional. If ICC data is read from the Card, the Acquirer must send the data in this field.

Network to Issuer: Conditional. If a certified Acquirer sends this bit and the Issuer is certified to receive it, then the Network sends it to the Issuer.

This table shows that this data extends across more than 1 subfield structure.

Maximum Length	Fixed/ Variable	Source	Field Name
4	F	Acquirer	ICC Header Version Name
2	F	Acquirer	ICC Header Version Number
8	F	Chip Card	Application Cryptogram
33	V	Chip Card	Issuer Application Data (IAD)
4	F	Terminal	Unpredictable Number
2	F	Chip Card	Application Transaction Counter (ATC)
5	F	Terminal	Terminal Verification Results (TVR)
3	F	Terminal	Transaction Date
1	F	Terminal	Transaction Type
6	F	Terminal	Amount, Authorized
2	F	Terminal	Transaction Currency Code

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2	F	Terminal	Terminal Country Code
2	F	Chip Card	Application Interchange Profile (AIP)
6	F	Terminal	Amount, Other
1	F	Chip Card	Application PAN Sequence Number
1	F	Chip Card	Cryptogram Information Data

Bit 55 Layout

The Bit 55 layout contains these subfields, which are mandatory for Acquirer to Network and Network to Issuer.

EMV Tag	Subfield	Format	Notes
	1 ICC Header Version Name	AN(4) EBCDIC	Required Value: AGNS
	2 ICC Header Version Number	(2) Binary (numeric)	Required Value: 0001
9F26	3 Application Cryptogram	(8) Binary Hexadecimal	
9F10	4 Issuer Application Data (IAD)	(33) max LLVAR Binary Hexadecimal (includes length LL indicator)	As Subfield 4 is a variable length field of type LLVAR, the "LL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLVAR length byte is in addition to the 32 byte maximum data length of the actual data (i.e., total 33 bytes). Example: ICC Structure Subfield 4, Issuer Application Data formatted according to the LLVAR and structure described above is 07 06 01 01 03 A0 00 00 where: • 07 = overall length indicator for Subfield 4 LLVAR • 06 = IAD Data Length (1 byte) • 01 = Derivation Key Index (1 byte) • 01 = Cryptogram Version Number (1 byte) • 03A00000 = CVR (4 bytes)
9F37	5 Unpredictable Number	(4) Binary Hexadecimal	
9F36	6 Application Transaction Counter (ATC)	(2) Binary Hexadecimal	
95	7 Terminal Verification Results (TVR)	(5) Binary Hexadecimal	Refer to the <i>AEIPS Terminal Technical Manual</i> or the <i>AEIPS Card Specification</i> on Knowledge Base. Use the Search function on Knowledge Base to locate these documents and get additional information on Terminal Verification Results (TVR).
9A	8 Transaction Date	(3) Binary (numeric)	Date Format: YYMMDD
9C	9 Transaction Type	(1) Binary (numeric)	Valid Values: • 00 = Debit • 01 = Cash • 20 = Credit • 30 = Debit (Restricted use - valid only for Transactions originating from certain markets)
9F02	10 Amount Authorized	(6) Binary (numeric)	• This amount can be different from the amount in Bit 4 or Bit 5. The decimal point is determined by the Transaction Currency Code, Subfield 11. • For Expresspay-enabled Card/device Application Transaction Counter (ATC) value requests, this sub-field contains a zero (0.00) amount.

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EMV Tag	Subfield	Format	Notes
5F2A	11 Transaction Currency Code	(2) Binary (numeric)	This subfield should match the contents of Bit 49, Currency Code, Transaction.
9F1A	12 Terminal Country Code	(2) Binary (numeric)	
82	13 Application Interchange Profile (AIP)	(2) Binary Hexadecimal	Expresspay Mobile Transactions Valid Values: Byte 2, Bit 7 0 = Expresspay Mobile not supported 1 = Expresspay Mobile supported
9F03	14 Amount Other	(6) Binary (numeric)	Should be zero-filled if cash back is not supported.
5F34	15 Application PAN Sequence Number	(1) Binary (numeric)	Card Acceptors (S/Es) must supply mandatory Authorization data fields unaltered when the chip is read by the POS Device. If the Application PAN Sequence Number is a numeric value (for example, 01), the value must be transmitted unaltered to the Network.
9F27	16 Cryptogram Information Data	(1) Binary Hexadecimal	
	AEIPS and Expresspay Cryptogram Validation Results		AEIPS and Expresspay Cryptogram Validation results will be passed in POS Authorization Request (1100), Bit 55, Subfields 17-19 to Issuers who want to get this data as part of the AEIPS and Expresspay Cryptogram Validation Service. Issuers certified for this service should use the data in the POS Authorization Request (1100), Bit 55, Subfields 17-19 to populate Bit 55 of the POS Authorization Response (1110) message. The Network edits or truncates Bit 55 data to support Issuers certified for AEIPS and Expresspay Cryptogram Validation. Data sent by the Acquirer in the POS Authorization Request (1100), Bit 55, Subfields 17-19 can result in Issuer declines. Issuers should not use this data as part of the chip authentication.
	17 ICC Header Version Name		Required Value: AGNS
	18 ICC Header Version Number		Required Value: 0001
	19 Issuer Authentication Data		If Card authentication was performed and was successful, this subfield must be present. If Card authentication could not be performed and was not successful (i.e., missing or incorrect cryptogram in the 1100 message), this subfield is not included. As Subfield 19 is a variable length field of type LLVAR, the "LL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLVAR length byte is in addition to the 16 byte maximum data length of the actual data (i.e., total 17 bytes). - Sub-Subfield 1 Authorization Response Cryptogram (ARPC) (8) - Sub-Subfield 2 ARPC Response Code N(2) (ASCII)

Bit 60

Market Specific Data

Definition

This bit provides additional authorization data.

Optionality

Acquirer to Network: Conditional.

- Payment Facilitator:** Market Specific Data with Additional Authorization Data format, Subfield 2, must contain the Seller ID of the Sponsored Merchant.

Note	• Bill Payment Provider: Market Specific Data with Additional Authorization Data format, Subfield 2, must contain the Seller ID of the End Beneficiary. • Terminal Classification Code is required for Transactions originating from a Mobile Device.
	Network to Issuer: Conditional. OTID must be received for Incremental Authorization Transactions. Certification is required for Acquirers to send Bit 60 and for Issuers to receive Bit 60.
	Market Specific Data: • There are two supported formats of Market Specific Data. • The structure and contents of this data is specified by market representatives. • Each market uses this data differently. • To use Market Specific Data, Acquirers and Issuers must be in the same market that supports this data.
	Contact your AEGNS Representative for more information.
Market Specific Data	
Format	AN(210 Bytes) Used by Participants in regions supporting Market Specific Data, which do not have a need to support Market Specific Data with Additional Authorization Data.
Notes	Used by Participants in regions that support Market Specific Data and do not have a need to support Additional Authorization Data.

Market Specific Data Layout Table

Position ¹	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality	Network to Issuer Optionality	Valid Value	Description
1-3	Variable Length Indicator (VLI)	3 Bytes	Numeric (EBCDIC)	M	M		VLI is the total length of the variable data in Bit 60 (not including VLI).
4-5	Market Specific Data Indicator	2 Bytes	Alphanumeric	C	O	MS	Market Specific Data Indicator is the constant literal "MS" for Market Specific Data Indicator.
6-8	Market Specific Data ISO Country Code of the Acquirer	3 Bytes	Numeric	C	O		ISO Country Code of the Acquirer must be populated if SF1 is present. Refer to the <i>Codes Reference Guide</i> for a complete list of codes.
9-10	Market Specific Data Version Indicator	2 Bytes	Numeric	C	O		Version Indicator must be populated if SF1 is present.
11-210	Market Specific Data Elements	200 Bytes Max	ANS	O	O		Market Specific Data Elements

1. The Network validates only the contents of Positions 4-5 and Positions 6-8.

Market Specific Data with Additional Authorization Data

Format AN(353) max LLLVAR

Used by Participants in regions supporting Market Specific Data and Additional Authorization Data elements identified in the table below. Certified Acquirers will send the appropriate data field(s), depending on their specific needs.

Notes

Seller ID:

- **Payment Facilitator:** Subfield 2, must contain the Seller ID of the Sponsored Merchant.
- **Bill Payment Provider:** Subfield 2, must contain the Seller ID of the End Beneficiary.

Token Requestor ID (TRID):

- Certification is required for Acquirers and Issuers to receive the TRID in Subfield 5.
- Acquirers requesting Subfield 5 - Token Requestor ID returned in the POS Authorization Response (1110) message must provide a required value in the POS Authorization Request (1100) message.

Original Transaction Identifier (Original TID)

- Issuers must receive the OTID for Incremental Authorization Transactions.

Terminal Classification Code

- Acquirers must send the Terminal Classification code for Transactions originating from a mobile device.

Market Specific Data with Additional Authorization Data Layout Table

Position	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality	Network to Issuer Optionality	Valid Value	Description
1-3	Variable Length Indicator (VLI)	3 Bytes	Numeric (EBCDIC)	M	M		VLI is the total length of the variable data in Bit 60 (not including VLI).
4-5	Primary ID	2 Bytes	Alphanumeric	M	M	AX	Primary ID is constant literal "AX" for Market Specific Data/Additional Authorization Data Indicator.
6-8	Secondary ID	3 Bytes	Alphanumeric	M	M	AAD	Secondary ID is constant literal "AAD" for Additional Authorization Data.
9-12	Bitmap Identifier	4 Bytes	Binary (Hexadecimal configuration)	M	M		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the bitmap, the layout consists of at least 1 of the following subfields. Each bit position of the 32 bit/4 byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, the corresponding subfield will be present.
SF1	Market Specific Data						
	Market Specific Data VLI	3 Bytes	Numeric (EBCDIC)	C	O		Market Specific VLI is mandatory if the Bitmap Identifier indicates that SF1-Market Specific Data is enabled.
	Market Specific Data Indicator	2 Bytes	Alphanumeric	C	O	MS	Data Indicator of the constant literal "MS" must be populated if SF1 is present.
	Market Specific Data ISO Country Code of the Acquirer	3 Bytes	Numeric	C	O		ISO Country Code of the Acquirer must be populated if SF1 is present. Refer to the <i>Codes Reference Guide</i> for a complete list of codes.
	Market Specific Data Version Indicator	2 Bytes	Numeric	C	O		Version Indicator must be populated if SF1 is present.
	Market Specific Data Elements	200 Bytes Max	ANS	O	O		Market Specific Data Elements
SF2	Seller ID	20 Bytes	AN	C	O		Combination of letters and/or numbers that uniquely identifies a Sponsored Merchant, or End Beneficiary. Bill Payment Provider: SF2 must contain the Seller ID of the End Beneficiary. Payment Facilitator: SF2 must contain the Seller ID of the Sponsored Merchant.

Market Specific Data with Additional Authorization Data Layout Table (continued)

Position	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality	Network to Issuer Optionality	Valid Value	Description
SF3	Reserved	42 Bytes	ANS	NA	NA		
SF4	Reserved	20 Bytes	AN	NA	NA		
SF5	Token Requestor ID	11 Bytes	AN	O	O		An 11-byte code pairing Token Requestor with Token Domain
SF6	Reserved	1 Byte	AN	NA	NA		
SF7	Original Transaction Identifier (Original-TID)	15 Bytes	N	O	C		Original Transaction Identifier (Original-TID) Issuers must receive the OTID for incremental Authorization Transactions.
SF8	Terminal Classification Code	2 Bytes	AN	C	O		Code that uniquely classifies the architecture of the terminal associated with the Transaction. Acquirers must send the Terminal Classification code for Transactions originating from a mobile device. Refer to the <i>Codes Reference Guide</i> for a complete list of codes.
SF9	National Identification Document (ID)	20 Bytes	AN	O	O		ID of the Cardholder as defined by the government of their nationality.

Bit 61

National Use Data

Definition	This is an ISO field that supplies Cardmember authentication data during Authorization.
Format	4 bytes min, 103 bytes max LLLVAR
Optionality	Acquirer to Network: Conditional. This bit must be present for a Digital Wallet application-initiated Transaction or a SafeKey Transaction. Network to Issuer: Conditional. If this bit is present, the Network sends it to the certified Issuer.
Bit Layout Usage	Refer to Table 5-1 for SafeKey Transactions. Refer to Table 5-2. Digital Wallet Transaction Layout on page 5-27 for Digital Wallet application-initiated Transactions.
Notes	Bit 61 Layout The information below provides SafeKey Bit 61 layout details. Refer to Knowledge Base for additional SafeKey information.

Table 5-1. American Express SafeKey Transaction Layout

Subfield	Format	Optionality		Notes
		A to N	N to I	
1 Variable Length Indicator (VLI)	(3) (numeric EBCDIC)	Mandatory	Mandatory	
2 Primary ID	(2) (alphanumeric)	Mandatory	Mandatory	Required Value: AX
3 Secondary ID	(3) (alphanumeric)	Mandatory	Mandatory	Valid Value: ASK
4 Electronic Commerce Indicator (ECI)	(2) (alphanumeric)	Mandatory	Mandatory	Valid Values: 05 = Authenticated with AEVV 06 = Attempted with AEVV 07 = Not Authenticated
5 American Express Verification Value (AEVV) ID	(4) (alphanumeric)	Conditional	Conditional	If Subfield 6 is present, AEVV ID must be populated. Valid Value: AEVV

Table 5-1. American Express SafeKey Transaction Layout (continued)

Subfield	Format	Optionality		Notes
		A to N	N to I	
6 American Express Verification Value (AEVV)	(20) (binary)	Conditional	Conditional	This subfield is necessary if ECI is 05 or 06. Do not use spaces. Do not use zeros (0) in all positions. The AEVV value is proof of the SafeKey Authentication and will be supplied from the SafeKey Authentication process. Sub-Subfield 1 Authentication Results Code (1): 0 = Authenticate Successful 7 = Attempt non participating 8 = Attempt Access Control Server (ACS) not available Sub-Subfield 2 Second Factor Authentication Code (1): Default value '00' Sub-Subfield 3 AEVV Key Indicator (1): Card Issuers ACS = 01, 02 Attempts ACS = 98, 99 Sub-Subfield 4 iCSC-3 Output (2) Sub-Subfield 5 Unpredictable Number (2) Sub-Subfield 6 Authentication Tracking Number (8) Sub-Subfield 7 Additional AEVV Information (5) - Values are determined by ACS. Can be Zero filled or: Version (4 bits): 0=Authentication Action and Cardmember IP address not present 1=Authentication Action and Cardmember IP address present 2-9=Reserved Authentication Action (4 bits): 0=Standard authentication performed 1=Activated During Shopping registration/authentication performed 2=Forgot your password (reregistration/re-authentication performed) Cardmember IP Address (4 bytes)
7 SafeKey Transaction ID	(3) (alphanumeric)	Conditional	Conditional	This subfield is necessary if the SafeKey Transaction ID Value is present. Valid Value: XID
8 SafeKey Transaction ID Value	(20) (binary)	Conditional	Conditional	SafeKey Transaction ID (XID) value is required if XID is present in the SafeKey authentication.

Table 5-2. Digital Wallet Transaction Layout

Subfield	Format	Optionality		Notes
		A to N	N to I	
1 Variable Length Indicator (VLI)	(3) (numeric EBCDIC)	Mandatory	Mandatory	
2 Primary ID	(2) (alphanumeric)	Mandatory	Mandatory	Required Value: AX
3 Secondary ID	(3) (alphanumeric)	Mandatory	Mandatory	Valid Value: TKN

Table 5-2. Digital Wallet Transaction Layout (continued)

Subfield	Format	Optionality		Notes
4 Electronic Commerce Indicator (ECI)	(2) (alphanumeric)	Mandatory	Mandatory	Valid Value: 20 = Payment Token Data Present
5 Token Data Block A ID	(4) (alphanumeric)	Mandatory	Mandatory	Valid Value: TDBA If the Secondary ID = TKN, then this subfield is required.
6 Token Data Block A	(20) (binary)	Mandatory	Mandatory	If the Secondary ID = TKN, then this subfield is required. Refer to the Expresspay Mobile documents on Knowledge Base for additional information. Use the Search function to locate these documents.
7 Token Data Block B ID	(3) (alphanumeric)	Conditional	Conditional	Valid Value: DBB If Subfield 8 - Token Data Block B is provided, this subfield is required with a value of DBB.
8 Token Data Block B	(20) (binary)	Conditional	Conditional	If Subfield 7 - Token Data Block B ID = DBB, then this subfield is required. Refer to the Expresspay Mobile documents on Knowledge Base for additional information. Use the Search function to locate these documents.

Bit 62

Fraud Indicator

Definition

The Fraud Indicator field is used to provide fraud mitigation indicators for a variety of situations.

Format

AN(5) min, AN(60) max LLLVAR

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Conditional. The Network puts the validation results in this bit and sends it only to certified Issuers.

Notes

Two formats are used for Bit 62: Fixed Length and Variable Length as described below.

Fixed Length Format

This field is populated by the Network with the 5CSC validation results, Fraud Probability Score, and Recommendation Code. This field is passed as Fraud Indicators only to certified Issuers. For Request Type Identifiers FD1 and FD2, only 18 of the 60 bytes are being used.

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD1 FD2	Request Type Identifier FD1 = 5CSC checking results FD2 = 5CSC validation results, Recommendation Code, and Fraud Probability Score If FD1 value is listed, then the field will end at 10 bytes. If FD2 value is listed, a combination of codes and spaces will be included in positions 9-18, dependent on the features for which the Participant certified.
9-10 ¹	XX	Two digit 5CSC Indicator Refer to the Fraud Indicators section of the <i>Codes Reference Guide</i> for a complete listing of codes and descriptions.

11-15 ¹	XXXXX	Fraud Probability Score This is an implied leading zero and decimal. A Fraud Probability Score of 12345 should be interpreted as 0.12345.
16-18 ¹	XXX	Recommendation Codes. Refer to the DFMNS Recommendation Codes of the <i>Codes Reference Guide</i> .

1. If GNSAS is not available, it is possible that positions 9-18 can be blank for Issuers certified for Fraud Probability Score and Recommendation Code.

FD3 Request Type Identifier

Bit 62 FD3 Request Type is supplied for SafeKey Transactions where the Network created the AEV when the Issuer ACS was not available.

Certification

When the Participant certifies for the FD3 Request Type Identifier, FD1 and FD2 are no longer supported.

It is not necessary for existing Issuers to certify for Field 62, Fraud Indicators, unless they re-certify their Authorizations capabilities.

American Express SafeKey requires Participants to certify for Bit 62 and support the FD3 Request Type Identifier.

Refer to the Fraud Indicators section of the *Codes Reference Guide* for a list of valid codes and descriptions.

Valid Values:

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD3	Request Type Identifier FD3 = Bitmap Variable Length

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9-12	32 bit Binary	Bitmap Indicator Identifies the fields in the structure. After the bitmap, the layout consists of at least one of the subfields as defined below. Each bit position of the 32-bit / 4-byte bitmap represents which fraud indicators are present. If a particular bit is "ON" in the bitmap, that corresponding sub-field will be present in positions 13-60. • Subfield 1 5CSC Indicator (2 Bytes) • Subfield 2 Fraud Probability Score (5 Bytes) • Subfield 3 Recommendation Code(3 Bytes) • Subfield 4 AEVV Validation Result (1 Byte) • Subfield 5 AEIPS and Expresspay Cryptogram Validation Result (2 Bytes) • Subfield 6 PIN Validation Results (2 Bytes) • Subfield 7 EMV Data Results (4 Bytes) • Subfield 8 Reserved for AEGNS use • Subfield 9 ATC Validation Results (2 Bytes) • Subfield 10 Reserved for Future Use • Subfield 11 ATC Value (2 Bytes - hexadecimal) • Subfield 12 Cardmember Traveling Indicator (1 Byte) • Subfield 13 4CSC Indicator (2 Bytes) • Subfield 14 Payment Token Expiration Validation Result (2 Bytes) • Subfield 15 Digital Capabilities Indicator (4 Bytes) Note: Cryptogram Validation (Subfield 5) will use the Fraud Indicator Code of 02 - HSM Not Available. Refer to the <i>Codes Reference Guide</i> for valid codes and definitions.
13-60	Fraud Value	Describes the results of the Bitmap Indicators that are set.
Example: Positions 9 - 12 		

Example: If Bit 62 includes Fraud Probability Score, Recommendation Code, and AEVV Validation Result:

Position	Value
1-3	018
4-5	AX
6-8	FD3
9-12	01110000000000000000000000000000 X'70000000'
13-60	555553331

Bit 63

Definition

Format

Optionality

Automated Address Verification

A message option that allows an S/E to verify the Cardmember address information.

AN(4) min, AN(103) max LLLVAR

Acquirer to Network: Conditional. Acquirers must be certified to send this bit.

Notes

Network to Issuer: Conditional. If a certified Acquirer sends this bit and the Issuer is certified to receive it, then the Network sends it to the Issuer.

This field is necessary if the Bit 3 - Processing Code is 004800, 174800, or 200800. For more instruction, refer to:

- [Table 2-23. Automated Address Verification Procedure with Credit Authorization Request on page 2-29](#)
- [Table 2-24. Automated Address Verification Only Procedure on page 2-30](#)

If the Cardmember address is available but the Postal Code is not, the Postal Code subfield must be zero-filled. If the Postal Code is available but the Cardmember address is not, the address subfield must be character space-filled.

Position	Value	Use
1-3	033	Variable Length Indicator
4-5	AX	Service Identifier, AX for American Express
6-7	AD or AP	Request Type Identifier AD = Standard Address Verification (alphanumeric) AP = Address Verification (numeric) If an AAV code (AXAP) is received by the Network for an Issuer not certified for AXAP Address Verification, the Network will change the code from AXAP to AXAD.
8-16		Postal Code The Postal Code is left justified, and the subfields should have trailing zeros. Other alphanumeric Postal Codes can be padded with zeros or character spaces. If the Request Type Identifier is "AXAP," or if the Network has changed the code from AXAP to AXAD, this subfield can contain the numeric values from the Cardmember's postal code.
17-36		First 20 Characters of Cardmember Address (20 characters, space-filled). Alpha characters must be upper case. Leading or trailing zeros or slashes (/) are not permitted. This subfield can also contain only the numeric values from the Cardmember's address.

Bit 111

Definition

Format

Optionality

Encryption Data

Encryption Data contains the data for various encryption activities.

AN(44) max LLLLVAR

Acquirer to Network: Conditional. The Acquirer must be certified to send Bit 111 to the Network. Acquirers submitting PIN data in Bit 111 must not submit PIN data in Bit 52 - Personal Identification Number (PIN).

Network to Issuer: Conditional. The Issuer must be certified to receive Bit 111 from the Network. The Network will pass PIN data received in either Bit 52 OR Bit 111. PIN data will not be passed in both Bit 52 and Bit 111 in the same POS Authorization Request (1100) message.

Notes

This field supports multiple data elements identified in the table below. Certified Participants will send/receive the appropriate data, depending on their specific needs.

Bit 52 - Personal Identification Number (PIN) must not be present when PIN data is sent in Bit 111.

American Express DKE PIN Encryption Data Transactional Layout

Bit	Position	Data Element	Subfield Length	Subfield Type	Optionality		Valid Value	Description
					A to N	N to I		
1	1-4	Variable Length Indicator (VLI)	4 bytes	Numeric (EBCDIC) LLLVAR	M	M		VLI is the total length of variable data in this data field (not including VLI).
2	5-6	Primary ID	2 bytes	Alphanumeric	M	M		Primary ID (Card Type Code) is constant literal "AX" (American Express)
3	7-9	Secondary ID	3 bytes	Alphanumeric	M	M	DKE = Dynamic Key Encryption	Secondary ID (Data Type Code) is variable based on Encryption Type
4	10-11	PIN Block VLI	2 bytes	Numeric (EBCDIC LLVAR)	M	M		Total length of variable data in the PIN Block (not including VLI)
5	12-27	PIN Block	16 bytes Max LLVAR	Binary (Hexadecimal configuration)	M	M		PIN encryption data block
6	28-29	Key Check VLI	2 bytes	Numeric (EBCDIC LLLVAR)	M	M		Total length of variable data in the Key Check Value (not including VLI)
7	30-34	Key Check Value	5 bytes Max LLVAR	Binary (Hexadecimal configuration)	M	M		Key check value for PIN encryption key
8	35-42	Reserved for Future Use	8 bytes	Numeric	M	M		Unused numeric fields should be zero filled

American Express DUKPT PIN Encryption Data Transactional Layout

Bit	Position	Data Element	Subfield Length	Subfield Type	Optionality		Valid Value	Description
					A to N	N to I		
1	1-4	Variable Length Indicator (VLI)	4 bytes	Numeric (EBCDIC) LLLVAR	M	M		VLI is the total length of variable data in this data field (not including VLI).
2	5-6	Primary ID	2 bytes	Alphanumeric	M	M		Primary ID (Card Type Code) is constant literal "AX" (American Express)
3	7-9	Secondary ID	3 bytes	Alphanumeric	M	M	DUK = DUKPT	Secondary ID (Data Type Code) is variable based on Encryption Type

American Express DUKPT PIN Encryption Data Transactional Layout (continued)

4	10-11	PIN Block VLI	2 bytes	Numeric (EBCDIC LLVAR)	M	M		Total length of variable data in the PIN Block (not including VLI)
5	12-27	PIN Block	16 bytes Max LLVAR	Binary (Hexadecimal configuration)	M	M		PIN encryption data block
6	28-29	Key Serial Number VLI	2 bytes	Numeric (EBCDIC LLVAR)	M	M		Total length of variable data in the Key Serial Number Value (not including VLI)
7	30-41	Key Serial Number (KSN)	12 bytes Max	Numeric (Hexadecimal configuration)	M	M		A unique key for each DUKPT Transaction.
8	42-44	DUKPT Format	3 bytes	Alphanumeric	M	M		DUKPT format indicator.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Request (1100)
Bit 112 Payment Account Reference (PAR) Message Layout

Bit 112 **Payment Account Reference (PAR)**

Definition

Unique, non-financial data element used to identify payment accounts across multiple form factors.

Format X(82) LLLVAR

Optionality **Acquirer to Network:** Conditional

Network to Issuer: Conditional

Notes Acquirers must provide the requested flag in the POS Authorization Request (1100) message in order for the PAR to be returned in the POS Authorization Response (1110) message.

The Network will populate Bit 112 - Payment Account Reference (PAR) of the POS Authorization Request (1100) and send it only to certified Issuers.

Table 5-3. Bit 112 - Payment Account Reference (PAR) Layout

Position	Data Element	Subfield Length	Subfield Type	Optionality (M/O/C)	Valid Values	Description
1-3	Variable Length Indicator	3 bytes	Numeric (EBCDIC)	M		VLI indicates total length of variable data in this data field (not including VLI).
4-5	Primary ID	2 bytes	Alphanumeric	M	AX	Primary ID (Card Type Code) is constant literal "AX" (American Express)
6-8	Secondary ID	3 bytes	Alphanumeric	M	AE1	Secondary ID (Data Type Code) is constant literal "AE1" (Payment Account Data)
9-12	Bitmap Identifier	4 bytes	Binary (Hexadecimal configuration)	M		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the bitmap, the layout consists of at least (1) of the following subfields. Each bit position of the 32 bit/4 byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, the corresponding subfield will be present.
SF1	Payment Account Reference (PAR)	35 byte	Alphanumeric	C		A unique identifier associated with a specific Cardholder PAN. This identifier can be used in place of sensitive consumer identification fields and transmitted across the payments ecosystem to facilitate consumer identification.
SF2	Reserved for Future Use	35 bytes	Alphanumeric	N/A		Reserved for future American Express use. Bit designation will of "OFF".

Bit 113 **Acceptance Environment Data**

Definition This bit identifies the initiating party and PSD2 elements.

Format X(18) LLLVAR

Optionality **Acquirer to Network:** Optional

Network to Issuer: Optional

Notes The Network populates Bit 113 - Acceptance Environment Data in the POS Authorization Request (1100) message and sends it **only** to certified Issuers.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Request (1100)
Bit 113 Acceptance Environment Data Message Layout

Table 5-4. Bit 113 - Acceptance Environment Data Layout

Position	Data Element	Subfield Length	Subfield Type	Optionality (M/O/C)	Valid Values	Description
1-3	Variable Length Indicator (VLI)	3 bytes	Numeric (EBCDIC)	M		VLI is the total length of variable data in this data field (not including VLI)
4-5	Primary ID	2 bytes	Alphanumeric	M	AX	Primary ID (Card Type Code) is constant literal "AX" (American Express)
6-8	Secondary ID	3 bytes	Alphanumeric	M	AED	Secondary ID (Data Type Code) is constant literal "AED" (Acceptance Environment Data)
9	Initiating Party Indicator	1 byte	Alphanumeric	O	0=Customer 1=Merchant	Indicator that communicates if the Authorization was initiated by the Cardholder (Cardmember-Initiated Transaction (CIT) or the S/E (Merchant-initiated Transaction (MIT)
10-13	Bitmap Identifier	4 bytes	Binary (Hexadecimal configuration)	M		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the Bitmap, the layout consists of at least (1) of the following subfields. Each bit position of the 32 bit/4 byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, then the corresponding subfield will be present.
SF1	Unattended terminal for transport fares and parking fees exemption	1 byte	Alphanumeric	C	0=Not claimed 1=Claimed	Indicator that communicates if an exemption to PSD2 Strong Customer Authentication (SCA) is being claimed based on the Unattended Terminal Transport Fare and Parking Fee Exemption.
SF2	Secure corporate payment process and protocol exemption	1 byte	Alphanumeric	C	0=Not claimed 1=Claimed	Indicator that communicates if an exemption to PSD2 Strong Customer Authentication (SCA) is being claimed based on Secure Corporate Payment Process and Protocol Exemption.
SF3	Transaction risk analysis exemption	1 byte	Alphanumeric	C	0=Not claimed 1=Claimed	Indicator that communicates if an exemption to PSD2 Strong Customer Authentication (SCA) is being claimed based on the Transaction Risk Analysis Exemption.

Table 5-4. Bit 113 - Acceptance Environment Data Layout (continued)

Position	Data Element	Subfield Length	Subfield Type	Optionality (M/O/C)	Valid Values	Description
SF4	Delegated Authority	1 byte	Alphanumeric	C	0=False 1=True	Indicator that communicates if PSD2 Strong Customer Authentication (SCA) requirements have already been applied to the Transaction on the basis of Delegated Authority.
SF5	Authentication Outage Indicator	1 byte	Alphanumeric	C	0=Not Claimed 1=Claimed	Indicator that communicates if an exception to Strong Customer Authentication (SCA) requirements is being claimed when an E-commerce Transaction is unable to reach SafeKey systems. Authentication Outage Indicator is intended for E-commerce Transactions based out of the United Kingdom.

POS Authorization Response (1110)

Description

This message is used to respond to a Transaction Authorization Request to purchase goods or services at an S/E.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Response (1110). Refer to the [Programming Notes for POS Authorization Response \(1110\) on page 5-39](#) for more instruction.

POS Authorization Response (1110) Message Layout

EMV Tag	Bit Position	Format	Optionality		Field Name
			I to N	N to A	
	N/A	N(4)	M	M	Message Type Identifier
	N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
	1	8 bytes, 64 bits for each bitmap Binary (Hex)	C	C	Bitmap Secondary
	2	N(21) max LLVAR	ME	ME	Primary Account Number (PAN)
	3	N(6)	ME	ME	Processing Code
	4	N(12)	CE	CE	Amount, Transaction
	5	N(12)	CE	N/A	Amount, Reconciliation
	7	N(10)	N/A	OE	Date and Time, Transmission
	11	N(6)	ME	ME	Systems Trace Audit Number
	12	N(12)	ME	ME	Date and Time, Local Transaction
	24	AN(3)	CE	CE	Function Code
	30	N(24)	C	C	Amount, Original
	31	N(40) max LLVAR	ME	M	Acquirer Reference Data, Transaction Identifier (TID)
	32	N(13) LLVAR	ME	ME	Acquiring Institution Identification (AIN) Code
	34	AN(30) Max LLVAR (with length indicator)	C	C	Primary Account Number, Extended
	37	AN(12)	OE	OE	Acquirer Reference Number (ARN)

POS Authorization Response (1110) Message Layout (continued)

EMV Tag	Bit Position	Format	Optionality		Field Name
			I to N	N to A	
	38	AN(6)	C	C	Approval Code
	39	N(3)	M	M	Action Code
	41	ANS(8)	OE	OE	Card Acceptor Terminal Identification
	42	AN(15)	ME	ME	Card Acceptor Identification Code
	44	AN(4) max LLVAR	C	C	Additional Response Data
	48	AN(353) LLLVAR	C	C	Deferred Payment Plan (DPP)
	49	N(3)	ME	ME	Currency Code, Transaction
	50	N(3)	ME	N/A	Currency Code, Reconciliation
	54	ANS(123) 4 bytes min 123 bytes max LLLVAR	C	C	Amount, Additional
	55	AN(259) max (includes length indicator) LLLVAR	C	C	ICC System Related Data
		AN(4) EBCDIC	ME	ME	Subfield 1 ICC Header Version Name
		(2) Binary (numeric)	ME	ME	Subfield 2 ICC Header Version Number
91		(17) max LLVAR Binary Hexadecimal (includes length LL indicator)	C	C	Subfield 3 Issuer Authentication Data Sub-Subfield 1 Authorization Response Cryptogram (ARPC) (8) Sub-Subfield 2 ARPC Response Code N(2) (ASCII)
		(129) max LLLVAR Binary Hexadecimal (includes length LLL indicator)	C	C	Subfield 4 Issuer Script Data
	60	AN(210) max LLLVAR	CE	CE	Market Specific Data
		AN(353) max LLLVAR			Market Specific Data with Additional Authorization Data

POS Authorization Response (1110) Message Layout (continued)

EMV Tag	Bit Position	Format	Optionality		Field Name
			I to N	N to A	
	61	4 bytes min, 103 bytes max LLLVAR	C	C	National Use Data
		(3) (numeric EBCDIC)	ME	ME	Subfield 1 Variable Length Indicator (VLI)
		(2) (alphanumeric)	ME	ME	Subfield 2 Primary ID
		(3) (alphanumeric)	ME	ME	Subfield 3 Secondary ID
		(1) (alphanumeric)	CE	CE	Subfield 4 AEVV Validation Result
	112	X(82) LLLVAR	CE	C	Payment Account Reference (PAR)
	113	X(18) LLLVAR	CE	CE	Acceptance Environment Data

Programming Notes for POS Authorization Response (1110)

Bit N/A

Message Type Identifier

Definition

This code identifies the type of message sent by the forwarding institution.

Format

N(4)

Optionality

Issuer to Network: Mandatory

Notes

Network to Acquirer: Mandatory

Required Value: 1100

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

Format

8 bytes, 64 bits for each bitmap, Binary (Hex)

Optionality

Issuer to Network: Mandatory

Notes

Network to Acquirer: Mandatory

If the field is mandatory or optional and the Participant uses that field, its assigned bitmap position must contain a value of 1. If the field is optional and is not used, its assigned bitmap position must contain a value of 0.

Bit 1

Bitmap Secondary

Definition

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

Format

8 bytes, 64 bits for each bitmap, Binary (Hex)

Optionality

Acquirer to Network: Conditional

Notes

Network to Issuer: Conditional

If the field is mandatory or optional and the Participant uses that field, its assigned bitmap position must contain a value of 1. If the field is optional and is not used, its assigned bitmap position must contain a value of 0.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Response (1110)
Bit 2 Primary Account Number (PAN) Message Layout

Bit 2

Definition

Primary Account Number (PAN)

A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.

Format

N(21) max LLVAR

Optionality

Issuer to Network: Mandatory Echo

Network to Acquirer: Mandatory Echo

Bit 3

Definition

Processing Code

This is the service being requested.

Format

N(6)

Optionality

Issuer to Network: Mandatory Echo

Network to Acquirer: Mandatory Echo

Notes

Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 4

Definition

Amount, Transaction

The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.

Format

N(12)

Optionality

Issuer to Network: Conditional Echo.

- Fully Authorized Transaction: If the requested amount is approved, this field is echoed back to the Network.
- Partial Authorization: If a partial amount is approved, only the partially approved amount is sent back to the Network.

Network to Acquirer: Conditional Echo. This bit is sent to the Acquirer with only the approved Transaction amount.

Bit 5

Definition

Amount, Reconciliation

This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.

Format

N(12)

Optionality

Issuer to Network: Conditional Echo.

- Fully Authorized Transaction: If the requested amount is approved, this field is echoed back to the Network.
- Partial Authorization: If a partial amount is approved, only the partially approved amount is sent back to the Network in the currency defined by the Issuer (Bit 50).

Network to Acquirer: Not applicable

Bit 7

Definition

Date and Time, Transmission

This is the local date and time when the Acquirer transmits the request to the Network.

Format

N(10) MMDDHHMMSS

Optionality

Issuer to Network: Not Applicable

Network to Acquirer: Optional Echo

Notes

The Network does not edit for a valid date and time.

Bit 11

Definition

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format

N(6)

Optionality

Issuer to Network: Mandatory Echo

Network to Acquirer: Mandatory Echo

Bit 12

Definition

Date and Time, Local Transaction

This is the local date and time when a Transaction occurred.

Format N(12) YYMMDDHHMMSS
Optionality **Issuer to Network:** Mandatory Echo
Network to Acquirer: Mandatory Echo

Bit 24

Function Code

Definition This code shows the reason for a message in its Message Type.
Format AN(3)
Optionality **Issuer to Network:** Conditional Echo. If the Issuer receives a Function Code from the Network, it must be echoed.
Network to Acquirer: Conditional Echo. If the Network receives a Function Code from the Issuer, it must be echoed.
Notes Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 30

Amount, Original

Definition The Transaction Amount from the POS Authorization Request (1100) message, when a partial amount is approved.

Format N(24)
Optionality **Issuer to Network:** Conditional
Network to Acquirer: Conditional
This bit is required if the Action Code = 002 Partial Approval

If the participating S/E requested a Partial Authorization and a partial amount was approved, Bit 30 contains the first Transaction Amount of the Authorization Request.

Position	Use
1-12	First Transaction Amount from Bit 4 Amount, Transaction of the 1100 POS Authorization Request message
13-24	Zero-filled

Bit 31

Acquirer Reference Data, Transaction Identifier (TID)

Definition This unique number is given by the Network and follows a Transaction during its life cycle.
Format N(40) max LLVAR
Optionality **Issuer to Network:** Mandatory Echo. Issuers must echo the TID received from the POS Authorization Request (1100) message.
Network to Acquirer: Mandatory
Notes **Network to Acquirer:** The Network will pass the TID to the Acquirer.
Refer to [The Role of the Transaction Identifier on page 2-2](#) for more instruction about the TID and the authorization life cycle.

Bit 32

Acquiring Institution Identification (AIN) Code

Definitions This is a unique identification code given to an Acquirer by the Network.
Format N(13) LLVAR
Optionality **Issuer to Network:** Mandatory Echo
Network to Acquirer: Mandatory Echo
Notes The variable length indicator is not included in the length of the bit.
For example, Acquiring Institution Identification code 45678912345 appears as 1145678912345.
Required Value: 06370100 American Express

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POS Authorization Response (1110)
Bit 34 Primary Account Number, Extended Message Layout

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

Bit 34

Definitions

Format

Optionality

Notes

Primary Account Number, Extended

This field is a multipurpose field which provides the Primary Account Number or associated data.

AN(30) Max LLVAR (with length indicator)

Issuer to Network: Conditional. This bit is sent by Issuers certified for Expresspay Translation (PAN Only request or PAN request extended - PAN with PAN Expiry Date).

Currently, the Network supports Function Code 194 - Expresspay Translation (PAN Only request) when sending a POS Authorization Request (1100) message to certified Issuers, or when receiving a POS Authorization Response (1110) message from certified Issuers.

Network to Acquirer: Conditional. This bit is sent to Acquirers certified for Expresspay Translation (PAN Only request or PAN request extended - PAN with PAN Expiry Date).

For an Expresspay Translation, this bit may contain data identifying a reply to an Expresspay Transaction (PAN Only request or PAN request extended - PAN with PAN Expiry Date).

Expresspay Translation (PAN Only request)

Expresspay Translation (PAN Request) - this format is supported when Function Code 194 is received. When the Primary Account Number (PAN) is supplied, Bit 34 is populated as follows:

- Two (2) bytes numeric Variable Length Indicator (VLI)
- One (1) byte alpha Alias to PAN Request Result shows request results:
 - Y = PAN returned
 - N = PAN not found/Does not exist
 - R = Attempt PAN request again
- Examples of possible bit values:
 - PAN returned: 16Y271234567890123
 - PAN not found/Does not exist: 01N
 - Attempt PAN request again: 01R

Expresspay Translation (PAN Request Extended - PAN with PAN Expiry Date)

This format is provided when Function Code 196 is received. When the PAN Expiry Date with Primary Account Number (PAN) is supplied, Bit 34 is populated as follows:

- Two (2) bytes numeric Variable Length Indicator (VLI)
- One (1) byte alpha Alias to PAN Request Result shows request results:
 - E = PAN Expiry Date¹ returned with PAN
 - N = PAN not found/Does not exist
 - R = Attempt PAN request again
- Examples of possible field values:
 - PAN returned: 20E1601271234567890123
 - PAN not found/Does not exist: 01N
 - Attempt PAN request again: 01R

1. PAN Expiry Date Format: YYYY

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POS Authorization Response (1110)
Bit 37 Acquirer Reference Number (ARN) Message Layout

Bit 37	Acquirer Reference Number (ARN)
Definition	A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.
Format	AN(12)
Optionality	Issuer to Network: Optional Echo Network to Acquirer: Optional Echo
Bit 38	Approval Code
Definition	A unique code assigned by the Card Transaction authorizing institution to indicate the disposition for the Transaction.
Format	AN(6)
Optionality	Issuer to Network: Conditional - If the POS Authorization Request (1100) message is approved, this bit must contain a number. The first number of the left justified value cannot be 0 (zero). - If the POS Authorization request (1100) message is denied, this bit can be present but contains spaces. - This bit is space-filled for Prepaid Balance Return requests. Network to Acquirer: Conditional. If this bit is present, it is sent to the Acquirer.
Notes	The Network makes the decision to send the left 2 or the full 6 digits to the Acquirer depending on the Acquirer capabilities.
Bit 39	Action Code
Definition	The type of action done or the type of action to do for a Transaction.
Format	N(3)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Action Code 002 is returned to the Network only if the Issuer approved an amount less than the requested amount. Refer to the Action Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 41	Card Acceptor Terminal Identification
Definition	This code identifies a POS terminal at a Card Acceptor location.
Format	ANS(8)
Optionality	Issuer to Network: Optional Echo Network to Acquirer: Optional Echo
Bit 42	Card Acceptor Identification Code
Definition	The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.
Format	AN(15)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 44	Additional Response Data
Definition	This field contains the Issuer response to address verification requests or PCSC validation requests in the POS Authorization Request (1100) message.
Format	AN(4) max LLVAR
Optionality	Issuer to Network: Conditional. This bit must contain a value if Bit 63 and/or 53 are present in the POS Authorization Request (1100) to the Issuer. Network to Acquirer: Conditional. If a certified Issuer sends this bit and the Acquirer is certified to receive it, then the Network sends it to the Acquirer.
Notes	Positions

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POS Authorization Response (1110)
Bit 44 Additional Response Data Message Layout

This field has four positions. The first two positions designate the length, the third position is used for address verification response when requested in Bit 63 Automated Address Verification (AAV), and the fourth position is used for PCSC, DCSC or OTP validation when requested in Bit 53 Security Related Control Information of the 1100 message. Use this field only when Bit 53, Bit 63, or the two are in the 1100 message.

Do not use the fourth position to return a validation response for 5CSC. Refer to the *Business and Operational Policies* manual for PCSC validation policy.

Positions 1 and 2 of the field designate the length as follows:

Position	Value	Use
1-2	01, 02	Variable Length Indicator (01 = AAV, 02 = AAV and PCSC/DCSC/OTP, or only PCSC/DCSC/OTP)

Position 3 is used by certified Issuers to supply the verification response for AAV requests (1100 message, Bit 63 Automated Address Verification). Acquirers receive the verification response when the address is supplied in Bit 63.

Position	Value	Use
3	*	Automated Address Verification Response

The Automated Address Verification Response Codes table gives the value and description for each AAV Code:

Automated Address Verification Response Codes (Position 3)	
Value	Description
A	Address is correct, Postal Code is incorrect
N	Neither Address nor Postal Code is correct
R	Retry, temporarily unavailable
S ¹	Service not allowed, Acquirer not certified
U	Information is unavailable; AAV is not supported by the Issuer
Y	Yes, Address and Postal Code are correct
Z	Postal Code is correct, Address incorrect or Address not included in request
~2	Non-response: AAV is not present and PCSC is present

1. S is for use by AEGNS only
2. "~" represents a space.

Position 4 is used by Issuers to supply the verification response for PCSC, DCSC, or OTP requests (1100 message, Bit 53 Security Related Control Information). Certified Acquirers receive the verification response when the PCSC, DCSC, or OTP is supplied in Bit 53.

Position	Value	Use
4	#	PCSC/DCSC/OTP Validation Match Results

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Response (1110)
Bit 48 Deferred Payment Plan (DPP) Message Layout

The Card Security Code Validation Codes table gives the value and description for each PCSC or DCSC Validation Code:

Card Security Code or OTP Validation Code (Position 4)	
Value	Description
N	PCSC, DCSC, or OTP not matched
U	PCSC, DCSC, or OTP unchecked
Y	PCSC, DCSC, or OTP matched

Examples:

As this field can be used to return the results for AAV and/or PCSC/DCSC/OTP validation, these examples show how to format the field correctly.

- **AAV Request Only:** 01A = Address is correct, Postal Code is incorrect
- **AAV Request with PCSC, DCSC, or OTP:** 02AY = AAV shows Address is correct, Postal Code is incorrect, and PCSC/DCSC/OTP shows matched
- **PCSC, DCSC, or OTP Request Only:**
 - 02~Y = No AAV requested and PCSC/DCSC/OTP shows matched
 - "~" represents a space.

Bit 48

Definition

Deferred Payment Plan (DPP)

This is the Deferred Payment Plan data, permitting charges to be billed in equal installments rather than in one (1) total billing.

Format

AN(353) LLLVAR

Optionality

Issuer to Network: Conditional. This bit must be populated for a DPP Transaction.

Network to Acquirer: Conditional. If this bit is present, it is sent to the Acquirer.

Notes

Issuer DPP Pre-Authorization

This is a pre-authorization request for deferred or extended payments, including the plan code data.

Bit Layout (Issuer DPP Pre-Authorization Response)

270DP05bbbb7777REQDTL1ABDTL2ABDTL1BLDTL2BLDTLccyyymmdd

A three-digit variable length indicator is before the Issuer DPP Pre-Authorization data. This indicator is not included in the length of the bit. If the Issuer does not support the function, or if the Issuer is offline, this field contains an echo of the contents.

Valid Values:

Position	Value	Use
1-3	270	Variable Length Indicator
4-7	DP05	Issuer DPP identifier
8-11		Binary (32-bit) bitmap in hexadecimal configuration identifying the fields in the DPP Structure. The bitmap value for Issuer DPP Pre-Authorization Responses is x "401F8000." After the bitmap, Bit 48 has seven (7) subfields.

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POS Authorization Response (1110)
Bit 48 Deferred Payment Plan (DPP) Message Layout

Subfield	Format	Notes
2 AN(4) Plan Type		Required Value: 0005 Issuer-sponsored plan
12 Requested Plan Details	AN(50) Group	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49). AN(50) Group: • Interest Rate - N(10) • Number of Installments - N(4) • Amount of Installment - N(12) • Finance Charges - N(12) • Amount Plus Finance Charges - N(12)
13 One Installment Above Requested Plan Details		Subfield must be character zeros if the installment plan is not offered. Use the same format as Subfield 12.
14 Two Installments Above Requested Plan Details		Subfield must be character zeros if the installment plan is not offered. Use the same format as Subfield 12.
15 One Installment Below Requested Plan Details		Subfield must be character zeros if the installment plan is not offered. Use the same format as Subfield 12.
16 Two Installments Below Requested Plan Details		Subfield must be character zeros if the installment plan is not offered. Use the same format as Subfield 12.
17 Date of First Installment Payment	N(8)	Date Format: CCYYMMDD

Issuer DPP Authorization

The deferred payment plan data for the Issuer Deferred Payment Plan.

A three-digit variable length indicator if before the Issuer DPP Authorization data. This indicator is not included in the length of the bit.

Bit Layout (Issuer DPP Authorization Response)

116DP05bbbb77779999iiiiiiiiiiddddddddddddddmmmmmmmmaaaaaaaagggggggggg
gssssssssssffffffftttttttt

Valid Values:

Position	Value	Use
1-3	116	Variable Length Indicator
4-7	DP05	Issuer DPP identifier
8-11		Binary (32-bit) bitmap in hexadecimal configuration identifying the fields in the DPP Structure. The bitmap value for Issuer DPP Authorization Responses is x"7FE00000."
		After the bitmap, Bit 48 has ten subfields.

Subfield	Format	Notes
2 Plan Type	AN(4)	Required Value: 0005 Issuer-sponsored plan

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POS Authorization Response (1110)
Bit 48 Deferred Payment Plan (DPP) Message Layout

Subfield	Format	Notes
3 Number of Installments	N(4)	
4 Amount of Installment	N(12)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49).
5 Plan Description	AN(20)	
6 Monthly Interest Rate	N(10)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49).
7 Annual Interest Rate	N(10)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49)
8 Tip	N(12)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49)
9 Surcharges	N(12)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49)
10 Finance Charges	N(12)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49)
11 Amount Plus Finance Charges	N(12)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49)

Acquirer DPP Authorization

This is the deferred payment plan data for the Acquirer Deferred Payment Plan.

A three-digit variable length indicator is before the Acquirer DPP Authorization data. This indicator is not included in the length of the bit.

Bit Layout (Acquirer DPP Authorization Response)

[illegible]

Valid Values:

Position	Value	Use
1-3	048	Variable Length Indicator
4-7	DP03	Acquirer DPP identifier
8-11		Binary (32-bit) bitmap in hexadecimal configuration identifying the fields in the DPP Structure. The bitmap value for Acquirer DPP Authorization Responses is x"78000000."
		After the bitmap, Bit 48 has four subfields.

Subfield	Format	Notes
2 Plan Type	AN(4)	Required Value: 0003 Acquirer-sponsored plan
3 Number of Installments	N(4)	
4 Amount of Installment	N(12)	The position of the implied decimal point is determined by the Currency Code, Transaction (Bit 49).
5 Plan Description	AN(20)	

Bit 49

Definition

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Issuer to Network: Mandatory Echo

Notes

Network to Acquirer: Mandatory Echo

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition

This is the numeric code of the currency defined by the Issuer.

Format

N(3)

Optionality

Issuer to Network: Mandatory Echo

Notes

Network to Acquirer: Not Applicable

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 54

Definition

The available amount of funds in the Card currency remaining on a Card.

Format

ANS(123), 4 bytes min, 123 bytes max, LLLVAR

Optionality

Issuer to Network: Conditional. This bit may only be sent when a Partial Authorization and/or Balance Return Function Code is used (Bit 24).

- **Required:** Anonymous, Non-reloadable Prepaid products
- **Optional:** Reloadable Prepaid products
- **Not Allowed:** Debit Card products

Network to Acquirer: Conditional

- If a Partial Authorization and/or Balance Return Function Code is used (Bit 24), and a certified Issuer sends this bit and the Acquirer is certified to receive it, then the Network sends it to the Acquirer.
- Bit 54 will be dropped by AEGNS and not sent to the Acquirer if available balance is returned from the Issuer on Debit Card products.

Notes

S/Es supporting balance return may show the Card currency and available balance on the POS terminal and/or print it on the Cardmember receipt.

Transaction Layout:

Position	Value	Use
1-3	020	Variable Length Indicator
4-5	00	Required Value: 00=Account Type Code
6-7	05	Required Value: 05=Amount Type Code
8-10		Numeric Currency Code of the Card
11	C	Required Value: C = Credit
12-23		Prepaid Card balance, right justified, zero-filled, decimal implied

Bit 55

Definition

This is the data sent to and from the Integrated Circuit Card and used in an AEIPS or Expresspay Transaction.

Format

AN(259) max (includes length indicator) LLLVAR

Optionality

Issuer to Network: Conditional. This bit must be populated for Transactions with Bit 55 populated in the POS Authorization Request (1100).

Network to Acquirer: Conditional. If this bit is present, it is sent to the Acquirer.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Response (1110)
Bit 55 ICC System Related Data Message Layout

Notes

Response data must be sent in this field if Bit 55 ICC System Related Data is provided in the POS Authorization Request (1100) message.

Do not use Bit 55 to determine if the Transaction was approved or declined. Use Bit 39 - action Code in the POS Authorization Response (1110) message.

The following table shows the format of Bit 55 when it is included in a POS Authorization Response (1110) message.

Subfield	Name	Length	Source
1	ICC Header Version Name	4	Issuer (echo from POS Authorization Request (1100) message)
2	ICC Header Version Number	2	Issuer (echo from POS Authorization Request (1100) message)
3	Issuer Authentication Data (IAD)	up to 17 bytes	Issuer
4	Issuer Script Data	up to 129 bytes	Issuer

The Issuer is required to create this bit and include the data within it. The header is simply a reflection of the header contained in the corresponding request. Refer to the *EMV Chip Issuer Implementation Guide* for instructions on how to create the Issuer Authentication Data (IAD).

Issuer Scripts are not supported for Expresspay Transactions. Issuer Script Data sent in an Expresspay POS Authorization Response (1110) message will not be processed by the chip on the Card.

Bit 55 Layout

The layout contains these subfields.

EMV Tag	Subfield	Format	Optionality		Notes
			I to N	N to A	
	1 ICC Header Version Name	AN(4) EBCDIC	Mandatory Echo	Mandatory Echo	Required Value: AGNS
	2 ICC Header Version Number	(2) Binary (numeric)	Mandatory Echo	Mandatory Echo	Required Value: 0001

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POS Authorization Response (1110)
Bit 60 Market Specific Data Message Layout

EMV Tag	Subfield	Format	Optionality		Notes
91	3 Issuer Authentication Data	(17) max LLVAR Binary Hexadecimal (includes length LL indicator)	Conditional	Conditional	If Card authentication was correct, include this subfield. If Card authentication is not done or is not correct (i.e., missing or invalid cryptogram in the 1100 message), do not return this subfield. When this subfield is not present, it is referred to as a 'downgraded response' and is a valid Transaction response. As Subfield 3 is a variable length field of type LLVAR, the "LL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLVAR length byte is in addition to the 16 byte maximum data length of the actual data (i.e., total 17 bytes). - Sub-Subfield 1 Authorization Response Cryptogram (ARPC) (8) - Sub-Subfield 2 ARPC Response Code N (2) (ASCII) Note: Do not use IAD to determine if the transaction was approved or declined. Use Bit 39 - Action Code in the POS Authorization Response (1110) message.
	4 Issuer Script Data	(129) max LLLVAR Binary Hexadecimal (includes length LLL Indicator)	Conditional	Conditional	This field can only be used if Subfield 3, Issuer Authentication Data, is present. As Subfield 4 is a variable length field of type LLLVAR, the "LLL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLLVAR length byte is in addition to the 128 byte maximum data length of the actual data (i.e., total 129 bytes).

Bit 60

Market Specific Data

Definition

This bit provides additional authorization data.

Market Specific Data

Format

AN(210) max LLLVAR

Optionality

Issuer to Network: Conditional Echo. This optionality can be different by market. Contact your AEGNS Representative for more instruction.

Network to Acquirer: Conditional Echo. This optionality can be different by market. Contact your AEGNS Representative for more instruction.

Market Specific Data with Additional Authorization Data

Format

AN(353) max LLLVAR

Optionality

Issuer to Network: Conditional Echo. If a certified Issuer receives Bit 60 from the Network in the POS Authorization Request (1100) message, then the Issuer must echo Bit 60. The optionality for Subfield 1 - Market Specific Data can be different by market. Contact your AEGNS Representative for more instruction.

Network to Acquirer: Conditional Echo. If the Network receives Bit 60 from the Issuer, it is sent to the Acquirer. The optionality for Subfield 1 - Market Specific Data can be different by market.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Response (1110)
Bit 61 National Use Data Message Layout

For Payment Token Transactions, the Network will send Subfield 5 – Token Requestor ID in the POS Authorization Response (1110) message if the Acquirer has provided the required value in the POS Authorization Request (1100) message, and the Transaction has been approved. Contact your AEGNS Representative for more instruction.

Bit 61

National Use Data

Definition This is an ISO field that supplies Cardmember authentication data during Authorization.

Format 4 bytes min, 103 bytes max LLLVAR

Optionality **Issuer to Network:** Conditional. This bit is required for SafeKey Transactions.

Network to Acquirer: Conditional. If this bit is present, it is sent to the Acquirer.

Notes **Bit 61 Layout**

The layout contains these subfields.

Subfield	Format	Optionality		Notes
		I to N	N to A	
1 Variable Length Indicator (VLI)	(3) (numeric EBC-DIC)	Mandatory Echo	Mandatory Echo	
2 Primary ID	(2) (alphanumeric)	Mandatory Echo	Mandatory Echo	Required Value: AX
3 Secondary ID	(3) (alphanumeric)	Mandatory Echo	Mandatory Echo	Valid Value: ASK
4 AEVV Validation Result	(1) (alphanumeric)	Conditional Echo	Conditional Echo	The AEVV Validation Result shows that the Issuer validated the AEVV. If the AEVV Validation Result is present in Bit 62 of the Authorization Request (1100) message, then the same AEVV Validation result value must be echoed in Bit 61 of the Authorization Response (1110) message. Refer to the Fraud Indicators section of the <i>Codes Reference Guide</i> for a list of valid codes and descriptions.

Bit 112

Payment Account Reference (PAR)

Definition

Unique, non-financial data element used to identify payment accounts across multiple form factors.

Format X(82) LLLVAR

Optionality **Issuer to Network:** Conditional Echo

Network to Acquirer: Conditional

Notes Acquirers must provide the requested flag in the POS Authorization Request (1100) message in order for the PAR to be returned in the POS Authorization Response (1110) message.

The Network will populate Bit 112 - Payment Account Reference (PAR) of the POS Authorization Response (1110) and send it only to certified Acquirers.

Field 113

Acceptance Environment Data

Definition This bit identifies the initiating party and PSD2 elements.

Format X(18) LLLVAR

Optionality **Issuer to Network:** Conditional Echo

Network to Acquirer: Conditional Echo

POS Authorization Advice (1120)

Description

This message informs Issuers of an Authorization Request that is approved or declined due to:

- Canceled status of the S/E
- Voice authorization referrals that the Acquirer resolves
- Transactions decisioned during Stand-In processing
- Online processing of refunds
- Fraud Probability Score and Recommendation Code (declined by Global Network Services Authorization System (GNSAS) on behalf of a certified Issuer)

In addition, Issuers may opt for GNSAS to decline digital wallet Authorization Requests on their behalf when key digital wallet Token data requirements are not met. This message notifies Issuers when a digital wallet Authorization Request has been declined on their behalf.

The Network does not send this message directly from the Acquirer to the Issuer. As an alternative, the Network responds to the Acquirer, keeps the Transaction in the Store and Forward queue, and sends the message along to the Issuer from the queue.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Advice (1120). Refer to the [Programming Notes for POS Authorization Advice \(1120\)](#) on page 5-53 for more instruction.

POS Authorization Advice (1120) Message Layout

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	M	M	Primary Account Number (PAN)
3	N(6)	M	M	Processing Code
4	N(12)	M	M	Amount, Transaction
5	N(12)	N/A	M	Amount, Reconciliation
11	N(6)	M	M	Systems Trace Audit Number
12	N(12)	M	M	Date and Time, Local Transaction
14	N(4)	O	N/A	Card Expiration Date

POS Authorization Advice (1120) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
22	AN(12)	M	M	Point of Service Data Code
23	N(3)	N/A	C	PAN Sequence Number
24	AN(3)	M	M	Function Code
26	N(4)	M	M	Card Acceptor Business Code
30	N(24)	N/A	C	Amount, Original
31	N(40) max LLVAR	N/A	M	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	O	O	Acquiring Institution Identification (AIN) Code
34	AN(30) max LLVAR (including length indicator)	N/A	C	Primary Account Number, Extended
35	ANS(39) max LLVAR	C	N/A	Track 2 Data
38	AN(6)	M	M	Approval Code
39	N(3)	M	M	Action Code
42	AN(15)	M	M	Card Acceptor Identification Code
43	ANS(101) max LLVAR	C	C	Card Acceptor Name and Location
45	ANS(78) max LLVAR	C	N/A	Track 1 Data
48	AN(353) max LLLVAR	C	C	Stand-In Parameter Information for Non-DPP Requests Stand-In Parameter Information for DPP Requests
49	N(3)	M	M	Currency Code, Transaction
50	N(3)	N/A	M	Currency Code, Reconciliation
56	N(37) max LLVAR	N/A	O	Original Data Elements
60	AN(210) max LLLVAR	C	C	Market Specific Data
	AN(316) max LLLVAR			Market Specific Data with Additional Authorization Data
61	4 bytes min, 103 bytes max LLLVAR	C	C	National Use Data
62	AN(5) min AN(60) max LLLVAR	N/A	C	Fraud Indicators

Programming Notes for POS Authorization Advice (1120)

Bit N/A

Message Type Identifier

Definition

This code identifies the type of message sent by the forwarding institution.

Format

N(4)

Optionality

Acquirer to Network: Mandatory

Notes	Network to Issuer: Mandatory Required Value: 1120 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit N/A	Bitmap Primary
Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap, Binary (Hex)
Optionality	Acquirer to Network: Mandatory
	Network to Issuer: Mandatory
Bit 2	Primary Account Number (PAN)
Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Acquirer to Network: Mandatory
	Network to Issuer: Mandatory
Notes	A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit. For example, a Cardmember Account Number of: 3759-876543-21001 appears as: 15375987654321001
Bit 3	Processing Code
Definition	This is the service being requested.
Format	N(6)
Optionality	Acquirer to Network: Mandatory
	Network to Issuer: Mandatory
Notes	Cash Disbursement Transaction: The Processing Code must be 014008. Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 4	Amount, Transaction
Definition	The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.
Format	N(12)
Optionality	Acquirer to Network: Mandatory
	Network to Issuer: Mandatory
Notes	If Processing Code = 220000 (Authorization Adjustment) and Function Code = 202 (Authorization Adjustment) then this bit is populated with the last Transaction amount. The decimal point is determined by the Currency Code, Transaction (Bit 49). For example, \$100.00 in U.S. Dollars shows as 000000010000.
Bit 5	Amount, Reconciliation
Definition	This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.
Format	N(12)
Optionality	Acquirer to Network: Not Applicable
	Network to Issuer: Mandatory

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POS Authorization Advice (1120)
Bit 11 Systems Trace Audit Number Message Layout

Notes	Authorization and Presentment can occur on different days. The reconciliation amounts can be different in the Authorization and Presentment messages. The decimal point is determined by the Currency Code, Reconciliation (Bit 50). For example, \$100.00 in USD shows as 000000010000.
Bit 11	Systems Trace Audit Number
Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.
Format	N(6)
Optionality	Acquirer to Network: Mandatory The Network returns this number in the Network to Acquirer piece of the POS Authorization Response (1130) message. Network to Issuer: Mandatory The Network gives this value. It is different than the value the Acquirer gives.
Bit 12	Date and Time, Local Transaction
Definition	This is the local date and time when a Transaction occurred.
Format	N(12) YYMMDDHHMMSS
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	This is the date and time when the Network performs Stand-In, and when the Acquirer received the Transaction from a canceled S/E.
Bit 14	Card Expiration Date
Definition	This is the Card Expiration Date embossed on the face of the Card.
Format	N(4) YYMM
Optionality	Acquirer to Network: Optional Network to Issuer: Not Applicable
Notes	The Network edits this field to make sure that it is a valid number.
Bit 22	Point of Service Data Code
Definition	This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).
Format	AN(12)
Optionality	Acquirer to Network: Mandatory
Notes	Network to Issuer: Mandatory The Network sets this field for Stand-In Advice messages. Refer to the Point of Service Data Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 23	PAN Sequence Number
Definition	A number distinguishing between separate Cards with the same Primary Account Number.
Format	N(3)
Optionality	Acquirer to Network: Not Applicable Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.
Bit 24	Function Code
Definition	This code shows the reason for a message in its Message Type.
Format	AN(3)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory

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Network Specifications - Authorization
POS Authorization Advice (1120)
Bit 26 Card Acceptor Business Code Message Layout

Notes Issuers supporting Network Stand-In must support all Estimated and Incremental Authorization Function Codes.
Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 26

Card Acceptor Business Code

Definition Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).

Format N(4)

Optionality **Acquirer to Network:** Mandatory

Notes **Network to Issuer:** Mandatory
Cash Disbursement: The Card Acceptor Business Code (MCC) must be 6010.

Payment Facilitator:

The Network will pass the MCC originally received in Bit 26 of the POS Authorization Request (1100), which must contain the MCC of the Sponsored Merchant. The MCC can be different for each Transaction dependent on the category of the Sponsored Merchant.

Bill Payment Provider:

The Network will pass the MCC originally received in Bit 26 of the POS Authorization Request (1100) which, as defined, should contain the MCC of the End Beneficiary. The MCC can be different for each Transaction dependent on the category of the End Beneficiary.

Refer to the Merchant Category Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 30

Amount, Original

Definition The Authorized Transaction Amount from the previous POS Authorization Response (1110) message.

Format N(24)

Optionality **Acquirer to Network:** Not Applicable

Network to Issuer: Conditional

- Issuers must certify for the POS Authorization Adjustment Financial Advice (1220) and the POS Authorization Adjustment Financial Advice Response (1230) messages to receive this bit.
- This bit contains a value **only** if Processing Code = 220000 and Function Code = 202.

Notes Positions 1-12 of this bit contains the value from the POS Authorization Response (1110), Bit 4 - Amount, Original.

Positions 13-24 contain zeros (0) and are reserved for future use.

Bit 31

Acquirer Reference Data, Transaction Identifier (TID)

Definition This unique number is given by the Network and follows a Transaction during its life cycle.

Format N(40) max LLVAR

Optionality **Acquirer to Network:** Not Applicable

Network to Issuer: Mandatory. All Issuers must be able to receive the TID.

Notes Refer to [The Role of the Transaction Identifier on page 2-2](#) for more instruction about the TID and the authorization life cycle.

Bit 32

Acquiring Institution Identification (AIN) Code

Definitions This is a unique identification code given to an Acquirer by the Network.

Format N(13) max LLVAR

Optionality **Acquirer to Network:** Optional

Network to Issuer: Optional

Notes A two-digit, variable length indication is before this bit. This indicator is not included in the length of the bit.

American Express Global Network Services
Network Specifications - Authorization
 POS Authorization Advice (1120)
 Bit 34 Primary Account Number, Extended Message Layout

For example, Acquiring Institution Identification code 45678912345 appears as 1145678912345.

Required Values:

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

Bit 34

Definitions

Format

Optionality

Notes

Primary Account Number, Extended

This field is a multipurpose field which provides the Primary Account Number or associated data.

AN(30) Max LLVAR (with length indicator)

Acquirer to Network: Not Applicable

Network to Issuer: Conditional. If this is a tokenization Transaction, this bit will be sent to the Issuer.

This field is populated by the Network with the Token, Wallet Provider ID, Token Expiration Date, and Wallet Provider ID.

Subfield	Format
1	Two Byte Variable Length Indicator (VLI)
2	Token
3	Two Byte Wallet Provider ID (ID)
4	Token Expiration Date

Data Example:

Subfield 1 is the VLI and must be 2 bytes. The Token immediately follows and is a variable length between 15 and 19 bytes. After the Token is a back slash (“\”) delimiter and then a 2 byte ID. After the Two Byte Wallet Provider ID is a back slash (“\”) delimiter and then the Token expiration date in the format MMY.

The symbol “~” is used to represent a space.

Example: 18371234567890123\AB\0114

Bit 35

Definition

Important!

Format

Optionality

Notes

Track 2 Data

This is the data encoded in a Track 2 magnetic stripe read or the data stored in the Track 2 field on the chip of a Chip Card.

S/Es cannot keep track data from the Authorization Request. Do not send Track data to the Issuer.

ANS(39) max LLVAR

Acquirer to Network: Conditional

Network to Issuer: Not Applicable

Point of Service Data Code (Acquirer to Network)

If Point of Service Data Code, Position 7 = “2”, “W”, or “5”, then the **Full Track Data read must be present**.

Bit 35 must be present if Bit 45 is not present and Bit 22 Point of Service Data Code Position 7 is a “W”, “2”, or “5”. Bit 22 describes the interaction between Bit 22, Bit 35, and Bit 45.

Position 7, Bit 22	Bit 35	Bit 45
W	Used	Not used
2	Used	Not used
5	Used	Not used

Track Data (Acquirers)

Track data must be sent as captured. Participants must support the ANSI track layouts and the ISO 7813 layouts

The Acquirer should:

- Strip the Start Sentinel and End Sentinel
- Translate from ASCII to EBCDIC
- Add the length of data to the beginning of the field

Refer to the ISO 7813 Standard Track 2 Layout or ANSI X4.16 Track 2 Layout in *Network Specifications - Instructions* for more instruction.

Bit 38

Definition

Format

Optionality

Notes

Approval Code

A unique code assigned by the Card Transaction authorizing institution to indicate the disposition for the Transaction.

AN(6)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Acquirer to Network:

If the Transaction is approved, this bit must contain the unique Approval Code.

If the Transaction is not approved, this bit must contain all 0s.

The first digit cannot be 0 (zero), excluding Contactless (Expresspay) Transactions.

Network to Issuer:

If this advice message is for a Stand-In that was not approved or a refund, this bit must contain all 0s.

If the Transaction is approved, this bit must contain the unique Approval Code.

The first digit cannot be 0 (zero), excluding Contactless (Expresspay) Transactions.

Bit 39

Definition

Format

Optionality

Notes

Action Code

The type of action done or the type of action to do for a Transaction.

N(3)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Acquirer to Network:

Valid Values:

189 = Canceled S/E

000 = Approved (refund)

Network to Issuer: The decision made by the Network for this Transaction during Stand-In, in response to a fraud referral, as part of the Recurring Billing Cancellation Service, or the Acquirer's disposition for a canceled S/E.

Refer to the Action Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Bit 42

Definition

Format

Optionality

Notes

Card Acceptor Identification Code

The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.

AN(15)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

The Network rejects this message if the S/E number fails the check digit routine.

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Network Specifications - Authorization
POS Authorization Advice (1120)
Bit 43 Card Acceptor Name and Location Message Layout

Bit 43

Card Acceptor Name and Location

Definition

The name of the Card Acceptor as it is known to Cardmembers.

Important!

This bit is very important to Participants and Cardmembers and should contain all the available data.

Format

ANS(101) max LLVAR

Optionality

Acquirer to Network: Conditional

If the Card Acceptor is a Payment Facilitator, Subfield 1 - Name and Subfield 4 - Country Code must contain the Name and Country Code, respectively, of the Sponsored Merchant.

Network to Issuer: Conditional.

If the Network receives Bit 43 and the Issuer is certified to receive it, the Network sends it to the Issuer.

Notes

This bit is free form and Issuers must accept the data as presented by the Acquirers.

Length of Field: 3 bytes minimum, 101 bytes maximum LLVAR with a Variable Length Indicator of 2 bytes EBCDIC, right justified with leading zeros.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Advice (1120)
Bit 43 Card Acceptor Name and Location

Position	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality				Network to Issuer Optionality	Description
				Payment Facilitators	Bill Payment Providers	Manual Cash Disbursement w/ 014008 Processing Code	All Others		
1-2	Variable Length Indicator (VLI)	2 Bytes	Numeric (EBCDIC)	M	M	M	C	C	VLI is the total length of the variable data in Bit 43 (not including VLI).
Subfield 1	Name\Street\City	83 Bytes	Alphanumeric	M	M	M	O	C	Name, Street and City of the Card Acceptor as it is known to Cardmembers. - Payment Facilitator: - Must contain the Name of the Sponsored Merchant. - If Provided, the Street and City must be that of the Sponsored Merchant. - Bill Payment Provider: - Must contain the Name of the End Beneficiary.
Subfield 2	Postal Code	10 Bytes	Alphanumeric	O	O	M	O	C	Postal Code where the Transaction took place. Payment Facilitator: IF provided, Postal Code must be that of the Sponsored Merchant.
Subfield 3	Region	3 Bytes	Alphanumeric	O	O	M	O	C	Region where the Transaction took place. Payment Facilitator: IF provided, Region must be that of the Sponsored Merchant.
Subfield 4	Country Code	3 Bytes	Alphanumeric	M	O	M	O	C	Country Code where the Transaction took place. Payment Facilitator: Must contain the Country Code of the Sponsored Merchant.

Data Examples

The elements of subfield 1 (subfield 1: name\street\city\) are variable length and use a back slash to identify the end of the element. If the data for an element is missing, a back slash is still used to identify the end of the element (e.g., name\city\, street is missing). Subfields 2, 3, and 4 are fixed format, and a back slash is only used for each of these subfields for which data is not present.

The symbol "~" is used to represent a space.

- Name and City are present: Merchant~One\\Anytown\\
- Name, City, and Region (State) are present: Merchant~One\\Anytown\\AZ~
- Name, City, Postal Code, and Region (State) are present:
Merchant~One\\Anytown\\85000~~~~~AZ~
- Name, City, Postal Code, Region (State), and Country Code are present:
Merchant~One\\Anytown\\85000~~~~~AZ~840
- No fields are present \\
- Name and Postal Code are present Merchant~One\\85000~~~~~\\
- Postal Code only present \\85000~~~~~\\

Payment Facilitator

- Subfield 1 must contain the Card Acceptor Name*DBA of the Sponsored Merchant. If street and city is supplied, it must be the street and city of the Sponsored Merchant.
Example: If the Payment Facilitator is "Pay4All" and a Sponsored Merchant is "ezhats" the Card Acceptor Name is sent as: Pay4All*ezhats.
- Subfield 2 - If a postal code is supplied, it must be the postal code of the Sponsored Merchant.
- Subfield 3 - If region is supplied, it must be the region of the Sponsored Merchant.
- Subfield 4 must contain the Country Code of the Sponsored Merchant.

Bill Payment Provider:

- Subfield 1 - **Must** contain the Bill Payment Provider Name*DBA of the End Beneficiary.
Example: If the Bill Payment Provider is "Pay4All" and an End Beneficiary is "ezhats" the Card Acceptor Name is sent as: Pay4All*ezhats.

Bit 45

Definition

Format

Optionality

Notes

Track 1 Data

This is the data encoded in a Track 1 magnetic stripe read or the data stored in the Track 1 field on the chip of a Chip Card.

ANS(78) max LLVAR

Acquirer to Network: Conditional

Network to Issuer: Not Applicable

Point of Service Data Code

If Point of Service Data Code, Position 7 = "2", "W" or "5", then the **Full Track Data read must be present.**

Bit 45 must be present if Bit 35 is not present and Bit 22, Point of Service Data Code Position 7 is a "W", "2", or "5". Bit 22 describes the interaction between Bit 22, Bit 35, and Bit 45

Position 7, Bit 22	Bit 45	Bit 35
W	Used	Not used
2	Used	Not used
5	Used	Not used

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Bit 48 Stand-In Parameter Information for Non-DPP Requests Message
Layout

Track Data

Track data must be sent as captured. Participants must support the ANSI track layouts and the ISO 7813 layouts. The Acquirer should strip the Start Sentinel and End Sentinels, translate from ASCII to EBCDIC, and add the length of data to the beginning of the field.

Refer to the ISO 7813 Standard Track 2 Layout or ANSI X4.16 Track 2 Layout in *Network Specifications - Instructions* for more instruction.

Bit 48

Stand-In Parameter Information for Non-DPP Requests

Definition (Non-DPP Requests)

Acquirer to Network:

This field is present for advice messages when the Acquirer or S/E supplied a Stand-In decision.

Network to Issuer:

This field is present for advice messages when the Network supplied a Stand-In decision for a request that does not contain DPP data.

For more instruction regarding Stand-In, refer to the Exhibits Chapter of the *Business and Operational Policies* manual.

Format

AN(353) max LLLVAR

Optionality

Acquirer to Network: Conditional

Network to Issuer: Conditional

Notes

Acquirer to Network:

Required Value: 008AA98s00s, where s is a space character.

Network to Issuer: If the Network supplied Stand-In for a request that did not contain DPP data, then bit 48 contains these values:

Position	Value	Use
1-3	008	Variable Length Indicator
4-5	AA	Authorization Advice Data Sentinel
6-7	01-95,99	Network Host Resolved Stand-In Table Number
	EM	Acquirer/SE Stand-In resolved, EMV Card
	DC	Network Stand-In for High Risk rule
	96	Authorizer Manual Entry
	97	Network Remote Stand-In Resolved
	98	Acquirer/SE Stand-In Resolved
	AR	Reserved for AEGNS
	IR	Reserved for AEGNS
	ND	Reserved for AEGNS
8	A-M space	Network Host S/E Type
9-10	00-99 space	Network Host Resolved Stand-In S/E Type
11	L, O, space	Network Host Resolved Local or Outbound Indicator

Voice Referral Handling Service: There is no Bit 48 for a Voice Referral Handling Service Transaction.

Bit 48

Stand-In Parameter Information for DPP Requests

Definition

This field is present for advice messages when the Network supplied a Stand-In decision for a request that contains DPP data. For more instruction regarding Stand-In, refer to the Exhibits Chapter of the *Business and Operational Policies* manual.

Format

AN(353) max LLLVAR

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Bit 49 Currency Code, Transaction Message Layout

Optionality

Acquirer to Network: Conditional

Notes

Network to Issuer: Conditional

Network to Issuer: If the Network supplied Stand-In for a request that contains DPP data, then bit 48 contains these values:

Position	Value	Use
1-3	012	Variable Length Indicator
4-5	AA	Authorization Advice Data Sentinel
6-7	01-95,99	Network Host Resolved Stand-In Table Number
	EM	Acquirer/SE Stand-In resolved, EMV Card
	DC	Network Stand-In for High Risk rule
	96	Authorizer Manual Entry
	97	Network Remote Stand-In Resolved
	98	Acquirer/SE Stand-In Resolved
	AR	Reserved for AEGNS
	IR	Reserved for AEGNS
	ND	Reserved for AEGNS
8	A-M space	Network Host S/E Type
9-10	00-99 space	Network Host Resolved Stand-In S/E Type
11	L, O, space	Network Host Resolved Local or Outbound Indicator
12-13	03, 05	DPP Acquirer based type indicator
14-15	01-99	Number of Installments for DPP plan

Voice Referral Handling Service: There is no Bit 48 for a Voice Referral Handling Service Transaction.

Bit 49

Currency Code, Transaction

Definition

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Currency Code, Reconciliation

Definition

This is the numeric code of the currency defined by the Issuer.

Format

N(3)

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Mandatory

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 56

Original Data Elements

Definition

Data from the Authorization Request (1100 or 1200) or POS/ATM Authorization Reversal Advice (1420) message.

Format

N(37) max LLVAR

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Optional

Notes

If data from the Network to Issuer POS Authorization Request (1100) is available, these positions can contain that data.

Example:

Position	Value	Use
1-2	30	Variable Length Indicator
3-6		Message Type from 1100 message
7-12		Trace Audit Number (Bit 11) from 1100 message
13-24		Data and Time, Local Transaction (Bit 12) from 1100 message
25-32	06370100	Acquiring Institution ID Code (Bit 32)

Bit 60

Definition
Optionality

Market Specific Data

This bit provides additional authorization data.

Acquirer to Network: Conditional.

- **Payment Facilitator:** If the Acquirer is certified to send the POS Authorization Advice (1120) message, then Market Specific Data with Additional Authorization Data format, Subfield 2, must contain the Seller ID of the Sponsored Merchant.
- **Bill Payment Provider:** If the Acquirer is certified to send the POS Authorization Advice (1120) message, then Market Specific Data with Additional Authorization Data format, Subfield 2, must contain the Seller ID of the End Beneficiary.

Network to Issuer: Conditional. Issuers must support all Estimated and Incremental Authorization Function Codes.

Notes

Certification is required for Acquirers to send Bit 60 and for Issuers to receive Bit 60.

Market Specific Data:

- There are two supported formats of Market Specific Data.
- The structure and contents of this data is specified by market representatives.
- Each market uses this data differently.
- To use Market Specific Data, Acquirers and Issuers must be in the same market that supports this data.
- Contact your AEGNS Representative for more information.

Market Specific Data

Format

AN(210) max LLLVAR

Notes

Used by Participants in regions supporting Market Specific Data, which do not have a need to support Market Specific Data with Additional Authorization Data.

Market Specific Data Layout Table

Position ¹	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality	Network to Issuer Optionality	Valid Value	Description
1-3	Variable Length Indicator (VLI)	3 Bytes	Numeric (EBCDIC)	M	M		VLI is the total length of the variable data in Bit 60 (not including VLI).
4-5	Market Specific Data Indicator	2 Bytes	Alphanumeric	C	O	MS	Market Specific Data Indicator is the constant literal "MS" for Market Specific Data Indicator.
6-8	Market Specific Data ISO Country Code of the Acquirer	3 Bytes	Numeric	C	O		ISO Country Code of the Acquirer must be populated if SF1 is present. Refer to the <i>Codes Reference Guide</i> for a complete list of codes.
9-10	Market Specific Data Version Indicator	2 Bytes	Numeric	C	O		Version Indicator must be populated if SF1 is present.
11-210	Market Specific Data Elements	200 Bytes Max	ANS	O	O		Market Specific Data Elements

1. The Network validates only the contents of Positions 4-5 and Positions 6-8.

Market Specific Data with Additional Authorization Data

Format

AN(316) max LLLVAR

Used by Participants in regions that support Market Specific Data and Additional Authorization Data elements identified in the table below. Certified Acquirers will send the appropriate data field(s) depending on their specific needs.

Seller ID:

- **Payment Facilitator:** Subfield 2 must contain the Seller ID of the Sponsored Merchant.
- **Bill Payment Provider:** Subfield 2 must contain the Seller ID of the End Beneficiary.

Token Requestor ID (TRID):

- Certification is required for Acquirers and Issuers to receive the TRID in Subfield 5.
- Acquirers requesting Subfield 5 - Token Requestor ID returned in the POS Authorization Advice Response (1130) message, must provide a required value in the POS Authorization Advice (1120) message.

Original Transaction Identifier (Original TID)

- Issuers must receive the OTID for Incremental Authorization Transactions.

Market Specific Data with Additional Authorization Data Layout Table

Position	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality	Network to Issuer Optionality	Valid Value	Description
1-3	Variable Length Indicator (VLI)	3 Bytes	Numeric (EBCDIC)	M	M		VLI is the total length of the variable data in Bit 60 (not including VLI).
4-5	Primary ID	2 Bytes	Alphanumeric	M	M	AX	Primary ID is constant literal "AX" for Market Specific Data/Additional Authorization Data Indicator.
6-8	Secondary ID	3 Bytes	Alphanumeric	M	M	AAD	Secondary ID is constant literal "AAD" for Additional Authorization Data.
9-12	Bitmap Identifier	4 Bytes	Binary (Hexadecimal configuration)	M	M		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the bitmap, the layout consists of at least 1 of the following subfields. Each bit position of the 32 bit/4 byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, the corresponding subfield will be present.
SF1	Market Specific Data						
	Market Specific Data VLI	3 Bytes	Numeric (EBCDIC)	C	O		Market Specific VLI is mandatory if the Bitmap Identifier indicates that SF1-Market Specific Data is enabled.
	Market Specific Data Indicator	2 Bytes	Alphanumeric	C	O	MS	Data Indicator of the constant literal "MS" must be populated if SF1 is present.
	Market Specific Data ISO Country Code of the Acquirer	3 Bytes	Numeric	C	O		ISO Country Code of the Acquirer must be populated if SF1 is present. Refer to the <i>Codes Reference Guide</i> for a complete list of codes.
	Market Specific Data Version Indicator	2 Bytes	Numeric	C	O		Version Indicator must be populated if SF1 is present.
	Market Specific Data Elements	200 Bytes Max	ANS	O	O		Market Specific Data Elements
SF2	Seller ID	20 Bytes	AN	C	O		Combination of letters and/or numbers that uniquely identifies a Sponsored Merchant, or End Beneficiary. Bill Payment Provider: SF2 must contain the Seller ID of the End Beneficiary. Payment Facilitator: SF2 must contain the Seller ID of the Sponsored Merchant.

Market Specific Data with Additional Authorization Data Layout Table (continued)

Position	Data Element	Subfield Length	Subfield Type	Acquirer to Network Optionality	Network to Issuer Optionality	Valid Value	Description
SF3	Reserved	42 Bytes	ANS	NA	NA		
SF4	Reserved	20 Bytes	AN	NA	NA		
SF5	Token Requestor ID	11 Bytes	AN	O	O		An 11-byte code pairing Token Requestor with Token Domain
SF6	Reserved	1 Byte	AN	NA	NA		
SF7	Original Transaction Identifier (Original TID)	15 Bytes	N	N/A	C		Original Transaction Identifier (Original TID) Issuers must receive the OTID for Incremental Authorization Transactions.

Bit 61

National Use Data

Definition	This is an ISO field that supplies Cardmember authentication data during Authorization.
Format	4 bytes min, 103 bytes max LLLVAR
Optionality	Acquirer to Network: Conditional. This bit must be present for a digital wallet application-initiated Transaction or a SafeKey Transaction.
	Network to Issuer: Conditional. If this bit is present, the Network sends it to the certified Issuer.
Bit Layout Usage	For SafeKey Transactions refer to Table 5-5 . For Digital Wallet application-initiated Transactions, refer to Table 5-2. Digital Wallet Transaction Layout on page 5-27 .
Notes	Bit 61 Layout The layout contains these subfields.

Table 5-5. American Express SafeKey Transaction Layout

Subfield	Format	Optionality		Notes
		A to N	N to I	
1 Variable Length Indicator (VLI)	(3) (numeric EBCDIC)	Mandatory	Mandatory	
2 Primary ID	(2) (alphanumeric)	Mandatory	Mandatory	Required Value: AX
3 Secondary ID	(3) (alphanumeric)	Mandatory	Mandatory	Valid Value: ASK
4 Electronic Commerce Indicator (ECI)	(2) (alphanumeric)	Mandatory	Mandatory	Valid Values: 05 = Authenticated with AEVV 06 = Attempted with AEVV 07 = Not Authenticated
5 American Express Verification Value (AEVV) ID	(4) (alphanumeric)	Conditional	Conditional	If Subfield 6 is present, AEVV ID must be populated. Valid Value: AEVV

Table 5-5. American Express SafeKey Transaction Layout

Subfield	Format	Optionality		Notes
		A to N	N to I	
6 American Express Verification Value (AEVV)	(20) (binary)	Conditional	Conditional	This subfield is necessary if ECI is 05 or 06. Do not use spaces. Do not use zeros (0) in all positions. The AEVV value is proof of the SafeKey Authentication and will be supplied from the SafeKey Authentication process. Sub-Subfield 1 Authentication Results Code (1): 0 = Authenticate Successful 7 = Attempt non participating 8 = Attempt Access Control Server (ACS) not available Sub-Subfield 2 Second Factor Authentication Code (1): Default value '00' Sub-Subfield 3 AEVV Key Indicator (1): - Card Issuers ACS = 01, 02 - Attempts ACS = 98, 99 Sub-Subfield 4 iCSC-3 Output (2) Sub-Subfield 5 Unpredictable Number (2) Sub-Subfield 6 Authentication Tracking Number (8) Sub-Subfield 7 Additional AEVV Information (5) - Values are determined by ACS. Can be Zero filled or: Version (4 bits): 0=Authentication Action and Cardmember IP address not present 1=Authentication Action and Cardmember IP address present 2-9=Reserved Authentication Action (4 bits): 0=Standard authentication performed 1=Activated During Shopping registration/authentication performed 2=Forgot your password (reregistration/re-authentication performed) Cardmember IP Address (4 bytes)
7 SafeKey Transaction ID	(3) (alphanumeric)	Conditional	Conditional	This subfield is necessary if the SafeKey Transaction ID Value is present. Valid Value: XID
8 SafeKey Transaction ID Value	(20) (binary)	Conditional	Conditional	SafeKey Transaction ID (XID) value is required if XID is present in the SafeKey authentication.

Table 5-6. Digital Wallet Transaction Layout

Subfield	Format	Optionality		Notes
		A to N	N to I	
1 Variable Length Indicator (VLI)	(3) (numeric EBCDIC)	Mandatory	Mandatory	
2 Primary ID	(2) (alphanumeric)	Mandatory	Mandatory	Required Value: AX
3 Secondary ID	(3) (alphanumeric)	Mandatory	Mandatory	Valid Value: TKN

Table 5-6. Digital Wallet Transaction Layout (continued)

Subfield	Format	Optionality		Notes
4 Electronic Commerce Indicator (ECI)	(2) (alphanumeric)	Mandatory	Mandatory	Valid Value: 20 = Payment Token Data Present
5 Token Data Block A ID	(4) (alphanumeric)	Mandatory	Mandatory	Valid Value: TDBA If the Secondary ID = TKN, then this subfield is required.
6 Token Data Block A	(20) (binary)	Mandatory	Mandatory	If the Secondary ID = TKN, then this subfield is required. Refer to the Expresspay Mobile documents on Knowledge Base for additional information. Use the Search function to locate these documents.
7 Token Data Block B ID	(3) (alphanumeric)	Conditional	Conditional	Valid Value: DBB If subfield 8 - Token Data Block B is provided, this subfield is required with a value of DBB.
8 Token Data Block B	(20) (binary)	Conditional	Conditional	If subfield 7 - Token Data Block B ID = DBB, then this subfield is required. Refer to the Expresspay Mobile documents on Knowledge Base for additional information. Use the Search function to locate these documents.

Bit 62

Fraud Indicators

Definition

The Fraud Indicator field is used to provide fraud mitigation indicators for a variety of situations.

Format

AN(5) min, AN(60) max LLLVAR

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Conditional. This field is populated by the Network with the 5CSC validation results, Fraud Probability Score, and Recommendation Code. This field is passed as Fraud Indicators only to certified Issuers.

Notes

Two formats are used for Bit 62: Fixed Length and Variable Length as described below.

Fixed Length Format:

For Request Type Identifiers FD1 and FD2, only 18 of the 60 bytes are being used.

Position ¹	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD1 FD2	Request Type Identifier FD1 = 5CSC checking results FD2 = 5CSC validation results, Recommendation Code, and Fraud Probability Score If FD1 value is listed, then the field will end at 10 bytes. If FD2 value is listed, a combination of codes and spaces will be included in positions 9-18, dependent on the features for which the Participant certified.
9-10	XX	Two digit 5CSC Indicator Refer to the Fraud Indicators section of the <i>Codes Reference Guide</i> for a complete listing of codes and descriptions.
11-15	XXXXX	Fraud Probability Score This is an implied leading zero and decimal. A Fraud Probability Score of 12345 should be interpreted as 0.12345.

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Bit 62 Fraud Indicators Message Layout

16-18	XXX	Recommendation Code. Refer to the DFMNS Recommendation Codes of the <i>Codes Reference Guide</i> .
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1. If GNSAS is not available, it is possible that positions 9-18 can be blank for Issuers certified for Fraud Probability Score and Recommendation Code.

When the Participant certifies for the FD3 Request Type Identifier, FD1 and FD2 are no longer supported.

SafeKey requires Participants to certify for Bit 62 and support the FD3 Request Type Identifier.

Refer to the Fraud Indicators section of the *Codes Reference Guide* for a list of valid codes and descriptions.

FD3 Request Type Identifier:

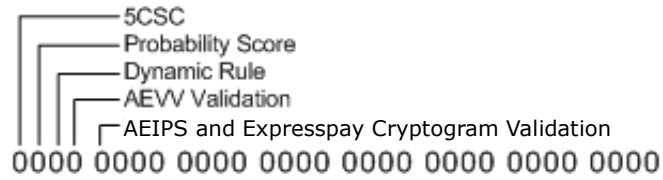
Bit 62 FD3 Request Type is supplied for SafeKey Transactions where the Network created the AEVV when the Issuer ACS was not available.

This table gives the data for FD3 Request Type Identifier.

Valid Values:

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD3	Request Type Identifier FD3 = Bitmap Variable Length
9-12	32 bit Binary	Bitmap Indicator Identifies the fields in the structure. After the bitmap, the layout has at least one of the subfields as defined below. These show which Fraud Indicators are supplied in variable data Positions 13-60. • Subfield 1 5CSC Indicator (2 Bytes) • Subfield 2 Fraud Probability Score (5 Bytes) • Subfield 3 Recommendation Code (3 Bytes) • Subfield 4 AEVV Validation Result (1 Byte) • Subfield 5 AEIPS and Expresspay Cryptogram Validation Result (2 Bytes) • Subfield 6 PIN Validation Results (2 Bytes) • Subfield 7 EMV Data Results (4 Bytes) • Subfield 8 Reserved for AEGNS use • Subfield 9 ATC Validation Results (2 Bytes) • Subfield 10 Reserved for Future Use • Subfield 11 ATC Value (2 Bytes - hexadecimal) • Subfield 12 Cardmember Traveling Indicator (1 Byte) • Subfield 13 4CSC Indicator (2 Bytes) • Subfield 14 Payment Token Expiration Validation Result (2 Bytes) Note: Cryptogram Validation (Subfield 5) will use the Fraud Indicator Code of 02 - HSM Not Available. Refer to the <i>Codes Reference Guide</i> for valid codes and definitions.
13-60	Fraud Value	Describes the results of the Bitmap Indicators that are set.

Example: Positions 9 - 12



Example: If Bit 62 includes Fraud Probability Score, Recommendation Code, and AEVV Validation Result:

Position	Value
1-3	018
4-5	AX
6-8	FD3
9-12	01110000000000000000000000000000 X'70000000'
13-60	555553331

POS Authorization Advice Response (1130)

Description

The POS Authorization Advice Response (1130) is sent in response to a POS Authorization Advice (1120) message.

The Network does not send this message from the Issuer to the Acquirer. The Network responds to the Acquirer, sends the advice to the Issuer, and accepts the POS Authorization Advice Response (1130) response from the Issuer.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Advice Response (1130). Refer to the [Programming Notes for POS Authorization Advice Response \(1130\) on page 5-75](#) for more instruction.

POS Authorization Advice Response (1130) Message Layout

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	ME	ME	Primary Account Number (PAN)
3	N(6)	ME	ME	Processing Code
4	N(12)	ME	ME	Amount, Transaction
5	N(12)	ME	N/A	Amount, Reconciliation
11	N(6)	ME	ME	Systems Trace Audit Number
12	N(12)	ME	ME	Date and Time, Local Transaction
31	N(40) max LLVAR	ME	ME	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	CE	CE	Acquiring Institution Identification (AIN) Code
39	N(3)	M	M	Action Code
42	AN(15)	ME	ME	Card Acceptor Identification Code
49	N(3)	ME	ME	Currency Code, Transaction
50	N(3)	ME	N/A	Currency Code, Reconciliation

POS Authorization Advice Response (1130) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
60	AN(210) max LLVAR	CE	CE	Market Specific Data
	AN(316) max LLVAR			Market Specific Data with Additional Authorization Data

Programming Notes for POS Authorization Advice Response (1130)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Required Value: 1130 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap, Binary (Hex)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 3

Processing Code

Definition	This is the service being requested.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 4

Amount, Transaction

Definition	The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 5

Amount, Reconciliation

Definition	This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.
Format	N(12)

Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Not Applicable
Bit 11	Systems Trace Audit Number
Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 12	Date and Time, Local Transaction
Definition	This is the local date and time when a Transaction occurred.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 31	Acquirer Reference Data, Transaction Identifier (TID)
Definition	This unique number is given by the Network and follows a Transaction during its life cycle.
Format	N(40) max LLVAR
Optionality	Issuer to Network: Mandatory Echo. Issuers must echo the TID received from the POS Authorization Advice (1120) message. Network to Acquirer: Mandatory Echo. If the Acquirer is certified and passes the TID in the 1120, the Network echoes the TID.
Notes	Refer to The Role of the Transaction Identifier on page 2-2 for more instruction about the TID and the authorization life cycle.
Bit 32	Acquiring Institution Identification (AIN) Code
Definition	This is a unique identification code given to an Acquirer by the Network.
Format	N(13) max LLVAR
Optionality	Issuer to Network: Conditional Echo Network to Acquirer: Conditional Echo
Bit 39	Action Code
Definition	The type of action done or the type of action to do for a Transaction.
Format	N(3)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Network to Acquirer: Valid Values: 900 = Advice Accepted 904 = Format Error Issuer to Network: Invalid values from the Issuer to Network message cause the GNSAS to queue the same message again. Refer to the Action Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 42	Card Acceptor Identification Code
Definition	The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.
Format	AN(15)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 49

Currency Code, Transaction

Definition	This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Currency Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 50

Currency Code, Reconciliation

Definition	This is the numeric code of the currency defined by the Issuer.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Not Applicable
Notes	Refer to the Currency Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 60

Market Specific Data

Definition	This bit provides additional authorization data.
Market Specific Data	
Format	AN(210) max, LLLVAR
Optionality	Issuer to Network: Conditional Echo. This optionality can be different by market. Contact your AEGNS Representative for more instruction. Network to Acquirer: Conditional Echo. This optionality can be different by market. Contact your AEGNS Representative for more instruction.

Market Specific Data with Additional Authorization Data

Format	AN(316) max LLLVAR
Optionality	Issuer to Network: Conditional Echo. If a certified Issuer receives Bit 60 from the Network in the POS Authorization Request (1100) message, then the Issuer must echo Bit 60. The optionality for Subfield 1 - Market Specific Data can be different by market. Contact your AEGNS Representative for more instruction. Network to Acquirer: Conditional Echo. If the Network receives Bit 60 from the Issuer, it is sent to the Acquirer. The optionality for Subfield 1 - Market Specific Data can be different by market. Contact your AEGNS Representative for more instruction.

POS Authorization Reversal Advice (1420)

Description

Acquirers can generate online reversals and voids with this message.

The 1420 is only an advice message. The POS Reversal Advice Response (1430) is required.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Reversal Advice (1420). Refer to the [Programming Notes for POS Authorization Reversal Advice \(1420\) on page 5-79](#) for more instruction.

POS Authorization Reversal Advice (1420) Message Layout

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	M	M	Primary Account Number (PAN)
3	N(6)	M	M	Processing Code
4	N(12)	M	M	Amount, Transaction
5	N(12)	N/A	M	Amount, Reconciliation
11	N(6)	M	M	Systems Trace Audit Number
12	N(12)	M	M	Date and Time, Local Transaction
23	N(3)	N/A	C	PAN Sequence Number
24	AN(3)	M	M	Function Code
25	N(4)	M	M	Message Reason Code
26	N(4)	N/A	C	Card Acceptor Business Code
30	N(24)	N/A	O	Amount, Original
31	N(40) max LLVAR	C	M	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	O	O	Acquiring Institution Identification (AIN) Code

POS Authorization Reversal Advice (1420) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
34	AN(30) max LLVAR (including length indicator)	N/A	C	Primary Account Number, Extended
37	AN(12)	O	O	Acquirer Reference Number (ARN)
38	AN(6)	O	O	Approval Code
42	AN(15)	M	M	Card Acceptor Identification Code
43	ANS(101) max LLVAR	N/A	O	Card Acceptor Name and Location
49	N(3)	M	M	Currency Code, Transaction
50	N(3)	N/A	M	Currency Code, Reconciliation
56	ANS(37) max LLVAR	O	O	Original Data Elements
60	AN(210) max LLLVAR	C	C	Market Specific Data
	AN(316) max LLLVAR			Market Specific Data with Additional Authorization Data

Programming Notes for POS Authorization Reversal Advice (1420)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Required Value: 1420 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap Binary (Hex)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Variable Length Indicator: A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Reversal Advice (1420)
Bit 3 Processing Code Message Layout

For example, a Cardmember Account Number of 3759-876543-21001
shows as 15375987654321001

Bit 3

Definition
Format
Optionality

Processing Code

This is the service being requested.

N(6)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Variable Authorizations: For Partial Reversals a 024000 is required.

Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Notes

Bit 4

Definition
Format
Optionality

Amount, Transaction

The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.

N(12)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

The decimal point is determined by the Currency Code, Transaction (Bit 49). For example, \$100.00 in U.S.D. is 000000010000.

Deferred Payment Plan (Airline): For Deferred Payment Plan airline Transactions, this amount includes the down payment and boarding fee.

Notes

Bit 5

Definition
Format
Optionality

Amount, Reconciliation

This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.

N(12)

Acquirer to Network: Not Applicable

Network to Issuer: Mandatory

The decimal point is determined by the Currency Code, Reconciliation (Bit 50). For example, \$100.00 in USD shows as 000000010000.

Authorization and Presentment can occur on different days. The reconciliation amounts can be different in the Authorization and Presentment messages.

Notes

Bit 11

Definition
Format
Optionality

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.

N(6)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Acquirer to Network: The Network returns this number in the Network to Acquirer piece of the POS Authorization Response (1430) message.

Network to Issuer: This value is assigned by the Network and is a different value than that generated by the Acquirer.

Notes

Bit 12

Definition
Format
Optionality

Date and Time, Local Transaction

This is the local date and time when a Transaction occurred.

N(12), YYMMDDHHMMSS

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Bit 23

Definition
Format

PAN Sequence Number

A number distinguishing between separate Cards with the same Primary Account Number.

N(3)

Optionality **Acquirer to Network:** Not Applicable

Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.

Bit 24

Function Code

Definition This code shows the reason for a message in its Message Type.

Format AN(3)

Optionality **Acquirer to Network:** Mandatory

Network to Issuer Mandatory. Issuers must support both Full and Partial Reversal Function Codes.

Notes **Required Value:**

- 400 = Full Reversal (Transaction did not complete as approved), voids, and reversals
- 401 = Partial Reversal

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 25

Message Reason Code

Definition This code gives the cause of a First and Final Chargeback, Retrieval Request Notification, File Acknowledgment, ATM and POS, and Reversal requests.

Format N(4)

Optionality **Acquirer to Network:** Mandatory

Network to Issuer: Mandatory

Notes This code shows if a sent file was accepted or rejected by the Network for a Network File Acknowledgment (1844) and a Participant File Acknowledgment (1824).

Refer to the Message Reason Code section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 26

Card Acceptor Business Code

Definition Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).

Format N(4)

Optionality **Acquirer to Network:** N/A

Network to Issuer: Conditional

- Issuers must receive MCC for Partial Reversals (Function Code 401 - Partial reversal (POS and ATM)).

Notes This bit must contain the Card Acceptor Business Code (MCC) sent in the Estimated Authorization from the initial POS Authorization Request (1100) message.

Bit 30

Amount, Original

Definition The Authorized Transaction Amount from the first Estimated Authorization in the POS Authorization Response (1110) message.

Format N(24)

Optionality **Acquirer to Network:** N/A

Network to Issuer: Optional

Notes **Network to Issuer:**

- If provided by the Acquirer, certified Issuers will receive the approved Transaction amount from Bit 4 of the POS Authorization Response (1110) message in positions 1-12
- Positions 13-24 are zero-filled reserved for future use by AEGNS.

American Express Global Network Services
Network Specifications - Authorization
 POS Authorization Reversal Advice (1420)
 Bit 31 Acquirer Reference Data, Transaction Identifier (TID) Message
 Layout

Position	Format	Use
1-12	N(12)	Approved Transaction amount from Bit 4 of the POS Authorization Response (1110) message
13-24	N(12)	Zero-filled. Reserved for AEGNS.

Bit 31

Acquirer Reference Data, Transaction Identifier (TID)

Definition

This unique number is given by the Network and follows a Transaction during its life cycle.

Format

N(40) max LLVAR

Optionality

Acquirer to Network: Conditional

Notes

Network to Issuer: Mandatory. All Issuers must be able to receive the TID.

Acquirer to Network: This Bit must contain the TID received in the POS Authorization Response (1110) message.

Refer to [The Role of the Transaction Identifier on page 2-2](#) for more instruction about the TID and the authorization life cycle.

Bit 32

Acquiring Institution Identification (AIN) Code

Definition

This is a unique identification code given to an Acquirer by the Network.

Format

N(13) max LLVAR

Optionality

Acquirer to Network: Optional

Notes

Network to Issuer: Optional

Acquirer to Network: The variable length indicator is not included in the length of the bit.

For example, Acquiring Institution Identification code 45678912345 shows as 1145678912345.

Network to Issuer:

Required Value: 06370100 American Express

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

Bit 34

Primary Account Number, Extended

Definitions

This field is a multipurpose field which provides the Primary Account Number or associated data.

Format

AN(30) Max LLVAR (with length indicator)

Optionality

Acquirer to Network: Not Applicable

Notes

Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.

This bit is populated by the Network with the Token, Wallet Provider ID, and Token Expiration Date.

Subfield	Format
1	Variable Length Indicator (VLI)
2	Token
3	Wallet Provider ID (ID)
4	Token Expiration Date

Data Example:

Subfield 1 is the VLI and must be 2 bytes. The token immediately follows and is a variable length between 15 and 19 bytes. After the token is a back slash ("\") delimiter and then a 2 byte ID. After the 2 byte ID is a back slash ("\") delimiter and then the token expiration date in the format YYMM.

American Express Global Network Services
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POS Authorization Reversal Advice (1420)
Bit 37 Acquirer Reference Number (ARN) Message Layout

The symbol "~" is used to represent a space.

Example: 23371234567890123\AB\1401

Bit 37

Acquirer Reference Number (ARN)

Definition

A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

Format

AN(12)

Optionality

Acquirer to Network: Optional

Network to Issuer: Optional

Bit 38

Approval Code

Definition

A unique code assigned by the Card Transaction authorizing institution to indicate the disposition for the Transaction.

Format

AN(6)

Optionality

Acquirer to Network: Optional

Network to Issuer: Optional

Notes

Acquirer to Network: The first Approval Code received in the Network to Acquirer POS Authorization Response (1110). This will be a six byte or two byte unique value (based on the capability of the Acquirer).

Network to Issuer: A code assigned by the authorizing institution to show the disposition for the Transaction.

- If the Transaction was approved, the Approval Code is a unique, six digit, left justified numeric value. The first digit cannot be "0" (zero). The Network makes a decision to send the left 2 digits or all 6 digits to the Acquirer based on the Acquirer's capability.
- If the Transaction was denied, this field can be present but space-filled.

Bit 42

Card Acceptor Identification Code

Definition

The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.

Format

AN(15)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

The Network rejects this message if the S/E number fails the check digit routine. This field must contain the S/E number from the first Transaction.

Bit 43

Card Acceptor Name and Location

Definition

The name of the Card Acceptor as it is known to Cardmembers.

Important

This bit is very important to Participants and Cardmembers and should contain all the available data.

Format

ANS(101) max LLVAR

Optionality

Acquirer to Network: N/A

Network to Issuer: Optional.

- This bit is free form and Issuers must accept the data as presented by the Acquirers.
- If the Network receives Bit 43 from a certified Acquirer and the Issuer is certified to receive it, the Network will send it to the Issuer as received.

Bit 49

Currency Code, Transaction

Definition

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

This field must contain the same currency code as the first Transaction.

American Express Global Network Services
Network Specifications - Authorization
 POS Authorization Reversal Advice (1420)
 Bit 50 Currency Code, Reconciliation Message Layout

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition
Format
Optionality

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.

N(3)

Acquirer to Network: Not Applicable

Network to Issuer: Mandatory

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 56

Definition
Format
Optionality

Original Data Elements

Data from the Authorization Request (1100 or 1200) or POS/ATM Authorization Reversal Advice (1420) message.

ANS(37) max LLVAR

Acquirer to Network: Optional

Network to Issuer: Optional

Notes

Example:

Position	Value	Use
1-2	30	Variable Length Indicator
3-6	1100	Message Type from 1100
7-12		Systems Trace Audit Number Acquirer to Network: Contains the STAN (Bit 11) from the 1100 to the Network Network to Issuer: Network System Generated Authorization Reversals: contains the STAN (Bit 11) from the 1100 to the Issuer Acquirer System Generated or S/E Initiated Reversals: zero-filled
13-24		Date and Time, Local Transaction Acquirer to Network: Contains the Date and Time (Bit 12) from the 1100 to the Network Network to Issuer: Network System Generated Authorization Reversals: - Contains the Date and Time (Bit 12) from the 1100 to the Issuer Acquirer System Generated or S/E Initiated Reversals: If valid: Will contain the date and time as it was sent by the Acquirer and will be passed to the Issuer as it was received by the Network. If not valid: Will contain the date and time when the Network sent the message to the Issuer.
25-32	06370100	Acquiring Institution ID Code Acquirer to Network: Contains the AIN (Bit 32) from the 1100 to the Network Network to Issuer: Network Generated Reversals: Contains the AIN (Bit 32) from the 1100 to the Issuer Acquirer System Generated or S/E Initiated Reversals: Contains the AIN (Bit 32) from the 1100 to the Issuer

Bit 60

Definition
Optionality

Market Specific Data

This bit provides additional authorization data.

Acquirer to Network: Conditional. The Acquirer must be certified to send Bit 60.

Network to Issuer: Conditional. The Issuer must be certified to receive Bit 60.

Bit 60 - Market Specific Data has two supported formats, as follows:

Market Specific Data

Format

AN(210) max LLLVAR

Used by Participants in regions supporting Market Specific Data, which do not have a need to support Market Specific Data with Additional Authorization Data.

Notes

The structure and contents of this data is specified by market representatives. Each market uses this data differently. In order to use Market Specific Data, Acquirers and Issuers must be in the same market that supports this data.

Contact your AEGNS Representative for instruction.

Position	Value	Use
1-3		Variable Length Indicator
4-5 ¹	MS	Market Specific Data Indicator
6-8 ¹		Market ISO Country Code
9-10		Market Specific Data Version Indicator
11-210		Market Specific Data Elements

1. The Network validates only the contents of Positions 4-5 and Positions 6-8.

Market Specific Data with Additional Authorization Data

Format

AN(316) max LLLVAR

Used by Acquirers and Issuers in regions supporting Market Specific Data (Subfield 1), that also have a need to support sending and receiving Seller ID (Subfield 2) and/or Token Requestor ID (Subfield 5).

Notes

This field supports multiple data elements identified in the table below. Certified Participants will send the appropriate data field(s), depending on their specific needs.

Certified Acquirers must use the format identified below. The Network will not pass Bit 60 to Issuers not certified to receive it.

For Market Specific Data, Subfield 1 - Market Specific Data, the structure and contents of this subfield is specified by market representatives. Each market uses this subfield differently. In order to use Market Specific Data (Subfield 1), Acquirers and Issuers must be in the same market that supports Market Specific Data.

Contact your AEGNS Representative for additional instruction.

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Market Specific Data/Additional Authorization Data Indicator
6-8	AAD	Additional Authorization Data

American Express Global Network Services
Network Specifications - Authorization
 POS Authorization Reversal Advice (1420)
 Bit 60 Market Specific Data Message Layout

9-12	32 Bit Binary	Bitmap Indicator Identifies the fields in the structure. Following the Bitmap, the layout consists of at least one of the subfields defined below. Each bit position of the 32 bit/byte bitmap represents which Market Specific /Additional Authorization Data element is present. If a particular bit is "ON" in the bitmap, that corresponding subfield is present. • Subfield 1 - Market Specific Data • Subfield 2 - Seller ID - Acquirer-assigned unique identifier of a Sponsored Merchant, e.g., up to twenty (20) byte combination of letters and/or numbers that uniquely identifies a Sponsored Merchant. • Subfield 3 - Reserved • Subfield 4 - Reserved • Subfield 5 - Token Requestor ID • Subfield 6 - Reserved
13-316	Market Specific/ Additional Authori- zation Data	Market Specific /Additional Authorization Data describes the results of the Bitmap indicators that are set to "ON." Refer to Market Specific/Additional Authorization Data Subfield Layout below.

Market Specific/Additional Authorization Data Subfield Layout

Subfield	Position	Value	Use
Subfield 1 - Market Specific Data	1-3	NNN	Variable Length Indicator
	4-5	MS	Market Specific Data Indicator
	6-8	NNN	Market Specific Data ISO Country Code of the Acquirer
	9-10	NN	Market Specific Data Version Indicator
	11-210		Market Specific Data Elements
Subfield 2	1-20		Seller ID
Subfield 3	1-42		Reserved
Subfield 4	1-20		Reserved
Subfield 5	1-11		Token Requestor ID
Subfield 6	1		Reserved

POS Authorization Reversal Advice Response (1430)

Description

This message is used to acknowledge the receipt of the POS Authorization Reversal Advice (1420).

The generation of the 1430 response confirms that the Network received the 1420. It does not mean that exposure on the account is reversed.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Reversal Advice Response (1430). Refer to the [Programming Notes for POS Authorization Reversal Advice Response \(1430\) on page 5-88](#) for more instruction.

POS Authorization Reversal Advice Response (1430) Message Layout

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	ME	ME	Primary Account Number (PAN)
3	N(6)	ME	ME	Processing Code
4	N(12)	ME	ME	Amount, Transaction
5	N(12)	ME	N/A	Amount, Reconciliation
11	N(6)	ME	ME	Systems Trace Audit Number
12	N(12)	ME	ME	Date and Time, Local Transaction
30	N(24)	CE	N/A	Amount, Original
31	N(40) max LLVAR	ME	CE	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	OE	OE	Acquiring Institution Identification (AIN) Code
37	AN(12)	N/A	OE	Acquirer Reference Number (ARN)
39	N(3)	M	M	Action Code
42	AN(15)	ME	ME	Card Acceptor Identification Code
49	N(3)	ME	ME	Currency Code, Transaction
50	N(3)	ME	N/A	Currency Code, Reconciliation

POS Authorization Reversal Advice Response (1430) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
60	AN(210) max LLVAR	CE	CE	Market Specific Data
	AN(316) max LLVAR			Market Specific Data with Additional Authorization Data

Programming Notes for POS Authorization Reversal Advice Response (1430)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Required Value: 1430 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap Binary (Hex)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 3

Processing Code

Definition	This is the service being requested.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 4

Amount, Transaction

Definition	The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 5

Amount, Reconciliation

Definition	This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.
Format	N(12)

Optionality

Issuer to Network: Mandatory Echo

Network to Acquirer: Not Applicable

Bit 11 **Systems Trace Audit Number**

Definition This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format N(6)

Optionality **Issuer to Network:** Mandatory Echo

Network to Acquirer: Mandatory Echo

Bit 12 **Date and Time, Local Transaction**

Definition This is the local date and time when a Transaction occurred.

Format N(12)

Optionality **Issuer to Network:** Mandatory Echo

Network to Acquirer: Mandatory Echo

Bit 30 **Amount, Original**

Definition The Authorized Transaction Amount from the first Estimated Authorization in the POS Authorization Response (1110) message.

Format N(24)

Optionality **Issuer to Network:** Conditional Echo

Network to Acquirer: N/A

Bit 31 **Acquirer Reference Data, Transaction Identifier (TID)**

Definition This unique number is given by the Network and follows a Transaction during its life cycle.

Format N(40) max LLVAR

Optionality **Issuer to Network:** Mandatory Echo. Issuers must echo the TID received from the POS Authorization Reversal Advice (1420) message.

Network to Acquirer: Conditional Echo. If the Acquirer is certified and passes the TID in the 1420, the Network echoes the TID.

Notes Refer to [The Role of the Transaction Identifier on page 2-2](#) for more instruction about the TID and the authorization life cycle.

Bit 32 **Acquiring Institution Identification (AIN) Code**

Definition This is a unique identification code given to an Acquirer by the Network.

Format N(13) max LLVAR

Optionality **Issuer to Network:** Optional Echo

Network to Acquirer: Optional Echo

Bit 37 **Acquirer Reference Number (ARN)**

Definition A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

Format AN(12)

Optionality **Issuer to Network:** Not Applicable

Network to Acquirer: Optional Echo

Bit 39 **Action Code**

Definition The type of action done or the type of action to do for a Transaction.

Format N(3)

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: Mandatory

Notes Refer to the Action Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

American Express Global Network Services
Network Specifications - Authorization
POS Authorization Reversal Advice Response (1430)
Bit 42 Card Acceptor Identification Code Message Layout

Bit 42

Card Acceptor Identification Code

Definition	The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.
Format	AN(15)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 49

Currency Code, Transaction

Definition	This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Currency Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 50

Currency Code, Reconciliation

Definition	This is the numeric code of the currency defined by the Issuer.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Not Applicable
Notes	Refer to the Currency Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 60

Market Specific Data

Definition	This bit provides additional authorization data.
Market Specific Data	
Format	AN(210) max LLLVAR
Optionality	Issuer to Network: Conditional Echo. This optionality can be different by market. Contact your AEGNS Representative for more instruction. Network to Acquirer: Conditional Echo. This optionality can be different by market. Contact your AEGNS Representative for more instruction.

Market Specific Data with Additional Authorization Data

Format	AN(316) max LLLVAR
Optionality	Issuer to Network: Conditional Echo. If a certified Issuer receives Bit 60 from the Network in the POS Authorization Request (1100) message, then the Issuer must echo Bit 60. The optionality for Subfield 1 - Market Specific Data can be different by market. Contact your AEGNS Representative for more instruction. Network to Acquirer: Conditional Echo. If the Network receives Bit 60 from the Issuer, it is sent to the Acquirer. The optionality for Subfield 1 - Market Specific Data can be different by market. Contact your AEGNS Representative for more instruction.

ATM Financial Transaction Request (1200)

Description

Use this message to request authorization to dispense funds from an Automated Teller Machine (ATM).

Length of Record

Variable

Message Layout

This is the message layout for the ATM Financial Transaction Request (1200). Refer to the [Programming Notes for ATM Financial Transaction Request \(1200\)](#) on page 5-93 for more instruction.

ATM Financial Transaction Request (1200) Message Layout

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
	N/A	N(4)		M	M	Message Type Identifier
	N/A	8 bytes, 64 bits for each bitmap Binary (Hex)		M	M	Bitmap Primary
	3	N(6)		M	M	Processing Code
	4	N(12)		M	M	Amount, Transaction
	5	N(12)		O	M	Amount, Reconciliation
	11	N(6)		M	M	Systems Trace Audit Number
	12	N(12)		M	M	Date and Time, Local Transaction
	19	N(3)		M	M	Country Code, Acquiring Institution
	22	AN(12)		M	M	Point of Service Data Code
	24	AN(3)		M	M	Function Code
	26	N(4)		O	M	Card Acceptor Business Code
	31	N(40) max LLVAR		N/A	C	Acquirer Reference Data, Transaction Identifier (TID)
	32	N(13) max LLVAR		M	M	Acquiring Institution Identification (AIN) Code
	33	N(13) max LLVAR		O	M	Forwarding Institution Identification Code
	35	ANS(39) max LLVAR		M	M	Track 2 Data
	37	AN(12)		O	M	Acquirer Reference Number (ARN)

ATM Financial Transaction Request (1200) Message Layout (continued)

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
	41	AN(8)		M	M	Card Acceptor Terminal Identification
	43	ANS(40), max LLVAR		M	M	Card Acceptor Name and Location
	49	N(3)		M	M	Currency Code, Transaction
	50	N(3)		O	M	Currency Code, Reconciliation
	52	8 bytes Binary		C	C	Personal Identification Number (PIN)
	53	9 bytes LLVAR Binary		C	C	Security Related Control Information
	55	AN(259) max (includes length indicator) LLLVAR		C	C	ICC System Related Data
		AN(4) EBCDIC	1	M	M	ICC Header Version Name
		(2) Binary (numeric)	2	M	M	ICC Header Version Number
9F26		(8) Binary Hexadecimal	3	M	M	Application Cryptogram
9F10		(33) max LLVAR Binary Hexadecimal (includes length LL indicator)	4	M	M	Issuer Application Data (IAD) (32) Sub-Subfield IAD Data Length (1) Derivation Key Index (1) Cryptogram Version Number (1) Card Verification Results (CVR) (4)
9F37		(4) Binary Hexadecimal	5	M	M	Unpredictable Number
9F36		(2) Binary Hexadecimal	6	M	M	Application Transaction Counter (ATC)
95		(5) Binary Hexadecimal	7	M	M	Terminal Verification Results (TVR)
9A		(3) Binary (numeric)	8	M	M	Transaction Date

ATM Financial Transaction Request (1200) Message Layout (continued)

EMV Tag	Bit Position	Format	Subfield	Optionality		Field Name
				A to N	N to I	
9C		(1) Binary (numeric)	9	M	M	Transaction Type
9F02		(6) Binary (numeric)	10	M	M	Amount Authorized
5F2A		(2) Binary (numeric)	11	M	M	Transaction Currency Code
9F1A		(2) Binary (numeric)	12	M	M	Terminal Country Code
82		(2) Binary Hexa- decimal	13	M	M	Application Interchange Profile (AIP)
9F03		(6) Binary (numeric)	14	M	M	Amount Other
5F34		(1) Binary (numeric)	15	M	M	Application PAN Sequence Number
9F27		(1) Binary Hexa- decimal	16	M	M	Cryptogram Information Data
			17	C	C	ICC Header Version Name
			18	C	C	ICC Header Version Number
			19	C	C	Issuer Authentication Data
	60	AN(210) max LLVAR		C	C	Market Specific Data
	62	AN(5) min AN(60) max LLVAR		N/A	C	Fraud Indicators
	111	AN(44) max LLLVAR		C	C	Encryption Data

Programming Notes for ATM Financial Transaction Request (1200)

Bit N/A

Definition

Format

Optionality

Message Type Identifier

This code identifies the type of message sent by the forwarding institution.

N(4)

Acquirer to Network: Mandatory

Notes	Network to Issuer: Mandatory Required Value: 1200 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit N/A	Bitmap Primary Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field. 8 bytes, 64 bits for each bitmap Binary (Hex) Acquirer to Network: Mandatory Network to Issuer: Mandatory
Definition Format Optionality	
Bit 3	Processing Code This is the service being requested. N(6) Acquirer to Network: Mandatory Network to Issuer: Mandatory Valid Values: • 014000 = Cash Request • 960000 = PIN Change • 970000 = PIN Unblock Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Definition Format Optionality	
Notes	
Bit 4	Amount, Transaction The cash amount at an ATM terminal. N(12) Acquirer to Network: Mandatory Network to Issuer: Mandatory The decimal point is determined by the Currency Code, Transaction (Bit 49). For example, \$100.00 in U.S.D. is 000000010000. PIN Change/Unblock: This field should be all zeros for PIN change and PIN unblock Transactions.
Definition Format Optionality	
Notes	
Bit 5	Amount, Reconciliation This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer. N(12) Acquirer to Network: Optional. This field must contain zeros if supplied by the Acquirer. The message must also contain Currency Code, Reconciliation (Bit 50). Network to Issuer: Mandatory. The Network puts the value into this field.
Definition Format Optionality	
Bit 11	Systems Trace Audit Number This is a unique (where applicable) trace number given to each Transaction or provisioning request. N(6) Acquirer to Network: Mandatory Network to Issuer: Mandatory
Definition Format Optionality	
Bit 12	Date and Time, Local Transaction This is the local date and time when a Transaction occurred. N(12), YYMMDDHHMMSS Acquirer to Network: Mandatory Network to Issuer: Mandatory
Definition Format Optionality	

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ATM Financial Transaction Request (1200)
Bit 19 Country Code, Acquiring Institution Message Layout

Bit 19	Country Code, Acquiring Institution
Definition	This is the Country Code of the acquiring institution.
Format	N(3)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Refer to the Country Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 22	Point of Service Data Code
Definition	This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).
Format	AN(12)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Refer to the Point of Service Data Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 24	Function Code
Definition	This code shows the reason for a message in its Message Type.
Format	AN(3)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Required Value: 200 = Original financial request/advice Refer to the Function Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 26	Card Acceptor Business Code
Definition	Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).
Format	N(4)
Optionality	Acquirer to Network: Optional Network to Issuer: Mandatory
Notes	Required Value: 6011 = Financial Institution Automatic Cash Dispense Refer to the Merchant Category Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 31	Acquirer Reference Data, Transaction Identifier (TID)
Definition	This unique number is given by the Network and follows a Transaction during its life cycle.
Format	N(40) max LLVAR
Optionality	Acquirer to Network: Not Applicable Network to Issuer: Conditional. The Network sends the TID to a certified Issuer whether the Acquirer is certified or not.
Notes	Refer to The Role of the Transaction Identifier on page 2-2 for more instruction about the TID and the authorization life cycle.
Bit 32	Acquiring Institution Identification (AIN) Code
Definition	This is a unique identification code given to an Acquirer by the Network.
Format	N(13) max LLVAR
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	This field uses the Routing and Transit Identifier assigned by the American Express Funds Access System (FAS).

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ATM Financial Transaction Request (1200)
Bit 33 Forwarding Institution Identification Code Message Layout

Example: An eight-byte Routing and Transit Identifier of "12345678" is formatted in the field with a prefix and suffix of "1," to which the length indicator is added to create 101123456781.

Bit 33

Definition
Format
Optionality

Forwarding Institution Identification Code

The institution identifier assigned to the processing institution that is sending the file.

N(13) max LLVAR

Acquirer to Network: Optional. **Required Value** = 1118000000511

Network to Issuer: Mandatory. **Required Value** = 1118000000511

Bit 35

Definition
Format
Optionality

Track 2 Data

This is the data encoded in a Track 2 magnetic stripe read or the data stored in the Track 2 field on the chip of a Chip Card.

ANS(39) max LLVAR

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Track 2 Rules: Track data must be sent as captured. Participants must support the ANSI track layouts and the ISO 7813 layouts. If applicable for sending other formats with no changes, Acquirer should strip the Start Sentinel and End Sentinel, translate from ASCII to EBCDIC, and add the length of data to the beginning of the field.

ATM Card Acceptor Terminals

The ATM Card Acceptor terminal must not keep track data from the Authorization Request. All track data must be erased when it is sent to the Issuer.

Refer to Chapter 4 Network Cards in the *Network Specifications - Instructions* for Track 2 Layout instruction.

Refer to the ISO 7813 Standard Track 2 Layout or ANSI X4.16 Track 2 Layout in *Network Specifications - Instructions* for more instruction.

Bit 37

Definition
Format
Optionality

Notes

Acquirer Reference Number (ARN)

A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

AN(12)

Acquirer to Network: Optional

Network to Issuer: Mandatory

If this value is less than 12 characters in length, it must be left justified and space-filled.

Bit 41

Definition
Format
Optionality

Card Acceptor Terminal Identification

This code identifies a POS terminal at a Card Acceptor location.

AN(8)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Bit 43

Definition
Format
Optionality

Card Acceptor Name and Location

The name of the Card Acceptor as it is known to Cardmembers.

ANS(40), max LLVAR

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

This bit is free form and Issuers must accept the data as presented by the Acquirers.

ATM:

Length of Field: ANS(40), max LLVAR

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 ATM Financial Transaction Request (1200)
 Bit 49 Currency Code, Transaction Message Layout

Position	Value	Use
1-2	40	Variable Length Indicator
3-17		Card Acceptor Name
18-40		Card Acceptor Location (may include address, city, postal code, etc.
41-42		Card Acceptor Country Code ¹

1. This country code is not defined in this manual. Contact your AEGNS Representative to get your two-digit country code. The Network does not compare the two-character country code to Bit 19, Country Code.

Bit 49

Definition

Currency Code, Transaction

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.

Format

N(3)

Optionality

Acquirer to Network: Optional

Network to Issuer: Mandatory

Notes

Network to Issuer:

- For a cash Transaction with Processing Code 014000, Bit 50 will show the Currency Code of the Issuer for the cash Transaction.
- For a Global PIN management Transaction with Processing Code 960000 (PIN Change) or 970000 (PIN Unblock), Bit 50 will show 000 as the default Currency Code.

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 52

Definition

Personal Identification Number (PIN)

A four (4) to twelve (12) personal code that lets an Issuer identify the Cardmember for approval of an ATM or POS Transaction.

Format

8 bytes binary

Optionality

Acquirer to Network: Conditional. Acquirers must submit PIN data in either Bit 52 or Bit 111 - Encryption Data. PIN data must not be submitted in both Bit 52 and Bit 111 - Encryption Data in the same ATM Financial Transaction Request (1200) message.

Network to Issuer: Conditional. The Network will pass PIN data received in either Bit 52 OR Bit 111 - Encryption Data. PIN data will not be passed in both Bit 52 and Bit 111 - Encryption Data in the same ATM Financial Transaction Request (1200) message.

Notes

It is the responsibility of the Issuer to let a Cardmember have multiple PIN attempts at an ATM.

Bit 53

Definition

Security Related Control Information

This field contains a variable length indicator and the Card Security Code (CSC) or One-Time Password (OTP) supplied by the Acquirer.

Card Security Code includes the 5CSC or the Printed Card Security Code (PCSC), which includes 3CSC and 4CSC (4DBC).

OTP is a dynamic 4-digit code provided by the Issuer to the Cardmember.

American Express Global Network Services
Network Specifications - Authorization
 ATM Financial Transaction Request (1200)
 Bit 55 ICC System Related Data Message Layout

Format 9 bytes LLVAR binary

Optionality **Acquirer to Network:** Conditional

Network to Issuer: Conditional

Notes **PIN Change Request:** Encrypted PIN data requesting a PIN change (if Processing Code = 960000).

Position	Value	Use
1	X'40'	One byte binary bitmap that shows Bit 53 contains a new PIN for Global PIN Management
2-9		New PIN block

Bit 55

ICC System Related Data

Definition This is the data sent to and from the Integrated Circuit Card and used in an AEIPS or Expresspay Transaction.

Format AN(259) max (includes length indicator) LLLVAR

Optionality **Acquirer to Network:** Conditional

Network to Issuer: Conditional

Notes If ICC data is read from the Card, you must send the data in this field. This table shows that this data extends across more than 1 subfield structure.

Maximum Length	Fixed/ Variable	Source	Field Name
4	F	Acquirer	ICC Header Version Name
2	F	Acquirer	ICC Header Version Number
8	F	Chip Card	Application Cryptogram
33	V	Chip Card	Issuer Application Data (IAD)
4	F	Terminal	Unpredictable Number
2	F	Chip Card	Application Transaction Counter (ATC)
5	F	Terminal	Terminal Verification Results (TVR)
3	F	Terminal	Transaction Date
1	F	Terminal	Transaction Type
6	F	Terminal	Amount, Authorized
2	F	Terminal	Transaction Currency Code
2	F	Terminal	Terminal Country Code
2	F	Chip Card	Application Interchange Profile (AIP)
6	F	Terminal	Amount, Other
1	F	Chip Card	Application PAN Sequence Number
1	F	Chip Card	Cryptogram Information Data

Bit 55 Layout

The layout contains these subfields. All subfields are mandatory from Acquirer to Network and Network to Issuer.

EMV Tag	Subfield	Format	Notes
	1 ICC Header Version Name	AN(4) EBCDIC	Required Value: AGNS
	2 ICC Header Version Number	(2) Binary (numeric)	Required Value: 0001
9F26	3 Application Cryptogram	(8) Binary Hexadecimal	

American Express Global Network Services
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ATM Financial Transaction Request (1200)
Bit 55 ICC System Related Data Message Layout

EMV Tag	Subfield	Format	Notes
9F10	4 Issuer Application Data (IAD) (32)	(33) max LLVAR Binary Hexadecimal (includes length LL indicator)	Sub-Subfields: - IAD Data Length (1) - Derivation Key Index (1) - Cryptogram Version Number (1) - Card Verification Results (CVR) (4) Subfield 4 Issuer Application Data (IAD) As Subfield 4 is a variable length field of type LLVAR, the "LL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLVAR length byte is in addition to the 32 byte maximum data length of the actual data (i.e., total 33 bytes). Example: ICC Structure Subfield 4, Issuer Application Data formatted with the LLVAR and structure above is 07 06 01 01 03 A0 00 00 where: 07 = overall length indicator for Subfield 4 LLVAR 06 = IAD Data Length (1 byte) 01 = Derivation Key Index (1 byte) 01 = Cryptogram Version Number (1 byte) 03A00000 = CVR (4 bytes)
9F37	5 Unpredictable Number	(4) Binary Hexadecimal	
9F36	6 Application Transaction Counter (ATC)	(2) Binary Hexadecimal	
95	7 Terminal Verification Results (TVR)	(5) Binary Hexadecimal	
9A	8 Transaction Date	(3) Binary (numeric)	Format: YYMMDD
9C	9 Transaction Type	(1) Binary (numeric)	Subfield 9 Transaction Type Valid Values: 00 = Debit 01 = Cash 20 = Credit
9F02	10 Amount Authorized	(6) Binary (numeric)	This amount can be different from the amount in Bit 4 or Bit 5. The decimal point is determined by the Transaction Currency Code, Subfield 11.
5F2A	11 Transaction Currency Code	(2) Binary (numeric)	This field should agree with the contents of Bit 49, Currency Code, Transaction.
9F1A	12 Terminal Country Code	(2) Binary (numeric)	
82	13 Application Interchange Profile	(2) Binary Hexadecimal	Expresspay Mobile Transactions Valid Values: Byte 2, Bit 7 0 = Expresspay Mobile not supported 1 = Expresspay Mobile supported
9F03	14 Amount Other	(6) Binary (numeric)	
5F34	15 Application PAN Sequence Number	(1) Binary (numeric)	Card Acceptors (S/Es) must supply mandatory Authorization data fields, with no changes, when the chip is read by the POS Device. If the Application PAN Sequence Number is a numeric value (for example, 01), the value must be transmitted with no changes to the Network.
9F27	16 Cryptogram Information Data	(1) Binary Hexadecimal	

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 ATM Financial Transaction Request (1200)
 Bit 60 Market Specific Data Message Layout

EMV Tag	Subfield	Format	Notes
	AEIPS and Expresspay Cryptogram Validation Results		AEIPS Cryptogram Validation results are passed in the ATM Financial Transaction Request Message (1200), Bit 55, Subfields 17-19 to Issuers who want to get this data as part of the AEIPS and Expresspay Cryptogram Validation Service. Issuers certified for this Service should use the ATM Financial Transaction Request (1200), Bit 55, Subfields 17-19 to populate Bit 55 of the ATM Financial Transaction Response (1210) message. The Network edits or truncates Bit 55 data to support Issuers certified for AEIPS and Expresspay Cryptogram Validation. Data sent by the Acquirer in Bit 55, Subfields 17-19 can result in Issuer declines. Issuers should not use this data as part of the chip authentication.
	17 ICC Header Version Name		Required Value: AGNS
	18 ICC Header Version Number		Required Value: 0001
	19 Issuer Authentication Data		This subfield must be present, if Card authentication is performed and is successful. This subfield is not present if Card authentication is not performed or is not successful (for example, a missing or incorrect cryptogram in the ATM Financial Transaction Request (1200) message). Subfield 19 is a variable length field (LLVAR). The "LL" length bytes must be present in Binary (Hexadecimal) format at the start of this field. The LLVAR length byte is in addition to the 16 byte maximum data length of the data for a total of 17 bytes. • Sub-Subfield 1 Authorization Response Cryptogram (ARPC) (8) • Sub-Subfield 2 ARPC Response Code N(2) (ASCII)

Bit 60

Market Specific Data

Definition

This bit provides additional authorization data.

Format

AN(210) max LLLVAR

Optionality

Acquirer to Network: Conditional. Certified Acquirers must send the field in the format shown below. The Network does not change or validate the contents.

Network to Issuer: Conditional

Notes

Valid Values:

Position	Value	Use
1-3		Variable Length Indicator
4-5	MS	Market Specific Data Indicator
6-8		Market Country Code
9-10		Market Specific Data Version Indicator
11-210		Market Specific Data Elements

Bit 62

Fraud Indicators

Definition

The Fraud Indicator field is used to provide fraud mitigation indicators for a variety of situations.

Format

AN(5) min AN(60) max LLLVAR

Optionality

Acquirer to Network: Not Applicable

Notes

Network to Issuer: Conditional. This field is populated by the Network with the 5CSC validation results and passed as fraud indicators only to certified Issuers.

Bit 62 is a 3-digit variable length field (LLLVAR). At this time, only 60 bytes are used.

Valid Values:

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD1	Request Type Identifier FD1 = 5CSC checking results If FD1 value is listed, then the field will end at 10 bytes.
9-10	XX	Two digit 5CSC Indicator

FD3 Request Type Identifier

Bit 62 FD3 Request Type is supplied for SafeKey Transactions where the Network created the AEVV when the Issuer ACS was not available.

Certification

When the Participant certifies for the FD3 Request Type Identifier, FD1 and FD2 are no longer supported.

It is not necessary for existing Issuers to certify for Field 62, Fraud Indicators, unless they re-certify their Authorizations capabilities.

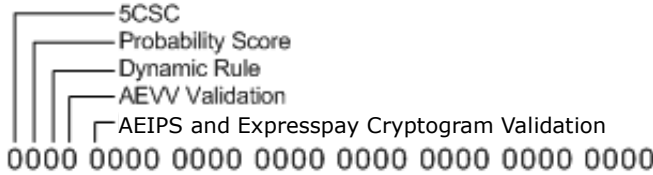
American Express SafeKey requires Participants to certify for Bit 62 and support the FD3 Request Type Identifier.

Refer to the Fraud Indicators section of the *Codes Reference Guide* for a list of valid codes and descriptions.

Valid Values:

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD3	Request Type Identifier FD3 = Bitmap Variable Length

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ATM Financial Transaction Request (1200)
Bit 111 Encryption Data Message Layout

9-12	32 bit Binary	Bitmap Indicator Identifies the fields in the structure. After the bitmap, the layout consists of at least one of the subfields as defined below. Each bit position of the 32 bit / 4-byte bitmap represents which fraud indicators are present. If a particular bit is "ON" in the bitmap, that corresponding sub-field will be present in positions 13-60. • Subfield 1 5CSC Indicator (2 Bytes) • Subfield 2 Fraud Probability Score (5 Bytes) • Subfield 3 Recommendation Code (3 Bytes) • Subfield 4 AEVV Validation Result (1 Byte) • Subfield 5 AEIPS and Expresspay Cryptogram Validation Result (2 Bytes) • Subfield 6 PIN Validation Results (2 Bytes) • Subfield 7 EMV Data Results (4 Bytes) • Subfield 8 Reserved for AEGNS use • Subfield 9 ATC Validation Results (2 Bytes) Refer to the <i>Codes Reference Guide</i> for valid codes and definitions.
13-60	Fraud Value	Describes the results of the Bitmap Indicators that are set.
Example: Positions 9 - 12 		

Example: If Bit 62 includes Fraud Probability Score, Recommendation Code, and AEVV Validation Result:

Position	Value
1-3	018
4-5	AX
6-8	FD3
9-12	01110000000000000000000000000000 X'70000000'
13-60	555553331

Bit 111

Definition

Format

Optionality

Notes

Encryption Data

Encryption Data contains the data for various encryption activities.

AN(44) max LLLLVAR

Acquirer to Network: Conditional. Acquirers must submit PIN data in either Bit 111 OR Bit 52 - Personal Identification Number (PIN). PIN data must not be submitted in both Bit 111 and Bit 52 - Personal Identification Number (PIN) in the same ATM Financial Transaction Request (1200) message.

Network to Issuer: Conditional. The Network will pass PIN data received in either Bit 52 OR Bit 111 - Encryption Data. PIN data will not be passed in both Bit 52 and Bit 111 - Encryption Data in the same ATM Financial Transaction Request (1200)

This field supports multiple data elements identified in the table below. Certified Acquirers will send the appropriate data, depending on their specific needs.

American Express DKE PIN Encryption Data Transactional Layout

Bit	Position	Data Element	Subfield Length	Subfield Type	Optionality		Valid Value	Description
					A to N	N to I		
1	1-4	Variable Length Indicator (VLI)	4 bytes	Numeric (EBCDIC) LLLLVAR	M	M		VLI is the total length of variable data in this data field (not including VLI).
2	5-6	Primary ID	2 bytes	Alphanumeric	M	M		Primary ID (Card Type Code) is constant literal "AX" (American Express)
3	7-9	Secondary ID	3 bytes	Alphanumeric	M	M	DKE = Dynamic Key Encryption	Secondary ID (Data Type Code) is variable based on Encryption Type
4	10-11	PIN Block VLI	2 bytes	Numeric (EBCDIC LLVAR)	M	M		Total length of variable data in the PIN Block (not including VLI)
5	12-27	PIN Block	16 bytes Max LLVAR	Binary (Hexadecimal configuration)	M	M		PIN encryption data block
6	28-29	Key Check VLI	2 bytes	Numeric (EBCDIC LLLVAR)	M	M		Total length of variable data in the Key Check Value (not including VLI)
7	30-34	Key Check Value	5 bytes Max LLVAR	Binary (Hexadecimal configuration)	M	M		Key check value for PIN encryption key
8	35-42	Reserved for Future Use	8 bytes	Numeric	M	M		Unused numeric fields should be zero filled

American Express DUKPT PIN Encryption Data Transactional Layout

Bit	Position	Data Element	Subfield Length	Subfield Type	Optionality		Valid Value	Description
					A to N	N to I		
1	1-4	Variable Length Indicator (VLI)	4 bytes	Numeric (EBCDIC) LLLLVAR	M	M		VLI is the total length of variable data in this data field (not including VLI).
2	5-6	Primary ID	2 bytes	Alphanumeric	M	M		Primary ID (Card Type Code) is constant literal "AX" (American Express)
3	7-9	Secondary ID	3 bytes	Alphanumeric	M	M	DUK = DUKPT	Secondary ID (Data Type Code) is variable based on Encryption Type

American Express DUKPT PIN Encryption Data Transactional Layout (continued)

4	10-11	PIN Block VLI	2 bytes	Numeric (EBCDIC LLVAR)	M	M		Total length of variable data in the PIN Block (not including VLI)
5	12-27	PIN Block	16 bytes Max LLVAR	Binary (Hexadecimal configuration)	M	M		PIN encryption data block
6	28-29	Key Serial Number VLI	2 bytes	Numeric (EBCDIC LLVAR)	M	M		Total length of variable data in the Key Serial Number Value (not including VLI)
7	30-41	Key Serial Number (KSN)	12 bytes Max	Numeric (Hexadecimal configuration)	M	M		A unique key for each DUKPT Transaction.
8	42-44	DUKPT Format	3 bytes	Alphanumeric	M	M		DUKPT format indicator.

ATM Financial Transaction Response (1210)

Description

Use this message to respond to an ATM Financial Transaction Request (1200) message.

Length of Record

Variable

Message Layout

This is the message layout for the ATM Financial Transaction Response (1210). Refer to the [Programming Notes for ATM Financial Transaction Response \(1210\)](#) on page 5-106 for more instruction.

ATM Financial Transaction Response (1210) Message Layout

EMV Tag	Bit Position	Format	Optionality		Field Name
			I to N	N to A	
	N/A	N(4)	M	M	Message Type Identifier
	N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
	2	N(21) max LLVAR	M	M	Primary Account Number (PAN)
	3	N(6)	ME	ME	Processing Code
	4	N(12)	M	M	Amount, Transaction
	5	N(12)	M	M	Amount, Reconciliation
	11	N(6)	ME	ME	Systems Trace Audit Number
	12	N(12)	ME	ME	Date and Time, Local Transaction
	19	N(3)	ME	ME	Country Code, Acquiring Institution
	22	AN(12)	ME	ME	Point of Service Data Code
	24	AN(3)	ME	ME	Function Code
	26	N(4)	ME	CE	Card Acceptor Business Code
	31	N(40) max LLVAR	CE	C	Acquirer Reference Data, Transaction Identifier (TID)
	32	N(13) max LLVAR	ME	ME	Acquiring Institution Identification (AIN) Code
	33	N(13) max LLVAR	ME	CE	Forwarding Institution Identification Code
	37	AN(12)	ME	CE	Acquirer Reference Number (ARN)
	38	AN(6)	M	M	Approval Code

ATM Financial Transaction Response (1210) Message Layout (continued)

EMV Tag	Bit Position	Format	Optionality		Field Name
			I to N	N to A	
	39	N(3)	M	M	Action Code
	41	AN(8)	ME	ME	Card Acceptor Terminal Identification
	49	N(3)	ME	ME	Currency Code, Transaction
	50	N(3)	ME	M	Currency Code, Reconciliation
	55	AN(259) max (includes length indica- tor) LLVAR	C	C	ICC System Related Data
		AN(4)	ME	ME	Subfield 1 ICC Header Version Name
		(2) Binary (numeric)	ME	ME	Subfield 2 ICC Header Version Number
91		(17) max LLVAR Binary Hexa- decimal (includes length LL indicator)	C	C	Subfield 3 Issuer Authentication Data Sub-Subfield 1 Authorization Response Cryptogram (ARPC) (8) Sub-Subfield 2 ARPC Response Code (2)
		(129) max LLVAR Binary Hexa- decimal (includes length LLL indicator)	C	C	Subfield 4 Issuer Script Data
	60	AN(210) max LLVAR	CE	CE	Market Specific Data

Programming Notes for ATM Financial Transaction Response (1210)

Bit N/A

Definition

Format

Optionality

Notes

Message Type Identifier

This code identifies the type of message sent by the forwarding institution.

N(4)

Issuer to Network: Mandatory

Network to Acquirer: Mandatory

Required Value: 1210

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Definition

Format

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

8 bytes, 64 bits for each bitmap Binary (Hex)

American Express Global Network Services
 Network Specifications - Authorization
 ATM Financial Transaction Response (1210)
 Bit 2 Primary Account Number (PAN) Message Layout

Optionality **Issuer to Network:** Mandatory
 Network to Acquirer: Mandatory

Bit 2 **Primary Account Number (PAN)**

Definition A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.

Format N(21) max LLVAR

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: Mandatory

Notes A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit.

For example, a Cardmember Account Number of 3759-876543-21001 shows as 15375987654321001.

Bit 3 **Processing Code**

Definition This is the service being requested.

Format N(6)

Optionality **Issuer to Network:** Mandatory Echo

Network to Acquirer: Mandatory Echo

Notes Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 4 **Amount, Transaction**

Definition The cash amount at an ATM terminal.

Format N(12)

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: Mandatory

Notes **Approved:** This value is echoed from the Network to Issuer 1200 ATM Financial Transaction Request message if approved.

Not Approved: The Issuer sets it to zeros if the Transaction is not approved.

This bit requires that Bit 49, Currency Code, Transaction contain a value.

Bit 5 **Amount, Reconciliation**

Definition This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.

Format N(12)

Optionality **Issuer to Network:** Mandatory

Network to Acquirer: Mandatory

Notes **Approved:** This is echoed from the ATM Financial Transaction Request (1200) message if the request is approved.

Not Approved: The Issuer must set this value to zeros if the Transaction is not approved.

This bit requires that Bit 50, Currency Code, Reconciliation contain a value.

Bit 11 **Systems Trace Audit Number**

Definition This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format N(6)

Optionality **Issuer to Network:** Mandatory Echo

Network to Acquirer: Mandatory Echo

Bit 12 **Date and Time, Local Transaction**

Definition This is the local date and time when a Transaction occurred.

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 ATM Financial Transaction Response (1210)
 Bit 19 Country Code, Acquiring Institution Message Layout

Format	N(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 19	Country Code, Acquiring Institution
Definition	This is the Country Code of the acquiring institution.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Country Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 22	Point of Service Data Code
Definition	This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).
Format	AN(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo Refer to the Point of Service Data Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 24	Function Code
Definition	This code shows the reason for a message in its Message Type.
Format	AN(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Function Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 26	Card Acceptor Business Code
Definitions	Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).
Format	N(4)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Conditional Echo. The value should be echoed from the value sent in the 1200 message.
Notes	Refer to the Merchant Category Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 31	Acquirer Reference Data, Transaction Identifier (TID)
Definition	This unique number is given by the Network and follows a Transaction during its life cycle.
Format	N(40) max LLVAR
Optionality	Issuer to Network: Conditional Echo. If the certified Issuer receives the TID from the Network, the Issuer echoes it in Bit 31. Network to Acquirer: Conditional. If the Acquirer is certified, the Network sends the TID to the Acquirer.
Notes	Refer to The Role of the Transaction Identifier on page 2-2 for more instruction about the TID and the authorization life cycle.
Bit 32	Acquiring Institution Identification (AIN) Code
Definition	This is a unique identification code given to an Acquirer by the Network.
Format	N(13) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

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ATM Financial Transaction Response (1210)
Bit 33 Forwarding Institution Identification Code Message Layout

Bit 33	Forwarding Institution Identification Code
Definition	The institution identifier assigned to the processing institution that is sending the file.
Format	N(13) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Conditional Echo Required Value: 1118000000511
Notes	
Bit 37	Acquirer Reference Number (ARN)
Definition	A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.
Format	AN(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Conditional Echo
Bit 38	Approval Code
Definition	A unique code assigned by the Card Transaction authorizing institution to indicate the disposition for the Transaction.
Format	AN(6)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Issuer to Network: • If the Transaction is approved, the Approval Code must be a unique, six digit, left justified numeric value. • Transactions not fully approved are not permitted. • If the Transaction is denied, this field must be zero-filled. Network to Acquirer: • Partial approvals are not permitted. • This field can contain alphanumeric characters if the Transaction was declined. • Acquirers should look at Bit 39, Action Code, to see if the Transaction was approved.
Bit 39	Action Code
Definition	The type of action done or the type of action to do for a Transaction.
Format	N(3)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Refer to the Action Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 41	Card Acceptor Terminal Identification
Definition	This code identifies a POS terminal at a Card Acceptor location.
Format	AN(8)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 49	Currency Code, Transaction
Definition	This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

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ATM Financial Transaction Response (1210)
Bit 50 Currency Code, Reconciliation Message Layout

Notes Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50 **Currency Code, Reconciliation**

Definition This is the numeric code of the currency defined by the Issuer.

Format N(3)

Optionality **Issuer to Network:** Mandatory Echo

Network to Acquirer: Mandatory

Notes Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 55 **ICC System Related Data**

Definition This is the data sent to and from the Integrated Circuit Card and used in an AEIPS or Expresspay Transaction.

Format AN(259) max (includes length indicator) LLLVAR

Optionality **Issuer to Network:** Conditional

Network to Acquirer: Conditional

Notes Response data must be sent in this field if Bit 55 ICC System Related Data is provided in the ATM Financial Transaction Request (1200) message.

The following table shows the format of Bit 55 when it is included in a ATM Financial Transaction Response (1210) message.

Subfield	Name	Length	Source
1	ICC Header Version Name	4	Issuer (echo from ATM Financial Transaction Request (1200) message)
2	ICC Header Version Number	2	Issuer (echo from ATM Financial Transaction Request (1200) message)
3	Issuer Authentication Data (IAD)	up to 17 bytes	Issuer
4	Issuer Script Data	up to 129 bytes	Issuer

The Issuer is required to create this bit and include the data within it. The header is simply a reflection of the header contained in the corresponding request. Refer to the *EMV Chip Issuer Implementation Guide* for instructions on how to create the Issuer Authentication Data (IAD).

Issuer Scripts are not supported for Expresspay Transactions. Issuer Script Data sent in an Expresspay ATM Financial Transaction Response (1210) message will not be processed by the chip on the Card.

EMV Tag	Subfield	Format	Optionality		Notes
			I to N	N to A	
	1 ICC Header Version Name	AN(4)	ME	ME	Required Value: AGNS
	2 ICC Header Version Number	(2) Binary (numeric)	ME	ME	Required Value: 0001 If Bit 55 data is not in the response, you must still echo with no changes from the Network to Issuer Request.

American Express Global Network Services
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ATM Financial Transaction Response (1210)
Field 60 Market Specific Data Message Layout

EMV Tag	Subfield	Format	Optionality		Notes
91	3 Issuer Authentication Data	(17) max LLVAR Binary Hexadecimal (includes length LL indicator)	C	C	- If Card authentication was successful, this subfield must be present. - If Card authentication is not done or is not correct (i.e., missing or invalid cryptogram in the 1200 message), do not return this subfield. - When this subfield is not present, it is referred to as a 'downgraded response' and is a valid Transaction response. Sub-Subfield 1 Authorization Response Cryptogram (ARPC) (8) Sub-Subfield 2 ARPC Response Code (2) Subfield 3 is a variable length field of type LLVAR, the "LL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLVAR length byte is in addition to the 16 byte maximum data length of the actual data (i.e., total 17 bytes).
	4 Issuer Script Data	(129) max LLLVAR Binary Hexadecimal (includes length LLL indicator)	C	C	This field is only used if Subfield 3, Issuer Authentication Data, is present. Subfield 4 is a variable length field of type LLLVAR, the "LLL" length bytes must be present at the start of this field in Binary (Hexadecimal) format. This LLLVAR length byte is in addition to the 128 byte maximum data length of the actual data (i.e., total 129 bytes).

Field 60

Market Specific Data

Definition	This bit provides additional authorization data.
Format	AN(210) max LLLVAR
Optionality	Issuer to Network: Conditional Echo Network to Acquirer: Conditional Echo

ATM Reversal Advice (1420)

Description

Acquirers can use the ATM Reversal Advice (1420) message to advise Issuers of a disposition change for a Transaction.

Length of Record

Variable

Message Layout

This is the message layout for the ATM Reversal Advice (1420). Refer to the [Programming Notes for ATM Reversal Advice \(1420\)](#) on page 5-113 for more instruction.

ATM Reversal Advice (1420) Message Layout

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bit-map Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	M	M	Primary Account Number (PAN)
3	N(6)	C	C	Processing Code
4	N(12)	M	M	Amount, Transaction
5	N(12)	O	M	Amount, Reconciliation
11	N(6)	M	M	Systems Trace Audit Number
12	N(12)	M	M	Date and Time, Local Transaction
19	N(3)	M	M	Country Code, Acquiring Institution
24	AN(3)	M	M	Function Code
25	N(4)	M	M	Message Reason Code
30	N(24)	M	M	Amount, Original
31	N(40) max LLVAR	C	C	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	ME	ME	Acquiring Institution Identification (AIN) Code
33	N(13) max LLVAR	O	M	Forwarding Institution Identification Code
37	AN(12)	O	M	Acquirer Reference Number (ARN)

ATM Reversal Advice (1420) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
41	AN(8)	ME	M	Card Acceptor Terminal Identification
49	N(3)	ME	ME	Currency Code, Transaction
50	N(3)	O	M	Currency Code, Reconciliation
56	ANS(37) max LLVAR	M	M	Original Data Elements
60	AN(210) max LLLVAR	C	C	Market Specific Data

Programming Notes for ATM Reversal Advice (1420)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Required Value: 1420 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap Binary (Hex)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit. For example, a Cardmember Account Number of 3759-876543-21001 shows as 15375987654321001.

Bit 3

Processing Code

Definition	This is the service being requested.
Format	N(6)
Optionality	Acquirer to Network: Conditional. This field is present only if it is sent for PIN change and PIN unblock Transactions. Network to Issuer: Conditional. This field is present only if it is sent for PIN change and PIN unblock Transactions.
Notes	Valid Values: 960000 = PIN Change 970000 = PIN Unblock

Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 4

Definition
Format
Optionality

Amount, Transaction

The cash amount at an ATM terminal.

N(12)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

This field contains zeros (non-dispense) or the partial amount dispensed by the Acquirer's ATM.

The decimal point is determined by the Currency Code, Transaction (Bit 49). For example, \$100.00 in U.S.D. is 000000010000.

Bit 5

Definition
Format
Optionality

Amount, Reconciliation

This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.

N(12)

Acquirer to Network: Optional

Network to Issuer: Mandatory

Notes

Acquirer to Network: If used, this field requires the Currency Code, Reconciliation (Bit 50) and must contain zeros.

Network to Issuer: If cash is not dispensed, the field will be zeros. If cash is not fully dispensed, the amount in Bit 4 is converted using the currency code contained in Bit 50.

Bit 11

Definition
Format
Optionality

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.

N(6)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

This number must be the same as the first ATM Financial Transaction Request (1200) message.

Bit 12

Definition
Format
Optionality

Date and Time, Local Transaction

This is the local date and time when a Transaction occurred.

N(12) YYMMDDHHMMSS

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Bit 19

Definition
Format
Optionality

Country Code, Acquiring Institution

This is the Country Code of the acquiring institution.

N(3)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Refer to the Country Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Bit 24

Definition
Format
Optionality

Function Code

This code shows the reason for a message in its Message Type.

AN(3)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Valid Values:

- 400 = Full reversal, Transaction did not complete as approved
- 401 = Partial reversal, Transaction did not complete for full amount

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 25

Definition

Format

Optionality

Notes

Message Reason Code

This code gives the cause of a First and Final Chargeback, Retrieval Request Notification, File Acknowledgment, ATM and POS, and Reversal requests.

N(4)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

This code shows if a sent file was accepted or rejected by the Network for a Network File Acknowledgment (1844) and a Participant File Acknowledgment (1824).

Refer to the Message Reason Code section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 30

Definition

Format

Optionality

Notes

Amount, Original

The Transaction Amount from the ATM Transaction Request (1200) message.

N(24)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Acquirer to Network:

Position	Use
1-12	Amount, Transaction from the ATM Financial Transaction (1200) message
13-24	Amount, Reconciliation. Must be zeros since the Network controls currency conversion.

Network to Issuer:

Position	Use
1-12	Amount, Transaction from the ATM Financial Transaction (1200) message
13-24	Amount, Reconciliation. The Network does currency conversion and puts the value in this field. Bit 50 must be present.

Bit 31

Definition

Format

Optionality

Notes

Acquirer Reference Data, Transaction Identifier (TID)

This unique number is given by the Network and follows a Transaction during its life cycle.

N(40) max LLVAR

Acquirer to Network: Conditional

Network to Issuer: Conditional. The Network puts the TID into this field if it is in the 1210 message.

Refer to [The Role of the Transaction Identifier on page 2-2](#) for more instruction about the TID and the authorization life cycle.

Bit 32

Definition

Format

Optionality

Acquiring Institution Identification (AIN) Code

This is a unique identification code given to an Acquirer by the Network.

N(13) max LLVAR

Acquirer to Network: Mandatory Echo

Network to Issuer: Mandatory Echo

Bit 33

Definition

Format

Forwarding Institution Identification Code

The institution identifier assigned to the processing institution that is sending the file.

N(13) max LLVAR

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ATM Reversal Advice (1420)
Bit 37 Acquirer Reference Number (ARN) Message Layout

Optionality

Acquirer to Network: Optional

Network to Issuer: Mandatory. Default Value = 1118000000511

Bit 37

Acquirer Reference Number (ARN)

Definition

A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

Format

AN(12)

Optionality

Acquirer to Network: Optional

Network to Issuer: Mandatory

Bit 41

Card Acceptor Terminal Identification

Definition

This code identifies a POS terminal at a Card Acceptor location.

Format

AN(8)

Optionality

Acquirer to Network: Mandatory Echo

Network to Issuer: Mandatory

Bit 49

Currency Code, Transaction

Definition

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Acquirer to Network: Mandatory Echo

Network to Issuer: Mandatory Echo

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Currency Code, Reconciliation

Definition

This is the numeric code of the currency defined by the Issuer.

Format

N(3)

Optionality

Acquirer to Network: Optional

Network to Issuer: Mandatory

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 56

Original Data Elements

Definition

Data from the Authorization Request (1100 or 1200) or POS/ATM Authorization Reversal Advice (1420) message.

Format

ANS(37) max LLVAR

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Valid Values:

Position	Value	Use
1-2	35	Variable Length Indicator
3-6		Message Type from 1200
7-12		Systems Trace Audit Number (Bit 11) from 1200
13-24		Date and Time, Local Transaction (Bit 12) from 1200
25-37		The first Acquiring Institution ID Code (Bit 32)

Bit 60

Market Specific Data

Definition

This bit provides additional authorization data.

Format

AN(210) max LLLVAR

Optionality

Acquirer to Network: Conditional

Notes

Network to Issuer: Conditional

Valid Values:

Position	Value	Use
1-3		Variable Length Indicator
4-5	MS	Market Specific Data Indicator
6-8		Market ISO Country Code
9-10		Market Specific Data Version Indicator
11-210		Market Specific Data Elements

ATM Reversal Advice Response (1430)

Description

The Network uses this message to acknowledge the receipt of the ATM Reversal Advice (1420) message. It does not mean that exposure on the account has been reversed.

Length of Record

Variable

Message Layout

This is the message layout for the ATM Reversal Advice Response (1430). Refer to the [Programming Notes for ATM Reversal Advice Response \(1430\)](#) on page 5-119 for more instruction.

ATM Reversal Advice Response (1430) Message Layout

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	ME	ME	Primary Account Number (PAN)
3	N(6)	CE	CE	Processing Code
4	N(12)	ME	ME	Amount, Transaction
5	N(12)	ME	M	Amount, Reconciliation
11	N(6)	ME	ME	Systems Trace Audit Number
12	N(12)	ME	ME	Date and Time, Local Transaction
19	N(3)	ME	ME	Country Code, Acquiring Institution
31	N(40) max LLVAR	CE	CE	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	ME	ME	Acquiring Institution Identification (AIN) Code
33	N(13) max LLVAR	ME	CE	Forwarding Institution Identification Code
37	AN(12)	ME	CE	Acquirer Reference Number (ARN)
39	N(3)	M	M	Action Code
41	AN(8)	ME	ME	Card Acceptor Terminal Identification
49	N(3)	ME	ME	Currency Code, Transaction
50	N(3)	ME	M	Currency Code, Reconciliation

ATM Reversal Advice Response (1430) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
56	ANS(37) max LLVAR	ME	ME	Original Data Elements
60	AN(210) max LLLVAR	CE	CE	Market Specific Data

Programming Notes for ATM Reversal Advice Response (1430)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Required Value: 1430 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap Binary (Hex)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 3

Processing Code

Definition	This is the service being requested.
Format	N(6)
Optionality	Issuer to Network: Conditional Echo Network to Acquirer: Conditional Echo
Notes	Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 4

Amount, Transaction

Definition	The cash amount at an ATM terminal.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 5

Amount, Reconciliation

Definition	This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.
Format	N(12)

Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory
Bit 11	Systems Trace Audit Number
Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 12	Date and Time, Local Transaction
Definition	This is the local date and time when a Transaction occurred.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	This is the same as the Date and Time, Local Transaction from the ATM Reversal Advice (1420) message.
Bit 19	Country Code, Acquiring Institution
Definition	This is the Country Code of the acquiring institution.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Country Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 31	Acquirer Reference Data, Transaction Identifier (TID)
Definition	This unique number is given by the Network and follows a Transaction during its life cycle.
Format	N(40) max LLVAR
Optionality	Issuer to Network: Conditional Echo. If the certified Issuer receives the TID from the Network, the Issuer echoes it in Bit 31. Network to Acquirer: Conditional Echo. If the Acquirer is certified, the Network sends the TID.
Notes	Refer to The Role of the Transaction Identifier on page 2-2 for more instruction about the TID and the authorization life cycle.
Bit 32	Acquiring Institution Identification (AIN) Code
Definition	This is a unique identification code given to an Acquirer by the Network.
Format	N(13) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Bit 33	Forwarding Institution Identification Code
Definition	The institution identifier assigned to the processing institution that is sending the file.
Format	N(13) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Conditional Echo
Notes	Required Value: 1118000000511
Bit 37	Acquirer Reference Number (ARN)
Definition	A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.
Format	AN(12)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Conditional Echo

Bit 39	Action Code
Definition	The type of action done or the type of action to do for a Transaction.
Format	N(3)
Optionality	Issuer to Network: Mandatory
	Network to Acquirer: Mandatory
Notes	Refer to the Action Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 41	Card Acceptor Terminal Identification
Definition	This code identifies a POS terminal at a Card Acceptor location.
Format	AN(8)
Optionality	Issuer to Network: Mandatory Echo
	Network to Acquirer: Mandatory Echo
Bit 49	Currency Code, Transaction
Definition	This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo
	Network to Acquirer: Mandatory Echo
Notes	Refer to the Currency Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 50	Currency Code, Reconciliation
Definition	This is the numeric code of the currency defined by the Issuer.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo
	Network to Acquirer: Mandatory
Notes	Refer to the Currency Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 56	Original Data Elements
Definition	Data from the Authorization Request (1100 or 1200) or POS/ATM Authorization Reversal Advice (1420) message.
Format	ANS(37) max LLVAR
Optionality	Issuer to Network: Mandatory Echo
	Network to Acquirer: Mandatory Echo
Bit 60	Market Specific Data
Definition	This bit provides additional authorization data.
Format	AN(210) max LLLVAR
Optionality	Issuer to Network: Conditional Echo
	Network to Acquirer: Conditional Echo

Network Management Request (1804)

Description

The Network uses this message to manage the flow of messages and to advise the Network of the availability of the Participant after scheduled or unscheduled outages.

Length of Record

Variable

Message Layout

This is the message layout for the Network Management Request (1804). Refer to the [Programming Notes for Network Management Request \(1804\)](#) on page 5-122 for more instruction.

Network Management Request (1804) Message Layout

Bit Position	Format	Optionality				Field Name
		A to N	I to N	N to A	N to I	
N/A	N(4)	M	M	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	M	M	Bitmap Primary
3	N(6)	M	M	M	M	Processing Code
11	N(6)	M	M	M	M	Systems Trace Audit Number
12	N(12)	M	M	M	M	Date and Time, Local Transaction
24	AN(3)	M	M	M	M	Function Code
33	N(11) max LLVAR	M	M	M	M	Forwarding Institution Identification Code
48	AN(353) max LLLVAR	N/A	C	N/A	N/A	Stand-In Parameter Information
111	AN(24) max LLLLVAR	C	C	N/A	N/A	Encryption Data Request

Programming Notes for [Network Management Request \(1804\)](#)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Acquirer to Network: Mandatory Issuer to Network: Mandatory Network to Acquirer: Mandatory Network to Issuer: Mandatory
Notes	Required Value: 1804

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Definition
Format
Optionality

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
8 bytes, 64 bits for each bitmap Binary (Hex)
Acquirer to Network: Mandatory
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Network to Issuer: Mandatory

Bit 3

Definition
Format
Optionality

Processing Code

This is the service being requested.
N(6)
Acquirer to Network: Mandatory
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Network to Issuer: Mandatory
Required Value: 000000 = Standard Network Management Processing
Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Notes

Bit 11

Definitions
Format
Optionality

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.
N(6)
Acquirer to Network: Mandatory
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Network to Issuer: Mandatory

Bit 12

Definition
Format
Optionality

Date and Time, Local Transaction

This is the local date and time when a Transaction occurred.
N(12) YYMMDDHHMMSS
Acquirer to Network: Mandatory
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Network to Issuer: Mandatory

Bit 24

Definition
Format
Optionality

Function Code

This code shows the reason for a message in its Message Type.
AN(3)
Acquirer to Network: Mandatory
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Network to Issuer: Mandatory
Valid Values:

- 801 = System condition sign on
- 802 = System condition sign off

Notes

American Express Global Network Services
Network Specifications - Authorization
Network Management Request (1804)
Bit 33 Forwarding Institution Identification Code Message Layout

- 811 = System security/key change
- 831 = System audit control/echo test

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 33

Definition
Format
Optionality

Forwarding Institution Identification Code

The institution identifier assigned to the processing institution that is sending the file.

N(11) max LLVAR

Acquirer to Network: Mandatory

Issuer to Network: Mandatory

Network to Acquirer: Mandatory

Network to Issuer: Mandatory

Notes

Valid Values:

Position	Value	Use
1-2	11	Variable Length Indicator
3-8	000000	Zero Filler
9-13		Organization ID

Bit 48

Definition
Format
Optionality

Stand-In Parameter Information

Issuer to Network: Stand-In parameter data if the Issuer signs off and requests Stand-In by the Network. This field is 3 bytes, plus 350 bytes of private use data.

AN(353) max LLLVAR

Acquirer to Network: Not Applicable

Issuer to Network: Conditional

Network to Acquirer: Not Applicable

Network to Issuer: Not Applicable

Notes

Issuer to Network: If Bit 24, Function Code, is equal to 802 (System Condition/Sign-Off), this field contains 7 bytes.

This is the layout.

Position	Value	Use
1-3	004	Variable Length Indicator
4-5	SI	Stand-In data sentinel
6-7	01-95, 99	Host Stand-In Authorization parameter for the Network to apply during Stand-In authorization

Bit 111

Definition
Format
Optionality

Encryption Data Request

The Encryption Data Request includes the data request for various encryption activities.

AN(24) max LLLLVAR

Acquirer to Network: Conditional

Issuer to Network: Conditional

Network to Acquirer: N/A

Network to Issuer: N/A

Notes

This field supports the exchange of dynamic key information. If Acquirers support DKE in Bit 111 of the 1100 or 1200 Messages, they must support Bit 111 in the 1804.

American Express DKE Encryption Data Request Transactional Layout

Data Element	Subfield Length	Subfield Type	Optionality				Description
			A to N	I to N	N to A	N to I	
Variable Length Indicator (VLI)	4 bytes	Numeric (EBCDIC) LLLLVAR	M	M	N/A	N/A	VLI is the total length of variable data in this data field (not including VLI)
Primary ID	2 bytes	Alphanumeric	M	M	N/A	N/A	Primary ID (Card Type Code) is constant literal "AX" (American Express)
Secondary ID	3 bytes	Alphanumeric	M	M	N/A	N/A	Secondary ID (Data Type Code) is constant literal "KXR"
Reserved for Future Use	8 bytes	Numeric	M	M	N/A	N/A	Numeric Reserved fields must contain zeros
Key Type	3 bytes	Alphanumeric	M	M	N/A	N/A	Identifies the type of key being used Valid Value: - PIN (PIN Encryption) - DTA (Data Encryption) - MAC (Message Authentication Code)
Key Usage	2 bytes	Alphanumeric	M	M	N/A	N/A	Provides information about the intended function of the protected key/sensitive data Valid Value: - P0 (P "Zero" - PIN Encryption) - D0 (D "Zero" - Data Encryption) - M6 (AESCMAC)
Algorithm	2 bytes	Alphanumeric	M	M	N/A	N/A	The approved algorithm for which the protected key may be used Valid Value: - T~ (with "~" representing a character space) - A1

Network Management Response (1814)

Description

Use this message to respond to the Network Management Request (1804).

Length of Record

Variable

Message Layout

This is the message layout for the Network Management Response (1814). Refer to the [Programming Notes for Network Management Response \(1814\) on page 5-126](#) for more instruction.

Network Management Response (1814) Message Layout

Bit Position	Format	Optionality				Field Name
		A to N	I to N	N to A	N to I	
N/A	N(4)	M	M	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bit-map Binary (Hex)	M	M	M	M	Bitmap Primary
3	N(6)	ME	ME	ME	ME	Processing Code
11	N(6)	ME	ME	ME	ME	Systems Trace Audit Number
12	N(12)	ME	ME	ME	ME	Date and Time, Local Transaction
24	AN(3)	ME	ME	ME	ME	Function Code
33	N(11) max LLVAR	ME	ME	ME	ME	Forwarding Institution Identification Code
39	N(3)	M	M	M	M	Action Code
111	AN(1055) max LLLLVAR	N/A	N/A	C	C	Encryption Data Response

Programming Notes for Network Management Response (1814)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Acquirer to Network: Mandatory Issuer to Network: Mandatory Network to Acquirer: Mandatory Network to Issuer: Mandatory
Notes	Required Value: 1814

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Definition
Format
Optionality

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
8 bytes, 64 bits for each bitmap Binary (Hex)
Acquirer to Network: Mandatory
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Network to Issuer: Mandatory

Bit 3

Definition
Format
Optionality

Processing Code

This is the service being requested.
N(6)
Acquirer to Network: Mandatory Echo
Issuer to Network: Mandatory Echo
Network to Acquirer: Mandatory Echo
Network to Issuer: Mandatory Echo
Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Notes

Bit 11

Definition
Format
Optionality

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.
N(6)
Acquirer to Network: Mandatory Echo
Issuer to Network: Mandatory Echo
Network to Acquirer: Mandatory Echo
Network to Issuer: Mandatory Echo

Bit 12

Definitions
Format
Optionality

Date and Time, Local Transaction

This is the local date and time when a Transaction occurred.
N(12)
Acquirer to Network: Mandatory Echo
Issuer to Network: Mandatory Echo
Network to Acquirer: Mandatory Echo
Network to Issuer: Mandatory Echo

Bit 24

Definition
Format
Optionality

Function Code

This code shows the reason for a message in its Message Type.
AN(3)
Acquirer to Network: Mandatory Echo
Issuer to Network: Mandatory Echo
Network to Acquirer: Mandatory Echo
Network to Issuer: Mandatory Echo
Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Notes

Bit 33

Definition
Format

Forwarding Institution Identification Code

The institution identifier assigned to the processing institution that is sending the file.
N(11) max LLVAR

Optionality

Acquirer to Network: Mandatory Echo

Issuer to Network: Mandatory Echo

Network to Acquirer: Mandatory Echo

Network to Issuer: Mandatory Echo

Bit 39

Action Code

Definition The type of action done or the type of action to do for a Transaction.

Format N(3)

Optionality

Acquirer to Network: Mandatory

Issuer to Network: Mandatory

Network to Acquirer: Mandatory

Network to Issuer: Mandatory

Notes

Valid Values:

- 800 = Accepted
- 912 = Host not available

Network to Issuer: 899 = Table not found, default used

Refer to the Action Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Bit 111

Encryption Data Response

Definition The Encryption Data Response houses the data response for various encryption activities.

Format AN(1055) max LLLLVAR

Optionality

Acquirer to Network: N/A

Issuer to Network: N/A

Network to Acquirer: Conditional

Network to Issuer: Conditional

Notes This field supports the exchange of dynamic key information and will be sent to certified Acquirers.

American Express DKE Encryption Data Response Transactional Layout

Data Element	Subfield Length	Subfield Type	Optionality				Description
			A to N	I to N	N to A	N to I	
Variable Length Indicator (VLI)	4 bytes	Numeric (EBCDIC) LLLLVAR	N/A	N/A	M	M	VLI is the total length of variable data in this data field (not including VLI)
Primary ID	2 bytes	Alphanumeric	N/A	N/A	M	M	Primary ID (Card Type Code) is constant literal "AX" (American Express)
Secondary ID	3 bytes	Alphanumeric	N/A	N/A	M	M	Secondary ID (Data Type Code) is constant literal "KXS"
Reserved for Future Use	8 bytes	Numeric	N/A	N/A	M	M	Numeric Reserved fields must contain zeros

American Express DKE Encryption Data Response Transactional Layout (continued)

Data Element	Subfield Length	Subfield Type	Optionality				Description
			A to N	I to N	N to A	N to I	
Key Type	3 bytes	Alphanumeric	N/A	N/A	M	M	Identifies the type of key being used Valid Value: - PIN (PIN Encryption) - DTA (Data Encryption) - MAC (Message Authentication Code)
KCV Import Key VLI	2 bytes	Numeric (EBCDIC LLLVAR)	N/A	N/A	M	M	Total length of variable data in the KCV Import Key (not including VLI)
KCV Import Key	5 bytes Max LLVAR	Binary (Hexadecimal Configuration)	N/A	N/A	M	M	Key check value (KCV) for PIN encryption key
Transport Key ID VLI	3 bytes	Numeric (EBCDIC) LLLVAR	N/A	N/A	M	M	Total length of variable data in the Transport Key ID (not including VLI)
Transport Key ID	20 bytes Max LLLVAR	Alphanumeric	N/A	N/A	M	M	Key identifier indicating what transport key is being used
Key Format Indicator	3 bytes	Alphanumeric	N/A	N/A	M	M	Format indicator for the transport key Valid Value: R31
Key Block Data VLI	3 bytes	Numeric (EBCDIC) LLLVAR	N/A	N/A	M	M	The total length of variable data in this the Key Block Data (not including VLI)
Key Block Data	999 bytes	Multiple Data Formats	N/A	N/A	M	M	Contains concatenated Key Block Data composed of the literal data as presented in the TR-31 format. The Data format will be Alphabetic ASCII and Numeric ASCII Refer to the <i>Online PIN Processing Implementation Guide</i> for additional details

Real Time Exception Request (1304)

Description

The Real Time Exception Request (1304) message is used by Issuers for one update done immediately to the status of the Card Account, as opposed to updating data in a batch. (Refer to Batch Exception Update Request (9344) in the *Network Specifications - Financial/Non-Financial* manual, the [Exception File on page 5-153](#), and the [Enhanced Exception File on page 5-158](#)). Multiple Card Account updates must be sent through either the Batch Exception Update Request (9344), the Exception File, or the Enhanced Exception File.

How To Submit

A Real Time Exception Update Request is submitted through the Authorization system and requires an response. This table shows an example of the submission process.

Table 5-7. Real Time Exception Submission Process

Step	Who	Does What
1	Issuer	Submits a Real Time Exception Request (1304) with a valid Function Code to add, update or delete the record, and a Message Reason Code to specify the account status to apply
2	Network	Replies with a Real Time Exception Response (1314) with an Action Code to communicate the results of the update to the Issuer

Certification

Participants who use the Real Time Exception process must also be certified for the Batch Exception Update Request (9344), the Exception File, or the Enhanced Exception File.

Length of Record

Variable

Message Layout

This is the message layout for the Real Time Exception Request (1304). Refer to the [Programming Notes for Real Time Exception Request \(1304\) on page 5-131](#) for more instruction.

Real Time Exception Request (1304) Message Layout

Bit Position	Format	Optionality	Field Name
N/A	N(4)	M	Message Type Identifier

Real Time Exception Request (1304) Message Layout (continued)

Bit Position	Format	Optionality	Field Name
N/A	8 bytes, 64 bits for each bit- map Binary (Hex)	M	Bitmap Primary
2	N(21) max LLVAR	M	Primary Account Number (PAN)
11	N(6)	M	Systems Trace Audit Number
12	N(12)	M	Date and Time, Local Transaction
14	N(4)	M	Record Expiration Date
24	AN(3)	M	Function Code
25	N(4)	M	Message Reason Code
33	N(11) max LLVAR	M	Forwarding Institution Identification Code

Programming Notes for Real Time Exception Request (1304)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Mandatory
Notes	Required Value: 1304 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap Binary (Hex)
Optionality	Mandatory

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(21) max LLVAR
Optionality	Mandatory
Notes	A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit. For example, a Cardmember Account Number of 3759-876543-21001 shows as 15375987654321001

Bit 11

Systems Trace Audit Number

Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.
Format	N(6)
Optionality	Mandatory

Bit 12

Date and Time, Local Transaction

Definition	This is the local date and time when a Transaction occurred.
Format	N(12) YYMMDDHHMMSS
Optionality	Mandatory

American Express Global Network Services
Network Specifications - Authorization
Real Time Exception Request (1304)
Bit 14 Record Expiration Date Message Layout

Bit 14

Definition

Format

Optionality

Notes

Record Expiration Date

The date that the exception record is set to expire on the Network Exception Database. This date often coincides with the date of the Card expiration.

N(4) YYMM or 9999

Mandatory

This date can not be later than seventy-two months from the date the message is received by the Network.

Valid Value: 9999 - The exception record will not expire from the Network Exception Database if this value is used. An Issuer must send a delete record with a Function Code of 303 = Delete in Bit 24 to remove this from the Network Exception Database.

Bit 24

Definition

Format

Optionality

Notes

Function Code

This code shows the reason for a message in its Message Type.

AN(3)

Mandatory

Valid Values:

- 301 = Add/update record
- 303 = Delete record

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 25

Definition

Format

Optionality

Notes

Message Reason Code

This code gives the cause of a First and Final Chargeback, Retrieval Request Notification, File Acknowledgment, ATM and POS, and Reversal requests.

N(4)

Mandatory

The value of the status that is added or removed from a Cardmember's Account on the Network.

This code shows if a sent file was accepted or rejected by the Network for a Network File Acknowledgment (1844) and a Participant File Acknowledgment (1824).

Refer to the Message Reason Code section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 33

Definitions

Format

Optionality

Notes

Forwarding Institution Identification Code

The institution identifier assigned to the processing institution that is sending the file.

N(11) max LLVAR

Mandatory

Valid Values:

Position	Value	Use
1-2	11	Variable Length Indicator
3-8	000000	Zero Filler
9-13		Organization ID

Real Time Exception Response (1314)

Description

The Real Time Exception Response (1314) message is an online message used by the Network to respond to the 1304 Real Time Exception Request to add, update, or delete Cardmember Account status data. The response message includes an Action Code to show the Network's disposition for the request.

Understanding Exception Response Action Codes

Issuers should react to the Action Code included in the Network's Real Time Exception Update Response (1314) or Batch Exception Update Response Count and Detail (9344) as shown in [Table 5-8](#).

Note: Some of these Action Codes are not used in the Batch Exception Update Response Count and Detail (9344).

Table 5-8. Action Codes for Network Batch Exception Responses

Action Code	Meaning	Present in Batch 9344 Response	Issuer Action
111	Invalid Account	Yes	Review the account number and make sure that it was transmitted correctly.
115	Requested Function Not Supported	Yes	Issuer is not certified to send Real Time Exception Requests.
300	Successful	No	No action required. Issuers can keep a log of the requests successfully sent to prevent sending duplicate or lower value exception records.
301	Not Supported by Receiver	Yes	The Message Reason Code is not valid for the message type.
302	Unable to Locate Record on File	No	For a delete request, this shows that the Network does not have a record on file for the account number. Issuers should review the account number and/or Message Reason Code to make sure that it was transmitted correctly.
303	Duplicate Record, Old Record Replaced	No	No action required. Issuers can keep a log of the requests successfully sent to prevent sending duplicate or lower value exception records.
304	Field Edit Error	Yes	An error in the message is not covered by a specific Action Code. Issuers should review the message data to make sure that it was constructed correctly and that the Record Expiration Date is not more than six years from the date sent. This code can also be used if a mandatory field is not present in the message.
380	File Update Not Accepted, Higher Priority Item on File	No	A more important record is in the Network Exception Database. The Issuer can delete the more important record and send a new update, or keep the more important level code.

Table 5-8. Action Codes for Network Batch Exception Responses (continued)

Action Code	Meaning	Present in Batch 9344 Response	Issuer Action
912	Host Unavailable	Yes	The Network is not online to accept the request. The Issuer can try to send the request again.

Certification

Participants must certify for the Batch Exception Request (9344) or the Exception File to use the Real Time Exception process.

Length of Record

Variable

Message Layout

This is the message layout for the Real Time Exception Response (1314). Refer to the [Programming Notes for Real Time Exception Response \(1314\) on page 5-134](#) for more instruction.

Real Time Exception Response (1314) Message Layout

Bit Position	Format	Optionality	Field Name
N/A	N(4)	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	Bitmap Primary
2	N(21) max LLVAR	ME	Primary Account Number (PAN)
11	N(6)	ME	Systems Trace Audit Number
12	N(12)	ME	Date and Time, Local Transaction
14	N(4)	ME	Record Expiration Date
24	AN(3)	ME	Function Code
25	N(4)	ME	Message Reason Code
33	N(11) max LLVAR	ME	Forwarding Institution Identification Code
39	N(3)	M	Action Code

Programming Notes for Real Time Exception Response (1314)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Mandatory
Notes	Required Value: 1314

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Definition
Format
Optionality

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
8 bytes, 64 bits for each bitmap Binary (Hex)
Mandatory

Bit 2

Definition
Format
Optionality

Primary Account Number (PAN)

A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
N(21) max LLVAR
Mandatory Echo

Bit 11

Definition
Format
Optionality

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.
N(6)
Mandatory Echo

Bit 12

Definition
Format
Optionality

Date and Time, Local Transaction

This is the local date and time when a Transaction occurred.
N(12)
Mandatory Echo

Bit 14

Definition
Format
Optionality

Record Expiration Date

The date that the exception record is set to expire on the Network Exception Database. This date often coincides with the date of the Card expiration.
N(4)
Mandatory Echo

Bit 24

Definition
Format
Optionality
Notes

Function Code

This code shows the reason for a message in its Message Type.
AN(3)
Mandatory Echo
Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 25

Definition
Format
Optionality
Notes

Message Reason Code

This code gives the cause of a First and Final Chargeback, Retrieval Request Notification, File Acknowledgment, ATM and POS, and Reversal requests.
N(4)
Mandatory Echo
This code shows if a sent file was accepted or rejected by the Network for a Network File Acknowledgment (1844) and a Participant File Acknowledgment (1824).
Refer to the Message Reason Code section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 33

Definition
Format
Optionality

Forwarding Institution Identification Code

The institution identifier assigned to the processing institution that is sending the file.
N(11) max LLVAR
Mandatory Echo

Bit 39	Action Code
Definition	The type of action done or the type of action to do for a Transaction.
Format	N(3)
Optionality	Mandatory
Notes	Refer to the Action Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.

POS Authorization Adjustment Financial Advice (1220)

Description

Acquirers send the POS Authorization Adjustment Financial Advice (1220) message when the last Transaction amount is less than the first authorized amount. Issuers can then release funds that are greater than the last Transaction amount.

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Adjustment Financial Advice (1220). Refer to the [Programming Notes for POS Authorization Adjustment Financial Advice \(1220\) on page 5-138](#) for more instruction.

POS Authorization Adjustment Financial Advice (1220) Message Layout

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(21) max LLVAR	M	M	Primary Account Number (PAN)
3	N(6)	M	M	Processing Code
4	N(12)	M	M	Amount, Transaction
5	N(12)	N/A	M	Amount, Reconciliation
11	N(6)	M	M	Systems Trace Audit Number
12	N(12)	M	M	Date and Time, Local Transaction
19	N(3)	M	M	Country Code, Acquiring Institution
22	AN(12)	M	M	Point of Service Data Code
23	N(3)	N/A	C	PAN Sequence Number
24	AN(3)	M	M	Function Code
26	N(4)	M	M	Card Acceptor Business Code
30	N(24)	M	M	Amount, Original
31	N(40) max LLVAR	M	C	Acquirer Reference Data, Transaction Identifier (TID)

POS Authorization Adjustment Financial Advice (1220) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		A to N	N to I	
32	N(13) max LLVAR	M	M	Acquiring Institution Identification (AIN) Code
33	N(3) min N(13) max LLVAR	O	O	Forwarding Institution Identification Code
34	AN(30) max LLVAR (including length indicator)	N/A	C	Primary Account Number, Extended
37	AN(12)	O	O	Acquirer Reference Number
41	ANS(8)	O	O	Card Acceptor Terminal Identification
42	AN(15)	M	M	Card Acceptor Identification Code
43	ANS(101) max LLVAR	C	C	Card Acceptor Name and Location
49	N(3)	M	M	Currency Code, Transaction
50	N(3)	N/A	M	Currency Code, Reconciliation
56	3 bytes min 37 bytes max LLLVAR	M	M	Original Data Elements
	N(2) EBCDIC	M	M	Subfield LL
	N(4)	M	M	Subfield 1 Message Type Identifier
	N(6)	M	M	Subfield 2 System Trace Audit Number
	N(12)	M	M	Subfield 3 Date and Time, Local Transaction
	N(13) max LLVAR	M	M	Subfield 4 Acquiring Institution Identification Code

Programming Notes for POS Authorization Adjustment Financial Advice (1220)

Bit N/A

Message Type Identifier

Definition

This code identifies the type of message sent by the forwarding institution.

Format

N(4)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

Required Value: 1220

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

Format

8 bytes, 64 bits for each bitmap Binary (Hex)

American Express Global Network Services
Network Specifications - Authorization
 POS Authorization Adjustment Financial Advice (1220)
 Bit 2 Primary Account Number (PAN) Message Layout

Optionality **Acquirer to Network:** Mandatory
 Network to Issuer: Mandatory

Bit 2 **Primary Account Number (PAN)**

Definition A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.

Format N(21) max LLVAR

Optionality **Acquirer to Network:** Mandatory

Notes **Network to Issuer:** Mandatory

A two-byte, variable length indicator is before the PAN. This indicator is not included in the length of the bit.

For example, a Cardmember Account Number of 3759-876543-21001 shows as

15375987654321001

Length of variable data - 19 bytes maximum, EBCDIC

Bit 3 **Processing Code**

Definition This is the service being requested.

Format N(6)

Optionality **Acquirer to Network:** Mandatory

Notes **Network to Issuer:** Mandatory

Required Value: 220000 = Authorization Adjustment

This bit is mandatory for processing this message. It will be echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.

Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 4 **Amount, Transaction**

Definition The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.

Format N(12)

Optionality **Acquirer to Network:** Mandatory

Notes **Network to Issuer:** Mandatory

The currency and decimal point are determined by the entry in Field 49 - Currency Code, Transaction.

For canceled (approved, but abandoned) Transactions, this field is zero-filled.

This bit is mandatory for processing this message. It will be echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.

Bit 5 **Amount, Reconciliation**

Definition This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.

Format N(12)

Optionality **Acquirer to Network:** Not Applicable

Notes **Network to Issuer:** Mandatory

The decimal point is determined by the Currency Code, Reconciliation (Bit 50). For example, \$100.00 in USD shows as 000000010000.

Authorization and Presentment can occur on different days. The reconciliation amounts can be different in the Authorization and Presentment messages.

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Network Specifications - Authorization
POS Authorization Adjustment Financial Advice (1220)
Bit 11 Systems Trace Audit Number Message Layout

Bit 11	Systems Trace Audit Number
Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.
Format	N(6)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Acquirer to Network: The Network returns this number in the Network to Acquirer piece of the POS Authorization Adjustment Financial Advice Response (1230) message. Network to Issuer: This value is assigned by the Network. It is different from the value of the Acquirer.
Bit 12	Date and Time, Local Transaction
Definition	This is the local date and time when a Transaction occurred.
Format	N(12) YYMMDDhhmmss
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	This bit is mandatory for processing this message. It will be echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.
Bit 19	Country Code, Acquiring Institution
Definition	This is the Country Code of the acquiring institution.
Format	N(3)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	This bit contains the Acquiring Institution Country Code that was sent in the first POS Authorization Request (1100) message. Refer to the Country Codes section of the <i>Codes Reference Guide</i> for a list of codes and descriptions.
Bit 22	Point of Service Data Code
Definition	This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).
Format	AN(12)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	This bit contains the Point of Service Data Code that was sent in the first POS Authorization Request (1100) message. Refer to the Point of Service Data Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.
Bit 23	PAN Sequence Number
Definition	A number distinguishing between separate Cards with the same Primary Account Number.
Format	N(3)
Optionality	Acquirer to Network: Not Applicable Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.
Bit 24	Function Code
Definition	This code shows the reason for a message in its Message Type.
Format	AN(3)
Optionality	Acquirer to Network: Mandatory Network to Issuer: Mandatory
Notes	Required Value: 202 - Authorization Adjustment

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POS Authorization Adjustment Financial Advice (1220)
Bit 26 Card Acceptor Business Code Message Layout

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 26

Definition

Format

Optionality

Notes

Card Acceptor Business Code

Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).

N(4)

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

This bit contains the Card Acceptor Business Code (Merchant Category Code (MCC)) sent in the first POS Authorization Request (1100) message.

Refer to the Merchant Category Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 30

Definition

Format

Optionality

Amount, Original

The Authorized Transaction Amount from the previous POS Authorization Response (1110) message.

N(24)

Acquirer to Network: Mandatory

- Positions 1-12 of this bit is the first Transaction Amount from Bit 4 = Amount, Transaction in the POS Authorization Response (1110) message.
- Positions 13-24 are zero-filled and reserved for future use.

Network to Issuer: Mandatory

- Positions 1-12 of this bit is the first Transaction amount in the currency defined by the Issuer from Bit 50 - Currency Code, Reconciliation in the POS Authorization Response (1110) message.
- Positions 13-24 are zero-filled and reserved for future use.
- Positions 1-24 will be zero-filled and passed to the Issuer if the original amount is not available.

Bit 31

Definition

Format

Optionality

Notes

Acquirer Reference Data, Transaction Identifier (TID)

This unique number is given by the Network and follows a Transaction during its life cycle.

N(40) max LLVAR

Acquirer to Network: Mandatory

Network to Issuer: Conditional. The Network sends the bit to the Issuer if the Acquirer and Issuer are certified to receive it.

Acquirer to Network: This bit must contain the TID received in the POS Authorization Response (1110) message.

Refer to [The Role of the Transaction Identifier on page 2-2](#) for more instruction about the TID and the authorization life cycle.

Bit 32

Definition

Format

Optionality

Notes

Acquiring Institution Identification (AIN) Code

This is a unique identification code given to an Acquirer by the Network.

N(13) max LLVAR

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Acquirer to Network: The variable length indicator is not included in the length of the bit. For example, Acquiring Institution Identification Code 45678912345 appears as 1145678912345.

Network to Issuer:

Required Value: 06370100 American Express

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

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POS Authorization Adjustment Financial Advice (1220)
Bit 33 Forwarding Institution Identification Code Message Layout

Bit 33

Forwarding Institution Identification Code

Definition The institution identifier assigned to the processing institution that is sending the file.

Format N(3) min, N(13) max LLVAR

Optionality **Acquirer to Network:** Optional

Network to Issuer: Optional

Bit 34

Primary Account Number, Extended

Definitions This field is a multipurpose field which provides the Primary Account Number or associated data.

Format AN(30) Max LLVAR (with length indicator)

Optionality **Acquirer to Network:** Not Applicable

Network to Issuer: Conditional. If this is a payment token Transaction, this bit will be sent to the certified Issuer.

Notes This bit is populated by the Network with the Token, Wallet Provider ID, and Token Expiration Date.

Subfield	Format
1	Variable Length Indicator (VLI)
2	Token
3	Wallet Provider ID (ID)
4	Token Expiration Date

Data Example:

Subfield 1 is the VLI and must be 2 bytes. The token immediately follows and is a variable length between 15 and 19 bytes. After the token is a back slash ("\") delimiter and then a 2 byte ID. After the 2 byte ID is a back slash ("\") delimiter and then the token expiration date in the format YYMM.

The symbol "~" is used to represent a space.

Example: 23371234567890123\AB\1401

Bit 37

Acquirer Reference Number

Definition A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

Format AN(12)

Optionality **Acquirer to Network:** Optional

Network to Issuer: Optional

Bit 41

Card Acceptor Terminal Identification

Definition This code identifies a POS terminal at a Card Acceptor location.

Format ANS(8)

Optionality **Acquirer to Network:** Optional

Network to Issuer: Optional

Notes **Acquirer to Network:** The Network returns this number in the Network to Acquirer piece of the POS Authorization Adjustment Financial Advice Response (1230) message.

Network to Issuer: This value is assigned by the Network. It is different from the value of the Acquirer.

Bit 42

Card Acceptor Identification Code

Definition The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.

Format AN(15)

Optionality **Acquirer to Network:** Mandatory

Network to Issuer: Mandatory

Notes This field contains the Card Acceptor Identification Code (Merchant ID) sent in the first POS Authorization Request (1100) message.

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 POS Authorization Adjustment Financial Advice (1220)
 Bit 43 Card Acceptor Name and Location Message Layout

This bit is mandatory for processing this message. It will be echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.

Bit 43

Card Acceptor Name and Location

Definition

The name of the Card Acceptor as it is known to Cardmembers.

Format

ANS(101) max LLVAR

Optionality

Acquirer to Network: Conditional

Network to Issuer: Conditional

Notes

This bit is free form and Issuers must accept the data as presented by the Acquirers.

Bit 49

Currency Code, Transaction

Definition

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

This bit contains the Transaction Currency Code sent in the first POS Authorization Request (1100) message.

This bit is mandatory for processing this message. It will be echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Currency Code, Reconciliation

Definition

This is the numeric code of the currency defined by the Issuer.

Format

N(3)

Optionality

Acquirer to Network: Not Applicable

Network to Issuer: Mandatory

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 56

Original Data Elements

Definition

Data from the Authorization Request (1100 or 1200) or POS/ATM Authorization Reversal Advice (1420) message.

Format

3 bytes min, 37 bytes max LLLVAR

Optionality

Acquirer to Network: Mandatory

Network to Issuer: Mandatory

Notes

This field contains four data elements from the first POS Authorization Request (1100) message.

Subfield	Format	Optionality		Notes
		A to N	N to I	
LL	N(2) EBCDIC	M	M	
1 Message Type Identifier	N(4)	M	M	
2 Systems Trace Audit Number	N(6)	M	M	
3 Date and Time, Local Transaction	N(12)	M	M	
4 Acquiring Institution Identification Code	N(13) max LLVAR	M	M	

POS Authorization Adjustment Financial Advice Response (1230)

Description

Issuers send the POS Authorization Adjustment Financial Advice Response (1230) message to Acquirers to acknowledge the Issuer's receipt of the POS Authorization Adjustment Financial Advice (1220).

Length of Record

Variable

Message Layout

This is the message layout for the POS Authorization Adjustment Financial Advice (1230) Response. Refer to the [Programming Notes for POS Authorization Adjustment Financial Advice Response \(1230\)](#) on page 5-145 for more instruction.

POS Authorization Adjustment Financial Advice Response (1230) Message Layout

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
N/A	N(4)	M	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	M	Bitmap Primary
2	N(3) min N(21) max LLVAR	ME	ME	Primary Account Number (PAN)
3	N(6)	ME	ME	Processing Code
4	N(12)	ME	ME	Amount, Transaction
5	N(12)	ME	N/A	Amount, Reconciliation
11	ANS(6)	ME	ME	Systems Trace Audit Number
12	N(12)	ME	ME	Date and Time, Local Transaction
30	N(24)	ME	ME	Amount, Original
31	N(40) max LLVAR	CE	ME	Acquirer Reference Data, Transaction Identifier (TID)
32	N(3) min N(13) max LLVAR	CE	CE	Acquiring Institution Identification (AIN) Code
37	ANS(12)	CE	CE	Acquirer Reference Number
39	N(3)	M	M	Action Code
41	ANS(8)	CE	CE	Card Acceptor Terminal Identification
42	ANS(15)	ME	ME	Card Acceptor Identification Code
49	N(3)	ME	ME	Currency Code, Transaction

POS Authorization Adjustment Financial Advice Response (1230) Message Layout (continued)

Bit Position	Format	Optionality		Field Name
		I to N	N to A	
50	N(3)	ME	N/A	Currency Code, Reconciliation
54	ANS(4) min ANS(123) max LLVAR)	O Prepaid Card	O Prepaid Card	Amount, Additional

Programming Notes for POS Authorization Adjustment Financial Advice Response (1230)

Bit N/A

Message Type Identifier

Definition	This code identifies the type of message sent by the forwarding institution.
Format	N(4)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	Required Value: 1230 Refer to the Message Type Identifier section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition	Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
Format	8 bytes, 64 bits for each bitmap Binary (Hex)
Optionality	Issuer to Network: Mandatory Network to Acquirer: Mandatory
Notes	If the field is Mandatory Echo or Conditional Echo (and the S/E uses it) then the bitmap position must contain a value of "1" to show that the field is present. If the field is Conditional Echo (and the S/E does not use it), then the bitmap position must contain the value of "0" to show that the field is absent.

Bit 2

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	N(3) min, N(21) max LLVAR
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo

Bit 3

Processing Code

Definition	This is the service being requested.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo Network to Acquirer: Mandatory Echo
Notes	Refer to the Processing Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Bit 4

Amount, Transaction

Definition	The POS sale or credit amount of the Transaction in the currency used at the Card Acceptor location.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo

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 POS Authorization Adjustment Financial Advice Response (1230)
 Bit 5 Amount, Reconciliation Message Layout

		Network to Acquirer: Mandatory Echo
Bit 5	Amount, Reconciliation	
Definition	This is the total amount of the Transaction after the Network changes the Amount, Transaction (Bit 4) to the currency defined by the Issuer (Bit 50). The Network sends this converted amount to the Issuer.	
Format	N(12)	
Optionality	Issuer to Network: Mandatory Echo	
	Network to Acquirer: Not Applicable	
Bit 11	Systems Trace Audit Number	
Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.	
Format	ANS(6)	
Optionality	Issuer to Network: Mandatory Echo	
	Network to Acquirer: Mandatory Echo	
Bit 12	Date and Time, Local Transaction	
Definition	This is the local date and time when a Transaction occurred.	
Format	N(12) YYMMDDhhmmss	
Optionality	Issuer to Network: Mandatory Echo	
	Network to Acquirer: Mandatory Echo	
Bit 30	Amount, Original	
Definition	The Authorized Transaction Amount from the previous POS Authorization Response (1110) message.	
Format	N(24)	
Optionality	Issuer to Network: Mandatory Echo	
	Network to Acquirer: Mandatory Echo	
Bit 31	Acquirer Reference Data, Transaction Identifier (TID)	
Definitions	This unique number is given by the Network and follows a Transaction during its life cycle.	
Format	N(40) max LLVAR	
Optionality	Issuer to Network: Conditional Echo	
	Network to Acquirer: Mandatory Echo	
Notes	Refer to The Role of the Transaction Identifier on page 2-2 for more instruction about the TID and the authorization life cycle.	
Bit 32	Acquiring Institution Identification (AIN) Code	
Definition	This is a unique identification code given to an Acquirer by the Network.	
Format	N(3) min, N(13) max LLVAR	
Optionality	Issuer to Network: Conditional Echo	
	Network to Acquirer: Conditional Echo	
Bit 37	Acquirer Reference Number	
Definition	A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.	
Format	ANS(12)	
Optionality	Issuer to Network: Conditional Echo. If this bit is in the POS Authorization Adjustment Financial Advice (1220) message, it is echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.	
	Network to Acquirer: Conditional Echo. If this bit is in the POS Authorization Adjustment Financial Advice (1220) message, it is echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.	

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POS Authorization Adjustment Financial Advice Response (1230)
Bit 39 Action Code Message Layout

Bit 39

Definition
Format
Optionality

Action Code

The type of action done or the type of action to do for a Transaction.
N(3)
Issuer to Network: Mandatory
Network to Acquirer: Mandatory
Refer to the Action Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Bit 41

Definition
Format
Optionality

Card Acceptor Terminal Identification

This code identifies a POS terminal at a Card Acceptor location.
ANS(8)
Issuer to Network: Conditional Echo. If this bit is in the POS Authorization Adjustment Financial Advice (1220) message, it is echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.
Network to Acquirer: Conditional Echo. If this bit is in the POS Authorization Adjustment Financial Advice (1220) message, it is echoed in the POS Authorization Adjustment Financial Advice Response (1230) message.

Bit 42

Definition
Format
Optionality

Card Acceptor Identification Code

The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.
ANS(15)
Issuer to Network: Mandatory Echo
Network to Acquirer: Mandatory Echo

Bit 49

Definition
Format
Optionality

Currency Code, Transaction

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
N(3)
Issuer to Network: Mandatory Echo
Network to Acquirer: Mandatory Echo
Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition
Format
Optionality

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.
N(3)
Issuer to Network: Mandatory Echo
Network to Acquirer: Not Applicable
Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 54

Definition
Format
Optionality

Amount, Additional

The available amount of funds in the Card currency remaining on a Card.
ANS(4) min, ANS(123) max LLLVAR
Issuer to Network: Conditional

- Conditional:** If supported, may be sent on Prepaid Card products
- Not Allowed:** Debit Card products

Network to Acquirer: Conditional

- If a certified Issuer sends this bit, and the Acquirer is certified to receive it, then the Network sends it to the Acquirer.
- Bit 54 will be dropped by AEGNS and not sent to the Acquirer if available balance is returned from the Issuer on Debit Card products.

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POS Authorization Adjustment Financial Advice Response (1230)
Bit 54 Amount, Additional Message Layout

Notes

S/Es supporting balance return may show the Card currency and available balance on the POS terminal and/or print it on the Cardmember receipt.

Card Transaction Layout

Variable Length Indicator: 3 bytes EBCDIC, right justified, zero-filled.

Variable data Length: 120 bytes maximum, EBCDIC

Bit Format: This bit has a 3-byte Variable Length Indicator (VLI) and 20 bytes of coded data that specifies the Account Type, Amount Type, Currency Code, Credit status, and the available amount of funds remaining on the Card.

The format is:

	Length	Position	Description
VVV	3 bytes	1-3	VLI (always 020)
AA	2 bytes	4-5	Account Type Code (always 00)
BB	2 bytes	6-7	Amount Type Code (always 05)
CCC	3 bytes	8-10	Numeric Currency Code Refer to the <i>Codes Reference Guide</i> for Currency Codes.
D	1 byte	11	Credit Code (C = Credit)
123...	12 bytes	12-23	12-digit, Available amount of funds right justified, zero-filled, with corresponding decimal implied (e.g., 840/USD = 2 decimal positions).

Preferred/Delinquent File

Description

Use the Preferred/Delinquent File message to update the Network with delinquency and preferred Cardmember status. The Preferred/Delinquent File is a 51-byte fixed block record length format with a block size of 15453 bytes. If a record is in the file in one transmission, and not the next, the Cardmember status returns to normal. There are no header and trailer records in this file. If the Issuer does not send a new file, the file from the day before is used to update the Network Preferred/Delinquent File. Refer to the [Network Preferred/Delinquent File on page 2-17](#) for more instruction.

Length of Record

51 bytes

Message Layout

This is the message layout for the Preferred/Delinquent File. Refer to the [Programming Notes for Preferred/Delinquent File on page 5-150](#) for more instruction.

Preferred/Delinquent File Message Layout

Field Number	Format	Positions	Optionality	Field Name
1	AN(1)	1	F	Filler
2	AN(2)	2-3	M	User Number
3	N(10)	4-13	M	Cardmember Account Number
4	AN(1)	14	F	Filler
5	AN(1)	15	F	Filler
6	N(1)	16	M	Repl Code
7	5 bytes S9(09) comp-3	17-21	M	Total Due
8	5 bytes S9(09) comp-3	22-26	M	30 Day Balance
9	5 bytes S9(09) comp-3	27-31	M	60 Day Balance
10	5 bytes S9(09) comp-3	32-36	M	90 Day Balance

Preferred/Delinquent File Message Layout (continued)

Field Number	Format	Positions	Optionality	Field Name
11	5 bytes S9(09) comp-3	37-41	M	New Charges
12	AN(1)	42	O	Credit Code Preferred
13	AN(1)	43	M	Member Type
14	AN(1)	44	F	Filler
15	AN(2)	45-46	F	Filler
16	AN(1)	47	F	Filler
17	4 bytes S9(09) comp-3	48-51	F	Filler

Programming Notes for Preferred/Delinquent File

Field 1

Definition

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

Format

AN(1)

Position

1

Optionality

Filler

Notes

Alphanumeric Filler fields must contain spaces.

Field 2

Definition

User Number

The approved person providing data to the Network.

Format

AN(2)

Position

2-3

Optionality

Mandatory

Notes

Required Value: 01

Field 3

Definition

Cardmember Account Number

Digits 3 through 11 from the original Cardmember (CM) Account Number and a calculated check digit.

Format

N(10)

Position

4-13

Optionality

Mandatory

Field 4

Definition

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

Format

AN(1)

Position

14

Optionality

Filler

Notes

Alphanumeric Filler fields must contain spaces.

Field 5

Definition

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

Format

AN(1)

American Express Global Network Services
Network Specifications - Authorization
Preferred/Delinquent File
Field 6 Repl Code Message Layout

Position	15
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.
Field 6	Repl Code
Definition	Digit 12 of the original Cardmember Account Number. This value represents the number of replacement Cards issued (e.g., a "4" in the twelfth position indicates four Cards have been issued to the Cardmember).
Format	N(1)
Position	16
Optionality	Mandatory
Field 7	Total Due
Definition	The total amount owed by the Cardmember including: (30 Day Balance) + (60 Day Balance) + (90 Day Balance) + (unbilled new Charges) + (Transactions billed, but not delinquent). Default is zero.
Format	5 bytes S9(09) comp-3
Position	17-21
Optionality	Mandatory
Field 8	30 Day Balance
Definition	Transactions billed to the Cardmember that are 30 days past due.
Format	5 bytes S9(09) comp-3
Position	22-26
Optionality	Mandatory
Field 9	60 Day Balance
Definition	Transactions billed to the Cardmember that are 60-89 days past due.
Format	5 bytes S9(09) comp-3
Position	27-31
Optionality	Mandatory
Field 10	90 Day Balance
Definition	Transactions billed to the Cardmember that are 90 or more days past due.
Format	5 bytes S9(09) comp-3
Position	32-36
Optionality	Mandatory
Field 11	New Charges
Definition	Transactions presented to the Issuer but not yet billed to the Cardmember.
Format	5 bytes S9(09) comp-3
Position	37-41
Optionality	Mandatory
Bit 12	Credit Code Preferred
Definition	A code that represents a set of changes to the CRIT Table's Cardmember charging thresholds. Each code maps to a unique set of indexes.
Format	AN(1)
Position	42
Optionality	Optional
Notes	Refer to the Credit Codes section of the <i>Codes Reference Guide</i> for a complete list of codes and descriptions.

Field 13

Definition The type of member.
Format AN(1)
Position 43
Optionality Mandatory
Notes **Required Value:** 1

Member Type

Field 14

Definition Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format AN(1)
Position 44
Optionality Filler
Notes Alphanumeric Filler fields must contain spaces.

Filler

Field 15

Definition Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format AN(2)
Position 45-46
Optionality Filler
Notes Alphanumeric Filler fields must contain spaces.

Filler

Field 16

Definition Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format AN(1)
Position 47
Optionality Filler
Notes Alphanumeric Filler fields must contain spaces.

Filler

Field 17

Definition Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format 4 bytes S9(09) comp-3
Position 48-51
Optionality Filler
Notes **Required Value:** Initialize to packed zeros, right justified

Filler

Exception File

Description

Use the Exception File to send exception updates from the Issuer to the Network. The Network uses this data to update the Network Exception Database, which stores data on Card accounts that should receive special handling.

The Exception File is a 60-byte fixed block record length format without a file header or trailer. The block size of the file should be 27960 bytes. This file should be sent when there are updates to report. The records in the Exception File are divided into fields numbered in the order they appear in the layout in this section. Include all fields in the Exception File.

The use of the Exception File is limited to these:

- Participants/Authorized Processors currently using the Exception File
- Insourced Issuers who use GNSAS to decision their Transactions
- Authorized Processors certified for the Exception File and who add new Issuers

Note: Issuers participating in the Recurring Billing Cancellation Service must use the Enhanced Exception File. Refer to the [Enhanced Exception File on page 5-158](#).

Length of Record

60 bytes

Message Layout

This is the message layout for the Exception File. Refer to the [Programming Notes for Exception File on page 5-154](#) for more instruction.

Exception File Message Layout

Field Number	Format	Positions	Optionality	Field Name
1	AN(15)	1-15	M	Primary Account Number (PAN)
2	AN(1)	16	F	Filler
3	AN(1)	17	M	Transaction Code
4	AN(1)	18	F	Filler
5	AN(2)	19-20	F	Filler
6	AN(1)	21	M	Source Identification Operating Center
7	AN(1)	22	F	Filler
8	AN(1)	23	F	Filler
9	N(2)	24-25	M	Exception Code
10	N(2)	26-27	M	Expiry Month
11	N(1)	28	M	Expiry Year

Exception File Message Layout (continued)

Field Number	Format	Positions	Optionality	Field Name
12	AN(5)	29-33	F	Filler
13	AN(1)	34	F	Filler
14	AN(2)	35-36	F	Filler
15	AN(1)	37	F	Filler
16	AN(1)	38	F	Reserved Field
17	AN(3)	39-41	F	Filler
18	AN(1)	42	C	Card Type Indicator
19	AN(18)	43-60	F	Filler

Programming Notes for Exception File

Field 1

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	AN(15)
Position	1-15
Optionality	Mandatory
Notes	Issuers must use a scramble routine for the Primary Account Number in transmissions to the Network. Refer to Scrambling Primary Account Numbers in the <i>Network Specifications - Instructions</i> manual.

Field 2

Filler

Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(1)
Position	16
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

Field 3

Transaction Code

Definition	This code requests that the Network perform an action.
Format	AN(1)
Position	17
Optionality	Mandatory
Notes	Valid Values: - A = Add record to file - D = Delete record from file - C = Change record on file

Field 4

Filler

Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(1)
Position	18
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

Field 5

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(2)

19-20

Filler

Alphanumeric Filler fields must contain spaces.

Field 6

Definition

Format

Position

Optionality

Notes

Source Identification Operating Center

The operating center servicing the Card.

AN(1)

21

Mandatory

Valid Values:

- E = European Participants
- W = All Other Participants

Field 7

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

22

Filler

Alphanumeric Filler fields must contain spaces.

Field 8

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

23

Filler

Alphanumeric Filler fields must contain spaces.

Field 9

Definition

Format

Position

Optionality

Notes

Exception Code

The disposition for the Card Account.

When the Network receives a Real Time Exception Update message, it applies the change only if the code is of the same or higher importance.

N(2)

24-25

Mandatory

Refer to [Table 2-12. Exception and Message Reason Code Priority on page 2-14](#) for exception status and priority value.

Refer to the Exception Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Field 10

Definition

Format

Position

Optionality

Notes

Expiry Month

A value from 01 to 12 that indicates the month the Exception File record will expire.

N(2) MM

26-27

Mandatory

If the Expiry Year is the same year the exception is reported, the Expiry Month must be greater than the month the exception is reported.

Valid Values: 01-12, 99

If this field = 99 then Field 11 (Expiry Year) must equal 9. The exception record will not expire from the Network Exception Database if this value is used. An Issuer must send a delete record with a Transaction Code of D in Field 3 to remove this from the Network Exception Database.

Field 11

Definition
Format
Position
Optionality
Notes

Expiry Year

The last digit of the year that the negative should expire; for example, 5=2005.

N(1) Y

28

Mandatory

The value cannot be a year that is more than six years from the current month and date. The value is the last digit of the Expiry Year.

Example: For July 2007, only the values 7 (with Expiry Month from 08 to 12 permitted), 8, 9, 10, 11, 12, and 13 are valid. In this example, 7 is valid only if the Expiry Month is greater than the current month.

Valid Values: 0-9

If this field = 9 and Field 10 (Expiry Month) = 99, the exception record will not expire from the Network Exception Database. An Issuer must send a delete record with a Transaction Code of D in Field 3 to remove this from the Network Exception Database.

Field 12

Definition

Format
Position
Optionality
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(5)

29-33

Filler

Alphanumeric Filler fields must contain spaces.

Field 13

Definition

Format
Position
Optionality
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

34

Filler

Alphanumeric Filler fields must contain spaces.

Field 14

Definition

Format
Position
Optionality
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(2)

35-36

Filler

Alphanumeric Filler fields must contain spaces.

Field 15

Definition

Format
Position
Optionality
Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

37

Filler

Alphanumeric Filler fields must contain spaces.

Field 16

Definition
Format
Position

Reserved Field

This field is for AEGNS internal use.

AN(1)

38

Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.
Field 17	Filler
Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(3)
Position	39-41
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.
Field 18	Card Type Indicator
Definition	When populated with the value of "C" by an Issuer, the Card Type Indicator identifies the account as an Expresspay-enabled Card.
Format	AN(1)
Position	42
Optionality	Conditional
Notes	Valid Values: C = Expresspay-enabled Card Expresspay Issuers must identify their Expresspay-enabled Card/device accounts. Expresspay: Expresspay Issuers must show a Card Type Indicator on the Exception File, the Batch Exception Update Request (9344), or the Enhanced Exception File messages to identify Expresspay-enabled Card/device accounts.
Field 19	Filler
Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(18)
Position	43-60
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

Enhanced Exception File

Description

Use the Enhanced Exception File to send exception data from the Issuer to the Network. The Network uses this data to update the Network Exception Database, which contains data about Card accounts that get special handling.

Note: The Enhanced Exception File contains fields that are specific to the Recurring Billing Cancellation Service.

Certification

Issuers who use the Exception File must certify for the Enhanced Exception File to use the Recurring Billing Cancellation Service.

Record Length

The Enhanced Exception File is a 150-byte fixed block record length format without a file header or trailer. The block size of the file should be 27900 bytes. If there are updates to report, send this file.

Message Layout

This is the message layout for the Enhanced Exception File. Refer to the [Programming Notes for Enhanced Exception File on page 5-159](#) for more instruction.

Enhanced Exception File Message Layout

Field Number	Format	Positions	Optionality	Field Name
1	AN(15)	1-15	M	Primary Account Number (PAN)
2	AN(1)	16	F	Filler
3	AN(1)	17	M	Transaction Code
4	AN(1)	18	F	Filler
5	AN(2)	19-20	F	Filler
6	AN(1)	21	M	Source Identification Operating Center
7	AN(1)	22	F	Filler
8	AN(1)	23	F	Filler
9	AN(2)	24-25	M	Exception Code
10	N(2)	26-27	M	Expiry Month
11	N(1)	28	M	Expiry Year
12	AN(5)	29-33	F	Filler
13	AN(1)	34	F	Filler
14	AN(2)	35-36	F	Filler
15	AN(1)	37	F	Filler
16	AN(1)	38	F	Reserved Field

Enhanced Exception File Message Layout (continued)

Field Number	Format	Positions	Optionality	Field Name
17	AN(3)	39-41	F	Filler
18	AN(1)	42	C	Card Type Indicator
19	AN(15)	43-57	C	Card Acceptor Identification Code
20	N(12)	58-69	C	Transaction Amount, Minimum
21	N(12)	70-81	C	Transaction Amount, Maximum
22	AN(3)	82-84	C	Currency Code, Transaction
23	AN(66)	85-150	F	Filler

Programming Notes for Enhanced Exception File

Field 1

Primary Account Number (PAN)

Definition	A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
Format	AN(15)
Position	1-15
Optionality	Mandatory
Notes	Issuers must use a scramble routine for the Primary Account Number in transmissions to the Network. Refer to scrambling Primary Account Numbers in the <i>Network Specifications - Instructions</i> manual.

Field 2

Filler

Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(1)
Position	16
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

Field 3

Transaction Code

Definition	This code requests that the Network perform an action.
Format	AN(1)
Position	17
Optionality	Mandatory
Notes	Valid Values: - A = Add record to file - D = Delete record from file - C = Change record on file

Field 4

Filler

Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(1)
Position	18
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

Field 5

Definition**Format****Position****Optionality****Notes**

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(2)

19-20

Filler

Alphanumeric Filler fields must contain spaces.

Field 6

Definition**Format****Position****Optionality****Notes**

Source Identification Operating Center

The operating center servicing the Card.

AN(1)

21

Mandatory

Valid Values:

- E = European Participants
- W = All Other Participants

Field 7

Definition**Format****Position****Optionality****Notes**

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

22

Filler

Alphanumeric Filler fields must contain spaces.

Field 8

Definition**Format****Position****Optionality****Notes**

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

23

Filler

Alphanumeric Filler fields must contain spaces.

Field 9

Definition**Format****Position****Optionality****Notes**

Exception Code

The disposition for the Card Account.

When the Network receives a Real Time Exception Update message, it applies the change only if the code is of the same or higher importance.

AN(2)

24-25

Mandatory

Refer to [Table 2-12. Exception and Message Reason Code Priority on page 2-14](#) for exception status and priority value.

Refer to the Exception Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Field 10

Definition**Format****Position****Optionality****Notes**

Expiry Month

A value from 01 to 12 that indicates the month the Exception File record will expire.

N(2) MM

26-27

Mandatory

If the Expiry Year is the same year the exception is being reported, the Expiry Month must be greater than the month the exception is being reported.

Valid Values: 01-12, 99

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Enhanced Exception File
Field 11 Expiry Year Message Layout

If this field = 99 then Field 11 (Expiry Year) must equal 9. The exception record will not expire from the Network Exception Database if this value is used. An Issuer must send a delete record with a Transaction Code of D in Field 3 to remove this from the Network Exception Database.

Field 11

Definition

Format

Position

Optionality

Notes

Expiry Year

The last digit of the year that the negative should expire; for example, 5=2005.

N(1) Y

28

Mandatory

The value cannot be a year that is more than six years from the current month and date The value is the last digit of the Expiry Year.

Example: For July 2007, only the values 7 (with Expiry Month allowed to be from 08 to 12), 8, 9, 10, 11, 12, and 13 are valid. In this example, 7 is valid only if the Expiry Month is greater than the current month.

Valid Values: 0-9

If this field = 9 and Field 10 (Expiry Month) = 99, the exception record will not expire from the Network Exception Database. An Issuer must send a delete record with a Transaction Code of D in Field 3 to remove this from the Network Exception Database.

Field 12

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(5)

29-33

Filler

Alphanumeric Filler fields must contain spaces.

Field 13

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

34

Filler

Alphanumeric Filler fields must contain spaces.

Field 14

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(2)

35-36

Filler

Alphanumeric Filler fields must contain spaces.

Field 15

Definition

Format

Position

Optionality

Notes

Filler

Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.

AN(1)

37

Filler

This code shows the reason for a message in its Message Type.

Field 16

Definition	This field is for AEGNS internal use.
Format	AN(1)
Position	38
Optionality	Filler
Notes	Alphanumeric Reserved Fields must contain spaces.

Reserved Field

Field 17

Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(3)
Position	39-41
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

Filler

Field 18

Definition	When populated with the value of "C" by an Issuer, the Card Type Indicator identifies the account as an Expresspay-enabled Card.
Format	AN(1)
Position	42
Optionality	Conditional. Expresspay Issuers must identify their Expresspay-enabled Card/device accounts.
Notes	Valid Value: C = Expresspay-enabled Card

Card Type Indicator

Expresspay: Expresspay Issuers must show a Card Type Indicator on the Exception File, the Batch Exception Update Request (9344), or the Enhanced Exception File messages to identify Expresspay-enabled Card/device accounts.

Field 19

Definition	The Service Establishment (S/E) number given to one or more locations of the Card Acceptor. The number is ten digits, followed by five spaces for a total of 15 bytes.
Format	AN(15)
Position	43-57
Optionality	Conditional
Notes	- This field is required when Exception Code = BZ. - This field is space-filled for all other Exception Codes. - This field must be a valid Card Acceptor Identification Code.

Card Acceptor Identification Code

Field 20

Definition	The minimum Transaction amount that the Network will block as part of the Recurring Billing Cancellation Service.
Format	N(12)
Position	58-69
Optionality	Conditional
Notes	- This field is required when Exception Code = BZ. - This field is space-filled for all other Exception Codes. - This Transaction amount is in the Transaction currency. Field 22, Currency Code, Transaction determines the decimal point location. For example, \$100.00 USD shows as 000000010000. - This field must be less than or equal to Field 21, Transaction Amount, Maximum.

Transaction Amount, Minimum

Field 21

Transaction Amount, Maximum

Definition	The maximum Transaction amount that the Network will block as part of the Recurring Billing Cancellation Service.
Format	N(12)
Position	70-81
Optionality	Conditional
Notes	• This field is required when Exception Code = BZ. • This field is space-filled for all other Exception Codes. • This Transaction amount is in the Transaction currency. Field 22, Currency Code, Transaction determines the decimal point location. For example, \$100.00 USD shows as 000000010000. • This field must be greater than or equal to Field 20, Transaction Amount, Minimum. • This field must be greater than zero (0).

Field 22

Currency Code, Transaction

Definition	This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
Format	AN(3)
Position	82-84
Optionality	Conditional
Notes	• This field is required when Exception Code = BZ • This field is space-filled for all other Exception Codes • This field must be a valid Currency Code

Field 23

Filler

Definition	Filler fields show positions in a message that are not used, but must contain default values described in the Notes section. Ignore data sent in this field.
Format	AN(66)
Position	85-150
Optionality	Filler
Notes	Alphanumeric Filler fields must contain spaces.

S/E Number Check Digit Computation

Description

The S/E number is a ten-digit number that identifies an S/E that accepts the American Express Card. This number is assigned to an S/E when the S/E signs up to accept the Card. The last digit to the right of the number is the check digit. The computation described here determines the validity of an S/E number by calculating a check digit and comparing that value to the given check digit. Matching check digits show a valid S/E number; non-matching check digits show an invalid S/E number.

Table 5-9. Check Digit Computation

Instructions	S/E Number = 9321011430, Check digit = 0									
1. Do not include the check digit (last number to the right) in the calculation.	9	3	2	1	0	1	1	4	3	
2. If the first three digits of the number are less than 930 or greater than 939, replace the first digit to the left with zero.	9	3	2	1	0	1	1	4	3	
3. Starting with the second digit from the left in Step 2, add together every other digit.		3			+1			+1		+4 =9
4. Starting with the first digit from the left, multiply every other digit by two.	9		2		0		1		3	
	x2		x2		x2		x2		x2	
	18		4		0		2		6	
5. If the product of Step 4 is a two-digit number, add the two digits of the product.	1+8 9		4		0		2		6	
6. Add the products from Step 5.	9		+4		+0		+2		+6 21	
7. Add the results of Steps 3 and 6.	9 + 21 = 30									
8. If the result of Step 7 is a multiple of 10 (10, 20, 30, etc.), the check digit is zero. If the result of Step 7 is not a multiple of 10, subtract that result from the next higher multiple of 10. The result is the check digit.	30 / 10 = 0 Given check digit is 0.									
9. If the calculated check digit matches the given check digit (last number to the right in the number), the S/E number is valid.										
10.If the check digits do not match, the S/E is invalid.	Calculated check digit is 0. The S/E number is valid.									

6 Payment Token Provisioning

Overview

Introduction

The American Express Token Service offers a suite of solutions that creates a secure payment method for online, digital, and mobile commerce. The American Express Token Service is available for Issuers, Acquirers, and S/Es on the American Express Network.

The American Express Token Service includes a Token Vault, Token provisioning and issuing, Token Life Cycle Management, and risk services to help prevent fraud.

Token provisioning replaces the Primary Account Number (PAN) with a unique digital identifier known as a Token.

Examples of Transaction types involving the use of Tokens include Card-on-file and Digital Wallet Transactions.

- Card-on-file Transactions involve the storing of the Card Number and associated data in support of billing service or recurring payments.
- Digital wallets allow Cardmembers to perform Mobile Device-initiated Transactions by using any of the following Digital Wallet Payment types:
 - Digital Wallet Contactless-initiated Transaction
 - Digital Wallet Application-initiated Transaction
 - Digital Wallet Magnetic Secure Transmission (MST) Transaction

Refer to the *Business and Operational Policies* manual for additional information on Card-on-file and Digital Wallet Program Transactions.

Important

The requirements in this document have been developed to support the global deployment of online, digital, and mobile commerce payments. Please contact your AEGNS Representative for additional information.

Benefits

The continued enhancements of these requirements supports Participants in the enablement and deployment of online, digital, and mobile commerce payments. The ongoing development of online, digital, and mobile commerce payments supports AEGNS objectives of providing new methods of payment leading to increased consumer choice, driving interoperability for these new payment methods, and increased speed, convenience, and security for Cardmembers and S/Es. It also maintains the AEGNS position as a leading payment network in emerging payments.

Participants may benefit from:

- Ability to offer new payment methods leading to increased consumer choice
- Increased speed, convenience, and security for Cardmembers and S/Es
- Defined policy and dispute requirements for online, digital, and mobile commerce payments

- Tokenized Transactions that can provide improved protection against unauthorized use of Cardmember data

Impact to Participants

Participants may participate in the development and deployment of Digital Wallet Payments, subject to AEGNS approval.

Participants who have AEGNS approval to enable and deploy Digital Wallet Payments must comply with the requirements in this document and any future AEGNS communication on Digital Wallet Payments.

Technical Impact to Participants

Issuers

Issuers should review the technical specifications in this document and review their internal systems to identify all impacts.

Issuers must complete AEGNS Network certification and beta testing for Digital Wallet and Payment Tokenization.

Refer to these sections for more instruction:

- [Payment Token Provisioning Request \(1100\) on page 6-3](#)
- [Payment Token Provisioning Response \(1110\) on page 6-16](#)
- [Payment Token Post Provisioning Notification \(1120\) on page 6-22](#)
- [Payment Token Post Provisioning Notification Response \(1130\) on page 6-31](#)

Acquirers

Acquirers should review their internal systems to identify all impacts.

Acquirers must complete AEGNS Network certification to enable Digital Wallet Payments and Payment Tokenization.

Payment Token Provisioning Request (1100)

Description

This message is passed by the Network to an Issuer to request that a Card be provisioned a Token by the Network.

Length of Record

Variable

Message Layout

This table shows the message layout for the [Payment Token Provisioning Request \(1100\) Message Layout](#). Refer to the [Programming Notes for Payment Token Provisioning Request \(1100\)](#) on page 6-4 for more instruction.

Payment Token Provisioning Request (1100) Message Layout

Bit Position	Format	Optionality	Field Name
		N to I	
N/A	N(4)	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	Bitmap Primary
2	N(21) max LLVAR	M	Primary Account Number (PAN)
3	N(6)	M	Processing Code
4	N(12)	M	Amount, Transaction
5	N(12)	M	Amount, Reconciliation
7	N(10)	N/A	Date and Time, Transmission
11	N(6)	M	Systems Trace Audit Number
12	N(12)	M	Date and Time, Local Transaction
13	N(4)	N/A	Card Effective Date
14	N(4)	C	Card Expiration Date
18	N(4)	N/A	Reserved Field
19	N(3)	N/A	Country Code, Acquiring Institution
22	AN(12)	M	Point of Service Data Code
24	AN(3)	N/A	Function Code
26	N(4)	M	Card Acceptor Business Code
31	N(40) max LLVAR	C	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	M	Acquiring Institution Identification (AIN) Code
35	ANS(39) max LLVAR	N/A	Track 2 Data

Payment Token Provisioning Request (1100) Message Layout (continued)

Bit Position	Format	Optionality	Field Name
		N to I	
37	AN(12)	N/A	Acquirer Reference Number (ARN)
41	ANS(8)	N/A	Card Acceptor Terminal Identification
42	AN(15)	M	Card Acceptor Identification Code
43	ANS(101) max LLVAR	M	Card Acceptor Name and Location
45	ANS(78) max LLVAR	N/A	Track 1 Data
48	AN(353) max LLLVAR	N/A	Deferred Payment Plan (DPP)
49	N(3)	M	Currency Code, Transaction
50	N(3)	M	Currency Code, Reconciliation
53	AN(8) max LLVAR	C	Security Related Control Information
55	AN(259) max (includes length indicator) LLLVAR	N/A	ICC System Related Data
60	ANS(520) max (includes length indicator) LLLVAR	M	Market Specific Data
61	4 bytes min, 103 bytes max LLLVAR	N/A	National Use Data
62	AN(5) min AN(60) max LLLVAR	C	Fraud Indicators
63	AN(4) min AN(103) max LLLVAR	C	Automated Address Verification

Programming Notes for Payment Token Provisioning Request (1100)

Bit N/A

Definition

Format

Optionality

Notes

Message Type Identifier

This code identifies the type of message sent by the forwarding institution.

N(4)

Network to Issuer: Mandatory

Required Value: 1100

Bit N/A

Definition

Format

Optionality

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

8 bytes, 64 bits for each bitmap, Binary (Hex)

Network to Issuer: Mandatory

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Payment Token Provisioning Request (1100)
Bit 2 Primary Account Number (PAN) Message Layout

Notes If the field is mandatory or optional and the Participant uses that field, the bitmap position must contain a value of 1. If the field is optional and is not used, its assigned bitmap position must contain a value of 0.

Bit 2 **Primary Account Number (PAN)**

Definition A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.

Format N(21) max LLVAR

Optionality **Network to Issuer:** Mandatory

Notes This field contains the primary funding account (PAN).

Bit 3 **Processing Code**

Definition This is the service being requested.

Format N(6)

Optionality **Network to Issuer:** Mandatory

Notes **Required Value:** 394800 = Payment Token Provisioning Request

Bit 4 **Amount, Transaction**

Definition The sale or credit amount of the provisioning request.

Format N(12)

Optionality **Network to Issuer:** Mandatory

Notes **Required Value:** 0 (zero dollar value)

Bit 5 **Amount, Reconciliation**

Definition This is the total amount of the provisioning request after the Network changes the Amount, Transaction (Bit 4) to the Issuer's currency (Bit 50).

Format N(12)

Optionality **Network to Issuer:** Mandatory

Notes This bit is populated by the Network.

Bit 7 **Date and Time, Transmission**

Definition This is the local date and time when the Acquirer transmits the request to the Network.

Format N(10) MMDDHHMMSS

Optionality **Network to Issuer:** Not Applicable

Bit 11 **Systems Trace Audit Number**

Definition This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format N(6)

Optionality **Network to Issuer:** Mandatory

Notes This value is assigned by the Network.

Bit 12 **Date and Time, Local Transaction**

Definition This is the current date and time supplied by the Network at the time of the provisioning request.

Format N(12) YYMMDDHHMMSS

Optionality **Network to Issuer:** Mandatory

Bit 13 **Card Effective Date**

Definition This is the Card Effective Date embossed on the face of the Card.

Format N(4) YYMM

Optionality **Network to Issuer:** Not Applicable

Bit 14

Definition
Format
Optionality

Card Expiration Date

This is the Card Expiration Date embossed on the face of the Card.
N(4) YYYY
Network to Issuer: Conditional. If this bit is present, it is sent to the Issuer.

Bit 18

Definition
Format
Optionality

Reserved Field

This field is for AEGNS internal use.
N(4)
Network to Issuer: Not Applicable

Bit 19

Definition
Format
Optionality

Country Code, Acquiring Institution

This is the Country Code of the acquiring institution.
N(3)
Network to Issuer: Not Applicable

Bit 22

Definition

Format
Optionality

Point of Service Data Code

This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).
AN(12)
Network to Issuer: Mandatory

Bit 24

Definition
Format
Optionality

Function Code

This code shows the reason for a message in its Message Type.
AN(3)
Network to Issuer: Not Applicable

Bit 26

Definition

Format
Optionality
Notes

Card Acceptor Business Code

Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).
N(4)
Network to Issuer: Mandatory
Refer to the Merchant Category Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 31

Definition
Format
Optionality

Notes

Acquirer Reference Data, Transaction Identifier (TID)

This unique number is given by the Network and follows a Transaction during its life cycle.
N(40) max LLVAR
Network to Issuer: Conditional. The Network sends this bit to the Issuer if both Acquirer and Issuer are certified to receive it.
The Network generates the TID and sends it to the Issuer.

Bit 32

Definitions
Format
Optionality
Notes

Acquiring Institution Identification (AIN) Code

This is a unique identification code given to an Acquirer by the Network.
N(13) max LLVAR
Network to Issuer: Mandatory
Required Value: 06370100 = American Express

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

Bit 35

Definition

Format

Optionality

Track 2 Data

This is the data encoded in a Track 2 magnetic stripe read or the data stored in the Track 2 field on the chip of a Chip Card.

ANS(39) max LLVAR

Network to Issuer: Not Applicable

Bit 37

Definition

Format

Optionality

Acquirer Reference Number (ARN)

A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.

AN(12)

Network to Issuer: Not Applicable

Bit 41

Definition

Format

Optionality

Card Acceptor Terminal Identification

This code identifies a POS terminal at a Card Acceptor location.

ANS(8)

Network to Issuer: Not Applicable

Bit 42

Definition

Format

Optionality

Card Acceptor Identification Code

The Service Establishment (S/E) number given to the provisioning requestor. The number is ten (10) digits, followed by five (5) spaces for a total of 15 bytes.

AN(15)

Network to Issuer: Mandatory

Bit 43

Definition

Format

Optionality

Card Acceptor Name and Location

For a Digital Wallet Payment Token Provisioning Request, this is the name of the Digital Wallet as it is known to the Cardmember.

ANS(101) max LLVAR

Network to Issuer: Mandatory

Bit 45

Definition

Format

Optionality

Track 1 Data

This is the data encoded in a Track 1 magnetic stripe read or the data stored in the Track 1 field on the chip of a Chip Card.

ANS(78) max LLVAR

Network to Issuer: Not Applicable

Bit 48

Definition

Format

Optionality

Deferred Payment Plan (DPP)

This is the Deferred Payment Plan data, permitting charges to be billed in equal installments rather than in one (1) total billing.

AN(353) max LLLVAR

Network to Issuer: Not Applicable

Bit 49

Definition

Format

Optionality

Notes

Currency Code, Transaction

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

N(3)

Network to Issuer: Mandatory

Refer to the *Codes Reference Guide* for a list of Currency Codes.

Bit 50

Definition

Format

Optionality

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.

N(3)

Network to Issuer: Mandatory

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Bit 53 Security Related Control Information Message Layout

Notes This bit is populated by the Network.

Bit 53 **Security Related Control Information**

Definition This field contains a variable length indicator and the Card Security Code (CSC) supplied by the digital wallet.

Card Security Code includes the 3CSC and 4CSC (4DBC).

Format AN(8) max LLVAR

Optionality **Network to Issuer:** Conditional

Notes Required for Digital Wallet Payment Token Provisioning Request.

Optional for Card-on-file Payment Token Provisioning Request.

Bit 55 **ICC System Related Data**

Definition This is the data sent to and from the Integrated Circuit Card and used in an AEIPS or Expresspay Transaction.

Format AN(259) max (includes length indicator) LLLVAR

Optionality **Network to Issuer:** Not Applicable

Bit 60 **Market Specific Data**

Definition This is the provisioning-specific data unique to and supplied by the digital wallet provider.

Format ANS(520) max (includes length indicator) LLLVAR

Optionality **Network to Issuer:** Mandatory

Notes This table shows the layout of Bit 60.

Subfield	Length	Use	Optionality	Notes
	3 bytes	Variable Length Indicator		
	2 bytes	Market Specific Data Indicator		Required Value: MS
	3 bytes	Market ISO Country Code		Refer to the <i>Codes Reference Guide</i> for a list of Country Codes.
	2 bytes	Market Specific Data Version Indicator		Required Value: 02
	4 bytes	Bitmap Identifier		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the Bitmap, the layout consists of at least one (1) of the subfields below. Each bit position of the 32 bit/4-byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, that corresponding subfield will be present.
Subfield 1	1 byte	Wallet Provider Recommended Provision Path	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Digital Wallet Valid Values: 0 = Recommended Approval 1 = Verify with One Time Password (OTP) 6 = Recommended Decline

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Subfield	Length	Use	Optionality	Notes
Subfield 2	6 bytes	Wallet Provider Recommended Provision Path Standards Version	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Format: xxxxxx Example: 123456 represents 1234.56
Subfield 3	30 bytes	Wallet Provider Provisioning Path Reason Codes	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Left justified, space-filled. Field Usage: Two (2) byte codes represent the reason the Wallet Provider chose the specific provisioning path. A limit of 15 Reason Codes can be sent in one Digital Wallet Provisioning Request (1100) message. A single Provisioning Request will not require more than 15 Reason Codes. Refer to <i>Codes Reference Guide</i> for a complete list of codes. Example: 01060C0E~~~~~ In the example, a '~' represents a space. 01 = ID too new relative to announcement 06 = The Account has not had activity in the last year 0C = Device Score is less than 3 0E = Device provisioning location outside of account home country
Subfield 4	1 byte	Wallet Provider Device Trust Level	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Valid Values: 1 = Improper activity noted 2 = Irregular activity noted 3 = Little or no history 4 = Short history of activity; No suspicious activity 5 = Long history of activity; No suspicious activity
Subfield 5	1 byte	Token Requestor Account Trust Level	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Valid Values: 1 = Improper activity noted 2 = Irregular activity noted 3 = Little or no history 4 = Short history of activity; No suspicious activity 5 = Long history of activity; No suspicious activity

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Subfield	Length	Use	Optionality	Notes
Subfield 6	2 bytes	Device Language Code	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Refer to the <i>Codes Reference Guide</i> for a list of Language Codes.
Subfield 7	64 bytes	Terms & Conditions Version	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Definition: Terms and Conditions Version accepted by the Cardmember
Subfield 8	12 bytes	Terms & Conditions Time Stamp	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Format: YYMMDDHHMMSS
Subfield 9	2 bytes	Wallet Provider ID	M	Definition: Two byte code, assigned by the Network, used to identify the digital wallet provider.
Subfield 10		Reserved for Future Use		
Subfield 11	23 bytes	Source IP	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Definition: IP address of device used to provision. Field Orientation: Left justified, space-filled. Example: - Example 1: 127.142.151.223..~~~~~ - Example 2: 127.142.5.56..~~~~~ - Example 3: 12.142.49.190..~~~~~ In the examples, a '~' represents a space.
Subfield 12	8 bytes	Longitude	O	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Left justified, zero-filled. Field Usage: The first byte represents positive or negative. Positive = '+'; Negative = '-' Format: xxxxxxxx Examples: +1234560 represents longitude +123.4560 +0892344 represents longitude +89.2344

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Bit 60 Market Specific DataMessage Layout

Subfield	Length	Use	Optionality	Notes
Subfield 13	8 bytes	Latitude	O	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Left justified, zero-filled. Field Usage: The first byte represents positive or negative. Positive = '+'; Negative = '-' Format: xxxxxxxx Examples: -9870000 represents latitude -987.0000 +0086500 represents latitude +8.6500
Subfield 14		Reserved for Future Use		
Subfield 15	26 bytes	Cardmember Name	M	The following characters are used to separate the elements of a Cardmember's Name: "/" (forward slash) = surname/first name separator " " (space) = first name/middle name or first name/middle initial separator (as required) "." (period) = name/title separator (as required) Examples: - Surname only: FROST/ - Surname and first name: FROST/CHARLES - Surname, first name, and middle initial: FROST/CHARLES F - Surname, first initial, and middle initial: FROST/C F - Surname, first name, middle name, and title: FROST/CHARLES FRANCIS.JR - The minimum encoded data shall be a single alpha character (as surname) and the surname separator.
Subfield 16	64 bytes	Wallet User ID	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Left justified, space-filled. Note: This may be a hash of the Wallet User ID.

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Subfield	Length	Use	Optionality	Notes
Subfield 17	2 bytes	Device Type	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Definition: A 2 byte code that represents the type of device that requested a Token to be provisioned. Field Orientation: Left justified, space-filled. Valid Values: 01 = Phone 02 = Tablet 03 = Watch 04 = Card 05 = Wearable 06 = Laptop 99= Other
Subfield 18	20 bytes	Recognizable Device Name	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Left justified, space-filled.
Subfield 19	15 bytes	Device Phone Number	O	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Right justified, space-filled.
Subfield 20	2 bytes	Token Count	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Field Orientation: Right justified, zero-filled.
Subfield 21	64 bytes	Device ID	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Definition: Distinctive device identifier associated with a mobile device or similar hand-held device. For example: International Mobile Equipment Identifier (MEID), International Mobile Security Identifier, or Serial ID (tablet devices).
Subfield 22	64 bytes	Conversation ID	M	Definition: A unique number given by the Network to identify a provision request throughout its life cycle. Field Orientation: Left justified, space-filled.

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Bit 61 National Use Data Message Layout

Subfield	Length	Use	Optionality	Notes
Subfield 23	16 bytes	PAN Source	M	Definition: A 16 byte code that identifies the source of the PAN used in the provisioning request. Valid Values: ONFILE = Card is on file with the Wallet Provider USERINPUT = User entered the PAN ISSUERAPP = Bank app has sent the PAN NFCREADER = Near Field Communication (wireless transmission) used to send the PAN STREAMLINE = PAN sourced from another device OTHER = Input source not listed above WEB = Bank web portal has sent the PAN
Subfield 24	11 bytes	Token Requestor ID	M	Definition: An 11-byte code pairing Token Requestor with Token Domain. Positions 1-3 - Token Service Provider Code, unique to each Token Service Provider Positions 4-11 - Assigned by the Token Service Provider for each requesting entity and Token Domain
Subfield 25	64 bytes	Wallet ID	C	- Conditional for a digital wallet Payment Token Provisioning Request. - Not applicable for a Card-on-file Payment Token Provisioning Request. Definition: Unique identifier representing a combination of data elements that typically designates a specific device and/or wallet user.

Bit 61

National Use Data

Definition

This is an ISO field that supplies Cardmember authentication data during Authorization.

Format

4 bytes min, 103 bytes max LLLVAR

Optionality

Network to Issuer: N/A

Bit 62

Fraud Indicator

Definition

The Fraud Indicator field is used to provide fraud mitigation indicators for a variety of situations.

Format

AN(5) min, AN(60) max LLLVAR

Optionality

Network to Issuer: Conditional. Issuers must be certified.

Notes

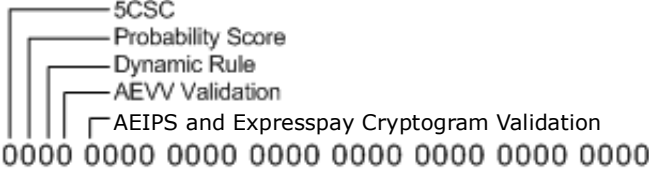
FD3 Request Type Identifier

Bit 62 FD3 Request Type is supplied for SafeKey Transactions where the Network created the AEVV when the Issuer ACS was not available.

Table 1: FD3 Request Type Identifier Layout

Position	Value	Use
1-3	XXX	Variable Length Indicator
4-5	AX	Service Identifier AX = American Express
6-8	FD3	Request Type Identifier FD3 = Bitmap Variable Length

Table 1: FD3 Request Type Identifier Layout (continued)

9-12	32 bit Binary	Bitmap Indicator Identifies the bits in the structure. Following the bitmap, the layout consists of at least one of the subfields as defined below. Each bit position of the 32 bit / 4-byte bitmap represents which fraud indicators are present. If a particular bit is "ON" in the bitmap, that corresponding subfield will be present in positions 13-60. • Subfield 1 5CSC Indicator (2 Bytes) • Subfield 2 Fraud Probability Score (5 Bytes) • Subfield 3 Recommendation Code (3 Bytes) Refer to <i>Codes Reference Guide</i> for valid codes and definitions for Subfields 1 through 5 and Subfield 11.
13-60	Fraud Value	Describes the results of the Bitmap Indicators that are set.
Example: Positions 9 - 12 		

Example: If Bit 62 includes Fraud Probability Score, Recommendation Code, and AEVV Validation Result:

Position	Value
1-3	018
4-5	AX
6-8	FD3
9-12	01110000000000000000000000000000 X'70000000'
13-60	555553331

Bit 63

Definition

Format

Optionality

Notes

Automated Address Verification

A message option that allows an S/E to verify the Cardmember address information.

AN(4) min, AN(103) max LLLVAR

Network to Issuer: Conditional. If the digital wallet provider sends this bit and the Issuer is certified to receive it, then the Network sends it to the Issuer.

If the Cardmember address is available but the Postal Code is not, the Postal Code subfield must be zero-filled. If the Postal Code is available but the Cardmember address is not, the address subfield must be character space-filled.

Position	Value	Use
1-3	033	Variable Length Indicator
4-5	AX	Service Identifier, AX for American Express
6-7	AD or AP	Request Type Identifier AD = Standard Address Verification (alphanumeric) AP = Address Verification (numeric) If an AAV code (AXAP) is received by the Network for an Issuer not certified for AXAP Address Verification, the Network will change the code from AXAP to AXAD.

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Bit 63 Automated Address Verification Message Layout

8-16		Postal Code The Postal Code is left justified, and the subfields should have trailing zeros. Other alphanumeric Postal Codes may be padded with zeros or character spaces. If the Request Type Identifier is "AXAP," or if the Network has changed the code from AXAP to AXAD, this subfield may contain the numeric values from the Cardmember's postal code.
17-36		First 20 Characters of Cardmember Address (20 characters, space-filled). Alpha characters must be upper case. Leading or trailing zeros or slashes (/) are not permitted. This subfield may also contain only the numeric values from the Cardmember's address.

Payment Token Provisioning Response (1110)

Description

This message is used by the Issuer to respond to a [Payment Token Provisioning Request \(1100\)](#) sent by the Network.

Length of Record

Variable

Message Layout

This table shows the message layout for the [Payment Token Provisioning Response \(1110\)](#). Refer to the [Programming Notes for Payment Token Provisioning Response \(1110\)](#) on page 6-17 for more instruction.

Payment Token Provisioning Response (1110) Message Layout

Bit Position	Format	Optionality	Field Name
		I to N	
N/A	N(4)	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	Bitmap Primary
2	N(21) max LLVAR	ME	Primary Account Number (PAN)
3	N(6)	ME	Processing Code
4	N(12)	ME	Amount, Transaction
5	N(12)	ME	Amount, Reconciliation
7	N(10)	N/A	Date and Time, Transmission
11	N(6)	ME	Systems Trace Audit Number
12	N(12)	ME	Date and Time, Local Transaction
23	N(3)	C	PAN Sequence Number
24	AN(3)	N/A	Function Code
30	N(24)	N/A	Amount, Original
31	N(40) max LLVAR	ME	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) LLVAR	ME	Acquiring Institution Identification (AIN) Code
34	AN(30) Max LLVAR (includes length indicator)	N/A	Primary Account Number, Extended
37	AN(12)	N/A	Acquirer Reference Number (ARN)
38	AN(6)	M	Approval Code
39	N(3)	M	Action Code
41	ANS(8)	N/A	Card Acceptor Terminal Identification

Payment Token Provisioning Response (1110) Message Layout (continued)

Bit Position	Format	Optionality	Field Name
		I to N	
42	AN(15)	ME	Card Acceptor Identification Code
44	AN(4) max LLVAR	M	Additional Response Data
48	AN(353) LLLVAR	N/A	Deferred Payment Plan (DPP)
49	N(3)	ME	Currency Code, Transaction
50	N(3)	ME	Currency Code, Reconciliation
54	ANS(123) 4 bytes min 123 bytes max LLLVAR	N/A	Amount, Additional
55	AN(259) max (includes length indicator) LLLVAR	N/A	ICC System Related Data
60	ANS(120) max (includes length indicator) LLLVAR	M	Market Specific Data
61	4 bytes min, 103 bytes max LLLVAR	N/A	National Use Data

Programming Notes for Payment Token Provisioning Response (1110)

Bit N/A

Message Type Identifier

Definition

This code identifies the type of message sent by the forwarding institution.

Format

N(4)

Optionality

Issuer to Network: Mandatory

Notes

Required Value: 1100

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Bitmap Primary

Definition

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

Format

8 bytes, 64 bits for each bitmap, Binary (Hex)

Optionality

Issuer to Network: Mandatory

Notes

If the field is mandatory or optional and the Participant elects to use that field, its assigned bitmap position must contain a value of 1. If the field is optional and is not used, its assigned bitmap position must contain a value of 0.

Bit 2

Primary Account Number (PAN)

Definition

A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.

Format

N(21) max LLVAR

Optionality	Issuer to Network: Mandatory Echo The Issuer must echo the primary funding account sent by the Network in the Digital Wallet Provisioning Request (1100) message.
Bit 3	Processing Code
Definition	This is the service being requested.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo
Bit 4	Amount, Transaction
Definition	The sale or credit amount of the provisioning request.
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo. The Issuer must echo the value of the Amount, Transaction sent by the Network in the Payment Token Provisioning Request (1100) message.
Bit 5	Amount, Reconciliation
Definition	This is the total amount of the provisioning request after the Network changes the Amount, Transaction (Bit 4) to the Issuer's currency (Bit 50).
Format	N(12)
Optionality	Issuer to Network: Mandatory Echo. The Issuer must echo the value of the Amount, Reconciliation sent by the Network in the Payment Token Provisioning Request (1100) message.
Bit 7	Date and Time, Transmission
Definition	This is the current date and time supplied by the Network at the time of the provisioning request.
Format	N(10) MMDDHHMMSS
Optionality	Issuer to Network: Not Applicable
Bit 11	Systems Trace Audit Number
Definition	This is a unique (where applicable) trace number given to each Transaction or provisioning request.
Format	N(6)
Optionality	Issuer to Network: Mandatory Echo
Bit 12	Date and Time, Local Transaction
Definition	This is the current date and time supplied by the Network at the time of the provisioning request.
Format	N(12) YYMMDDHHMMSS
Optionality	Issuer to Network: Mandatory Echo
Bit 23	PAN Sequence Number
Definition	A number distinguishing between separate Cards with the same Primary Account Number.
Format	N(3)
Optionality	Issuer to Network: Conditional. This bit is required if the Issuer supports PAN Sequence Number provisioning.
Bit 24	Function Code
Definition	This code shows the reason for a message in its Message Type.
Format	AN(3)
Optionality	Issuer to Network: Not Applicable
Bit 30	Amount, Original
Definition	The first Transaction Amount requested when a partial amount is approved.
Format	N(24)
Optionality	Issuer to Network: Not Applicable

Bit 31	Acquirer Reference Data, Transaction Identifier (TID)
Definition	This unique number is given by the Network and follows a Transaction during its life cycle.
Format	N(40) max LLVAR
Optionality	Issuer to Network: Mandatory Echo.
Bit 32	Acquiring Institution Identification (AIN) Code
Definitions	This is a unique identification code given to an Acquirer by the Network.
Format	N(13) LLVAR
Optionality	Issuer to Network: Mandatory Echo
Bit 34	Primary Account Number, Extended
Definitions	This field is a multipurpose field which provides the Primary Account Number or associated data.
Format	AN(30) Max LLVAR (includes length indicator)
Optionality	Issuer to Network: Not Applicable
Bit 37	Acquirer Reference Number (ARN)
Definition	A unique number assigned by the Acquirer. It contains data supplied by an Acquirer and used by an Acquirer to match Transactions.
Format	AN(12)
Optionality	Issuer to Network: Not Applicable
Bit 38	Approval Code
Definition	A unique code assigned by the Card provisioning request authorizing institution to indicate the disposition for the provisioning request.
Format	AN(6)
Optionality	Issuer to Network: Mandatory
Bit 39	Action Code
Definition	The type of action done or the type of action to do for a Transaction.
Format	N(3)
Optionality	Issuer to Network: Mandatory
Notes	Valid Values: • 000 = Approved • 001 = Approve with ID • 100 = Deny • 101 = Expired Card/Invalid Expiration Date • 107 = Please Call Issuer • 109 = Invalid Service Establishment • 111 = Invalid Account
Bit 41	Card Acceptor Terminal Identification
Definition	This code identifies a POS terminal at a Card Acceptor location.
Format	ANS(8)
Optionality	Issuer to Network: Not Applicable
Bit 42	Card Acceptor Identification Code
Definition	The Service Establishment (S/E) number given to the provisioning requestor. The number is ten (10) digits, followed by five (5) spaces for a total of 15 bytes.
Format	AN(15)
Optionality	Issuer to Network: Mandatory Echo

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Bit 44 Additional Response Data Message Layout

Bit 44	Additional Response Data
Definition	This field contains the Issuer response to address verification requests or PCSC validation requests in the POS Authorization Request (1100) message.
Format	AN(4) max LLVAR
Optionality	Issuer to Network: Mandatory
Bit 48	Deferred Payment Plan (DPP)
Definition	This is the Deferred Payment Plan data, permitting charges to be billed in equal installments rather than in one (1) total billing.
Format	AN(353) LLLVAR
Optionality	Issuer to Network: Not Applicable
Bit 49	Currency Code, Transaction
Definition	This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo
Bit 50	Currency Code, Reconciliation
Definition	This is the numeric code of the currency defined by the Issuer.
Format	N(3)
Optionality	Issuer to Network: Mandatory Echo
Bit 54	Amount, Additional
Definition	The available amount of funds in the Card currency remaining on a Card.
Format	ANS(123), 4 bytes min, 123 bytes max, LLLVAR
Optionality	Issuer to Network: Not Applicable
Bit 55	ICC System Related Data
Definition	This is the data sent to and from the Integrated Circuit Card and used in an AEIPS or Expresspay Transaction.
Format	AN(259) max (includes length indicator) LLLVAR
Optionality	Issuer to Network: Not Applicable
Bit 60	Market Specific Data
Definition	This is the provisioning-specific data unique to and supplied by the digital wallet provider.
Format	ANS(120) max (includes length indicator) LLLVAR
Optionality	Issuer to Network: Mandatory
Notes	Required Values:

Subfield	Length	Use	Optionality	Value
	3 bytes	Variable Length Indicator		
	2 bytes	Market Specific Data Indicator		Required Value: MS
	3 bytes	Market ISO Country Code		Refer to the <i>Codes Reference Guide</i> for a list of Country Codes.
	2 bytes	Market Specific Data Version Indicator		Required Value: 04

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 Bit 61 National Use DataMessage Layout

Subfield	Length	Use	Optionality	Value
	4 bytes	Bitmap Identifier		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the Bitmap, the layout consists of at least one (1) of the subfields below. Each bit position of the 32 bit/4-byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, that corresponding subfield will be present.
Subfield 1	2 bytes	PAN Sequence Number	O	Definition: A number distinguishing between separate Cards with the same Primary Account Number. Field Orientation: Right justified, zero-filled.
Subfield 2	64 bytes	Conversation ID	M	Definition: A unique number given by the Network to identify a provisioning request throughout its life cycle. Field Orientation: Left justified, space-filled.
Subfield 3	12 bytes	Asset ID	M	Definition: A code representing the appropriate card and product information to be displayed on the device.
Subfield 4	2 bytes	Return Code	M	Definition: The returned value associated with the provisioning request. Valid Values: 00 - Success 01 - Mandatory parameters missing 02 - CVV Verification failed 03 - MAX Token Limit 04 - Downstream system Issue 05 - Card account Lock-out 06 - User requires higher level of authentication
Subfield 5	26 bytes	Cardmember Name	O	The following characters are used to separate the elements of a Cardmember's Name: "/" (forward slash) = surname/first name separator " " (space) = first name/middle name or first name/middle initial separator (as required) "." (period) = name/title separator (as required) Examples: Surname only: FROST/ Surname and first name: FROST/CHARLES Surname, first name, and middle initial: FROST/CHARLES F Surname, first initial, and middle initial: FROST/C F Surname, first name, middle name, and title: FROST/CHARLES FRANCIS.JR The minimum encoded data shall be a single alpha character (as surname) and the surname separator.

Bit 61

National Use Data

Definition

This is an ISO field that supplies Cardmember authentication data during Authorization.

Format

4 bytes min, 103 bytes max LLLVAR

Optionality

Issuer to Network: N/A

Payment Token Post Provisioning Notification (1120)

Description

This message notifies Issuers of the result of a Provisioning Request.

The Network does not pass this message directly from the Token Requestor to the Issuer. Instead, the Network responds to the Token Requestor, stores the Transaction in the Store and Forward queue, and sends the message to the Issuer from the queue.

Length of Record

Variable

Message Layout

This is the message layout for the [Payment Token Post Provisioning Notification \(1120\)](#). Refer to the [Programming Notes for Payment Token Post Provisioning Notification \(1120\)](#) on page 6-23 for more instruction.

Payment Token Post Provisioning Notification (1120) Message Layout

Bit Position	Format	Optionality	Field Name
		N to I	
N/A	N(4)	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	Bitmap Primary
2	N(21) max LLVAR	M	Primary Account Number (PAN)
3	N(6)	M	Processing Code
4	N(12)	M	Amount, Transaction
5	N(12)	M	Amount, Reconciliation
11	N(6)	M	Systems Trace Audit Number
12	N(12)	M	Date and Time, Local Transaction
14	N(4)	C	PAN Expiration Date
22	AN(12)	M	Point of Service Data Code
23	N(3)	C	PAN Sequence Number
24	AN(3)	M	Function Code
26	N(4)	M	Card Acceptor Business Code
30	N(24)	N/A	Amount, Original
31	N(40) max LLVAR	C	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	O	Acquiring Institution Identification (AIN) Code

Payment Token Post Provisioning Notification (1120) Message Layout (continued)

Bit Position	Format	Optionality	Field Name
		N to I	
34	AN(30) max LLVAR (includes length indicator)	C	Primary Account Number, Extended
35	ANS(39) max LLVAR	N/A	Track 2 Data
38	AN(6)	M	Approval Code
39	N(3)	M	Action Code
42	AN(15)	M	Card Acceptor Identification Code
45	ANS(78) max LLVAR	N/A	Track 1 Data
48	AN(353) max LLLVAR	N/A	Stand-In Parameter Information for Non-DPP Requests Stand-In Parameter Information for DPP Requests
49	N(3)	M	Currency Code, Transaction
50	N(3)	M	Currency Code, Reconciliation
56	N(37) max LLVAR	N/A	Original Data Elements
60	ANS(416) max (includes length indicator) LLLVAR	M	Market Specific Data
61	4 bytes min, 103 bytes max LLLVAR	N/A	National Use Data
62	AN(5) min AN(60) max LLLVAR	C	Fraud Indicators

Programming Notes for Payment Token Post Provisioning Notification (1120)

Bit N/A

Definition

Format

Optionality

Notes

Message Type Identifier

This code identifies the type of message sent by the forwarding institution.

N(4)

Network to Issuer: Mandatory

Required Value: 1120

Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Definition

Format

Optionality

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.

8 bytes, 64 bits for each bitmap, Binary (Hex)

Network to Issuer: Mandatory

American Express Global Network Services
Network Specifications - Authorization
Payment Token Post Provisioning Notification (1120)
Bit 2 Primary Account Number (PAN) Message Layout

Bit 2

Primary Account Number (PAN)

Definition

A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.

Format

N(21) max LLVAR

Optionality

Network to Issuer: Mandatory

Notes

A two-digit, variable length indicator is before the PAN. This indicator is not included in the length of the bit.

For example, a Cardmember Account Number of:

3759-876543-21001

would appear as:

15375987654321001

Bit 3

Processing Code

Definition

This is the service being requested.

Format

N(6)

Optionality

Network to Issuer: Mandatory

Notes

Required Value: 394800 = Payment Token Provisioning Request

Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 4

Amount, Transaction

Definition

The sale or credit amount of the provisioning request.

Format

N(12)

Optionality

Network to Issuer: Mandatory

Notes

Required Value: \$0

Bit 5

Amount, Reconciliation

Definition

This is the total amount of the provisioning request after the Network changes the Amount, Transaction (Bit 4) to the Issuer's currency (Bit 50).

Format

N(12)

Optionality

Network to Issuer: Mandatory

Bit 11

Systems Trace Audit Number

Definition

This is a unique (where applicable) trace number given to each Transaction or provisioning request.

Format

N(6)

Optionality

Network to Issuer: Mandatory

The Network gives this value. It is different than the value the Acquirer gives.

Bit 12

Date and Time, Local Transaction

Definition

This is the current date and time supplied by the Network at the time of the provisioning request.

Format

N(12) YYMMDDHHMMSS

Optionality

Network to Issuer: Mandatory

Bit 14

PAN Expiration Date

Definition

This is the Card Expiration Date embossed on the face of the Card.

Format

N(4) YYMM

Optionality

Network to Issuer: Conditional

American Express Global Network Services
Network Specifications - Authorization
Payment Token Post Provisioning Notification (1120)
Bit 22 Point of Service Data Code Message Layout

Bit 22

Definition

Format

Optionality

Notes

Point of Service Data Code

This is a series of codes that shows the capability, security data, and conditions of a terminal (or wallet) when a Transaction occurs at the point of service (or during provisioning).

AN(12)

Network to Issuer: Mandatory

The Network sets this field for Stand-In Advice messages.

Refer to the Point of Service Data Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 23

Definition

Format

Optionality

PAN Sequence Number

A number distinguishing between separate Cards with the same Primary Account Number.

N(3)

Network to Issuer: Conditional. The Network sends this bit to the Issuer if the Issuer has provided this bit in the [Payment Token Provisioning Response \(1110\)](#) message.

Bit 24

Definition

Format

Optionality

Notes

Function Code

This code shows the reason for a message in its Message Type.

AN(3)

Network to Issuer: Mandatory

Refer to the Function Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 26

Definition

Format

Optionality

Notes

Card Acceptor Business Code

Also called the Merchant Category Code (MCC), this code classifies the business product or service for a Card Acceptor (S/E).

N(4)

Network to Issuer: Mandatory

Refer to the Merchant Category Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 30

Definition

Format

Optionality

Amount, Original

The first Transaction Amount requested when a partial amount is approved.

N(24)

Network to Issuer: Not Applicable

Positions 13-24 contain zeros (0) and are reserved for future use.

Bit 31

Definition

Format

Optionality

Notes

Acquirer Reference Data, Transaction Identifier (TID)

This unique number is given by the Network and follows a Transaction during its life cycle.

N(40) max LLVAR

Network to Issuer: Conditional. The Network sends this bit to the Issuer if the Issuer is certified to receive it.

Refer to [2 Authorization Processing on page 2-1](#) for information about the TID.

Bit 32

Definitions

Format

Optionality

Notes

Acquiring Institution Identification (AIN) Code

This is a unique identification code given to an Acquirer by the Network.

N(13) max LLVAR

Network to Issuer: Optional

A two-digit, variable length indication is before this bit. This indicator is not included in the length of the bit.

For example, Acquiring Institution Identification code 45678912345 would appear as 1145678912345.

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Payment Token Post Provisioning Notification (1120)
Bit 34 Primary Account Number, ExtendedMessage Layout

Required Values:

Position	Value	Use
1-2	06	Variable Length Indicator
3-8	370100	American Express

Bit 34

Definitions

Format

Optionality

Notes

Primary Account Number, Extended

This field is a multipurpose field which provides the Primary Account Number or associated data.

AN(30) Max (includes length indicator) LLVAR

Network to Issuer: Conditional. For Payment Token Provisioning Requests, Bit 34 may be sent by the Network to the Issuer based on provisioning status.

This field is populated by the Network with the Token, the Wallet Provider ID, and the Token Expiration Date.

Subfield	Format
1	Variable Length Indicator (VLI)
2	Token
3	Wallet Provider ID (ID)
4	Token Expiration Date

Data Example:

Subfield 1 is the VLI and must be 2 bytes. The Token immediately follows and is a variable length between 15 and 19 bytes. After the Token is a back slash (“\”) delimiter and then a 2 byte ID. After the 2 byte ID is a back slash (“\”) delimiter and then the Token expiration date in the format YYMM.

The symbol “~” is used to represent a space.

Example: 23371234567890123\AB\1401

Bit 35

Definition

Important!

Format

Optionality

Track 2 Data

This is the data encoded in a Track 2 magnetic stripe read or the data stored in the Track 2 field on the chip of a Chip Card.

S/Es cannot keep track data from the Authorization Request. Do not send Track data to the Issuer.

ANS(39) max LLVAR

Network to Issuer: Not Applicable

Bit 38

Definition

Format

Optionality

Notes

Approval Code

A unique code assigned by the Card provisioning request authorizing institution to indicate the disposition for the provisioning request.

AN(6)

Network to Issuer: Mandatory

Network to Issuer:

If this advice message is for a Stand-In that was not approved or a refund, this bit must contain all 0s.

The first digit cannot be 0 (zero), excluding Contactless (Expresspay) Transactions.

If this is a Post Provisioning Notification, this bit must contain all 0s.

Bit 39

Definition

Format

Optionality

Notes

Action Code

The type of action done or the type of action to do for a Transaction.

N(3)

Network to Issuer: Mandatory

Valid Values:

- 000 - Approved

American Express Global Network Services
Network Specifications - Authorization
Payment Token Post Provisioning Notification (1120)
Bit 42 Card Acceptor Identification Code Message Layout

- 001 - Approve with ID
- 100 - Deny

Bit 42

Definition

Format

Optionality

Notes

Card Acceptor Identification Code

The Service Establishment (S/E) number given to the provisioning requestor. The number is ten (10) digits, followed by five (5) spaces for a total of 15 bytes.

AN(15)

Network to Issuer: Mandatory

The Network rejects this message if the S/E number fails the check digit routine.

Bit 45

Definition

Format

Optionality

Track 1 Data

This is the data encoded in a Track 1 magnetic stripe read or the data stored in the Track 1 field on the chip of a Chip Card.

ANS(78) max LLVAR

Network to Issuer: Not Applicable

Bit 48

Definition (Non-DPP Requests)

Optionality

Format

Stand-In Parameter Information for Non-DPP Requests

Acquirer to Network:

This field is present for advice messages when the Acquirer or S/E supplied a Stand-In decision.

Network to Issuer:

This field is present for advice messages when the Network supplied a Stand-In decision for a request that does not contain DPP data.

For more instruction regarding Stand-In, refer to the Exhibits Chapter of the *Business and Operational Policies* manual.

Network to Issuer: Not Applicable

AN(353) max LLLVAR

Bit 49

Definition

Format

Optionality

Notes

Currency Code, Transaction

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

N(3)

Network to Issuer: Mandatory

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition

Format

Optionality

Notes

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.

N(3)

Network to Issuer: Mandatory

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 56

Definition

Format

Optionality

Original Data Elements

Data from the Authorization Request (1100 or 1200) or POS/ATM Authorization Reversal Advice (1420) message.

N(37) max LLVAR

Network to Issuer: Not Applicable

Bit 60

Definition

Format

Market Specific Data

This is the provisioning-specific data unique to and supplied by the digital wallet provider.

ANS(416) max (includes length indicator) LLLVAR

American Express Global Network Services
Network Specifications - Authorization
 Payment Token Post Provisioning Notification (1120)
 Bit 60 Market Specific DataMessage Layout

Optionality
Notes

Network to Issuer: Mandatory
 This table shows the layout of Bit 60.

Subfield	Length	Use	Optionality N-I	Notes
	3 bytes	Variable Length Indicator		
	2 bytes	Market Specific Data Indicator		Required Value: MS
	3 bytes	Market ISO Country Code		Refer to the <i>Codes Reference Guide</i> for a list of Country Codes.
	2 bytes	Market Specific Data Version Indicator		Required Value: 05
	4 bytes	Bitmap Identifier		Each bit in this data element identifies the presence (value 1) or absence (value 0) of a subfield. Following the Bitmap, the layout consists of at least one (1) of the subfields below. Each bit position of the 32 bit/4-byte bitmap represents which market specific data are present. If a bit is "ON" in the bitmap, that corresponding subfield will be present.
Subfield 1	64 bytes	Conversation ID	M	Definition: A unique number given by the Network to identify a provision request throughout its life cycle. Field Orientation: Left justified, space-filled.
Subfield 2	64 bytes	Request ID	M	Definition: A unique message identifier.
Subfield 3	64 bytes	Reference Identifier	M	Definition: A unique Token identifier.
Subfield 4	4 bytes	Token Expiration Date	M	Format: YYMM
Subfield 5	4 bytes	Token Effective Date	M	Format: YYMM
Subfield 6	2 bytes	Token Status	M	Definition: The status of the Token. Valid Values: • 00 = Active • 01 = Suspended • 02 = Pending Approval • 06 = Canceled
Subfield 7	26 bytes	Cardmember Name	M	The following characters are used to separate the elements of a Cardmember's Name: • "/" (forward slash) = surname/first name separator • " " (space) = first name/middle name or first name/middle initial separator (as required) • "." (period) = name/title separator (as required) Examples: • Surname only: FROST/ • Surname and first name: FROST/CHARLES • Surname, first name, and middle initial: FROST/CHARLES F • Surname, first initial, and middle initial: FROST/C F • Surname, first name, middle name, and title: FROST/CHARLES FRANCIS.JR • The minimum encoded data shall be a single alpha character (as surname) and the surname separator.

American Express Global Network Services
Network Specifications - Authorization
Payment Token Post Provisioning Notification (1120)
Bit 60 Market Specific DataMessage Layout

Subfield	Length	Use	Optionality N-I	Notes
Subfield 8	64 bytes	Terms & Conditions Version	C	- Conditional for: - Digital Wallet Payment Token Provisioning Request - SRC Token Provisioning - Not applicable for a Card-on-file Payment Token Provisioning Request Definition: Terms and Conditions Version accepted by the Cardmember
Subfield 9	12 bytes	Terms & Conditions Time Stamp	C	- Conditional for: - Digital Wallet Payment Token Provisioning Request - SRC Token Provisioning - Not applicable for a Card-on-file Payment Token Provisioning Request Format: YYMMDDHHMMSS
Subfield 10	1 byte	Terms and Conditions Acceptance	C	- Conditional for: - Digital Wallet Payment Token Provisioning Request - SRC Token Provisioning - Not applicable for a Card-on-file Payment Token Provisioning Request Definition: Whether the Cardmember has accepted the terms and conditions. Valid Values: 1 = Yes 0 = No
Subfield 11	4 bytes	Event Reason	M	Definition: If Subfield 12 Results = 0 (Failure) then this subfield will contain the reason why the action was a failure. 4001 = Account Locked, Max OTP retry exceeded 4002 = Max OTP generation limit reached. Next OTP can be generated after 24 hours 4444 = Card requires additional authentication 5001 = Provision Failed, Card could not be activated. 5002 = Provision Failed, Unable to send script to Wallet Provider 5003 = Provision Failed, RMAC validation failed. 5004 = Provision Failed, Issuer rules 5316 = Card lockout due to CID validation 5322 = Token provisioning count exceeded 5555 = Provision Failed, No Token available
Subfield 12	1 byte	Results	M	Definition: Whether the action requested was performed successfully. Valid Values: 1 = Success 0 = Failure

American Express Global Network Services
Network Specifications - Authorization
Payment Token Post Provisioning Notification (1120)
Bit 61 National Use Data Message Layout

Subfield	Length	Use	Optionality N-I	Notes
Subfield 13	64 bytes	Device ID	C	- Conditional for a Digital Wallet Payment Token Provisioning Request - Not applicable for: - Card-on-file Payment Token Provisioning Request - SRC Token Provisioning Definition: Distinctive device identifier associated with a mobile device or similar handheld device. For example: International Mobile Equipment Identifier (MEID), International Mobile Security Identifier, or Serial ID (tablet devices).
Subfield 14	15 bytes	Device Phone Number	C	- Conditional for a Digital Wallet Payment Token Provisioning Request - Not applicable for: - Card-on-file Payment Token Provisioning Request - SRC Token Provisioning Field Orientation: Right justified, space-filled.
Subfield 15	11 bytes	Token Requestor ID	M	Definition: An 11-byte code pairing Token Requestor with Token Domain. Positions 1-3 - Token Service Provider Code, unique to each Token Service Provider Positions 4-11 - Assigned by the Token Service Provider for each requesting entity and Token Domain
Subfield 16	2 bytes	Device Type	C	- Conditional for a Digital Wallet Payment Token Provisioning Request - Not applicable for: - Card-on-file Payment Token Provisioning Request - SRC Token Provisioning Definition: A 2 byte code that represents the type of device that requested a Token to be provisioned. Field Orientation: Left justified, space-filled. Valid Values: 01 = Phone 02 = Tablet 03 = Watch 04 = Card 05 = Wearable 06 = Laptop 99= Other

Bit 61

National Use Data

Definition This is an ISO field that supplies Cardmember authentication data during Authorization.

Format 4 bytes min, 103 bytes max LLLVAR

Optionality **Network to Issuer:** Not Applicable

Bit 62

Fraud Indicators

Definition The Fraud Indicator field is used to provide fraud mitigation indicators for a variety of situations.

Format AN(5) min, AN(60) max LLLVAR

Optionality **Network to Issuer:** Conditional

Payment Token Post Provisioning Notification Response (1130)

Description

The [Payment Token Post Provisioning Notification Response \(1130\)](#) is sent in response to a [Payment Token Post Provisioning Notification \(1120\)](#) message on [page 6-22](#).

The Network does not send this message from the Issuer to the Token Requestor. The Network responds to the Token Requestor, sends the advice to the Issuer, and accepts the [Payment Token Post Provisioning Notification Response \(1130\)](#) response from the Issuer.

Length of Record

Variable

Message Layout

This is the message layout for the [Payment Token Post Provisioning Notification Response \(1130\)](#). Refer to the [Programming Notes for Payment Token Post Provisioning Notification Response \(1130\)](#) on [page 6-32](#) for more instruction.

Payment Token Post Provisioning Notification Response (1130) Message Layout

Bit Position	Format	Optionality	Field Name
		I to N	
N/A	N(4)	M	Message Type Identifier
N/A	8 bytes, 64 bits for each bitmap Binary (Hex)	M	Bitmap Primary
2	N(21) max LLVAR	ME	Primary Account Number (PAN)
3	N(6)	ME	Processing Code
4	N(12)	ME	Amount, Transaction
5	N(12)	ME	Amount, Reconciliation
11	N(6)	ME	Systems Trace Audit Number
12	N(12)	ME	Date and Time, Local Transaction
23	N(3)	CE	PAN Sequence Number
31	N(40) max LLVAR	CE	Acquirer Reference Data, Transaction Identifier (TID)
32	N(13) max LLVAR	CE	Acquiring Institution Identification (AIN) Code
34	AN(30) max LLVAR (includes length indicator)	CE	Primary Account Number, Extended
39	N(3)	M	Action Code
42	AN(15)	ME	Card Acceptor Identification Code
49	N(3)	ME	Currency Code, Transaction
50	N(3)	ME	Currency Code, Reconciliation

Payment Token Post Provisioning Notification Response (1130) Message Layout (continued)

Bit Position	Format	Optionality	Field Name
		I to N	
60	AN(210) max LLVAR	N/A	Market Specific Data

Programming Notes for Payment Token Post Provisioning Notification Response (1130)

Bit N/A

Definition
Format
Optionality
Notes

Message Type Identifier

This code identifies the type of message sent by the forwarding institution.
 N(4)
Issuer to Network: Mandatory
Required Value: 1130
 Refer to the Message Type Identifier section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit N/A

Definition
Format
Optionality

Bitmap Primary

Each bit in this data element identifies the presence (value 1) or absence (value 0) of a field.
 8 bytes, 64 bits for each bitmap, Binary (Hex)
Issuer to Network: Mandatory

Bit 2

Definition
Format
Optionality

Primary Account Number (PAN)

A series of digits encoded into the magnetic stripe, stored on the chip of a Chip Card, and printed on the Card; used to identify the Card Issuer and the Cardmember (CM) account or relationship. Valid American Express Card account numbers begin with 34 or 37 and end in a valid check digit.
 N(21) max LLVAR
Issuer to Network: Mandatory Echo

Bit 3

Definition
Format
Optionality
Notes

Processing Code

This is the service being requested.
 N(6)
Issuer to Network: Mandatory Echo
 Refer to the Processing Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 4

Definition
Format
Optionality

Amount, Transaction

The sale or credit amount of the provisioning request.
 N(12)
Issuer to Network: Mandatory Echo

Bit 5

Definition
Format
Optionality

Amount, Reconciliation

This is the total amount of the provisioning request after the Network changes the Amount, Transaction (Bit 4) to the Issuer's currency (Bit 50).
 N(12)
Issuer to Network: Mandatory Echo

Bit 11

Definition
Format

Systems Trace Audit Number

This is a unique (where applicable) trace number given to each Transaction or provisioning request.
 N(6)

American Express Global Network Services
Network Specifications - Authorization
 Payment Token Post Provisioning Notification Response (1130)
 Bit 12 Date and Time, Local TransactionMessage Layout

Optionality **Issuer to Network:** Mandatory Echo

Bit 12 **Date and Time, Local Transaction**

Definition This is the current date and time supplied by the Network at the time of the provisioning request.

Format N(12)

Optionality **Issuer to Network:** Mandatory Echo

Bit 23 **PAN Sequence Number**

Definition A number distinguishing between separate Cards with the same Primary Account Number.

Format N(3)

Optionality **Issuer to Network:** Conditional Echo. If this is a digital wallet Transaction, this bit will be sent to the Issuer.

Bit 31 **Acquirer Reference Data, Transaction Identifier (TID)**

Definition This unique number is given by the Network and follows a Transaction during its life cycle.

Format N(40) max LLVAR

Optionality **Issuer to Network:** Conditional Echo. If a certified Issuer receives the TID from the Network, the Issuer must echo the TID.

Notes Refer to [2 Authorization Processing on page 2-1](#) for information about the TID.

Bit 32 **Acquiring Institution Identification (AIN) Code**

Definition This is a unique identification code given to an Acquirer by the Network.

Format N(13) max LLVAR

Optionality **Issuer to Network:** Conditional Echo

Bit 34 **Primary Account Number, Extended**

Definitions This field is a multipurpose field which provides the Primary Account Number or associated data.

Format AN(30) Max (includes length indicator) LLVAR

Optionality **Issuer to Network:** Conditional Echo. For Payment Token Provisioning Requests, Bit 34 may be sent by the Network to the Issuer based on provisioning status.

Notes This field is populated by the Network with the Token, the Wallet Provider ID, and the Token Expiration Date.

Subfield	Format
1	Variable Length Indicator (VLI)
2	Token
3	Wallet Provider ID (ID)
4	Token Expiration Date

Data Example:

Subfield 1 is the VLI and must be 2 bytes. The Token immediately follows and is a variable length between 15 and 19 bytes. After the Token is a back slash (“\”) delimiter and then a 2 byte ID. After the 2 byte ID is a back slash (“\”) delimiter and then the Token expiration date in the format YYYY.

The symbol “~” is used to represent a space.

Example: 23371234567890123\AB\1401

Bit 39 **Action Code**

Definition The type of action done or the type of action to do for a Transaction.

Format N(3)

American Express Global Network Services
Network Specifications - Authorization
Payment Token Post Provisioning Notification Response (1130)
Bit 42 Card Acceptor Identification CodeMessage Layout

Optionality
Notes

Issuer to Network: Mandatory

Issuer to Network: Invalid values from the Issuer to Network message cause the GNSAS to queue the same message again.

Refer to the Action Codes section of the *Codes Reference Guide* for a list of codes and descriptions.

Bit 42

Definition

Card Acceptor Identification Code

The Service Establishment (S/E) number given to the provisioning requestor. The number is ten (10) digits, followed by five (5) spaces for a total of 15 bytes.

Format

AN(15)

Optionality

Issuer to Network: Mandatory Echo

Bit 49

Definition

Currency Code, Transaction

This is the numeric code for the currency used in the Transaction or Payment Token Provisioning Request.

Format

N(3)

Optionality

Issuer to Network: Mandatory Echo

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 50

Definition

Currency Code, Reconciliation

This is the numeric code of the currency defined by the Issuer.

Format

N(3)

Optionality

Issuer to Network: Mandatory Echo

Notes

Refer to the Currency Codes section of the *Codes Reference Guide* for a complete list of codes and descriptions.

Bit 60

Definition

Market Specific Data

This is the provisioning-specific data unique to and supplied by the digital wallet provider.

Format

AN(210) max, LLLVAR

Optionality

Issuer to Network: Not Applicable - in response to a [Payment Token Post Provisioning Notification \(1120\)](#) message.

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